PViMS

User Manual

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SIAPS
Systems for Improved Access to Pharmaceuticals and Services

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About SIAPS

The goal of the Systems for Improved Access to Pharmaceuticals and Services (SIAPS) Program is to assure the availability of quality pharmaceutical products and effective pharmaceutical services to achieve desired health outcomes. Toward this end, the SIAPS result areas include improving governance, building capacity for pharmaceutical management and services, addressing information needed for decision-making in the pharmaceutical sector, strengthening financing strategies and mechanisms to improve access to medicines, and increasing quality pharmaceutical services.

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ACRONYMS AND ABBREVIATIONS

AE	Adverse Event
DR-TB	Drug Resistant Tuberculosis
MedDRA	Medical Dictionary for Regulatory Activities
NTP	National Tuberculosis Program
PViMS	Pharmacovigilance Monitoring System
TB	Tuberculosis
WHO	World Health Organization

1 Introduction

The Challenge

Active surveillance for monitoring the safety and effectiveness of medical products is increasingly recognized as an approach that complements passive (or spontaneous) reporting, which is the most common method used by countries' pharmacovigilance systems. Active surveillance is particularly important to support the introduction of new medicines in low- and middle- income countries whose regulatory systems are developing and need support. In resource-limited settings, active surveillance can help determine the real-life frequency, risk factors, and impact of clinically significant adverse drug events on treatment outcomes in the population. However, many of these countries lack the resources and capacity to implement active surveillance activities. One major resource constraint is the lack of a data collection and analysis tool to support active safety surveillance.

The Solution

The Pharmacovigilance Monitoring System (PViMS) is a web-based application used by clinicians, regulatory bodies, and implementing partners to monitor the safety and effectiveness of medicines.

The application can improve overall clinical documentation. It is designed to ensure completion of required fields, including clinical stage, concomitant medications, test results, co-morbid conditions, and treatment regimen initiation date to improve clinical documentation at participating sites. It provides for the use of common terms, checklists, and adoption of standard terminologies. Users enter the common terms or choose from pre-coded causality assessment lists and scales such as the Medical Dictionary for Regulatory Activities (MedDRA), the National Cancer Institute Common Terminology Criteria for Adverse Events, WHO, and Naranjo; or users can develop a local dictionary using standard terms.

PViMS provides for detailed description of adverse event (AE) outcomes and for generating safety signals. Description of AEs, severity and seriousness, laboratory values, AE outcomes, and AE management can be used to generate signals of increased incidence to inform for action or further evaluation.

It is interoperable with third-party clinical systems and statistical tools. PViMS can import and export data from third-party electronic medical record or dispensing tools in XML, CSV, and Excel. Analyses can be cross-checked by analyzing data with previously validated statistical tools. Additionally, PViMS has the ability to export case safety data in E2B interface, and is health level-7 (HL7) compliant.

For information about the MedDRA dictionary contact your system administrator.

1.1 Using the Manual

This document discusses functional requirements for the electronic pharmacovigilance system (PViMS) framework.

1.2 Purpose of the Document

A user manual defines the software program's functionalities. The document aims to ensure that any reader or user gains complete system knowledge of the product. The document should also function as a reference guide and training manual for new system users.

This User Manual will outline the system functionality that is currently included in the PViMS application framework and will be updated throughout the various incremental development iterations and any system upgrades.

1.3 Audience

The intended audiences for this document are identified as follows:

- All project stakeholders
- System super users
- General system users
- New system users

2 PViMS Structure

PViMS consists of five portals:

- Clinical
- Analytical
- Reporting
- Publishing
- Administration

The **Clinical** portal is the centralized hub for all patient and adverse drug event data collection, patient information and standardized patient care.

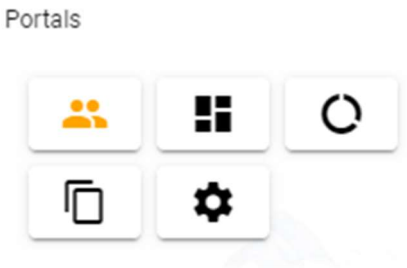
The **Analytical** portal is the centralized hub for causative drug assessment using traditional internationally recognized rating scales, standardized terminology and risk detection.

The **Reporting** portal allows the user to generate and print reports.

The **Publishing** portal is a centralized hub for report and document publication and presentation.

The **Administration** portal also allows the system administrator to manage the system to include, remove, and change users and manage the system structure. For information on the Administration portal, please see the *PViMS Administrator Manual.*

You use the icon bar to select the portal in which you want to work:



3 Using the System

PViMS is a web-based system, so you will need a web browser to run this application. Several Internet navigators (browsers) are available, and each one offers specific characteristics and resources. To have the system working properly, you must enable Java-script in your browser. If it's not enabled, please contact your system administrator.

PViMS has been tested using Google chrome and it is therefore recommended that Chrome be used as the preferred browser of choice when accessing PViMS.

3.1 Launching the Browser

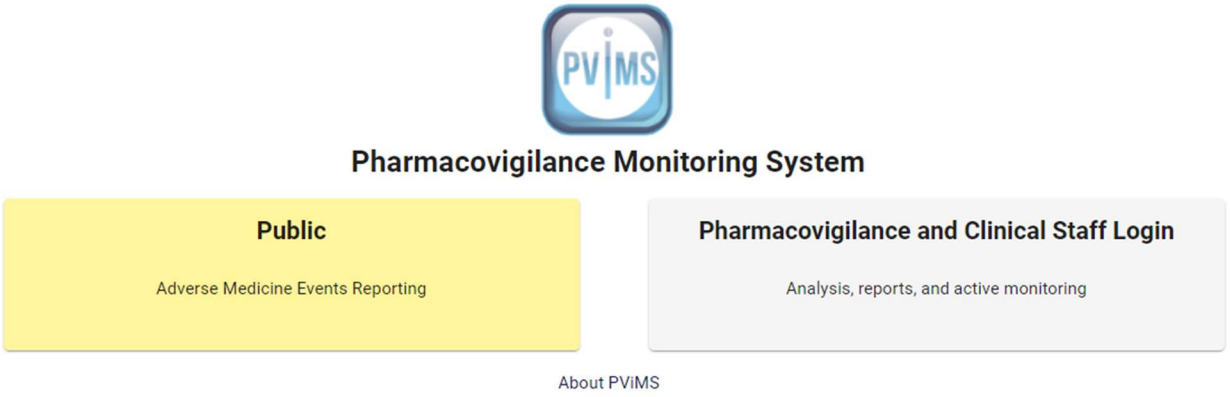
To start the application, open your browser and enter the system URL. If you don't know the system URL, contact the NTP representatives for instructions.

Note 1. What is a URL?

URL is the abbreviation for *uniform resource locator*. It's a global address of documents and other resources on the World Wide Web. The URL of PViMS depends on where it was installed. SIAPS maintains a demonstration version of the system at the URL http://dc-cpm-pvimsdemo.msh.org
Check with your technical support for the right URL of the system in use.

3.2 Logging in to PViMS

When you enter the correct URL, the system shows you the login page. The login page is used to authenticate the user in the system.

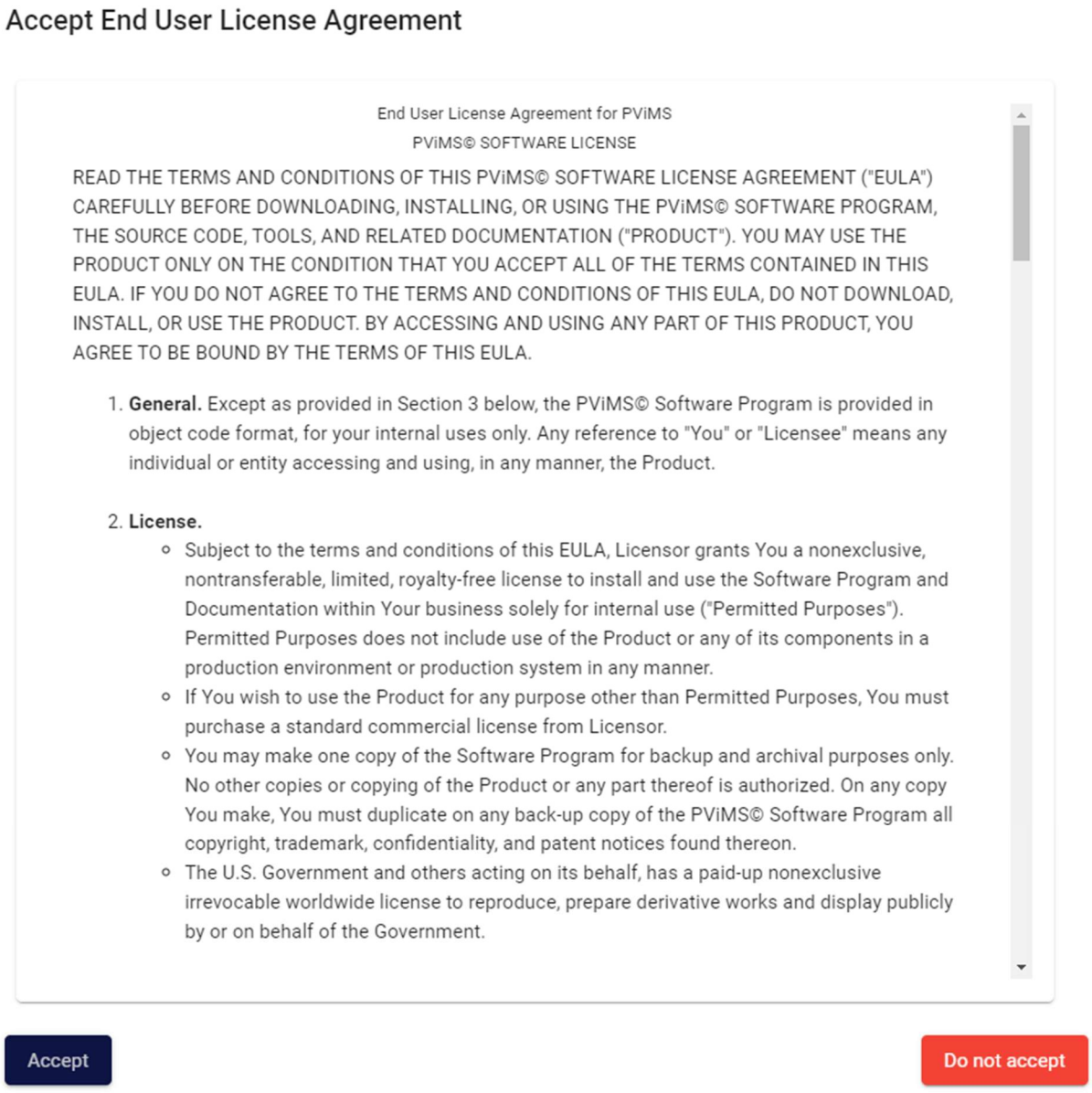


To access the system, you'll need a user login and password. If you don't have one, please contact your system administrator.

- Enter your assigned username and password.
- Click the log in button.

3.3 End User License Agreement

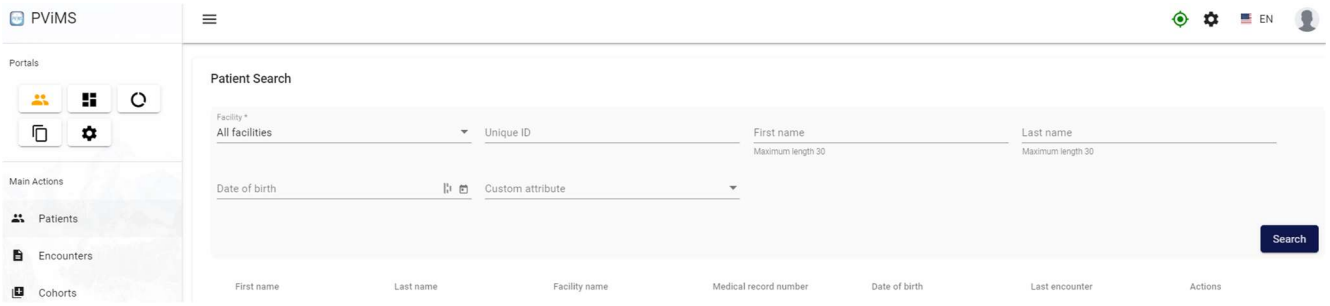
The first time you log on, you will be asked to read the terms and conditions of the PViMS software license agreement. You will not be able to log into PViMS unless the EULA is accepted.

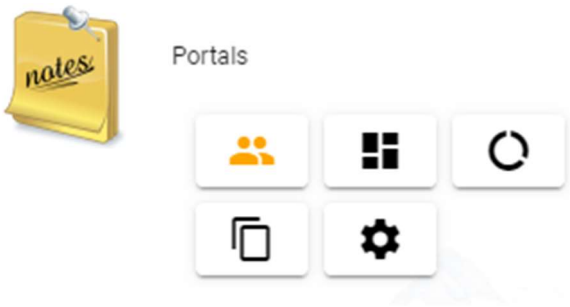


Click the **Accept** button to confirm the acceptance of the EULA and continue to access the system or click the **Do not accept** button to return to the login page.

3.4 The Home Page

After you complete the login page, the system will direct you to the system's **Home/Patient Search** page.



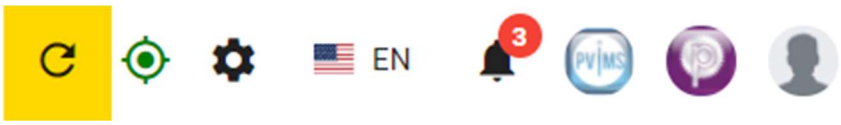


Use the portal icons listed above to navigate among the five different portals (**Clinical**, **Analytical**, **Reporting**, **Information,** and **Administration).**

3.5 System Toolbar

The system toolbar provides the user with the following abilities:

3.5.1 Refresh meta data



This icon allows the user to initiate a refresh of meta data. Please see the administrator manual for additional information.

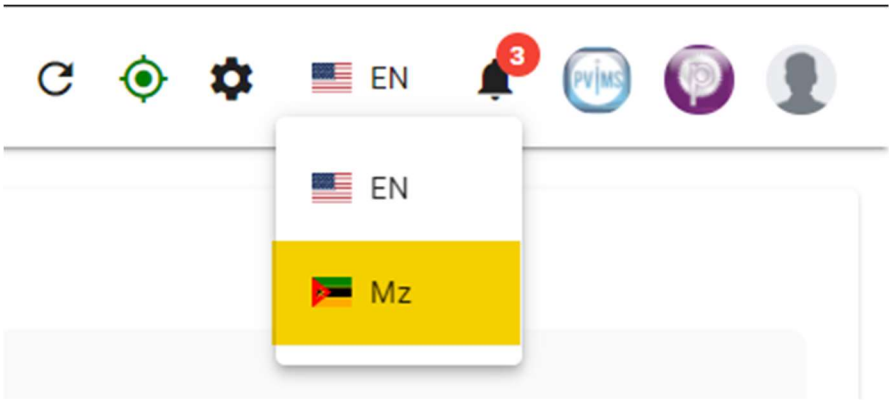
3.5.2 Online and offline status

This icon allows the user to check the offline status of PViMS. If the icon is green, PViMS is connected to the internet and is operational. If the icon is red, PViMS is not connected to the internet and can only function in an offline status when capturing forms. Clicking on the icon refreshes this status.

3.5.3 Change the display theme

This icon allows the user to change the colour theme for the site. The user is able to set header, sidebar, material themes and footer colours,

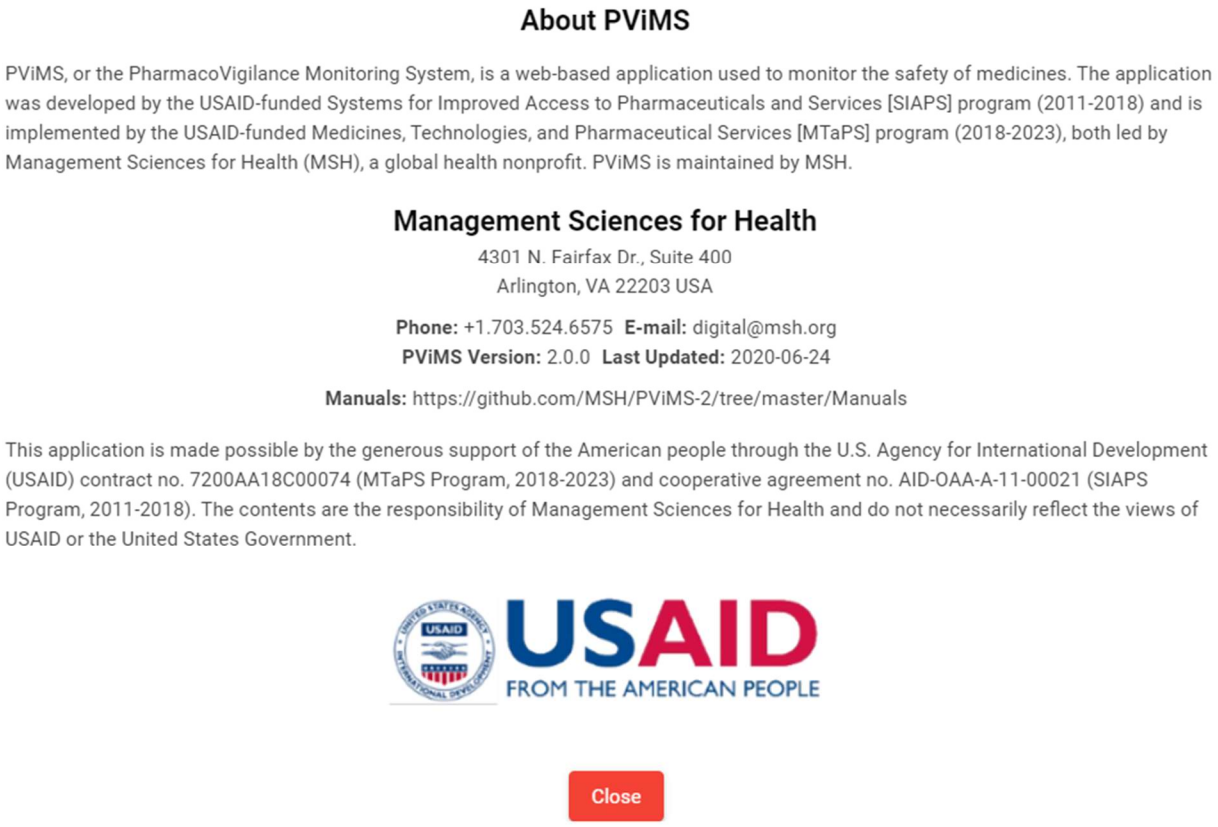
3.5.4 **Change the default language**



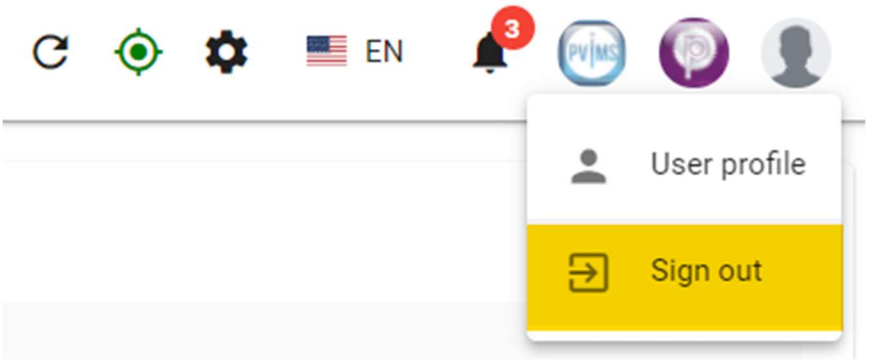
To change the default language that all text is displayed in, click on the flag in the top toolbar of PViMS. Once the flag has been selected, all system text will be translated to the language specified.

3.5.5 About PViMS

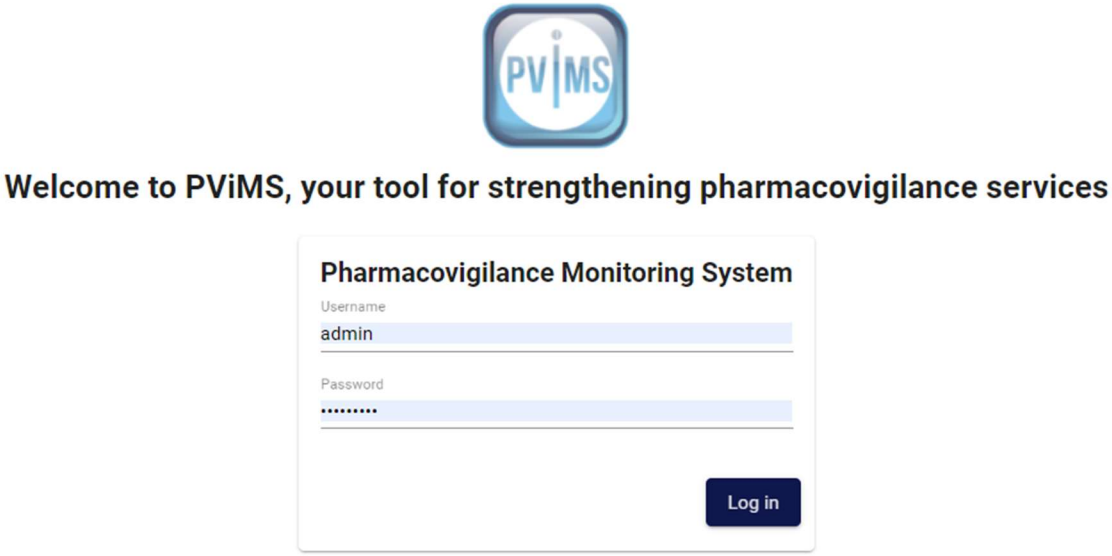
By clicking on this icon, the user will be about to view additional PViMS information.



3.5.6 **Logging out of PViMS**



To log out of PViMS, click on the **Sign Out** menu and the user will be redirected to the login screen.



4 Clinical Portal

At the **Home/Patient Search** page, you will be presented with the following options:

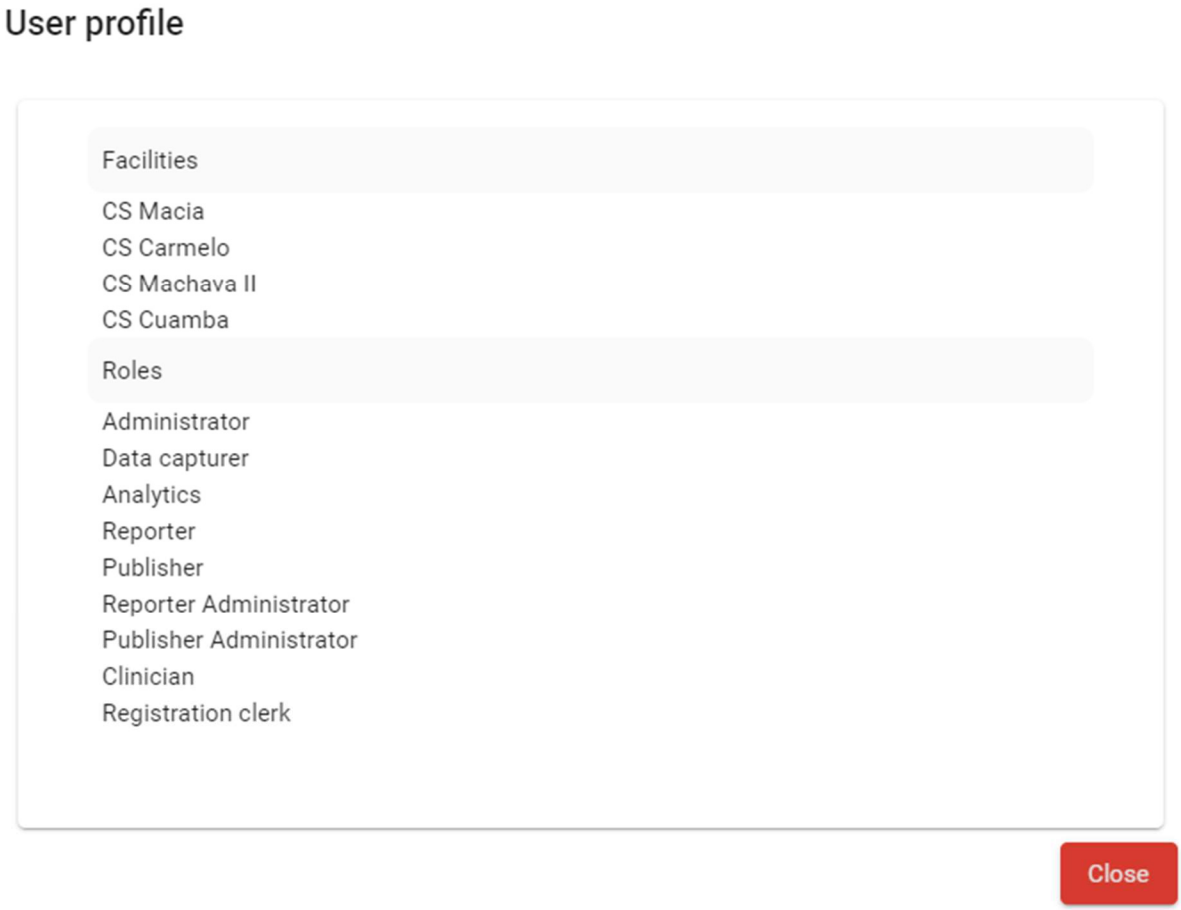
- Patients
- Encounters
- Cohorts
- Feedback
- Appointments
- Forms
- Synchronise

The clinical portal is the centralized hub for all patient and adverse drug event data collection, patient information and standardized patient care.

Note: the following roles have access to the clinical portal:

Administrator. The administrator has FULL permissions to the clinical portal.
Registration Clerk. The registration clerk is able to add and amend a patient record and create appointments.
Data Capturer. A data capturer is able to add and amend a patient record and add and amend an encounter record.
Clinician. A clinician is able to add and amend a patient record and add and amend an encounter record.

Click on the user profile menu option which appears when clicking on the profile icon to view roles and facilities you currently have access to.

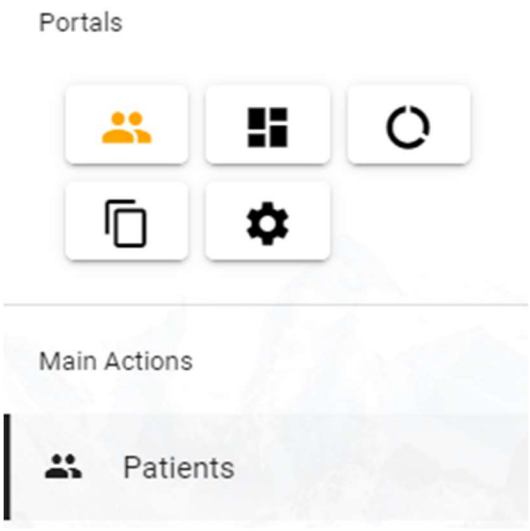


4.1 Patients

In the **Patients** function you can **Search** for patients, **Add** a new patient, and **Edit** patient information.

4.1.1 Search for Patients

The **Patient Search** function can be accessed through the **Patients** menu.

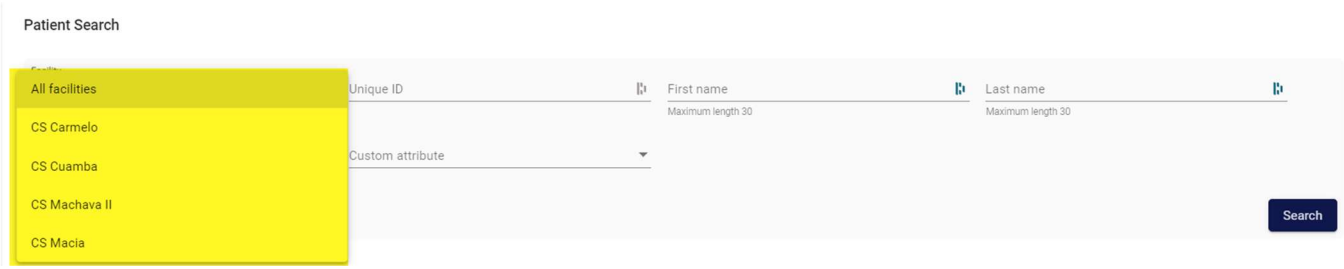


There are five ways to search for a patient. You can search by:

- Facility
- Patient Unique ID
- First Name and Last Name
- Date of Birth
- Custom Attribute

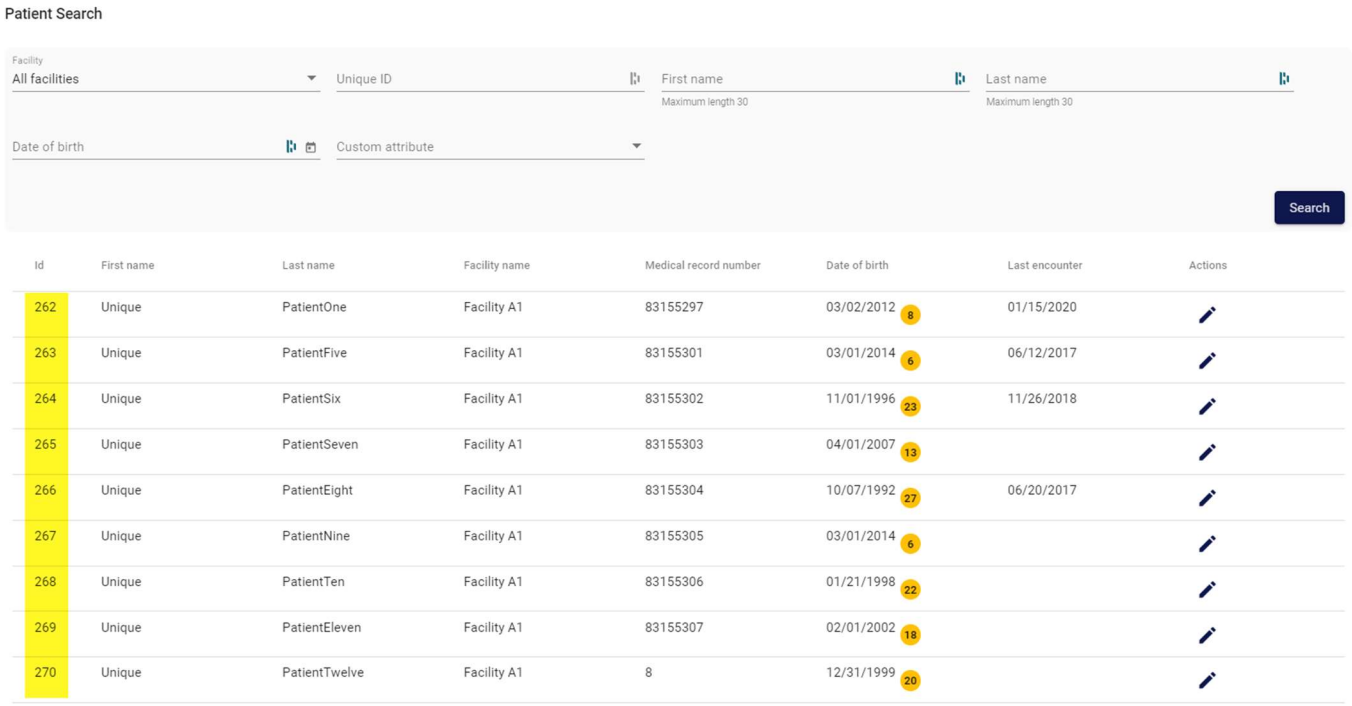
4.1.1.1 Search by Facility

- Click the **arrow** in the **Facility** field to select from the facility drop down list
- Select the facility you would like to search against specifically or select **All Facilities** if you would like to search against all facilities
- Click the **Search** button



You will only be able to search against facilities that you have been assigned access to. Please speak to your system administrator if you are unable to search against the necessary facility

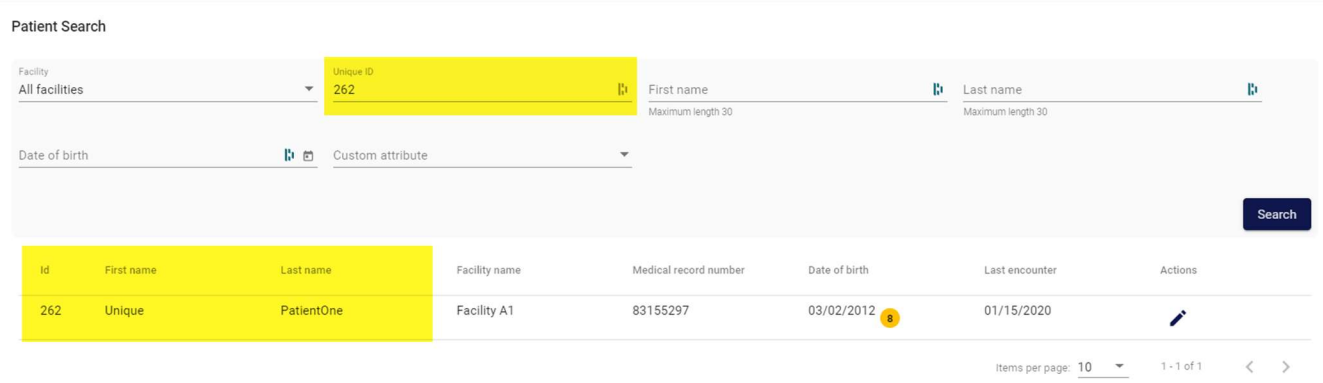
The system will display a list of patients according to the filter selected, please note the Unique ID of the patient in column 1.



4.1.1.2 Search by Patient Unique ID

Each patient is allocated a unique system id when they are created in the system. It is possible to search for this patient using this id.

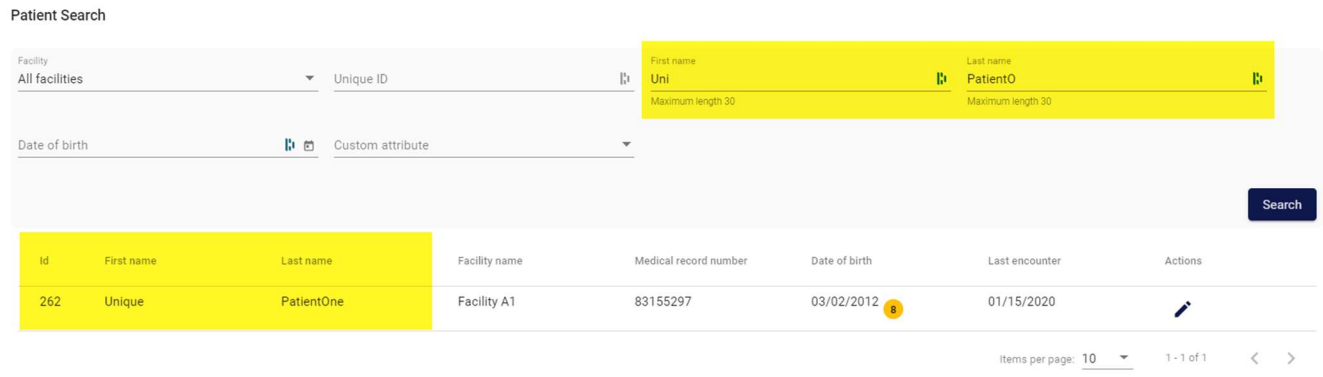
If you know the patient's unique ID, enter it in the **Unique ID** field and click **Search**.



4.1.1.3 Search by First Name or Last Name

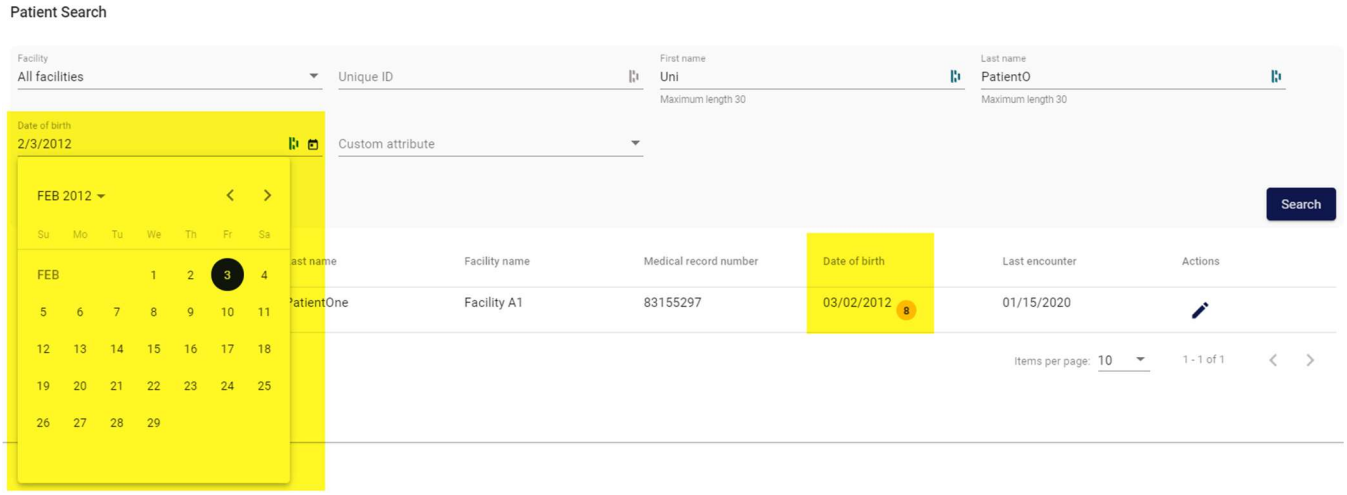
You can also search by the patient's **First name** or **Last Name**. Enter the name(s) in one or both of these areas and click the **Search** button.

It is possible to do a partial search by entering any three letters of the **First** or **Last names.** The system will return all matching records if a partial search is executed.



4.1.1.4 Search by Date of Birth

You can also search by the patient's **Date of Birth**. Select the date of birth and click the **Search** button.

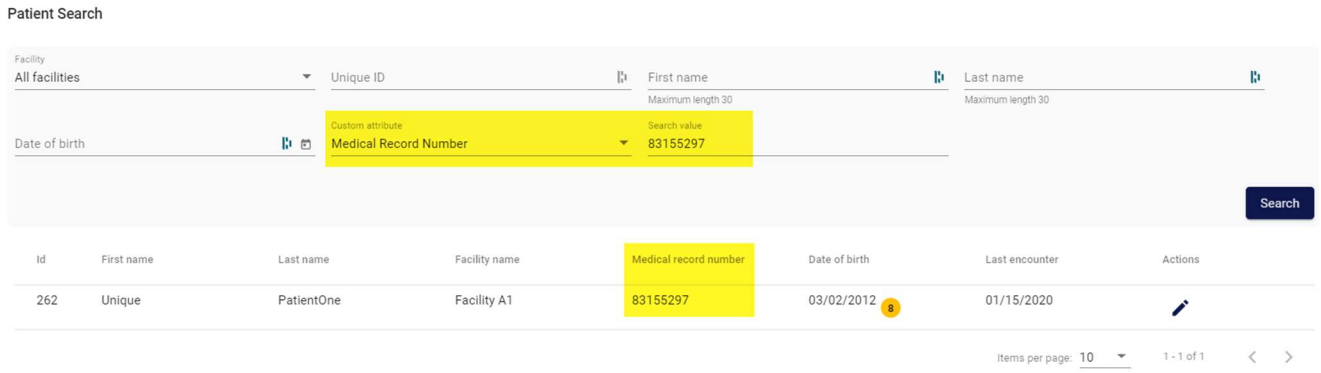


4.1.1.5 Search by Custom Attribute

The final search filter available is the ability to search by a **Custom Attribute**.

Custom attributes can be activated for filtering by the system administrator. Please consult your administrator if you would like to activate the ability to filter by a specific attribute.

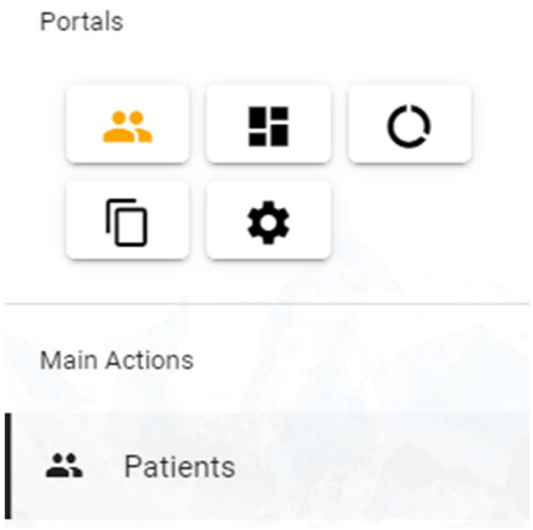
- Select the custom attribute variable that you would like to search against (e.g., Medical Record Number)
- Enter the search value you would like to filter against and click the **Search** button.



4.1.2 Return to the Patient Search Page

You can return to the **Patient Search** page from any place in the system by using the **Menu Bar**.

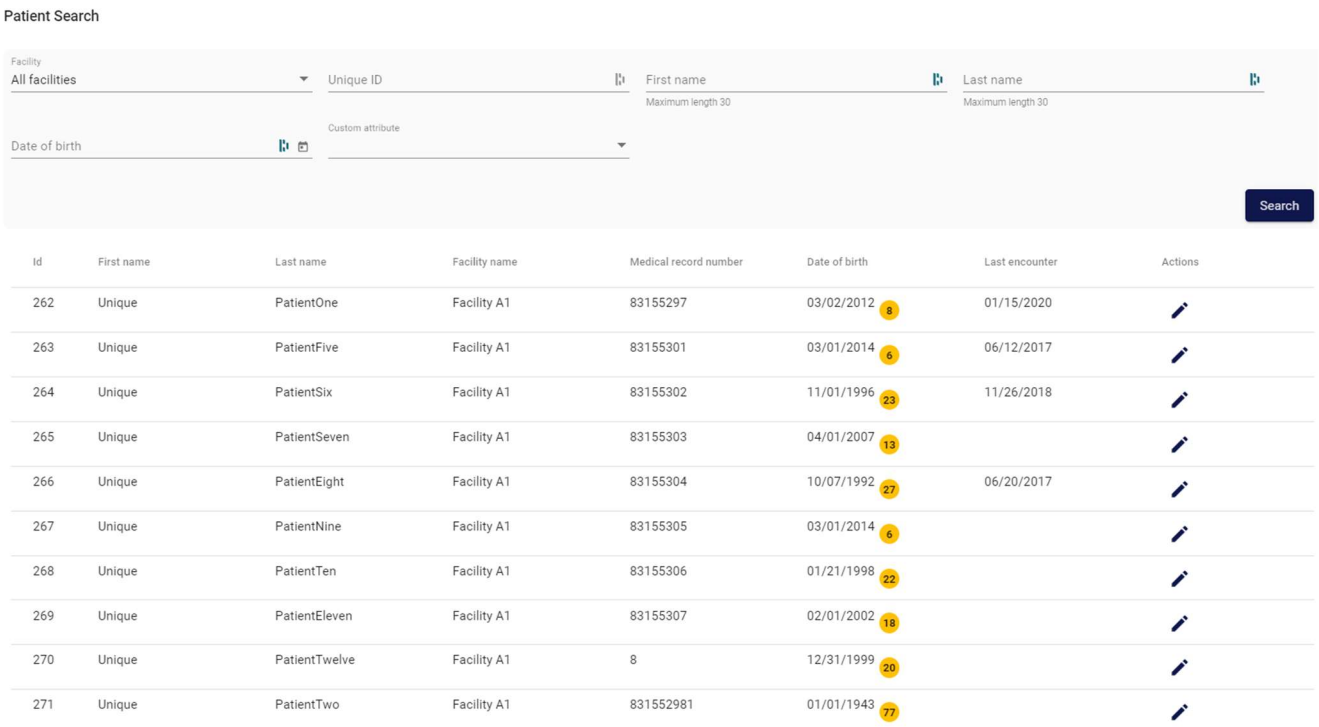
4.1.2.1 Menu Bar



The menu bar is located on the left-hand side of the page.

4.1.3 View an Existing Patient

After selecting the appropriate search filter and you have clicked the **Search** button, the system will present all matches as displayed in a table.

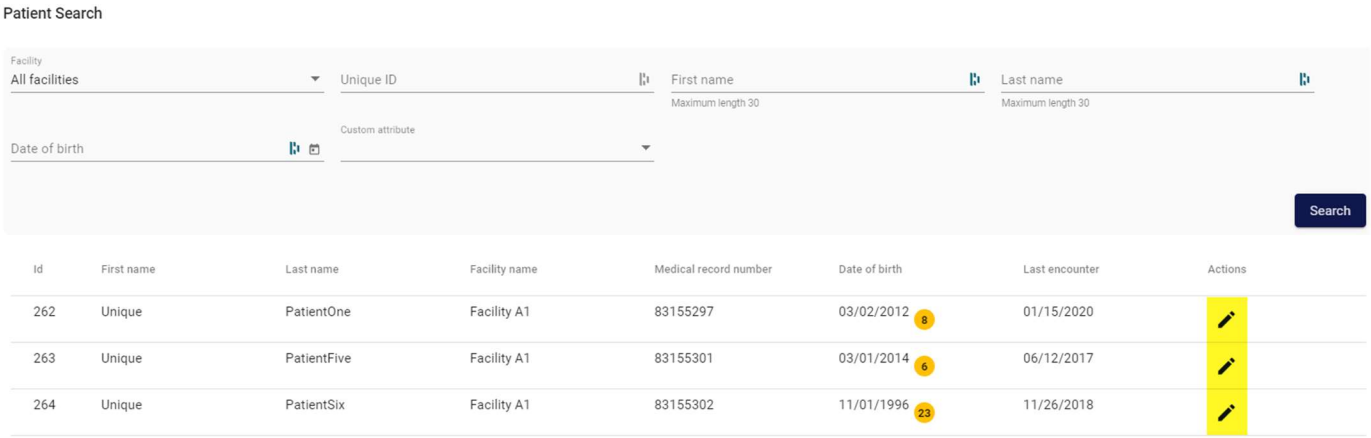


The columns in the table are described below:

ID	Unique identification number assigned by the system
First Name	Patient's first name as captured in the system
Last Name	Patient's last name as captured in the system
Facility name	Facility associated with the patient
Medical record number	ID number associated with the patient
Date of Birth (Age)	Patients date of Birth and Age indicator •
Last Encounter	Last encounter date, the date the patient last visited the

	facility
Action	Ability to view the patient's information

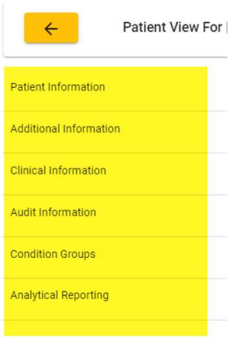
- To view a patient entered in the system, locate the patient in the patient table.
- Click the **View Patient** icon in the **Actions** column.



- The system will then open the **Patient View** page and allow you to view the demographics for this patient.

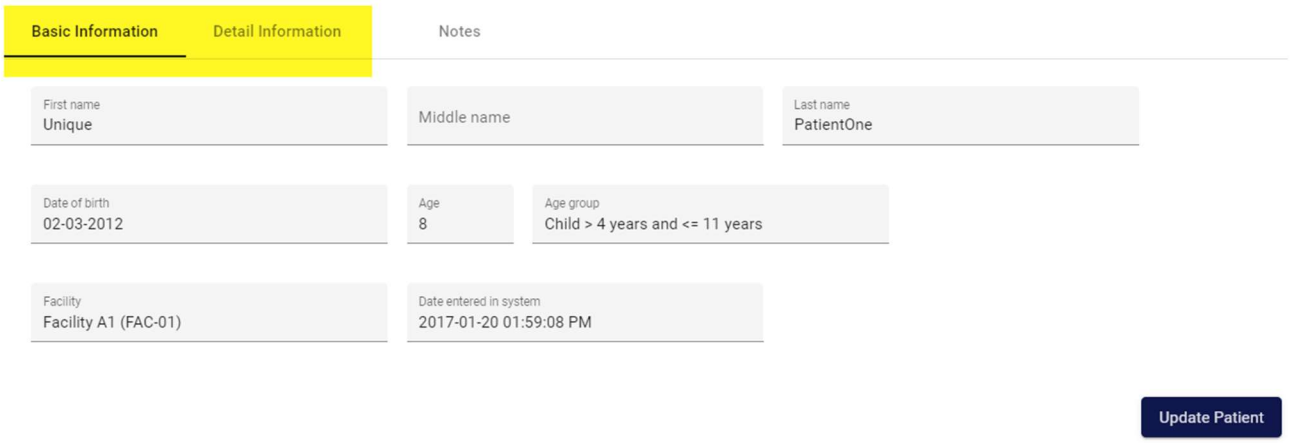
Patient View has been segregated into the following core sections:

- Patient Information
- Additional Information
- Clinical Information
- Audit Information
- Condition Groups
- Analytical Reporting



4.1.3.1 Patient Information – Basic and Detail Information

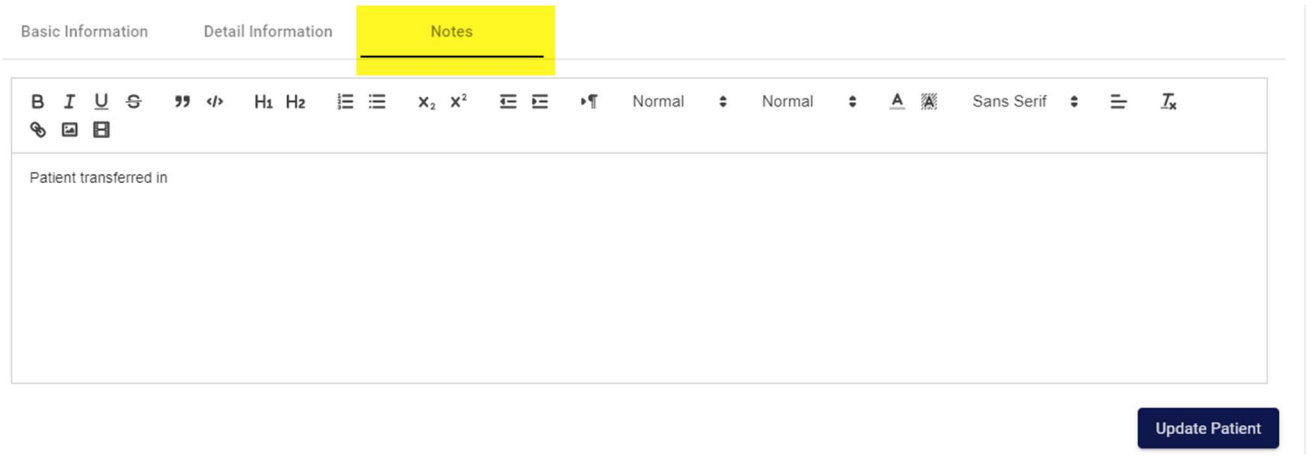
The **Patient Information** tab is divided into **Basic Information**, **Detail Information** and N



Patient demographic information will by enlarge remain rather static but should be verified and updated on a visit by visit basis to reflect up to date information. Various attributes defined as part of detail information can be used as risk factors when identifying signals in the analytical portal and therefore remain critical through the clinical portal data collection process.

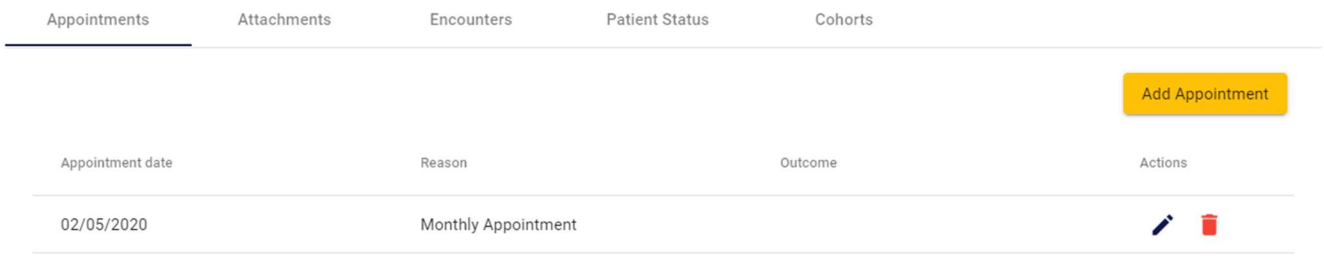
4.1.3.2 Patient Information – Notes

The **Notes** tab is where you can note generic information relating to the patient at the discretion of the clinician.



4.1.3.3 Additional information - Appointments

The **appointments** tab can be used to track upcoming appointments for the patient. This function can be leveraged to track additional clinical or demographic information if sufficient information was not collected in any of the patient's previous encounters.



The columns in the appointments table are described below:

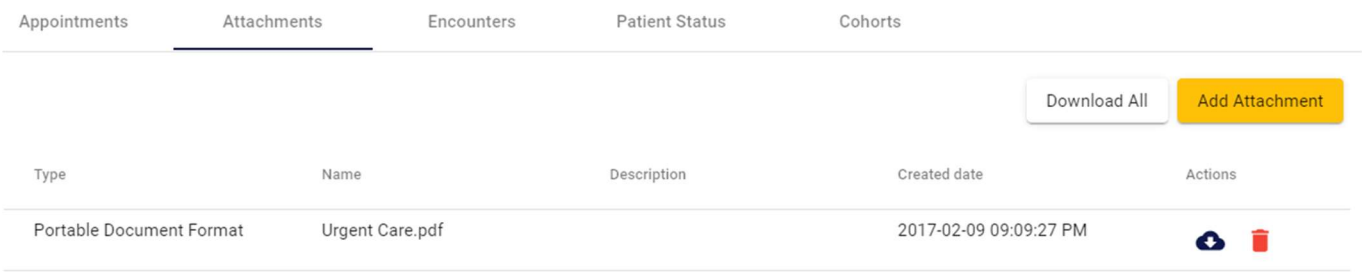
Appointment Date	Date of the appointment
Reason	Reason for the appointment
Outcome	Did the patient arrive for their appointment? Did the patient miss their appointment?
Actions	Ability to edit the appointment information or to delete the appointment from the calendar

4.1.3.4 Additional Information - Attachments

The **attachments** tab can be used to store physical file attachments for the associated patient. The number of attachments and size of attachments are configurable parameters within PViMS and can be adjusted based on your site's requirements.

The following file types are supported within PViMS:

- MS Word 2003-2007 Document
- MS Excel 2003-2007 Document
- MS Word Document
- MS Excel Document
- Portable Document Format
- Image | JPEG
- Image | PNG
- Image | BMP
- XML Document

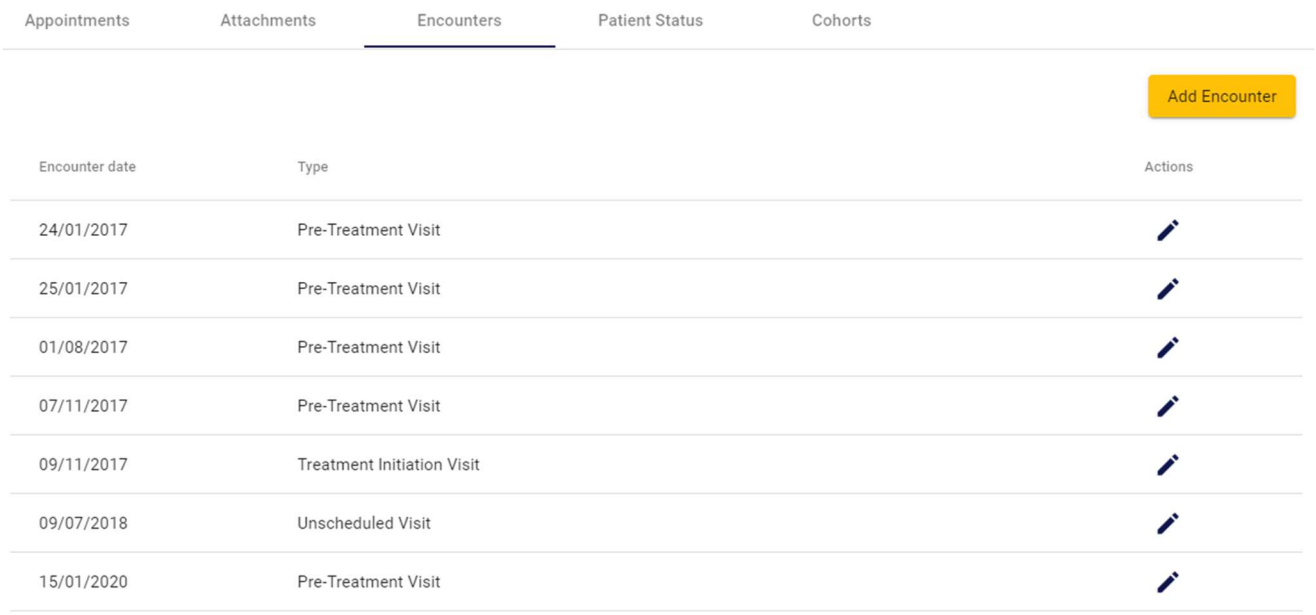


The columns in the attachments table are described below:

Type	Describes the file type (e.g., PDF, Word, Excel)
Name	Name of the file
Description	Description of the file entered
Created by	Name of the person who uploaded the file, and date of upload
Action	Ability to download or delete the file

4.1.3.5 Additional Information - Encounters

The **encounters** tab can be used to track all facility visits by the patient. Encounters effectively form part of the holistic longitudinal record for the patient and store contextual clinical data collected during that visit.

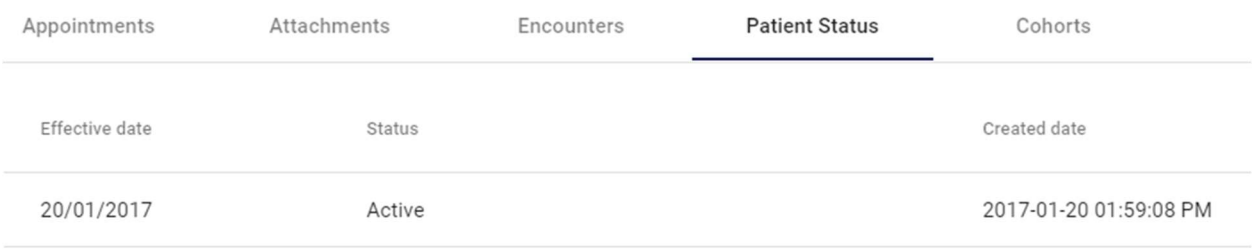


The columns in the encounters table are described below:

Date	Date of the encounter
Type	Type of encounter when the encounter was created (e.g., Pre-treatment Visit, Treatment initiation Visit, Unscheduled Visit)
Action	Ability to View an encounter

4.1.3.6 Additional Information - Patient Status

The **patient status** tab can be used to track if the patient is currently active or if the patient is now deceased. Status change is driven by an effective date for efficient accurate analysis.

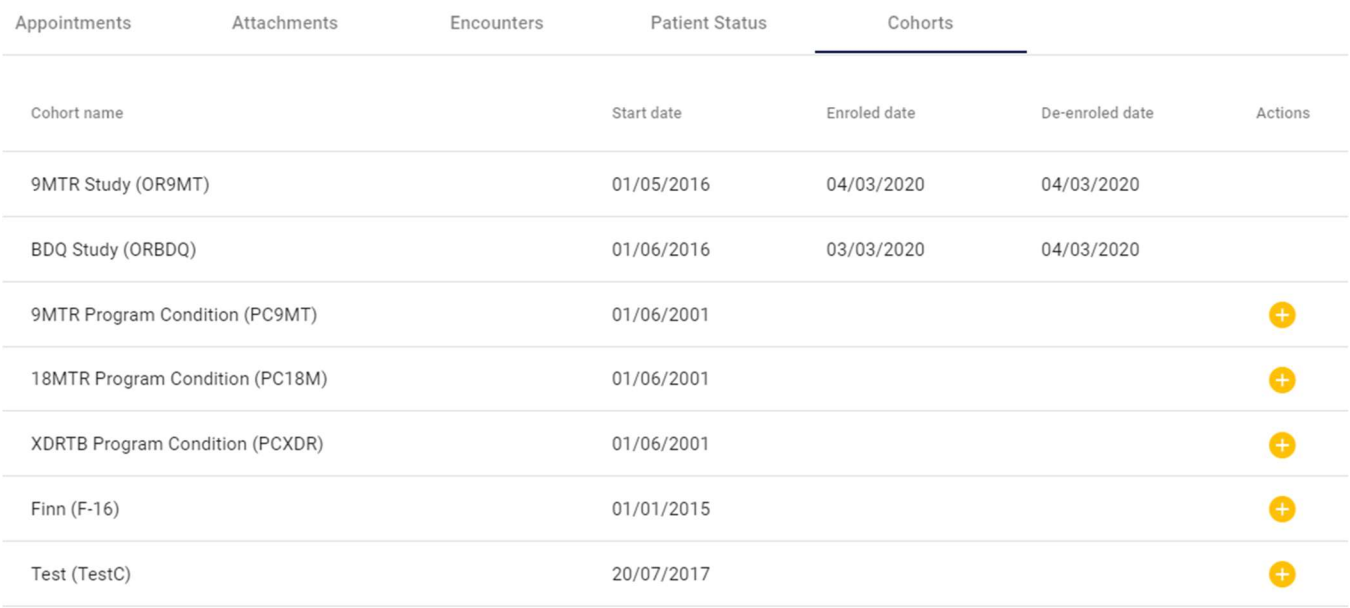


The columns in the status table are described below:

Effective Date	The effective date of the status change
Status	To indicate if the patient is active or inactive
Created	Name of the person who effected the status change, and date of this status change

4.1.3.7 Additional Information - Cohorts

The **cohort** tab can be used to track what cohorts a patient has been enrolled in. Analysis can be subdivided by cohort to target signal detection effectively.



The columns in the **Cohorts** table are described below:

Cohort	Name of cohort
Cohort Start	Date the cohort started
Enrolled Date	Date the patient was enrolled in the cohort
De-enrolled Date	Date the patient was de-enrolled from the cohort
Action	Ability to enroll, de-enroll, or remove patient from a cohort

4.1.3.8 Clinical information – Patient Conditions

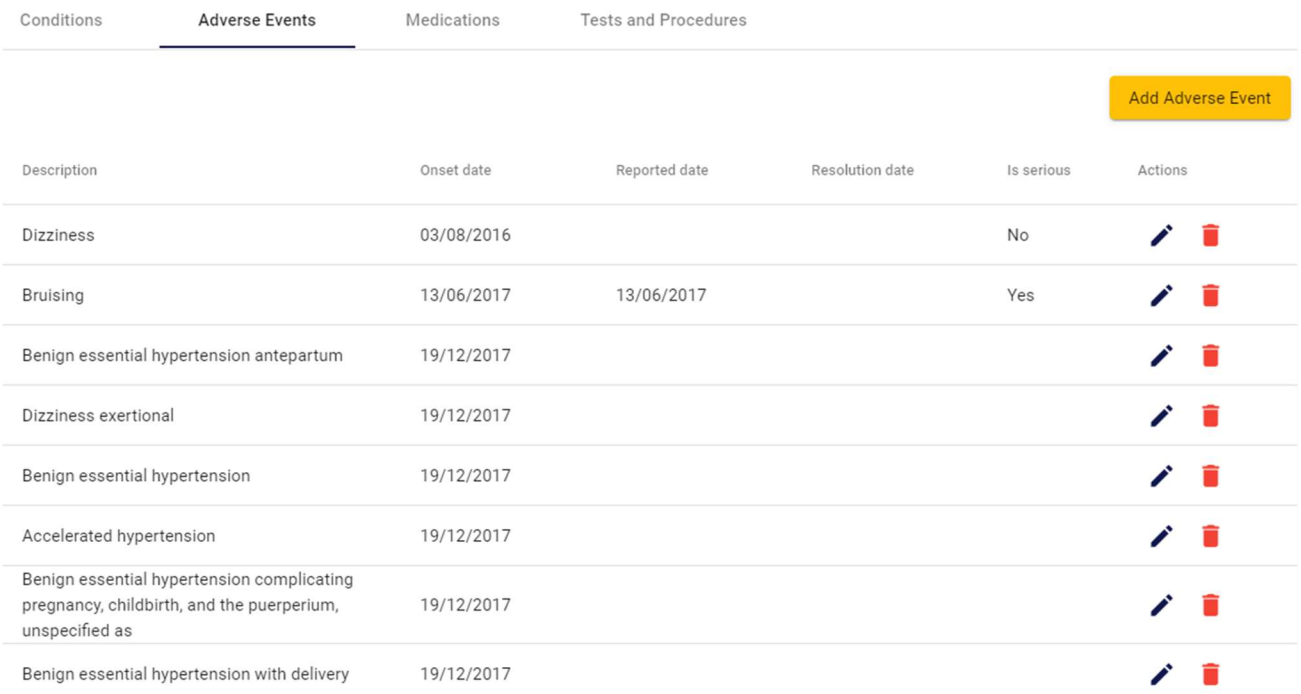
The **patient condition** tab can be used to track a history of concomitant conditions the patient has experienced. Being exposed to concomitant conditions as well as specific types of concomitant conditions can be used as risk factors to signal detection within the analytical portal.

The columns in the **Patient Conditions** table are described below:

Condition Name	Medical term for the patient's diagnosis (or symptoms if diagnosis is not available)
Start Date	Date the condition started
Outcome Date	Date the condition ended
Outcome	Outcome of the Condition
Actions	Ability to Edit or Delete the condition

4.1.3.9 Clinical information – Adverse Events

The **adverse events** tab can be used to track a history of adverse events the patient has experienced. The registration of an adverse event as part of the patient's longitudinal clinical record, results in the creation of a new adverse event report within the analytical portal for consumption by the designated Pharmacovigilance team. Progress against this registration can be tracked in the Analytical Reporting widget within the patient view.

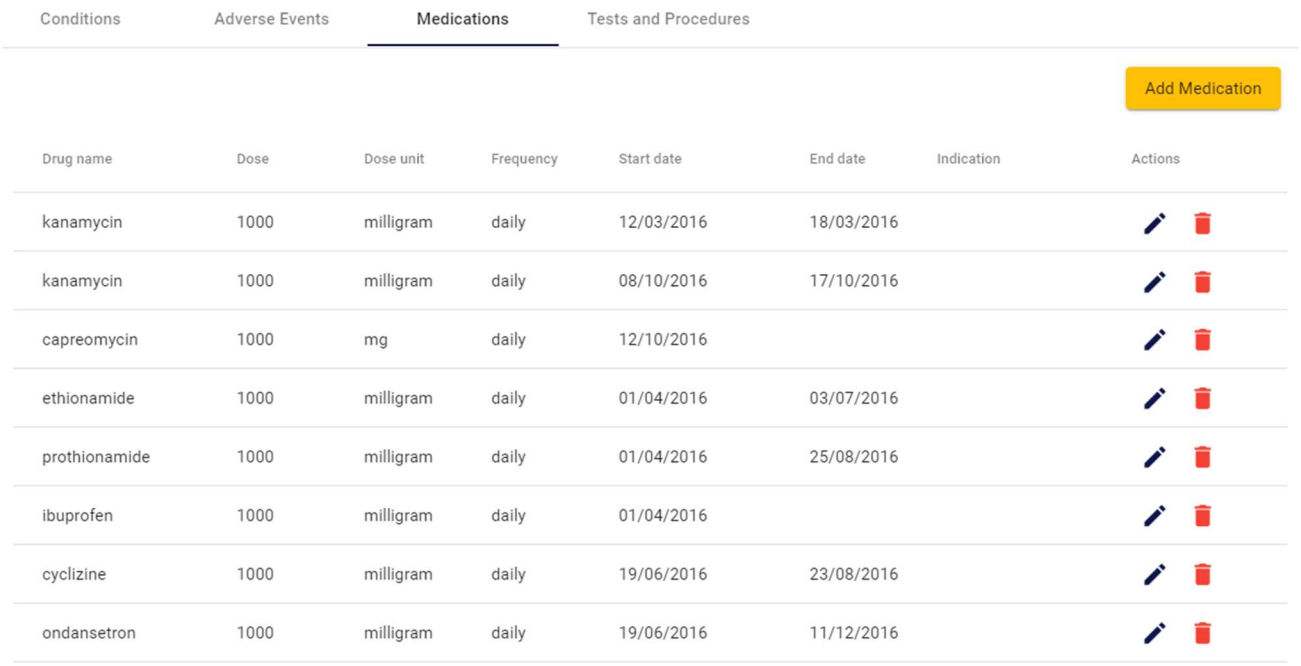


The columns in the **Adverse Events** table are described below:

Description	Description of the event from the MedDRA dictionary
Onset Date	Date the event started
Reported Date	Date the event was reported to the facility
Resolution Date	Date the event was resolved or stabilized
Is Serious	Is this a serious reaction?
Actions	Ability to Edit or Delete the adverse event

4.1.3.10 **Clinical information – Patient Medications**

The **patient medications** tab can be used to track a history of medications the patient has been exposed to. A comprehensive medications history is critical to ensure accurate signal detection within the analytical portal.

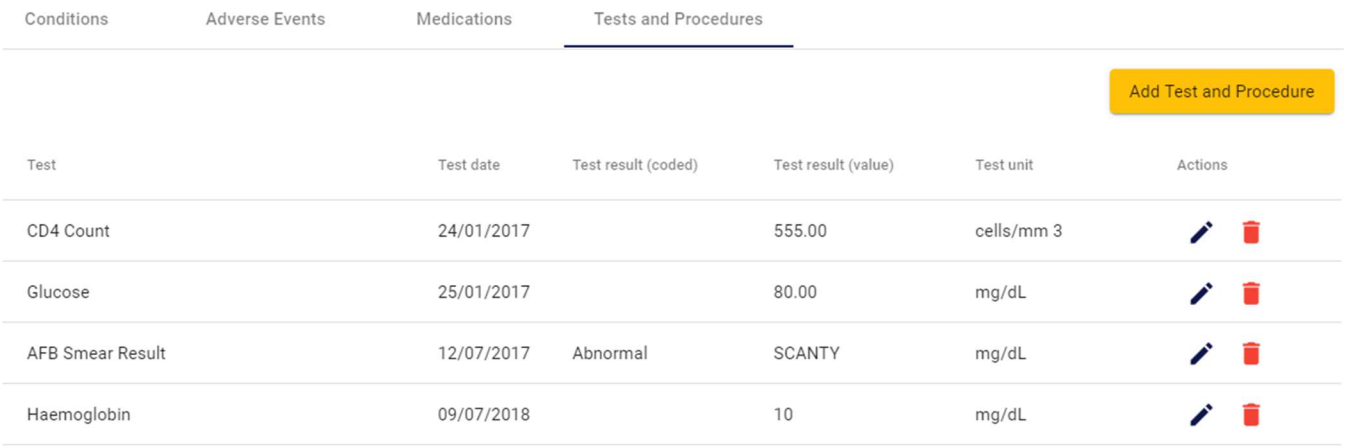


The columns in the **Patient Medication** table are described below:

Drug Name	Name of drug from the country drug dictionary
Dose	Number of units
Dose Unit	Unit of dose (e.g., mg, mEq, IU)
Dose Frequency	Number of times per day the dose is administered
Start Date	Date the patient started taking the medicine
End Date	Date the patient stopped taking the medicine
Indication Type	Purpose of medication (e.g., treat primary condition, treat pre-existing condition, or to tread and adverse event)
Actions	Ability to Edit or Delete the Patient Medication

4.1.3.11 **Clinical information – Tests and Procedures**

The **tests and procedures** tab can be used to track a history of tests and procedures the patient has been exposed to.



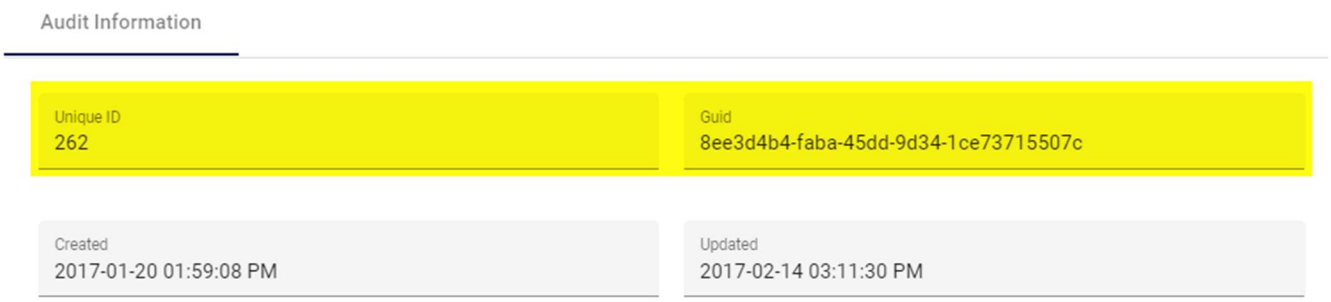
The columns in the **Tests and Procedures** table are described below:

Test	Name of lab test or clinical evaluation
Test Date	Date the test was conducted
Test Result (Coded)	Qualitative test result
Test Result (Value)	Quantitative test result - Number of units
Test Unit	Type of unit
Actions	Ability to Edit or Delete Tests and Procedures

4.1.3.12 **Identifiers and Audit information - Identifiers**

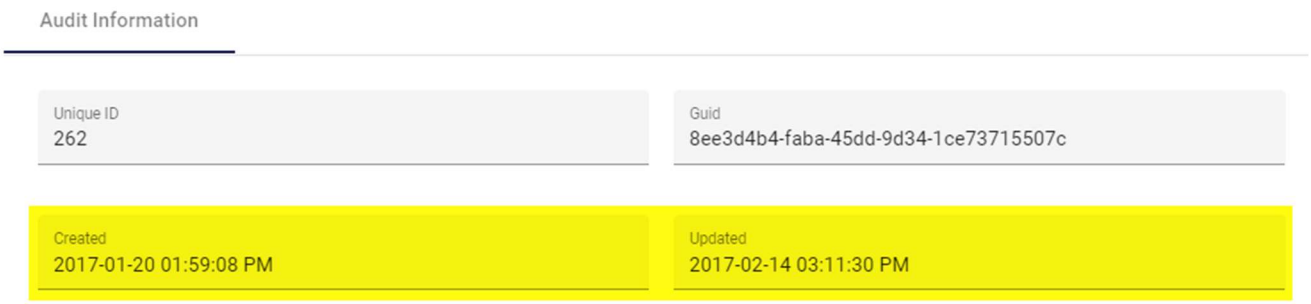
The **Identifiers** section displays the following unique identifiers stored per patient record:

- The patient's Unique ID assigned by the system
- A Globally Unique Identifier (GUID) assigned by the system



4.1.3.13 **Identifiers and Audit information – Audit Information**

The **Audit** section keeps a user record of any patient information changes.



Created	Gives the User Name of the person who created the file and the Date it was created
Updated	Gives the User Name of the person who last updated the information and the Date of the update

4.1.4 Add a New Patient

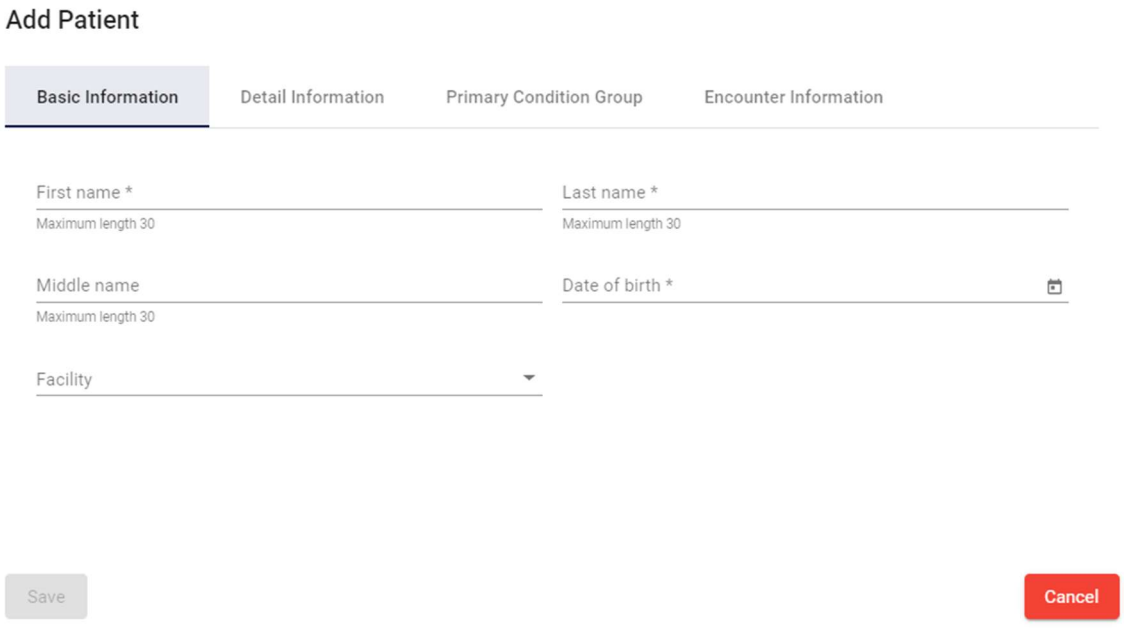
Adding a new patient to the PViMS database requires the completion of a patient search. This is to mitigate the potential risk of registering a patient more than once. If you are not able to find the patient in the existing database, you can add a new patient by clicking on **Add Patient** button.

The system will open a new **Patient** pop up form with several sections needing to be captured.

4.1.4.1 Basic Information

The **Basic Information** section captures basic patient demographic information.

To enter patient information, **enter text** in the corresponding fields (e.g., **First Name**, **Last Name**). Or click the **arrow** in a selected field to display a list of values and select one value from the list. Please ensure that all elements with an asterisk (mandatory) are captured.

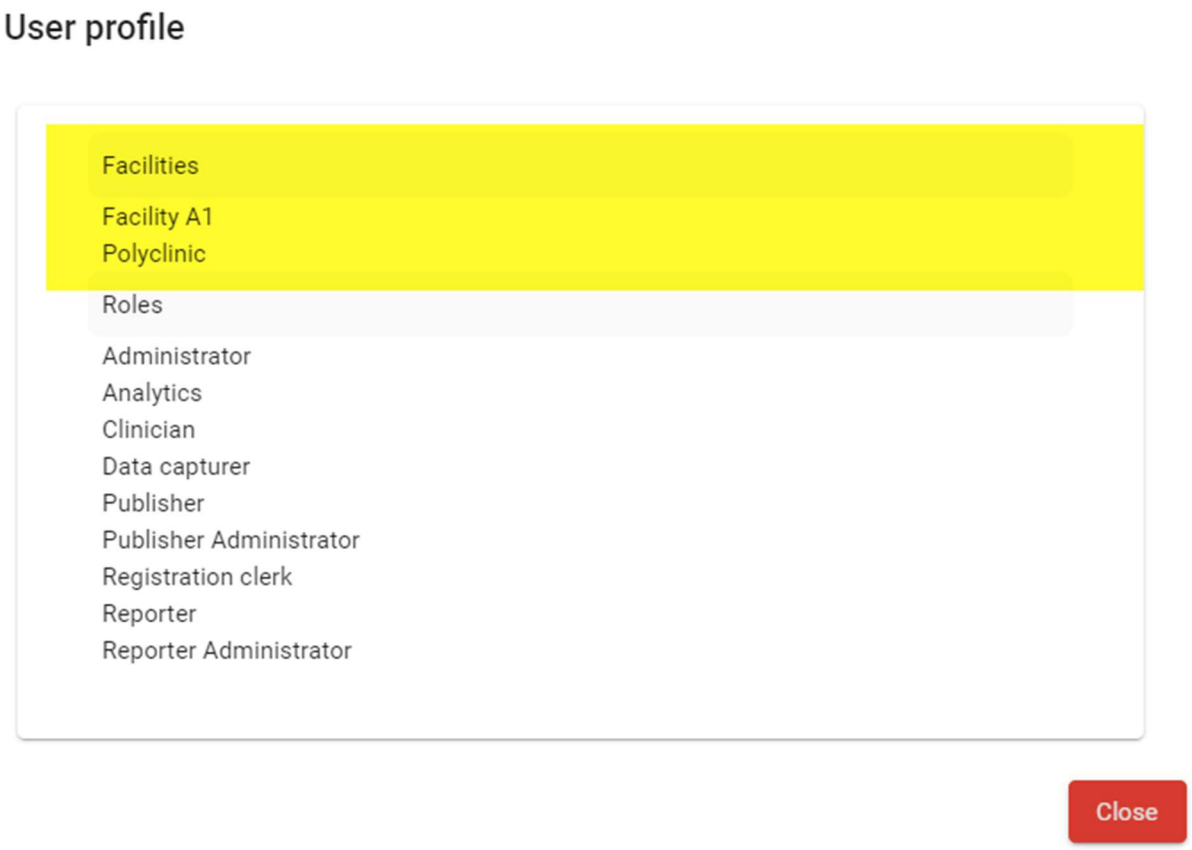


Fields in the **Basic Information** Section are described below:

First name	Text field to enter the patient's first name
Last name	Text field to enter the patient's last name
Middle name	Text field to enter the patient's first name
Facility	Dropdown list to select the patient's facility
Date of birth	Date field to select the patient's date of birth

All fields marked with an asterisk (*) are compulsory fields that must be completed before proceeding.

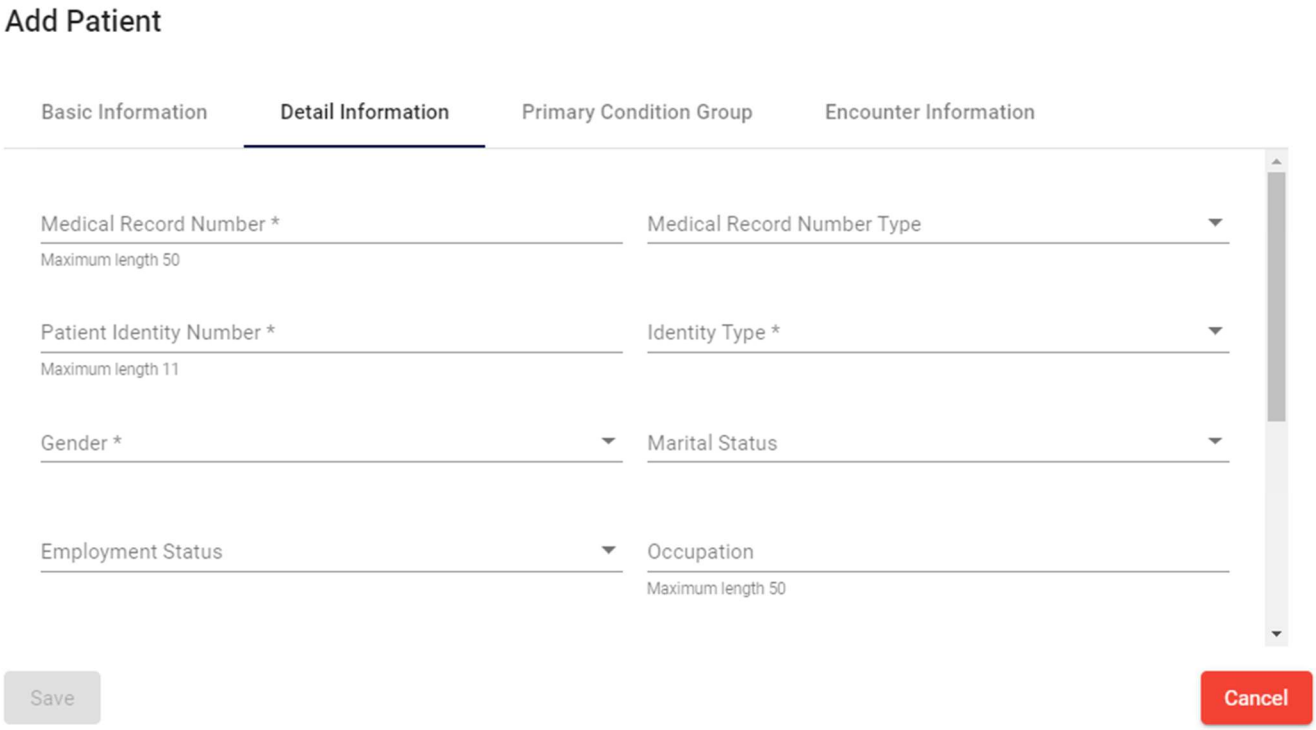
You will only be able to add patients to facilities you have been granted access to. To view which facilities, you have been granted access to, click on your user profile in the profile menu in the top right-hand corner of the page.



4.1.4.2 **Detail Information**

The **Detail Information** section captures comprehensive patient demographic information.

To enter patient information, **enter text** in the corresponding fields (e.g., **Medical Record Number, Medical Record Number Type, etc.**). Or click the **arrow** in a selected field to display a list of values, and select one value from the list (e.g., **Gender**).



Fields in the **Patient Demographic Information** section are described below:

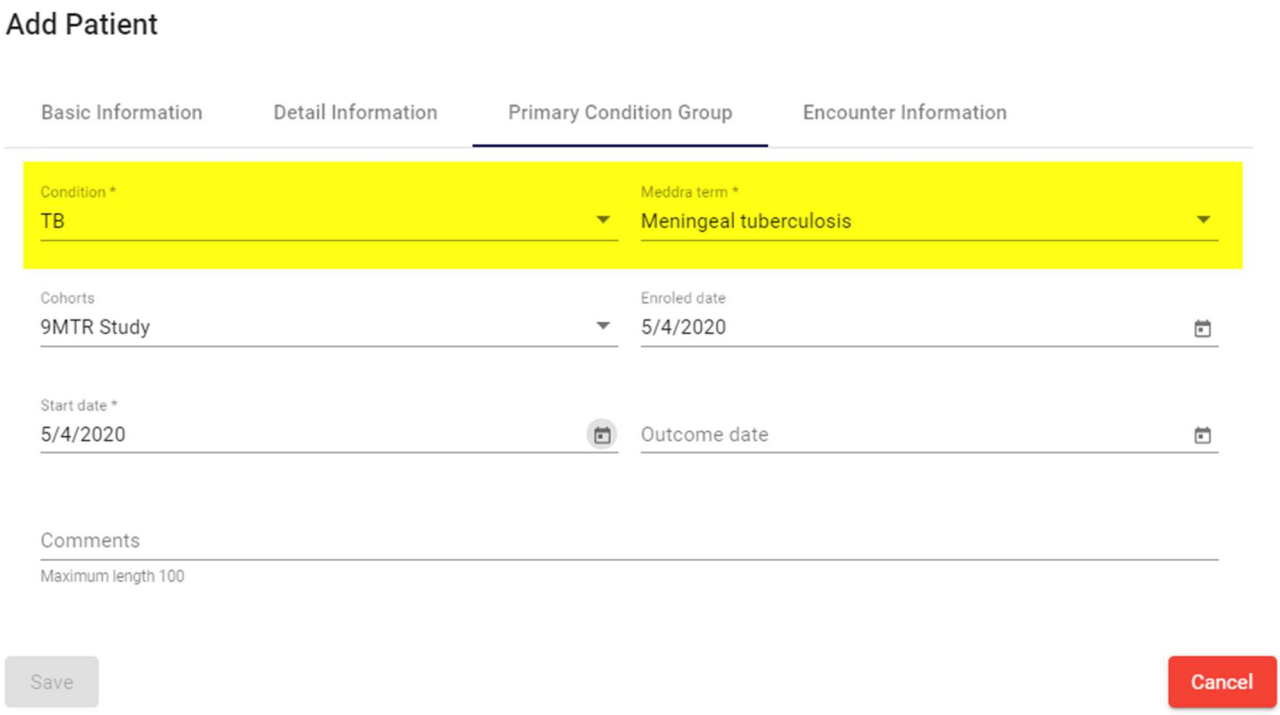
Medical Record Number	Text field to enter the patient's medical record number
Medical Record Number Type	Dropdown menu to select the medical record type
Patient Identity Number	Text field to enter the patient's identity number
Identity Type	Dropdown menu to select the identity type
Gender	Dropdown menu to select the patient's gender
Marital Status	Dropdown menu to select the patient's marital status
Employment Status	Dropdown menu to select the patient's employment status
Occupation	Text field to enter the patient's occupation
Language	Dropdown menu to select the patient's language
Address	Text field to enter the patient's address
Address Line 2	Text field to enter the patient's address
City	Text field to enter the patient's address
State	Text field to enter the patient's address
Postal Code	Text field to enter the patient's address
Patient Contact Number	Text field to enter the patient's contact number
Country of Birth	Dropdown menu to select the patient's country of birth

4.1.4.3 Primary Condition Group

The **Primary Condition Group** section allows you to assign a patient to a patient condition based on their medical condition (e.g., TB, HIV, and Malaria). The patient must be assigned to a patient condition group for their data is to be included when using the **Analytical Portal**.

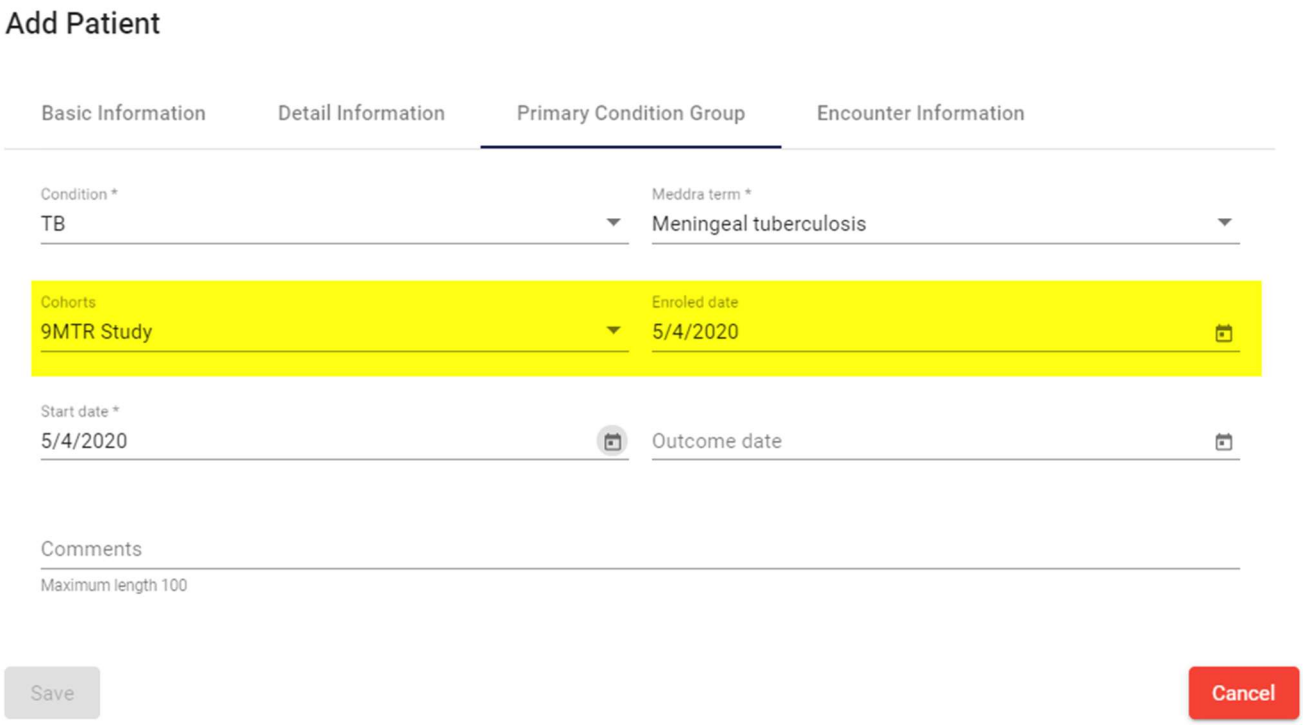
To assign a **Primary Condition Group** click the arrow in the **Condition** field**.** The system will display a list of conditions to choose from. Select the appropriate condition by clicking on the corresponding condition in the list.

The system will then prompt you to select the MedDRA term associated with the condition in the **MedDRA Terms** field.

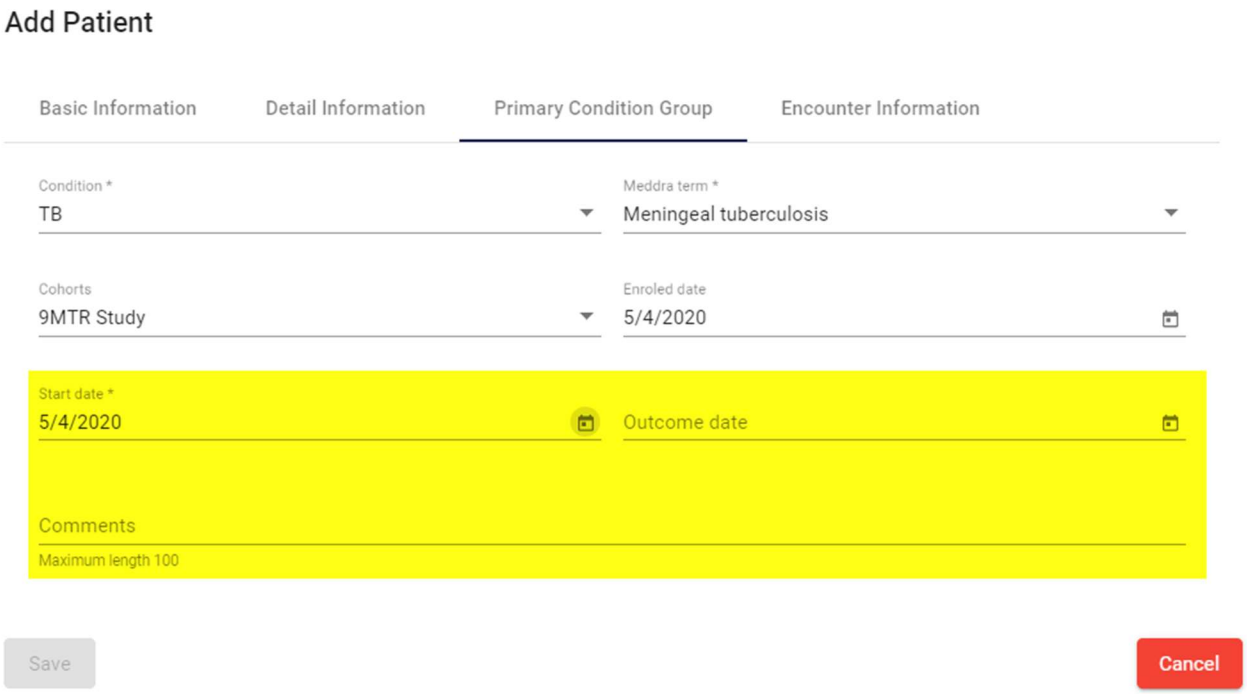


You also have the option to assign a patient to a cohort established by the public health program. To assign a patient to a cohort, click the **arrow** in the **Cohorts** field. The system will display a list of **Cohorts.** Click the cohort the patient should belong to in the list and enter the date the patient was enrolled in the cohort.

You will only be able to allocate patients to cohorts that are assigned to this specific condition group.



Enter the **Start and Outcome dates** (only enter the outcome date if one is applicable) for the condition and any **Comments** regarding the condition if appropriate.



4.1.4.4 Encounter information

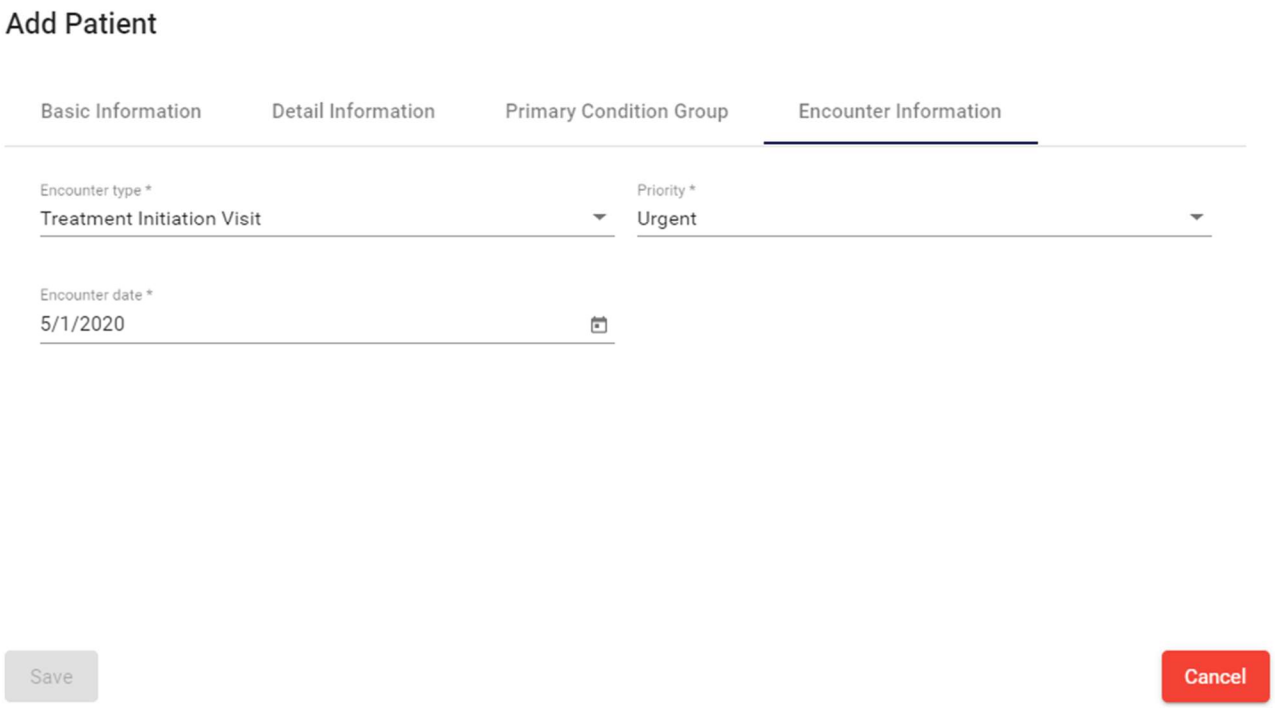
PViMS Term - Encounter

A patient's longitudinal health record is composed of multiple **encounters**. An encounter is effectively a signal that a patient has been seen by a health care provider such as a clinician and clinical data has been collected in context with this encounter.

Click the arrow in the **Encounter Type** field. The system will display an **Encounter Type** list. Click the appropriate **Encounter Type** from the list.

Set the priority for the encounter by clicking on the **arrow** in the **Priority** field. The system will display a **Priority** list. Select a **Priority** option from the list.

Finally enter the encounter date in the **Encounter Date** field. The encounter date will be the date the patient was registered at the facility.

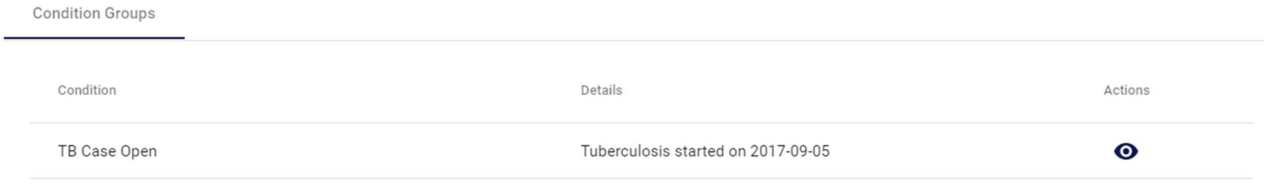


When all information for the page has been entered, click the **Save** button or click the **Cancel** button to cancel the action.

The system will then take you back to the to the **Patient View** page where you can **Add** or **Edit**, patient information described in Section 4.1.7.

4.1.5 Condition Groups

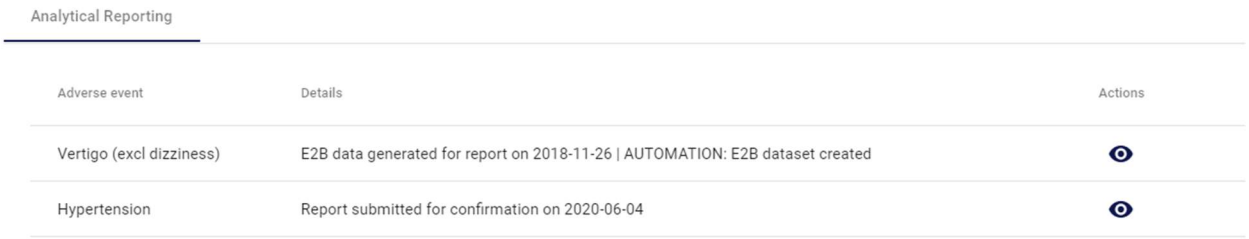
The **Condition Groups** tab which is accessible from within the patient view provides the name of the condition group the patient is assigned to, and the start of the condition. The condition column indicates whether the case is **Open** or **Closed.**



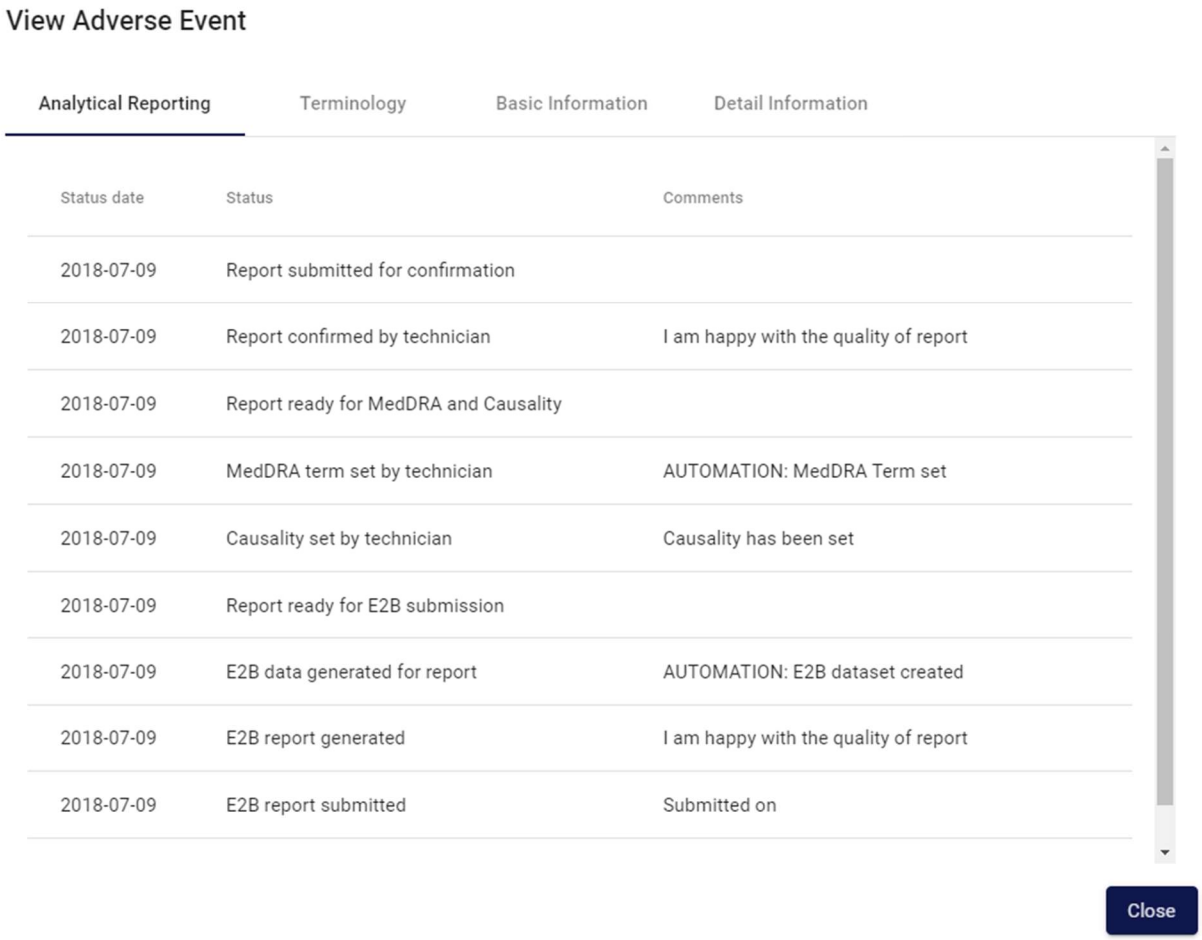
Only **Open** conditions will be displayed on the **Patient View** page.
A patient can be assigned to more than one condition group at the same time (e.g., the TB condition group and the HIV condition group).

4.1.6 Analytical Reporting

The **Analytical Reporting** tab which is accessible from within the patient view provides the current status of any pharmacovigilance activities that have been conducted within the analytical portal against adverse events that have been registered against this patient.



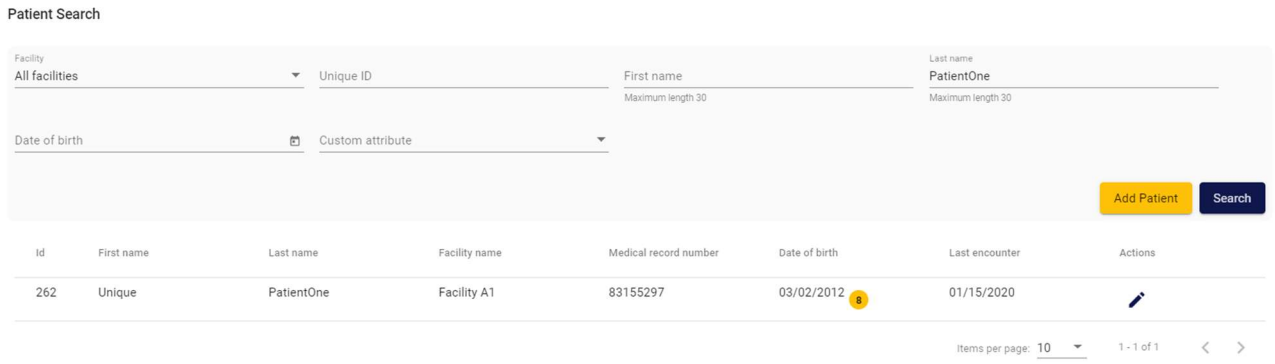
By clicking on the **View Adverse Event** icon, the system will display an adverse event pop up form for this event where you will be able to view a comprehensive history of pharmacovigilance activities and any terminology that has been set.



4.1.7 Add or Edit Patient Information

You can add or edit patient information at the **Patient View** page. But first, you need to locate the patient you would like to amend by searching for the patient using the Patient Search function. Click on the patient menu to access the **Patient Search** screen.

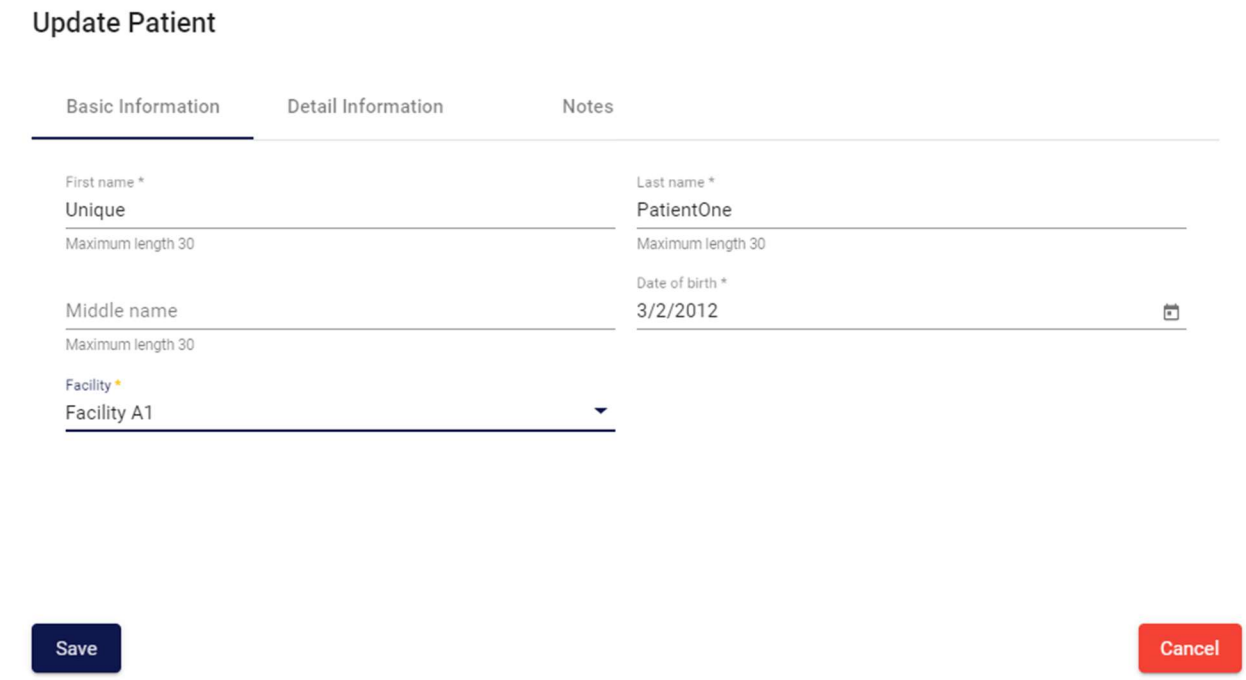
Enter the appropriate search criteria and click the search button. You will be presented with a list of patients that match the search criteria entered.



Click the **View Patient** icon and the system will display the **Patient View** page for the selected patient.

To update patient information, click the **Update Patient** button and the system will pop up a patient edit form where you can edit basic and detailed information as well as patient notes.

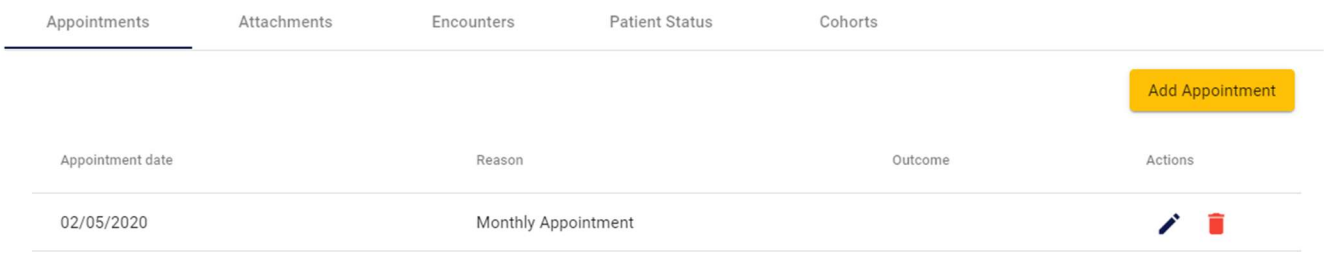
Make changes as appropriate then click the **Save** button to continue or click the **Cancel** button to undo the action and go back to the previous page.



After clicking the **Save** button the system will update the patient record and display the **Patient View** page with the updated information and a **Patient Saved Successfully** confirmation message.

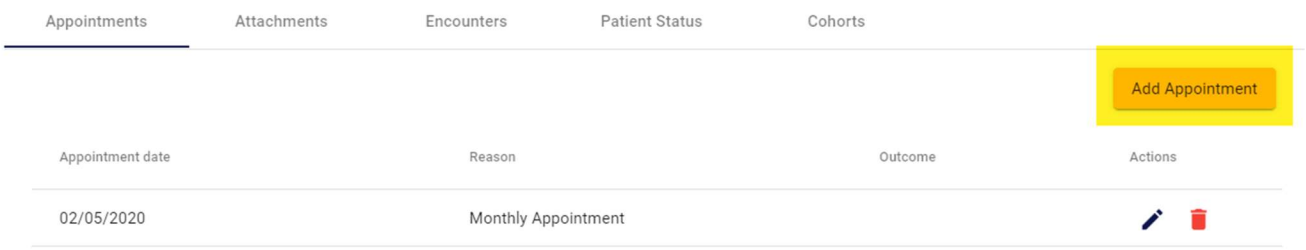
4.1.8 Add or Edit Additional Information

On the **Patient View** page, you can **Add** or **Edit** information in the **Additional Information** section.



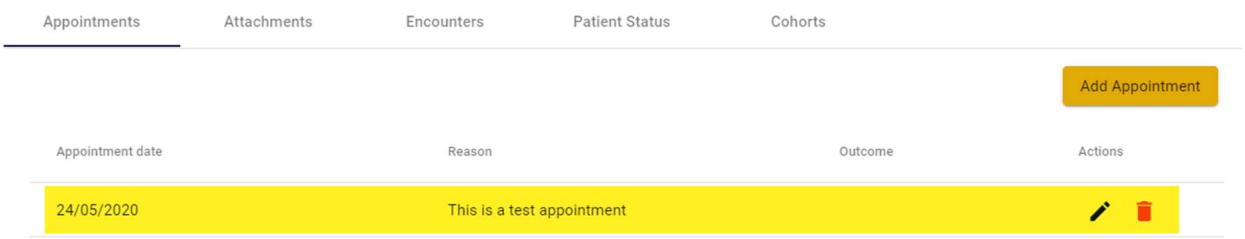
4.1.8.1 Add Appointments

At the **Appointments** tab click the **Add Appointments** button, after which the system will open the **Add Appointment** pop up form.



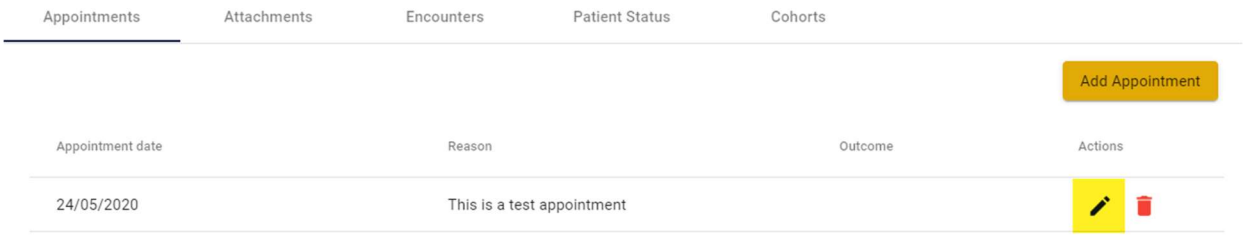
- Enter the **Appointment Date.** The appointment date may only be in the future and within the next 2 years
- Enter the **Reason** for the appointment
- Click the **Save** button to create the appointment or click the **Cancel** button to cancel the action and go back to the previous page

- After clicking on **Save** the system will confirm the update has completed successfully and will display the updated table under the **Appointments** tab



4.1.8.2 **Edit an Appointment**

To edit an existing appointment, locate the appointment in the table. Click the **Update Appointment** icon in the action column for the appointment to be edited after which the system will open the **Update Appointment** pop up form.

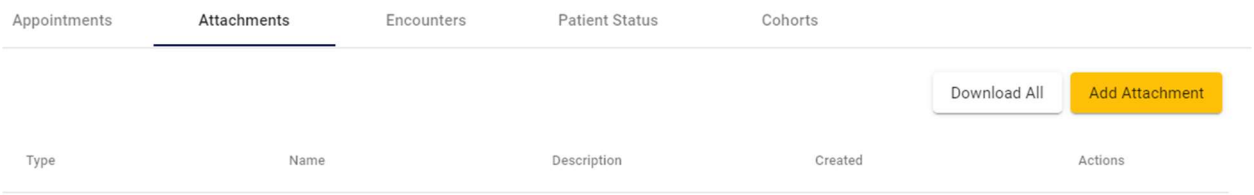


Make changes as needed then click the **Save** button to complete the edit or click the **Cancel** button to undo your changes and return to the previous page.

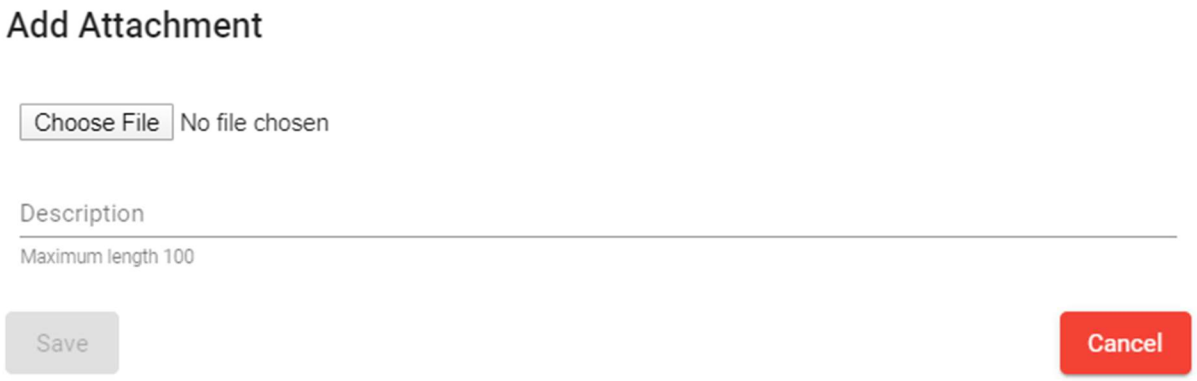
The system will display the updated **Appointment** Table.

4.1.8.3 Add an Attachment

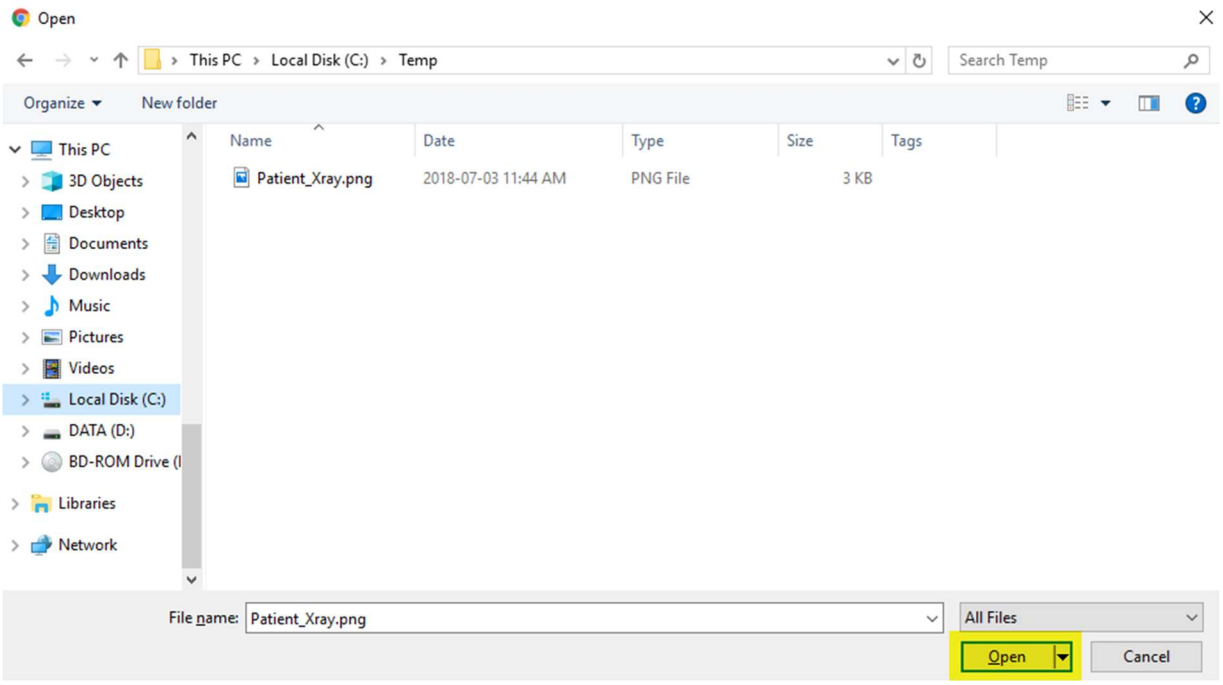
Select the **Attachments Tab** to view the list of attachments.



To add an attachment, click the **Add Attachment** button after which the system will open the **Add Appointment** pop up form. Click the **Choose File** button to search for the file to be attached.



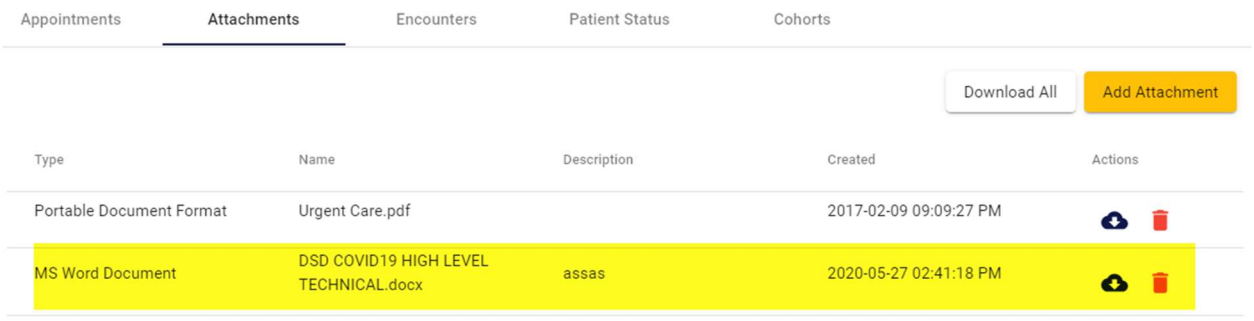
Select the file to upload and click the **Open** button.



The system will return to the **Add Attachment** form and will reflect that the file that was selected. You can add a description in the **Description** field.

Click the **Save** button to complete the attachment upload or click the **Cancel** button to undo your changes and return to the previous page.

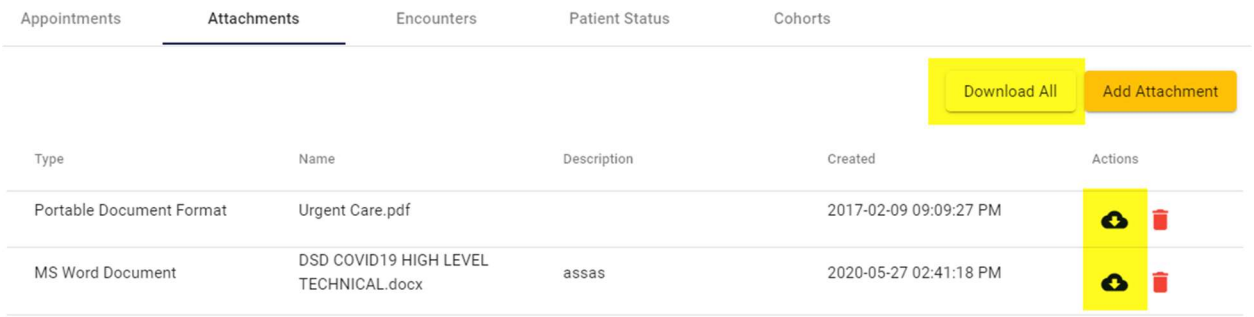
The system will show a confirmation message and an updated **Attachments Table** listing the newly added attachment.



4.1.8.4 Download an Attachment

There are two ways of downloading an attachment to your local computer for viewing. By clicking on the **Download All** button, all attachments associated with this patient will be compressed into a single zip file and downloaded to your local computer.

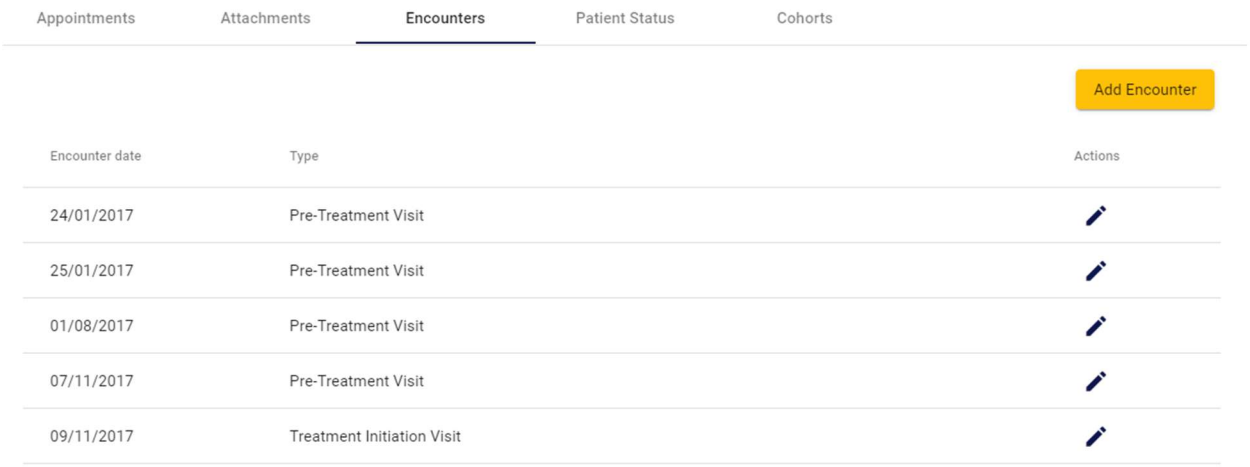
To download a single attachment, locate the attachment in the table and click the **Download Attachment** icon in the action column next to the attachment to download.



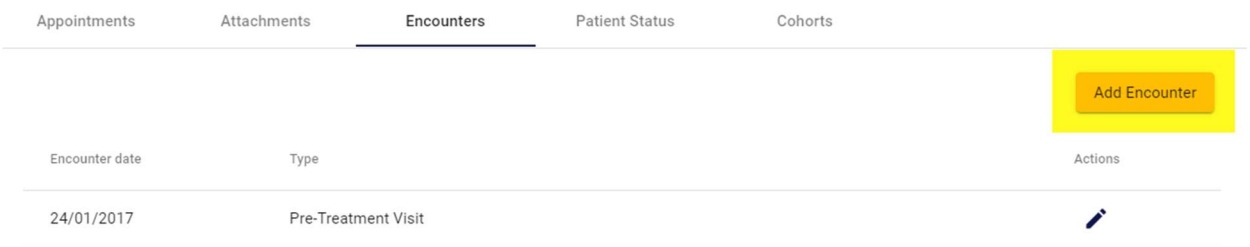
The system will show a message that the attachment has been **Downloaded Successfully.** The downloaded file will typically appear in your computer's **Downloads** or **My Documents** folder.

4.1.8.5 Add an Encounter

Select the **Encounters Tab** to view a list of encounters.

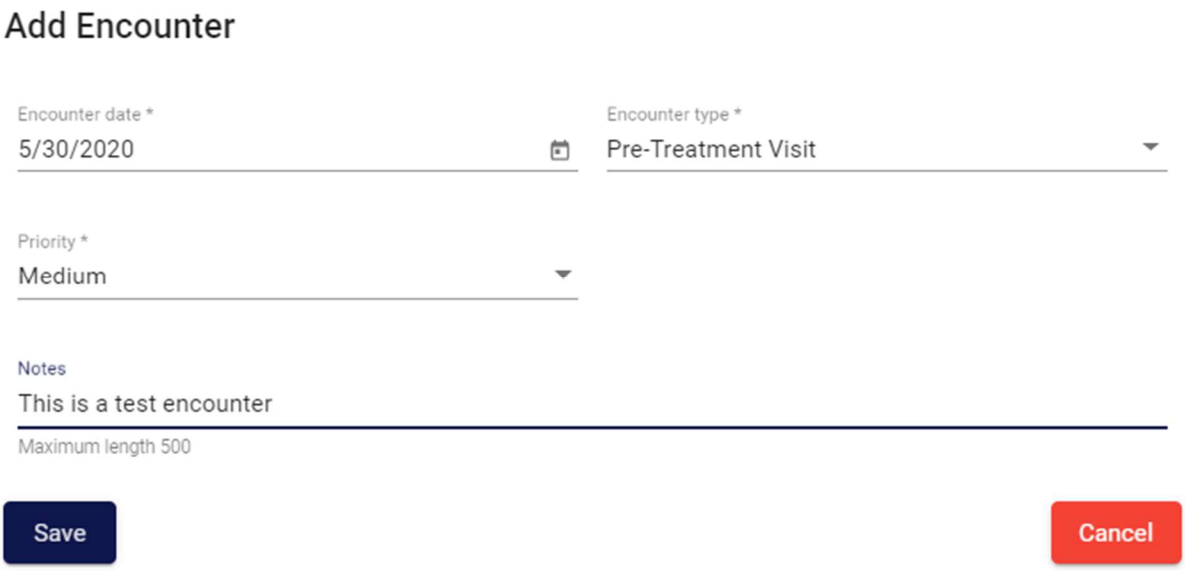


Click the **Add Encounter** button to add a new encounter for this patient.

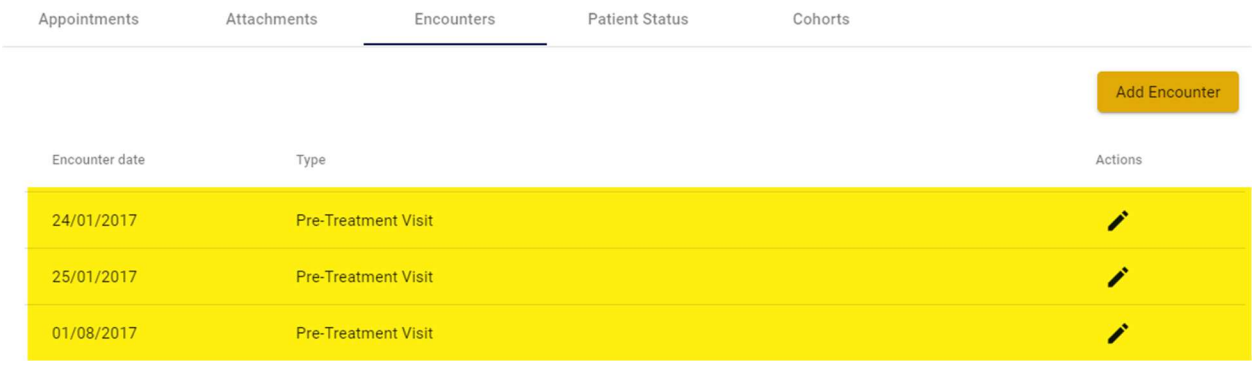


After clicking the **Add Encounter** button, the system will open the **Add Encounter** pop up form.

- Select the date of encounter using the date pop up control.
- Click the arrow in the **Encounter Type** field. The system will display an **Encounter Type** list to select from. Select the **Encounter Type**.
- Click the arrow in the **Priority** field. The system will display a **Priority** list to select from. Select the **Priority**.
- Enter any free format **Notes** regarding the encounter as appropriate.



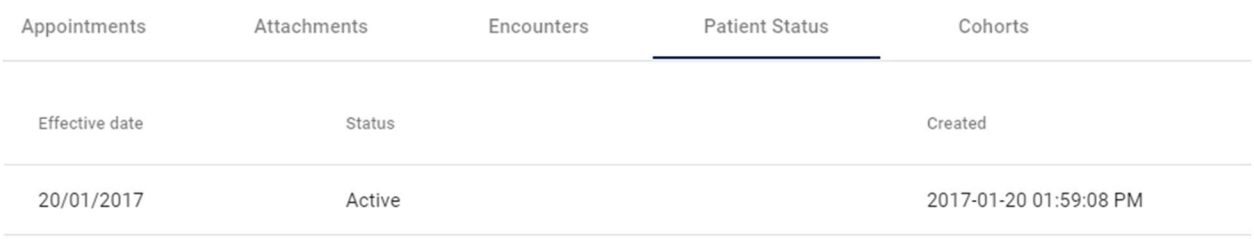
- Click the **Save** button to save the new encounter or click the **Cancel** button to undo the action and return to the previous page.
- After clicking on **Save** the system will confirm the save has completed successfully and will display the updated table under the **Encounters** tab.



Refer to the Encounters Section for details on how to **Add** or **Edit** Encounter information.

4.1.8.6 Patient Status – Read Only

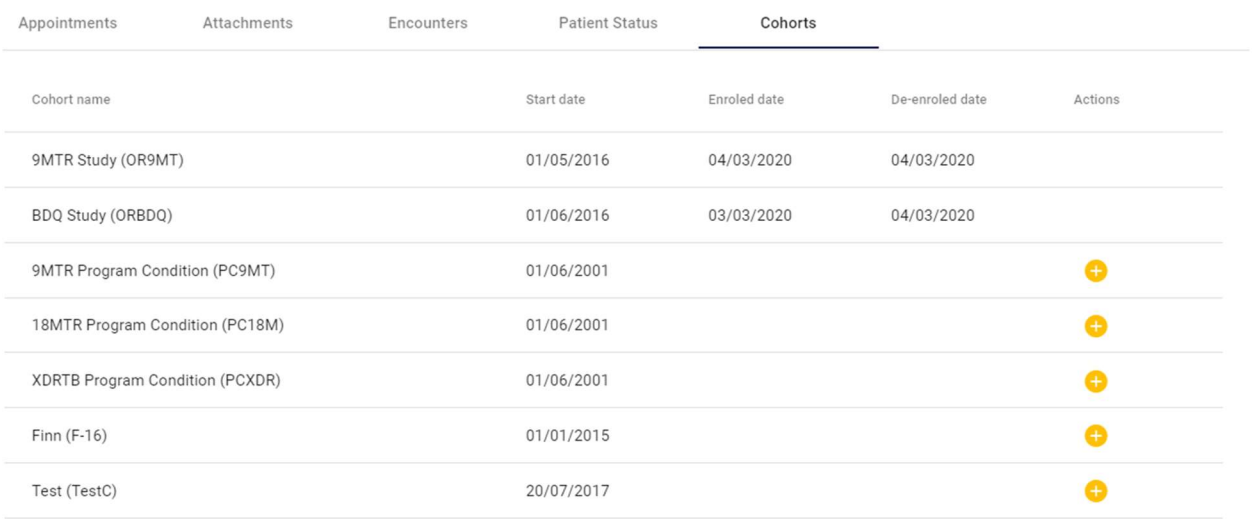
Select the **Patient Status Tab** to view a history of status changes.



The information in the **Status table** is read only and cannot be updated from this page. A patient's status will change from **Active** to **Deceased** when the system is updated in the **Condition Group** section in the **Encounter View** (e.g., the patient completed treatment, died, or lost to follow-up).

4.1.8.7 Cohort Enrolment

Click the **Cohorts** tab to a view a list of cohorts that the patient is or can be enrolled into.

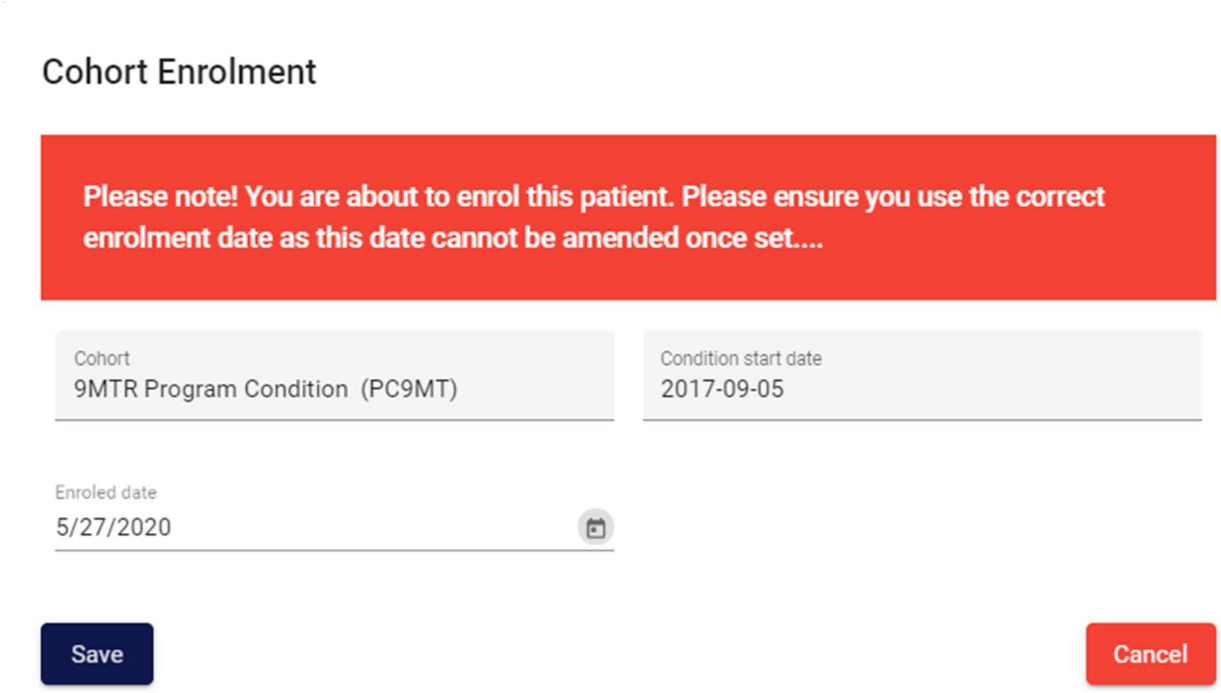


To enroll a patient in a cohort, first locate the **Cohort** in the table. Click the **Enroll Patient** icon in the action column for the cohort in which to enroll the patient.

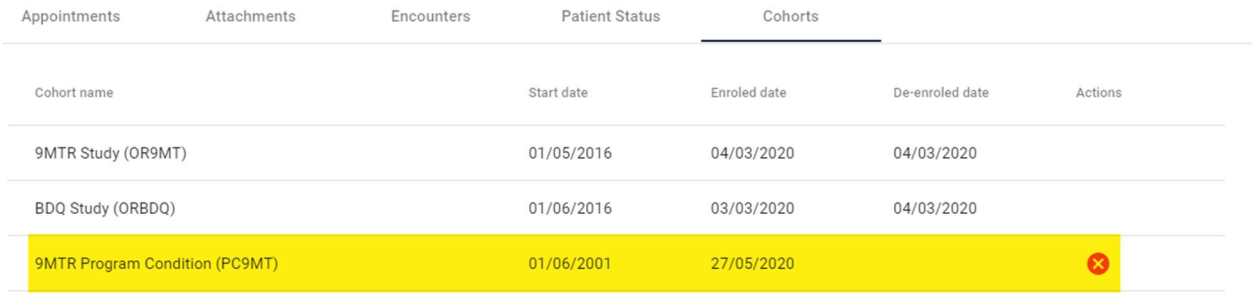
You may only enroll the patient into a cohort that is assigned to the same condition group the patient belongs to. So, for instance, an HIV patient may not be enrolled into a TB cohort.

The system will display a **Cohort Enrollment** pop up form.

Enter the date the patient was enrolled in the cohort. Click the **Save** button to confirm or click the **Cancel** button to undo the action and return to the previous page.



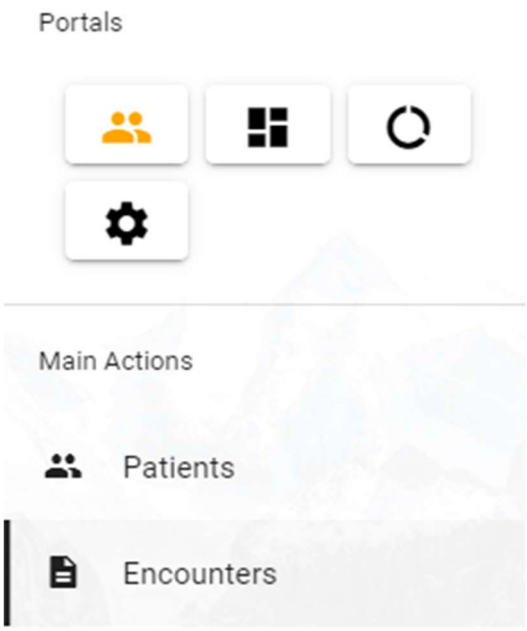
After clicking on the **Save** button the system will display the updated information in the Cohort table.



4.2 Encounters

4.2.1 Search for an Encounter

The **Encounter Search** function can be accessed through the **Encounters** menu.

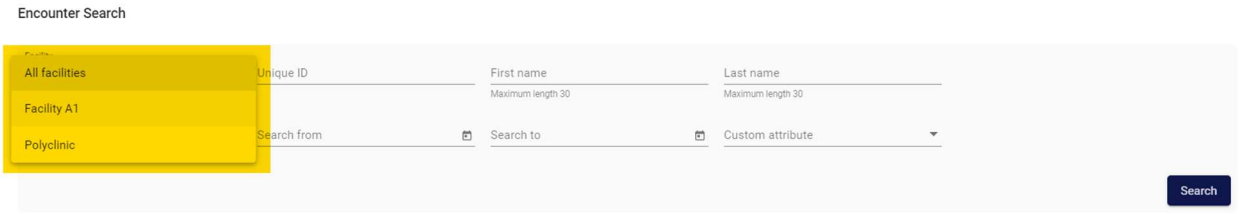


There are six ways to search for an encounter. You can search by:

- Facility
- Patient Unique ID
- First Name and Last Name
- Criteria
- Date Range
- Custom Attributes

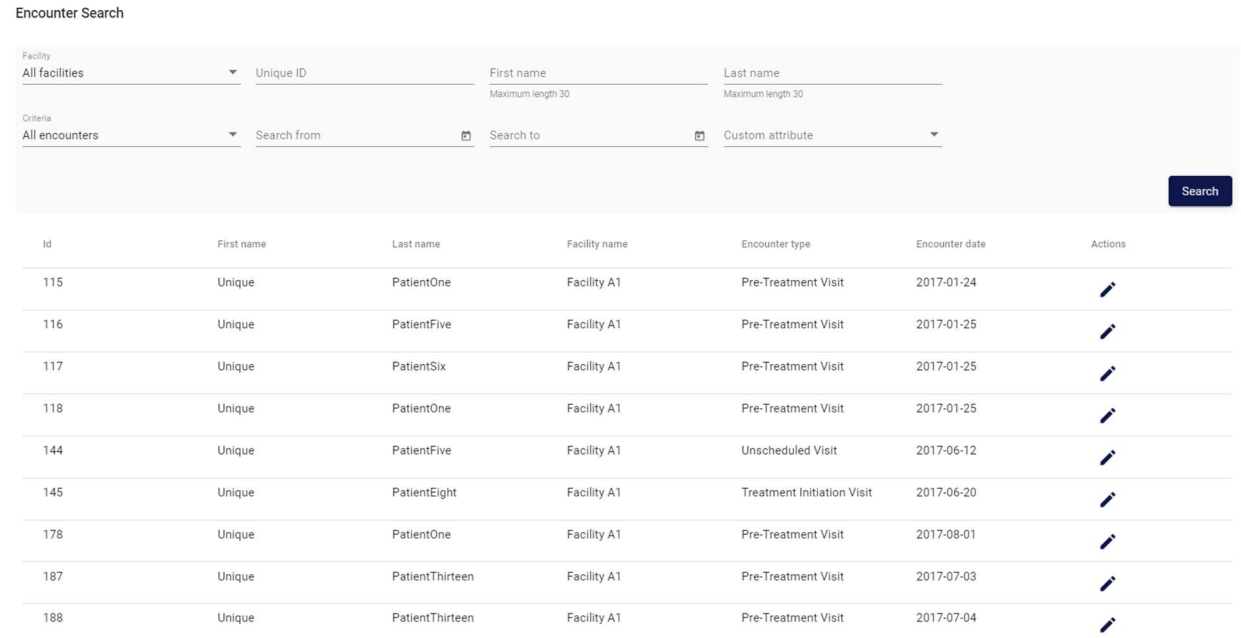
4.2.1.1 Search by Facility

- Click the **arrow** in the **Facility** field to select from the facility drop down list.
- Select the facility you would like to search against specifically or select **All Facilities** if you would like to search against all facilities.
- Enter the **Search From** date and the **Search To** for the date range to search as it is compulsory to enter a date range.
- Click the **Search** button.



You will only be able to search facilities that you have been assigned access to. Please speak to your system administrator if you are unable to search against the necessary facility.

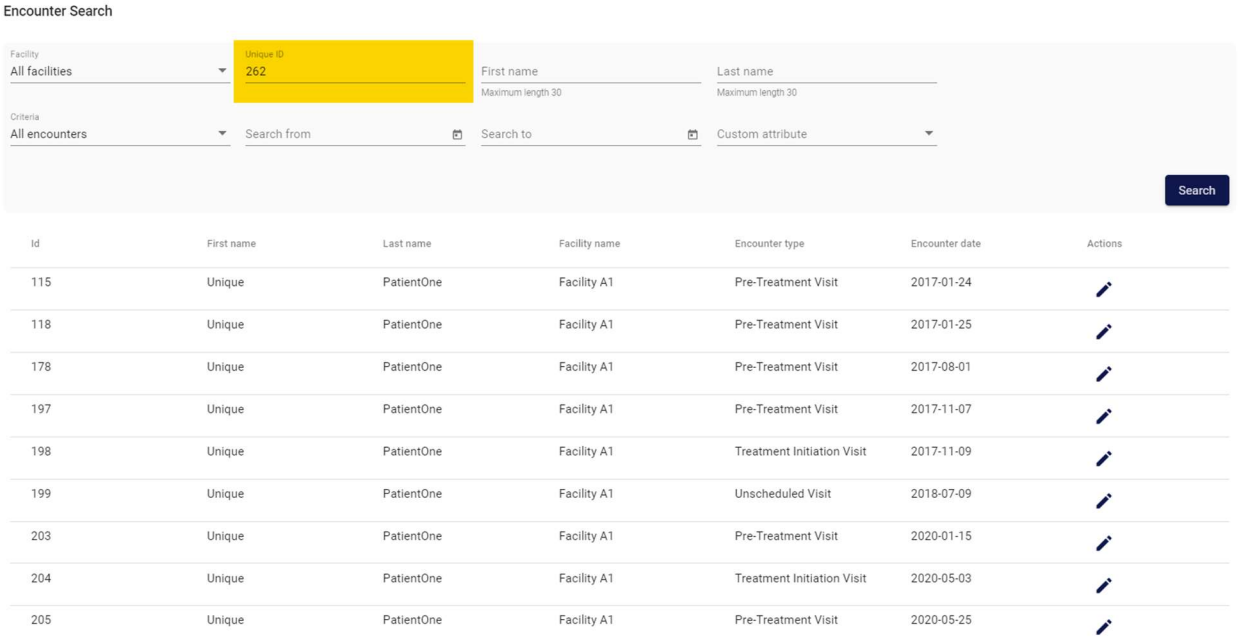
The system will display a list of encounters or appointments according to the filter selected.



4.2.1.2 Search by Patient Unique ID

Each patient is allocated a unique system ID when they are created in the system. It is possible to search for encounters using this ID.

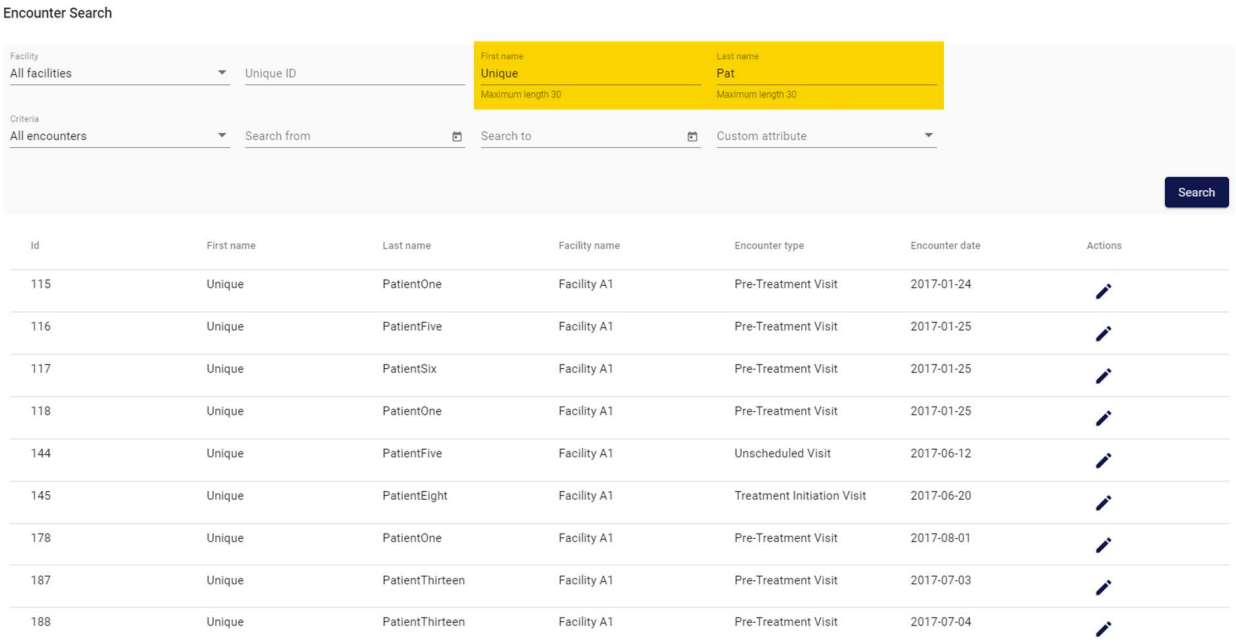
- If you know the patient's unique ID, enter it in the **Unique ID** field.
- Click the **Search** button.



4.2.1.3 Search by First Name or Last Name

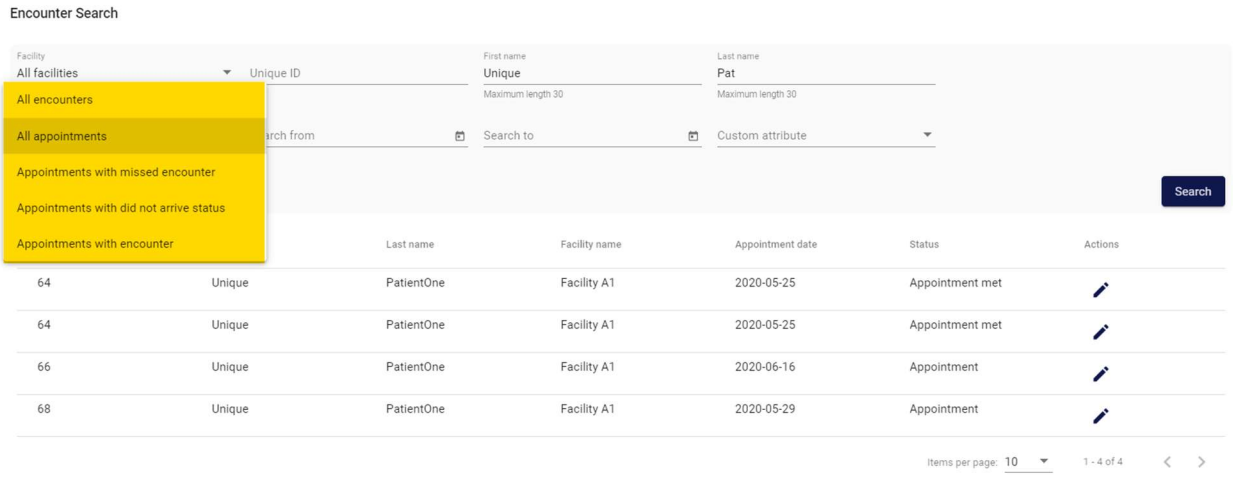
- You can also search by the patient's **First name** or **Last Name**. Enter the name(s) in one or both of these areas.
- Click the **Search** button.

It is possible to do a partial search by entering the first letters of the **First** or **Last names.** The system will return all matching records if a partial search is executed.



4.2.1.4 Search by Criteria

- You can also search by additional **Encounter Criteria**. Click the arrow in the **Criteria** field. The system will display a list of criteria to choose from. Select the **Criteria** you would like to filter on.
- Click the **Search** button.

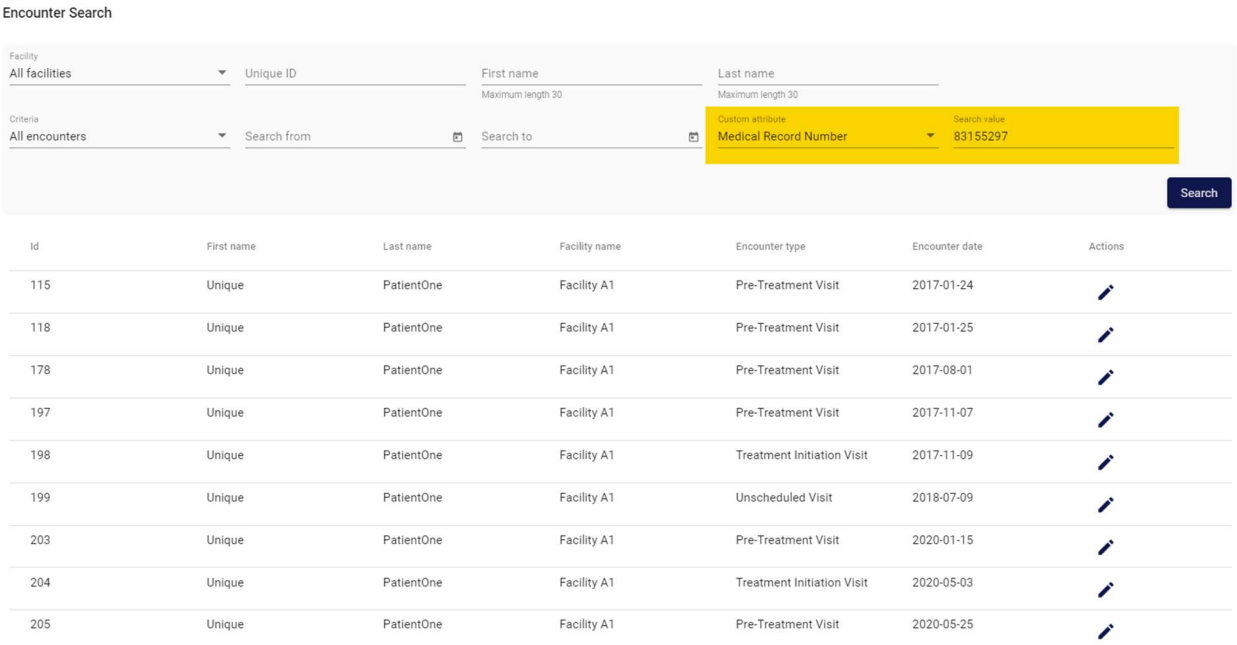


4.1.4.3 Search by Custom Attribute

The final search filter available is the ability to search by a **Custom Attribute**.

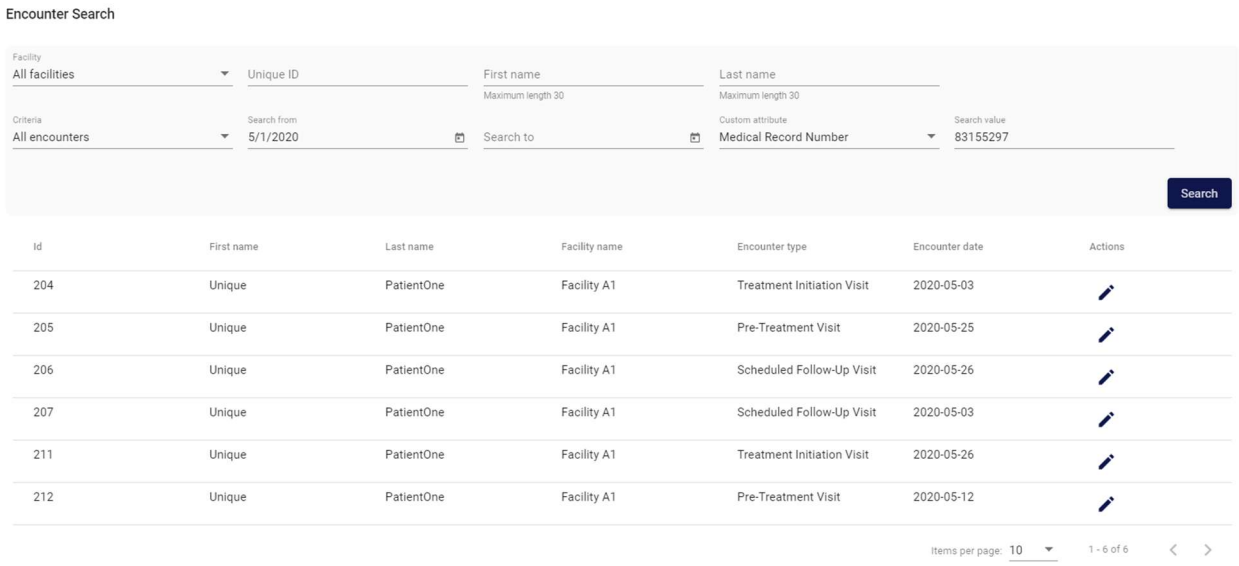
Custom attributes can be activated for filtering by the system administrator. Please consult your administrator if you would like to activate the ability to filter by a specific attribute.

- Select the custom attribute variable that you would like to search against (e.g., Medical Record Number).
- Enter the search value you would like to filter against and click the **Search** button.



4.2.2 View an Existing Encounter

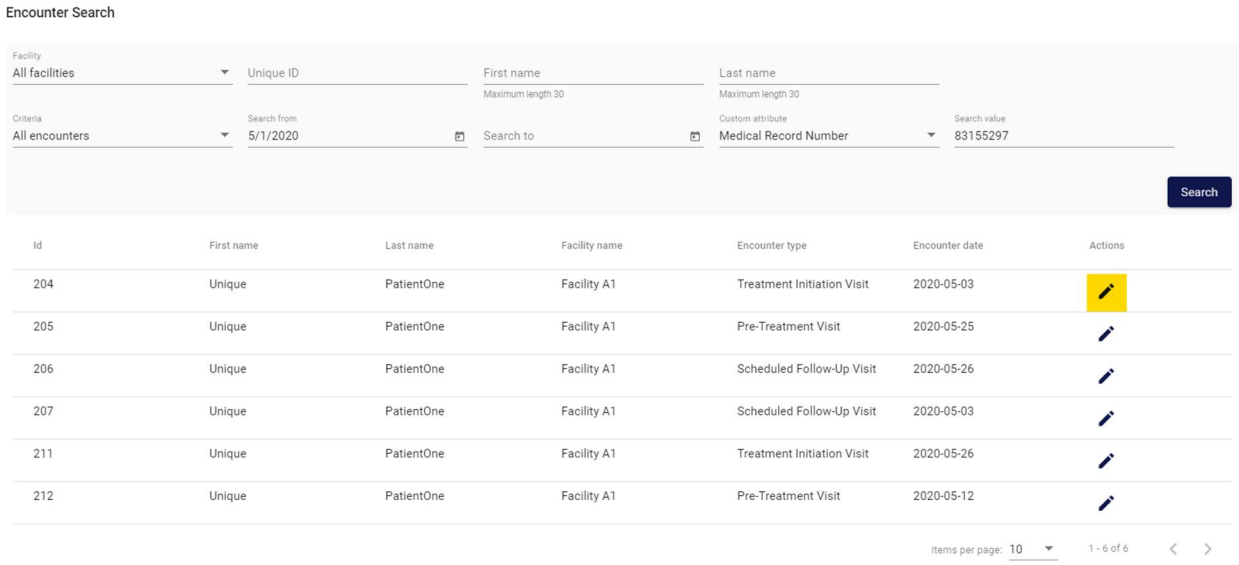
After selecting the appropriate search filter and clicking the **Search** button, the system will present all matches as displayed in a table.



The columns in the encounter table are described below:

ID	Unique encounter ID number assigned by the system
First Name	Patient's first name
Last Name	Patient's last name
Facility name	Facility where patient is registered
Encounter type	Type of encounter (e.g., pre-treatment, treatment initiation, scheduled follow-up or unscheduled visits
Encounter date	Date the encounter occurred
Action	Ability to view the encounter

- To view an encounter entered in the system, locate the encounter in the encounter table.
- Click the **View Encounter** icon in the **Action** column.

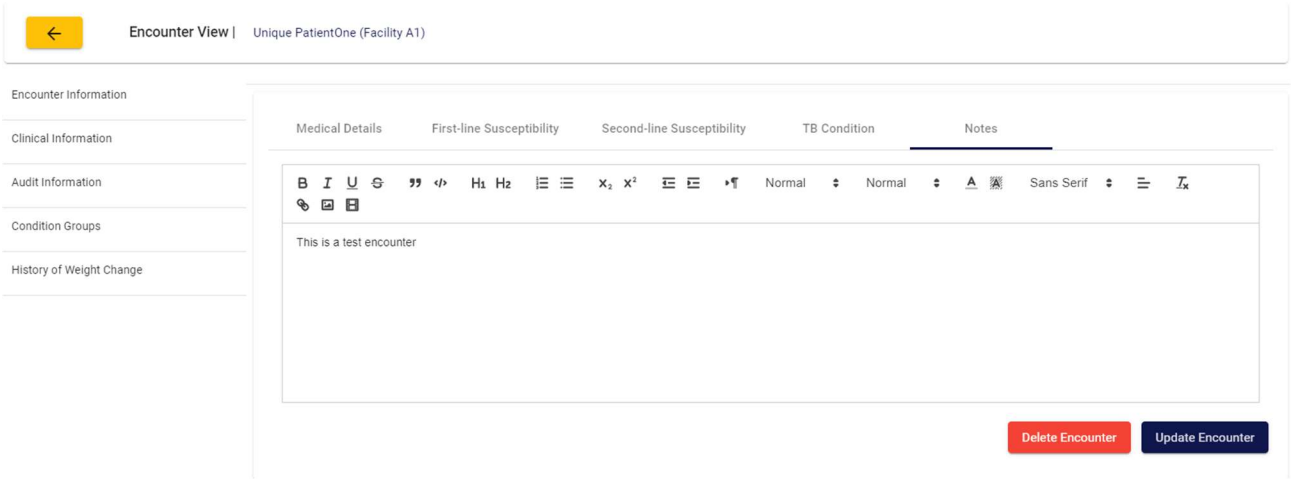


- The system will then open the **Encounter View.**

The **Encounter View** is sub-divided into the following sections:

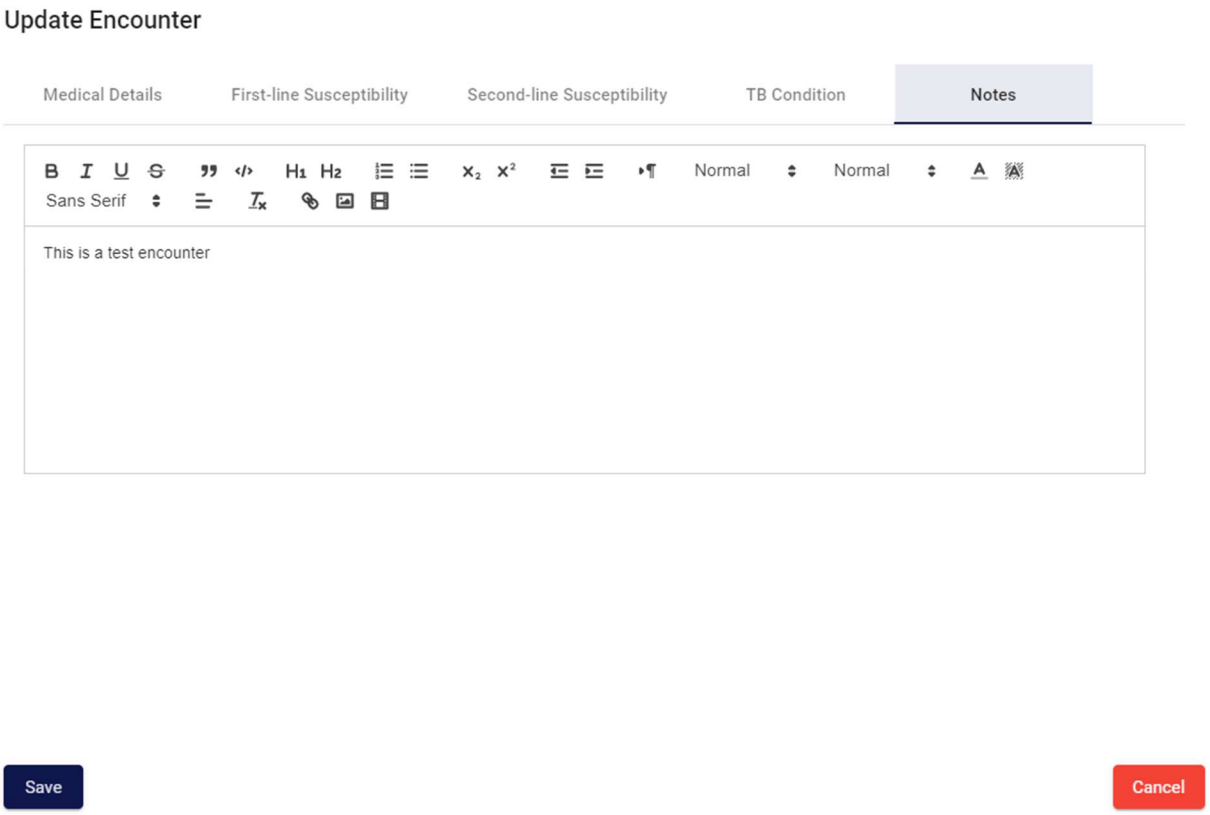
- Medical Details
- First-Line Susceptibility
- Second-Line Susceptibility
- TB Condition
- Notes

Medical Details and **Notes** tabs will be displayed for each patient.
TB Condition, **First-Line Susceptibility**, and **Second-Line Susceptibility** tabs will only appear for patients in the **TB Condition Group** as they are specific to tuberculosis.
The system administrator is responsible for managing the **Condition Group** specific tabs.



4.2.3 Add or Edit Encounter Information

To edit **Encounter Information,** click the **Update Encounter** button after which the system will display an encounter pop up form.



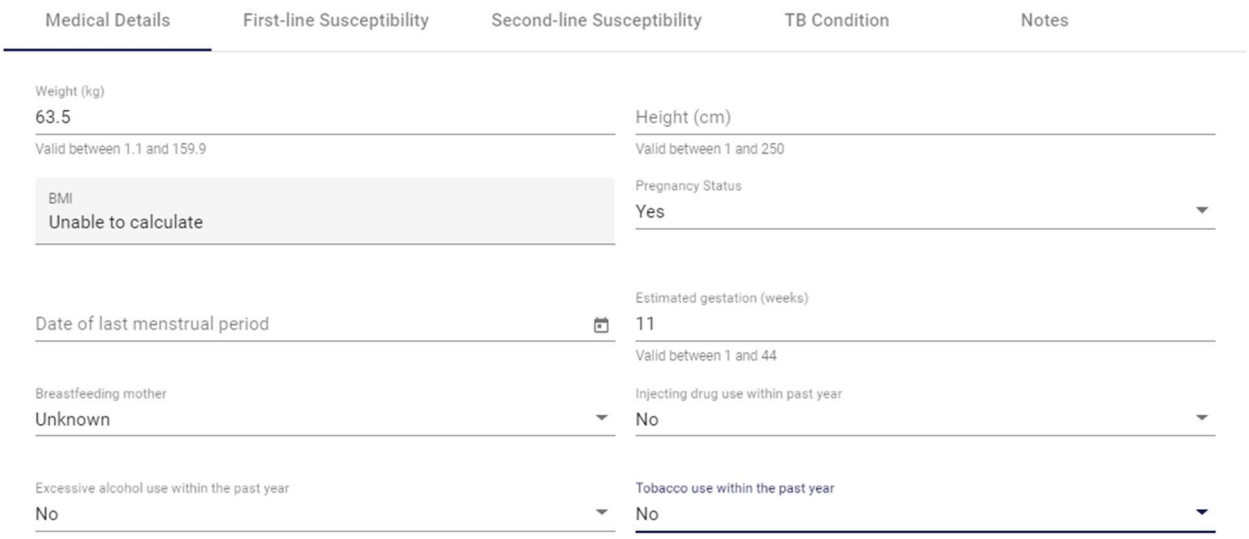
4.2.3.1 **Medical Details**

Fields on the **Medical Details** page are described below:

Weight	Numeric field to enter the patient's weight in kilograms
Height	Numeric field to enter the patient's height in centimeters
BMI	Auto calculated by the system
Pregnancy Status	Dropdown list to indicate yes, no, uncertain or NA
Date of last menstrual period	Date field to enter the patient's LMP
Estimated gestation (weeks)	Numeric field to enter patient's gestational period in weeks
Breastfeeding mother	Dropdown list to indicate yes, no, NA or unknown
Injecting drug use within the past year	Dropdown list to indicate yes, no, or unknown
Excessive alcohol use within the past year	Dropdown list to indicate yes, no, or unknown
Tobacco use within the past year	Dropdown list to indicate yes, no, or unknown

Add or **Edit** information on the page as appropriate. After all changes have been made, click the **Save** button to continue or click the **Cancel** button to undo the action and go back to the previous page.

After clicking the **Save** button, the system will update the **Medical Details** page.



4.2.3.2 First-line Susceptibility

Select the **First-line Susceptibility** tab, after which the system will display clinical data related to determining susceptibility for first-line drugs.

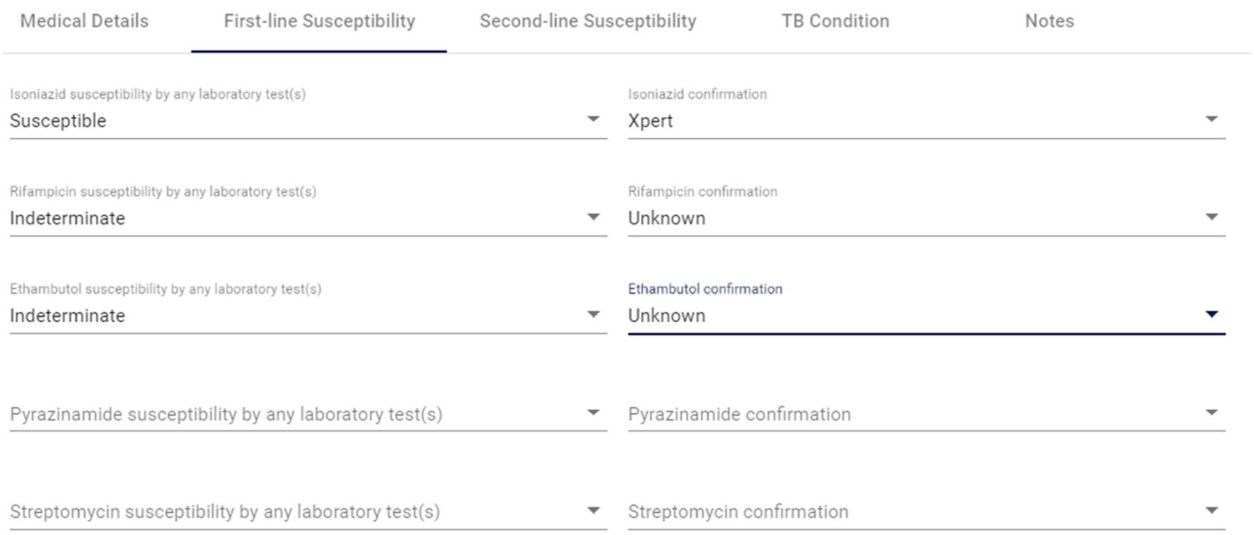
TB Condition, **First-Line Susceptibility**, and **Second-Line Susceptibility** tabs will only appear for patients in the **TB Condition Group** as they are specific to tuberculosis.

Fields on the **First-line Susceptibility** page for each medicine are described below:

Medicine susceptibility by any laboratory test(s)	Dropdown list of test results; Indeterminate, Resistant, Susceptible, Unknown
Medicine confirmation	Dropdown list of diagnostic tools; LPA, Unknown, Xpert, DST

Add information or make changes to the fields on the page as appropriate. Click the **Save** button to continue or click the **Cancel** button to undo the action and return to the previous page.

After clicking the **Save** button the system will update the **First-line Susceptibility** page accordingly.



4.2.3.3 Second-line Susceptibility

Select the **Second-line Susceptibility** tab, after which the system will display clinical data related to determining susceptibility for second-line drugs.

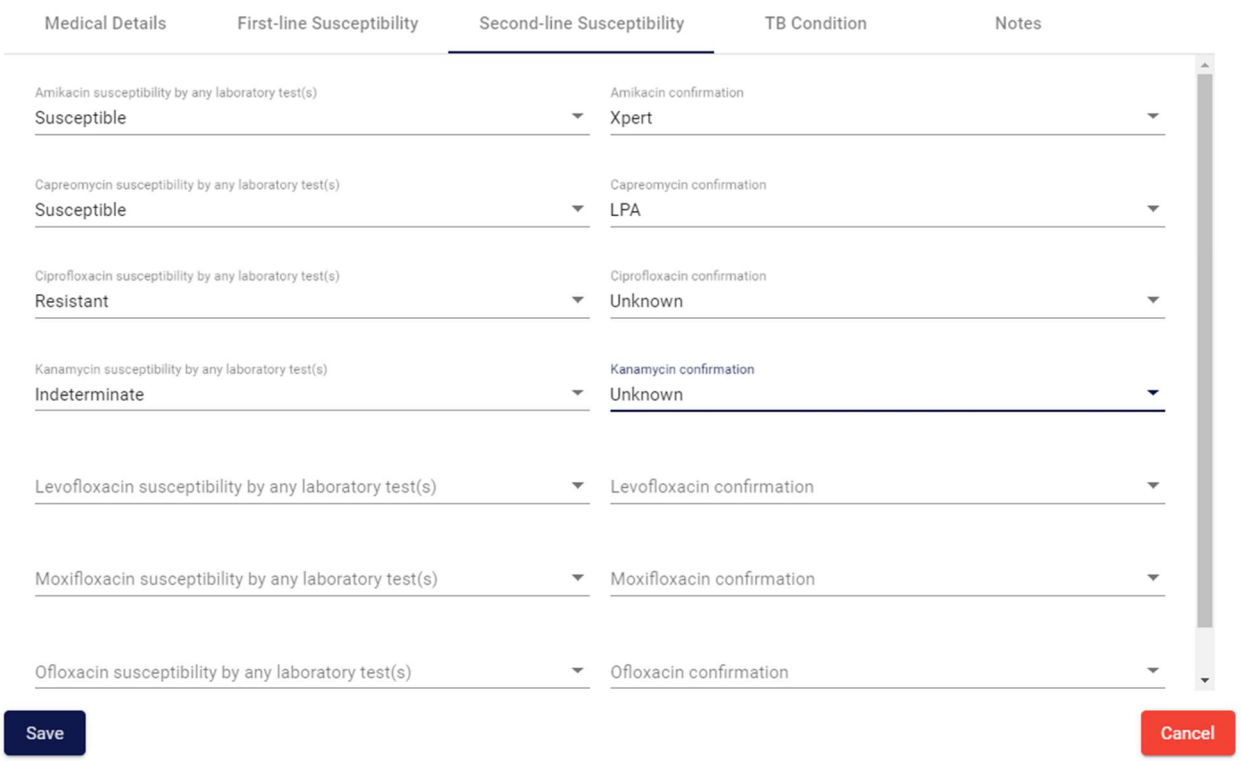
TB Condition, **First-Line Susceptibility**, and **Second-Line Susceptibility** tabs will only appear for patients in the **TB Condition Group** as they are specific to tuberculosis.

Fields on the **Second-line Susceptibility** page for each medicine are described below:

Medicine susceptibility by any laboratory test(s)	Dropdown list of test results; Indeterminate, Resistant, Susceptible, Unknown
Medicine confirmation	Dropdown list of diagnostic tools; LPA, Unknown, Xpert, DST

Add information or make changes to the fields on the page as appropriate. Click the **Save** button to continue or click the **Cancel** button to undo the action and return to the previous page.

After clicking the **Save** button the system will update the **Second-line Susceptibility** page accordingly.



4.2.3.4 TB Condition

Select the **TB Condition** tab, after which the system will display clinical data related to TB.

TB Condition, **First-Line Susceptibility**, and **Second-Line Susceptibility** tabs will only appear for patients in the **TB Condition Group** as they are specific to tuberculosis.

Fields on the **TB Conditions** page are described below:

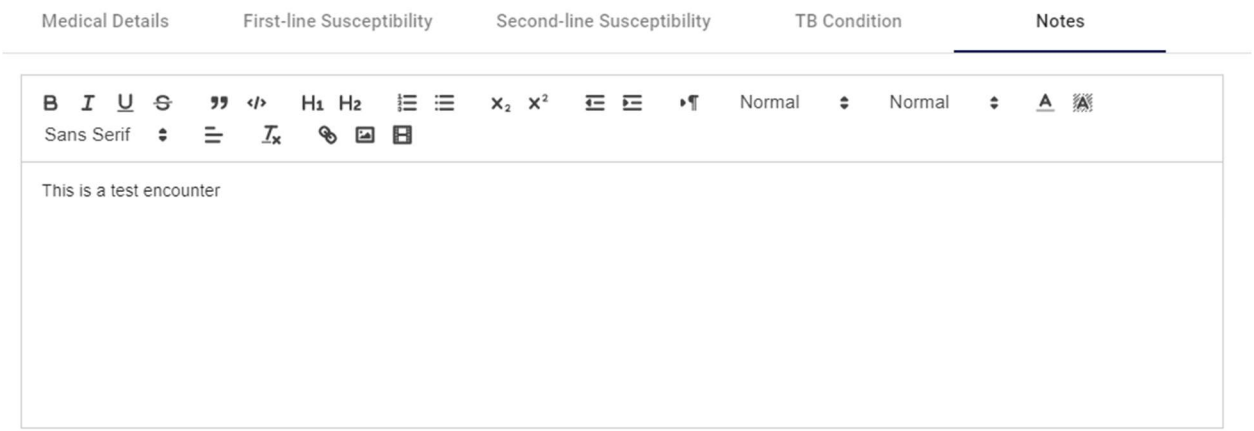
Previous TB treatment?	Dropdown list of responses; No, Unknown, Yes
Site of TB	Dropdown list of anatomical sites
Documented HIV infection	Dropdown list of responses; No, Unknown, Yes

Add information or make changes to the fields on the page as appropriate. Click the **Save** button to continue or click the **Cancel** button to undo the action and return to the previous page.

After clicking the **Save** button the system will update the **TB Conditions** page accordingly.

4.2.3.5 **Notes**

Select the **Notes** tab, after which the system will display the free format notes field for this patient's encounter.



Add information or make changes to the notes as appropriate. Click the **Save** button to continue or click the **Cancel** button to undo the action and return to the previous page.

After clicking the **Save** button the system will update the **Notes** page accordingly.

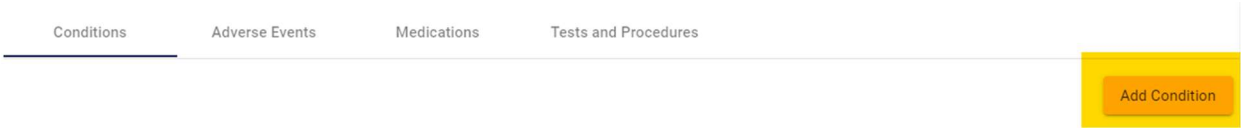
4.2.4 Add or Edit Clinical Information

The Clinical Information section is divided into four tabs:

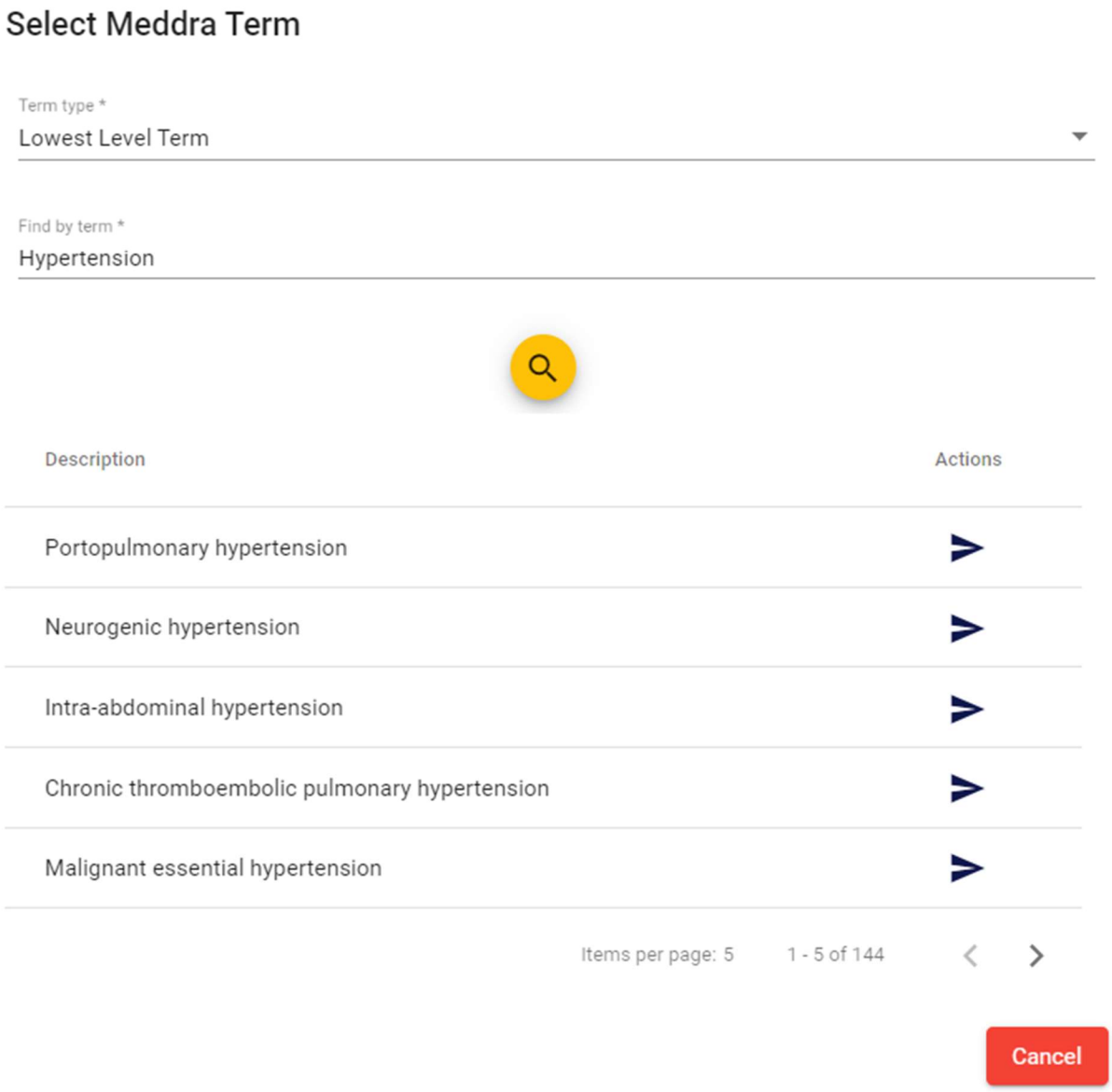
- Patient Conditions
- Adverse Events
- Patient Medication
- Tests and Procedures

4.2.4.1 Add Patient Condition

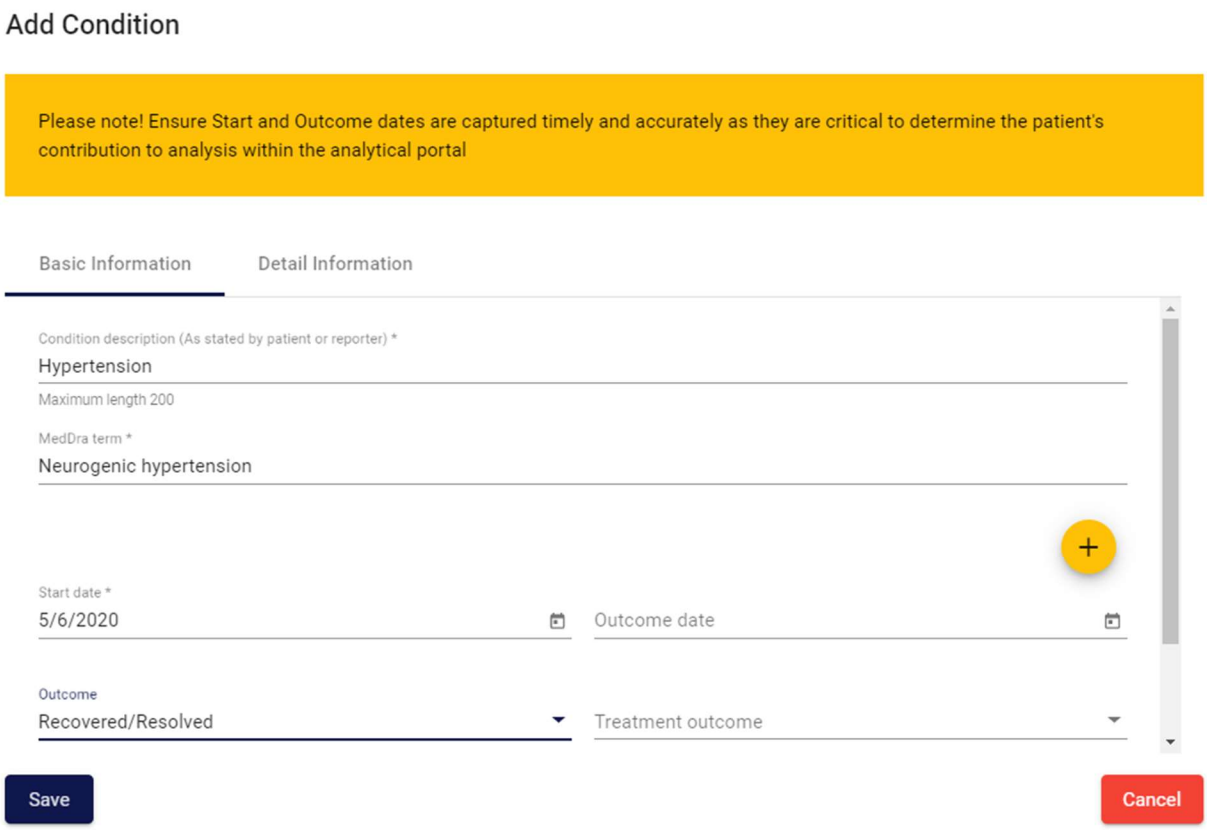
At the **Conditions** tab click the **Add Condition** button, after which the system will display an **Add Condition** pop up form.



- Select the MedDra term for the condition by clicking on the plus icon.
- Select the term type for the term and enter the condition name in the **Find by term** field. Click the **Search** button (magnifying glass) and the system with then provide a list of results from the MedDRA dictionary that match the term that you have searched on.



- Click the right arrow for the corresponding term that you would like to select.
- Enter a free format description of the condition as defined by the reporter or patient.
- Enter the **Start Date** and complete the remaining fields as appropriate.
- Click the **Save** button to create the condition or click the **Cancel** button to cancel the action and go back to the previous page.



- After clicking on **Save** the system will display the updated the table under the **Conditions** tab.

The fields on the **Add Condition** form are described below:

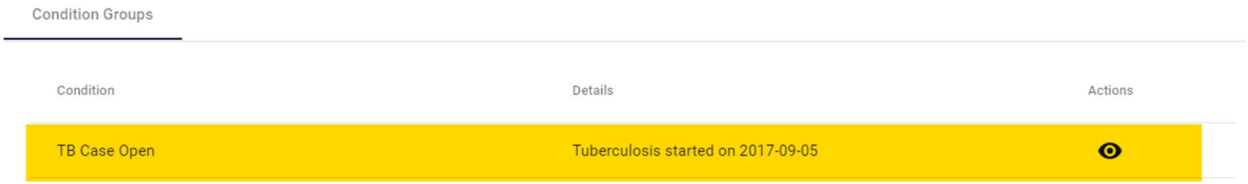
Condition description	A free format description of the condition by the reporter or patient
Term type	Dropdown list of MedDRA term hierarchy; Lowest level term, Preferred term, High level term, High level group term, or System organ class
Find by term	Text field; Enter name of condition
Term results	System generated list; Select appropriate term
Start date	Text field; Enter date condition started
Outcome	Dropdown list of Outcomes; Select either Fatal, Not Recovered/Not Resolved, Recovered/Resolved, Recovered/Recovered with Sequelae, Recovering/Resolving, or Unknown
Outcome Date	Date field; Enter condition outcome date
Treatment Outcome	Dropdown list of Outcomes; Select either Cured, Died, Lost to Follow-up, Not evaluated, Treatment Completed, or Treatment Failed
Comments	Text field; Enter comments about the condition not captured on the page
Condition Ongoing	Dropdown list of options; Select either No, Unknown, or Yes

The **Term Type** field displays the level of MedDRA hierarchy terms (from very general to very specific) to display. The table below describes the five levels.

Level	Example
System Organ Class	Gastrointestinal Disorders
High Level Group Term	Gastrointestinal Signs and Symptoms
High Level Term	Nausea and Vomiting Symptoms
Preferred Term	Nausea
Lowest Level Term	Feeling Queasy

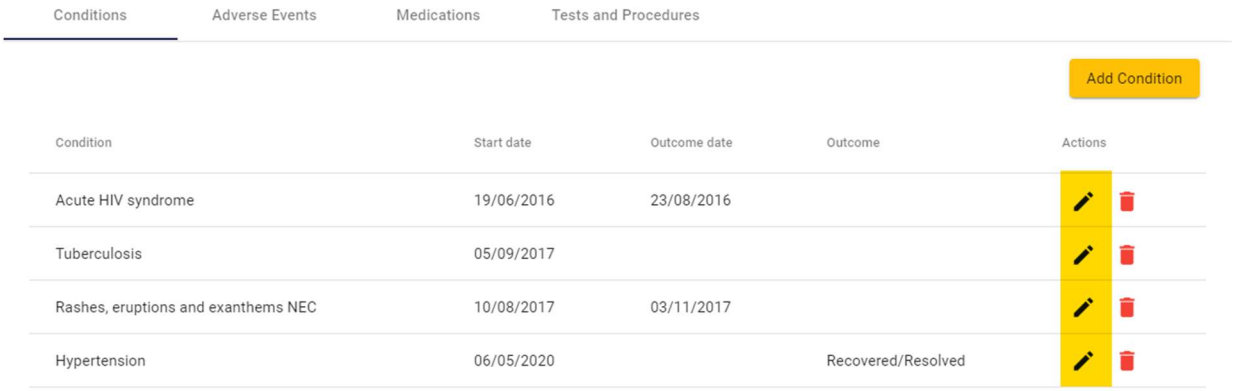
At the most specific level, called "Lowest Level Terms" (LLTs), there are more than 70,000 terms that parallel how information is communicated. These LLTs reflect how an observation might be reported in practice. This level directly supports assigning MedDRA terms within the PViMS database.

When the new condition is a **Condition Group Term**, a corresponding **Condition Group** button will appear in the encounter view.



4.2.4.2 Edit a Patient Condition

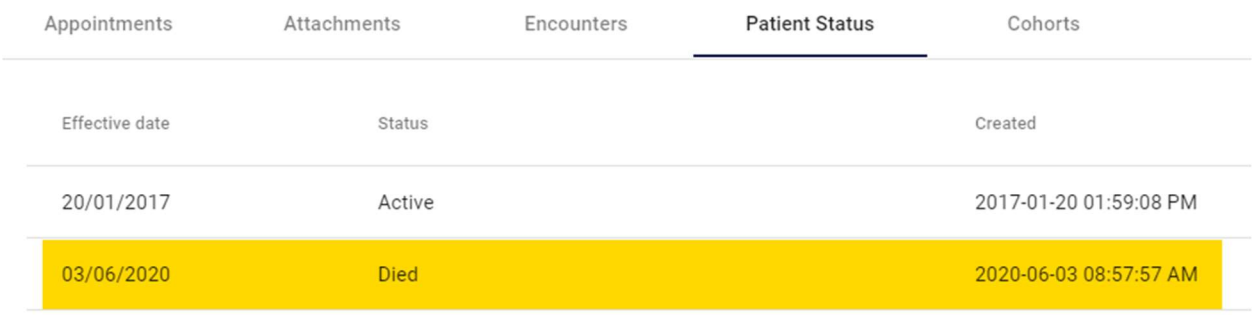
In either the patient or encounter view for the patient, navigate to the **Clinical Information** tab. Within the **Condition** tab, find the condition to edit in the table and click the **Edit Condition** icon after which the system will display the **Edit Condition** pop up form.



Add information or make changes to the page **Source Description, MedDra Term, Start Date**, **Outcome Date**, **Condition Outcome**, **Treatment Outcome**, **Comments**, or **Condition Ongoing** fields as appropriate.

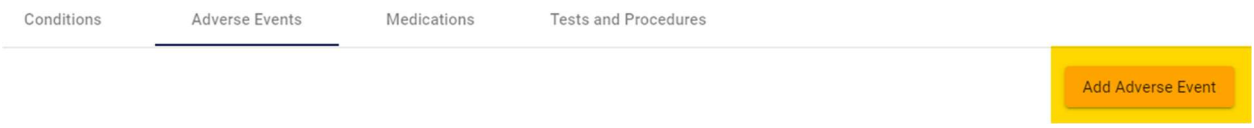
NOTE: The system will not allow you to change the Patient's Condition **MedDRA Term**. To change the **MedDRA Term** you will need to delete the record and enter the Patient Condition as a new entry.

Click the Save button or click the Cancel button to cancel the action and return to the previous page. If the patient's outcome is fatal, the patient's status will be updated to deceased.

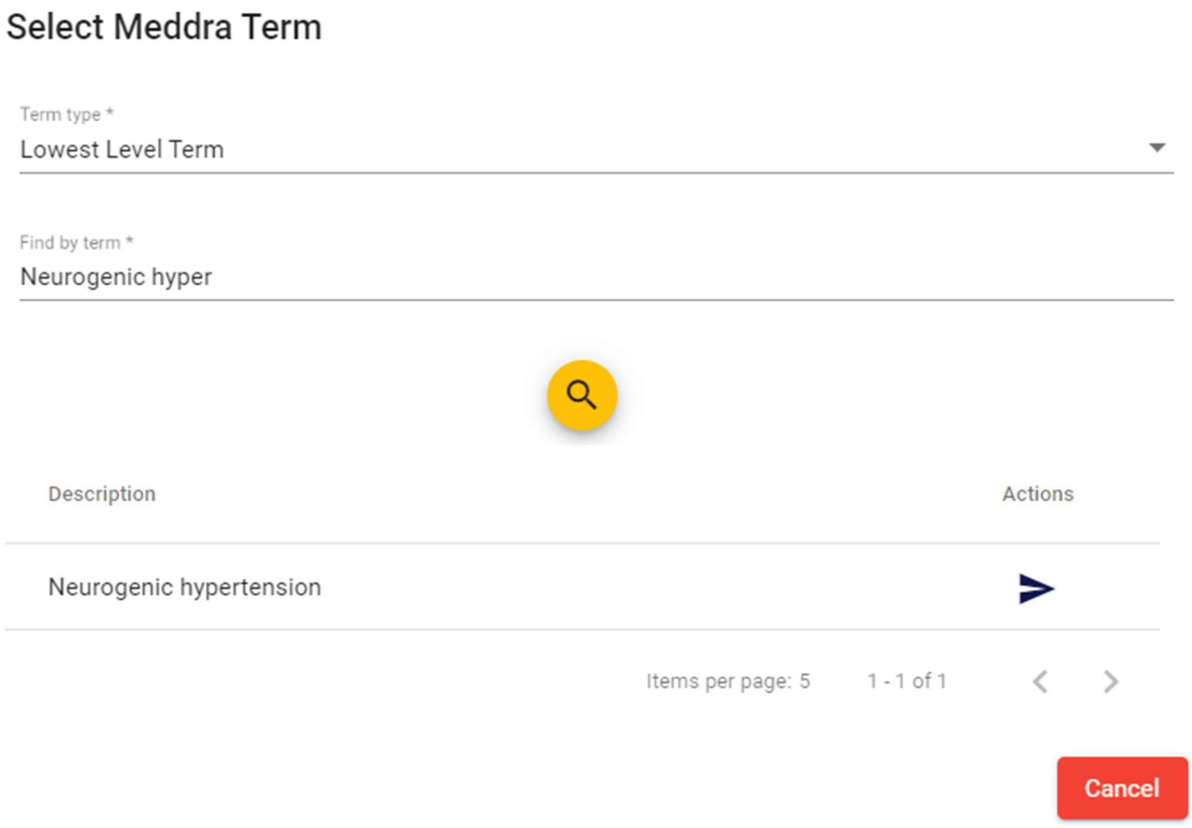


4.2.4.3 Add Adverse Event

At the **Adverse Events** tab click the **Add Adverse Event** button, after which the system will display an **Add Adverse Event** pop up form.

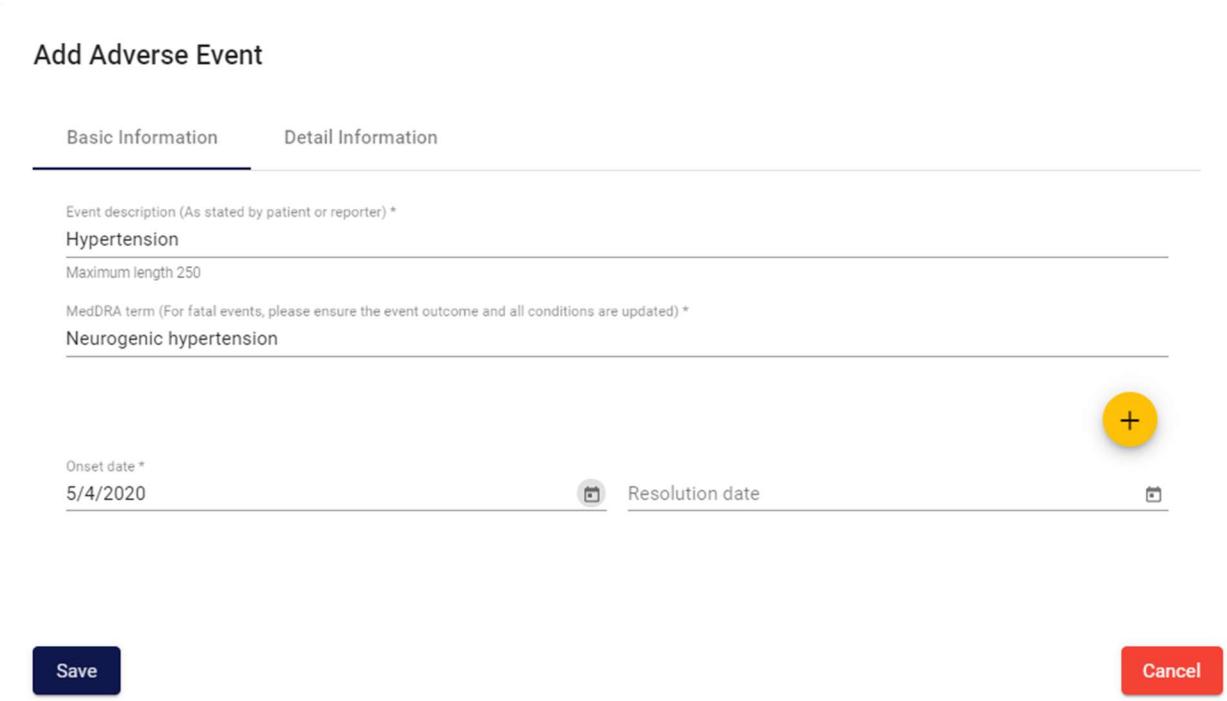


- Select the MedDra term for the adverse event by clicking on the plus icon.
- Select the term type for the term and enter the adverse event in the **Find by term** field. Click the **Search** button (magnifying glass) and the system with then provide a list of results from the MedDRA dictionary that match the term that you have searched on.

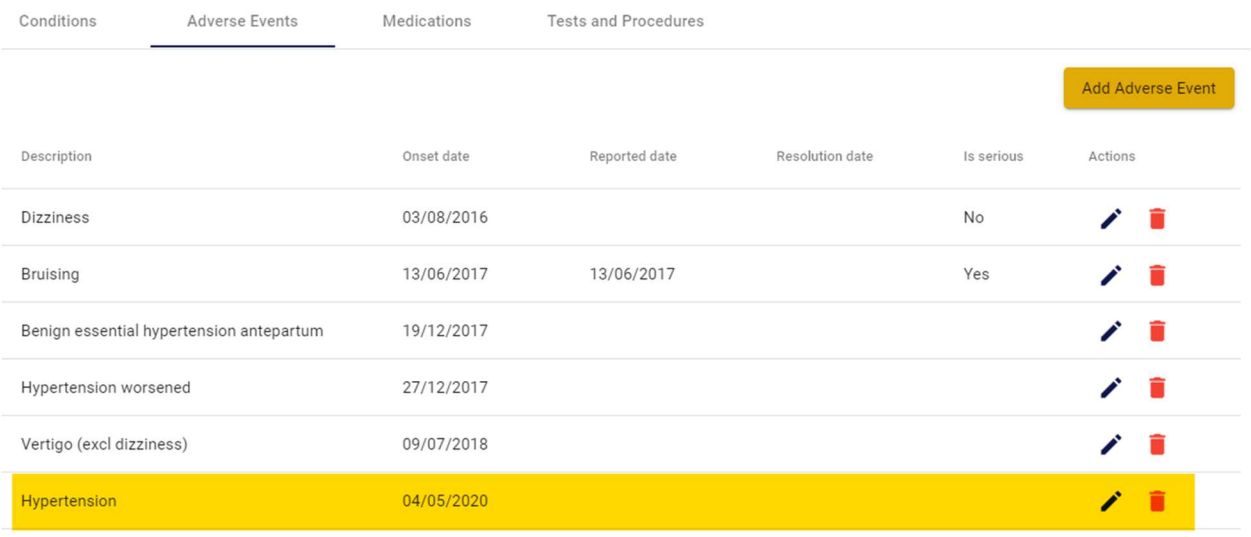


- Click the right arrow for the corresponding term that you would like to select.
- Enter a free format description of the adverse event as defined by the reporter or patient.
- Enter the **Onset Date** and complete the remaining fields as appropriate.

- Click the **Save** button to create the adverse event or click the **Cancel** button to cancel the action and go back to the previous page.



- After clicking on **Save** the system will display the updated the table under the **Adverse Events** tab.



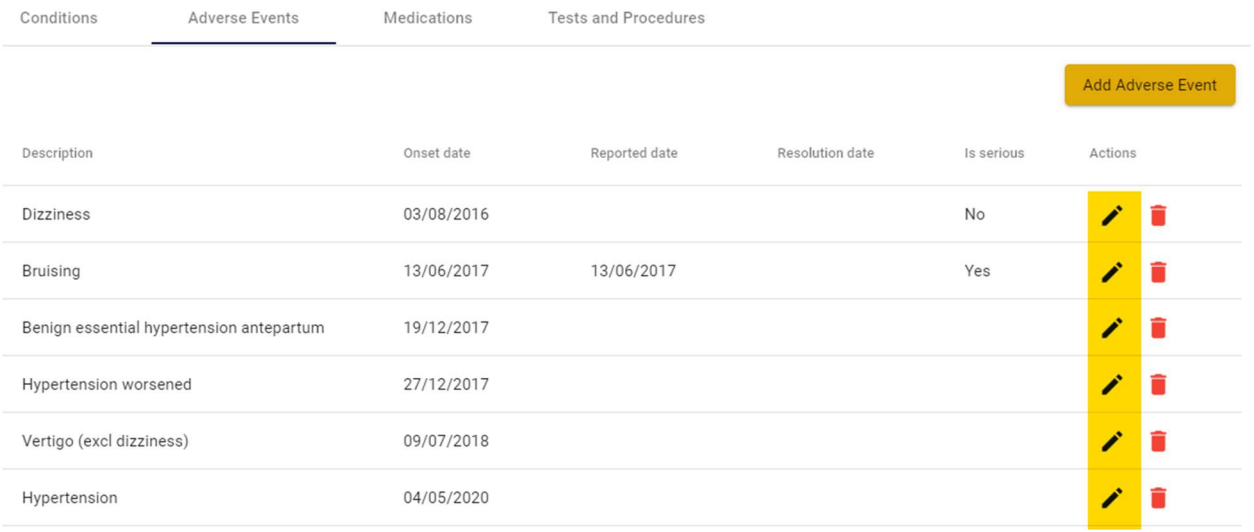
The fields on the **Add Adverse Event** form are described below:

Term type	Dropdown list of MedDRA term hierarchy; Lowest level term, Preferred term, High level term, High level group term, or System organ class
Find by term	Text field; Enter name of adverse event
Term results	System generated list; Select appropriate term from list
Event description (As stated by patient or reporter)	Text field; enter event term as stated in the medical records
Onset date	Date field; Enter date condition started
Resolution date	Date field; Enter condition outcome date
Intensity (Severity)	Dropdown list; Select from Mild, Moderate, or Severe
Treatment of Reaction	Dropdown list; Select from No Treatment, Non-Medical Treatment, Medical Treatment, Dialysis, Surgery, or Unknown
Was the AE attributed to one or more drugs?	Dropdown list; Select from Yes, No, or Unknown
Expected or Unexpected AE	Dropdown list; Select from Expected or Unexpected
Outcome	Dropdown list of Outcomes; Select either Fatal, Not Recovered/Not Resolved, Recovered/Resolved, Recovered/Recovered with Sequelae, Recovering/Resolving, or Unknown
Was the event reported to national PV?	Dropdown list; Select from Yes, No, or Unknown
Is the adverse event serious?	Dropdown list; Select from Yes, No, or Unknown
Seriousness	Dropdown list; Select from Congenital Anomaly or Birth Defect, Persistent or Significant Disability or Incapacity, Death, Initial or Prolonged Hospitalization, Life-threatening, or a Medically Important event
Admission Date (will only appear if Hospitalized)	Date field; Enter date patient was admitted
Discharge Date	Date field; Enter date patient was discharged

(will only appear if Hospitalized)	
Date of Death (will only appear if reason for Seriousness is Death)	Date field; Enter date patient died
Autopsy Done? (will only appear if reason for Seriousness is Death)	Dropdown list; Select from Yes or No
Severity Grade	Dropdown list; Select from Grade 1, Grade 2, Grade 3, Grade 4, or Grade 5
Severity Grading Scale	Dropdown list; Select the SAE Grading Reference (e.g., DAIDS, CTCAE)
Full Name of Reporter	Text field; Enter name of the person who reported the event
Date of Report	Date field; enter the date the event was first reported by the facility
Type of Reporter	Dropdown list; Select from Physician, Pharmacist, Other Health Professional, Lawyer, Consumer or Other Non-Health Professional
Reporter Contact Number	Text field; Enter a contact number for the reporter
FDA SAE Number (For use only by FDA officers)	Text field; Enter the SAE file number assigned by the FDA

4.2.4.4 Edit an Adverse Event

In either the patient or encounter view for the patient, navigate to the **Clinical Information** tab. Within the **Adverse Event** tab, find the adverse event you would like to edit within the table and click the **Update Adverse Event** icon. The system will display the **Update Adverse Event** pop up form.



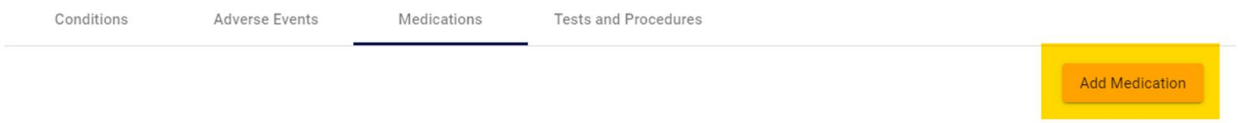
NOTE: The system will not allow you to change the Patient's Adverse Event **MedDRA Term**. To change the **MedDRA Term** you will need to delete the record and enter the Patient Adverse Event as a new entry.

Add information or make changes to the page as appropriate.

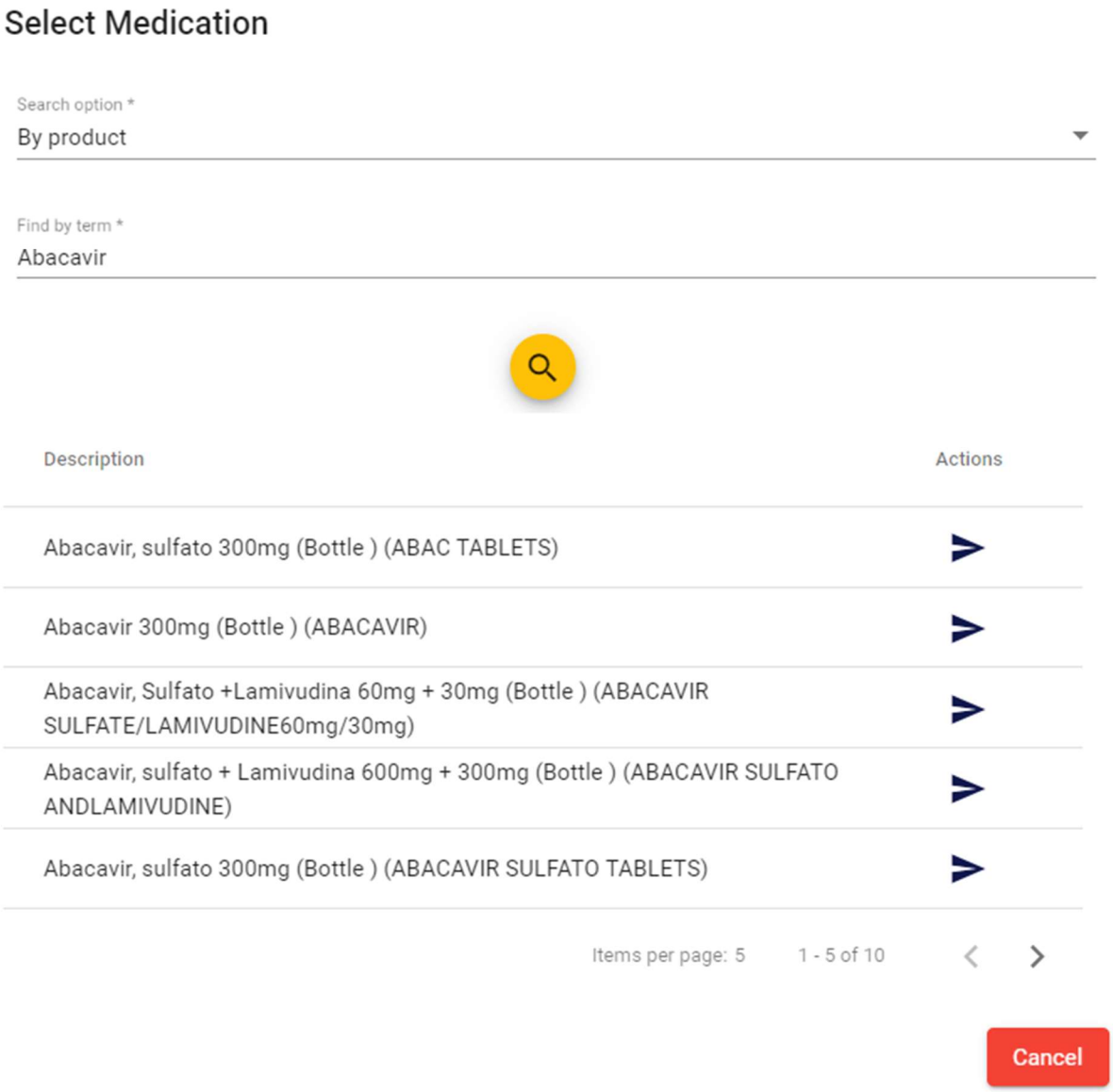
Click the Save button or click the Cancel button to cancel the action and return to the previous page

4.2.4.5 Add a Patient Medication

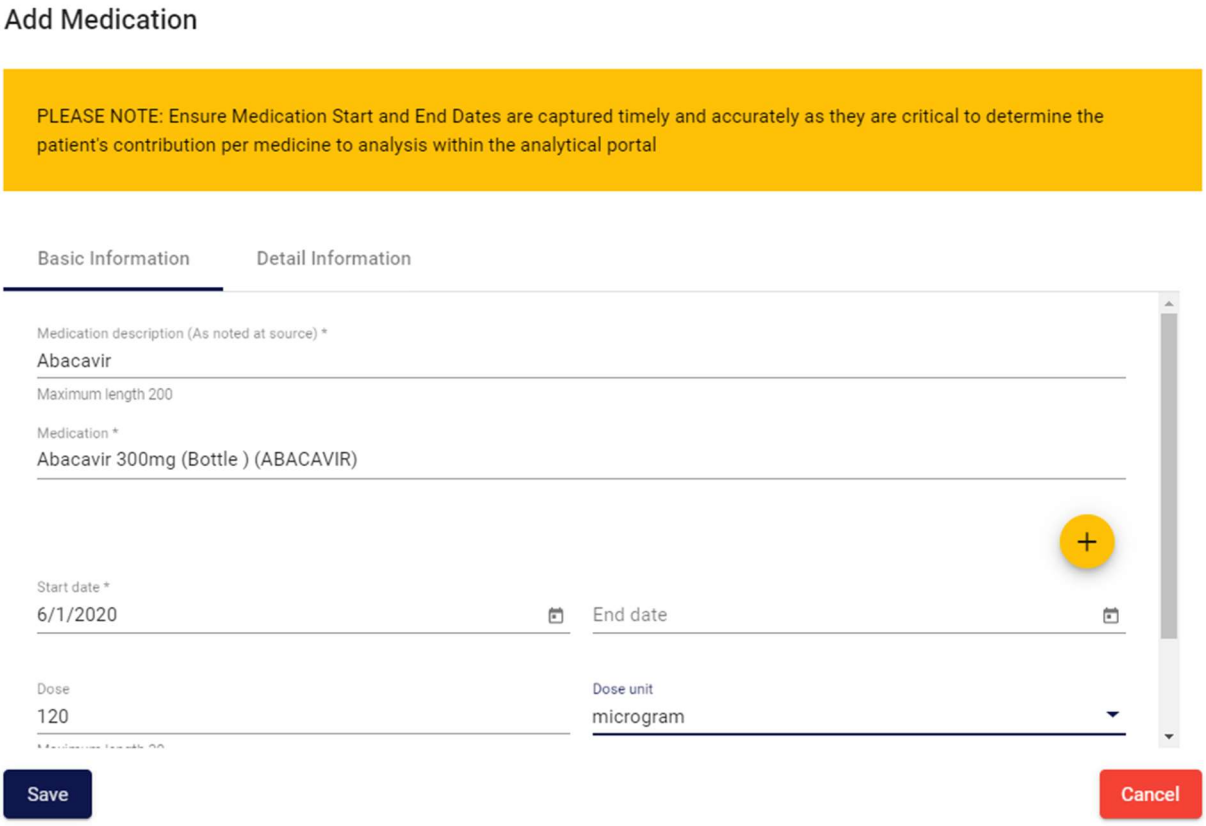
At the **Medications** tab click the **Add Medication** button, after which the system will display an **Add Medication** pop up form.



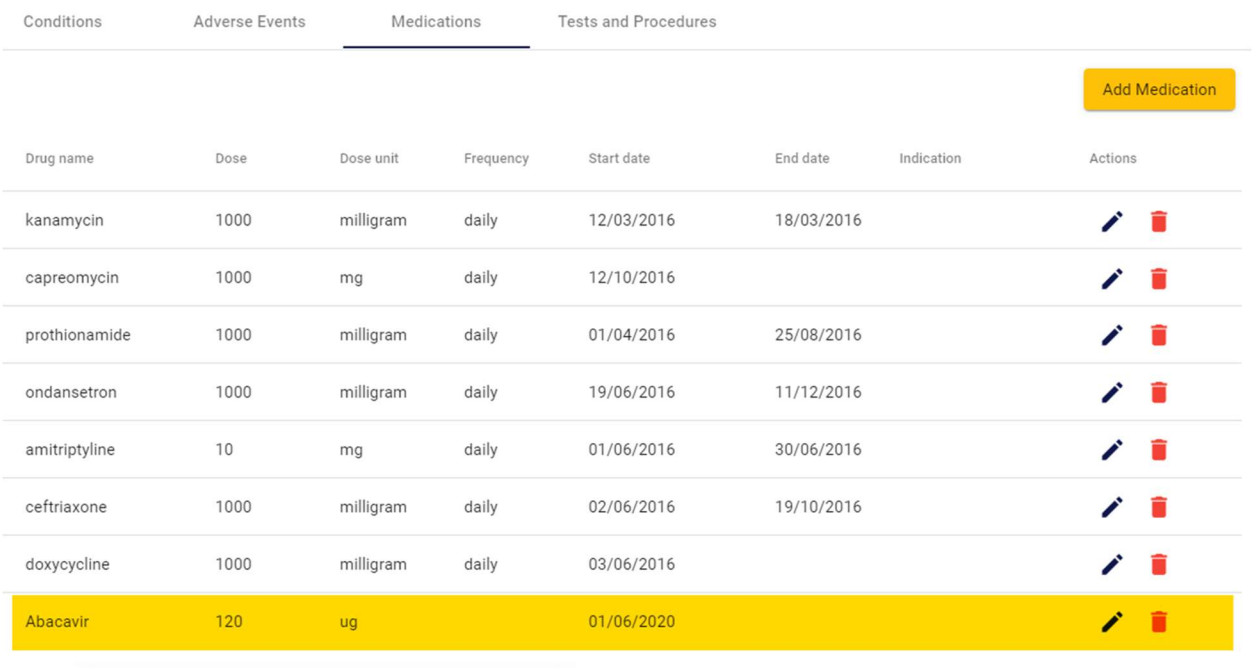
- Enter a free format description of the medication as defined by the reporter or patient.
- Select the medication by clicking on the plus icon.
- Select the option you would like to use for searching for the medication. You can either search by active ingredient or the product name
- Enter the active ingredient or product name in the **Find by term** field. Click the **Search** button (magnifying glass) and the system with then provide a list of results from the list of concepts that match the term that you have searched on.



- Click the right arrow for the corresponding medication that you would like to select.
- Enter the **Start Date** and complete the remaining fields as appropriate.
- Click the **Save** button to create the medication or click the **Cancel** button to cancel the action and go back to the previous page.



- After clicking on **Save** the system will display the updated the table under the **Medications** tab.

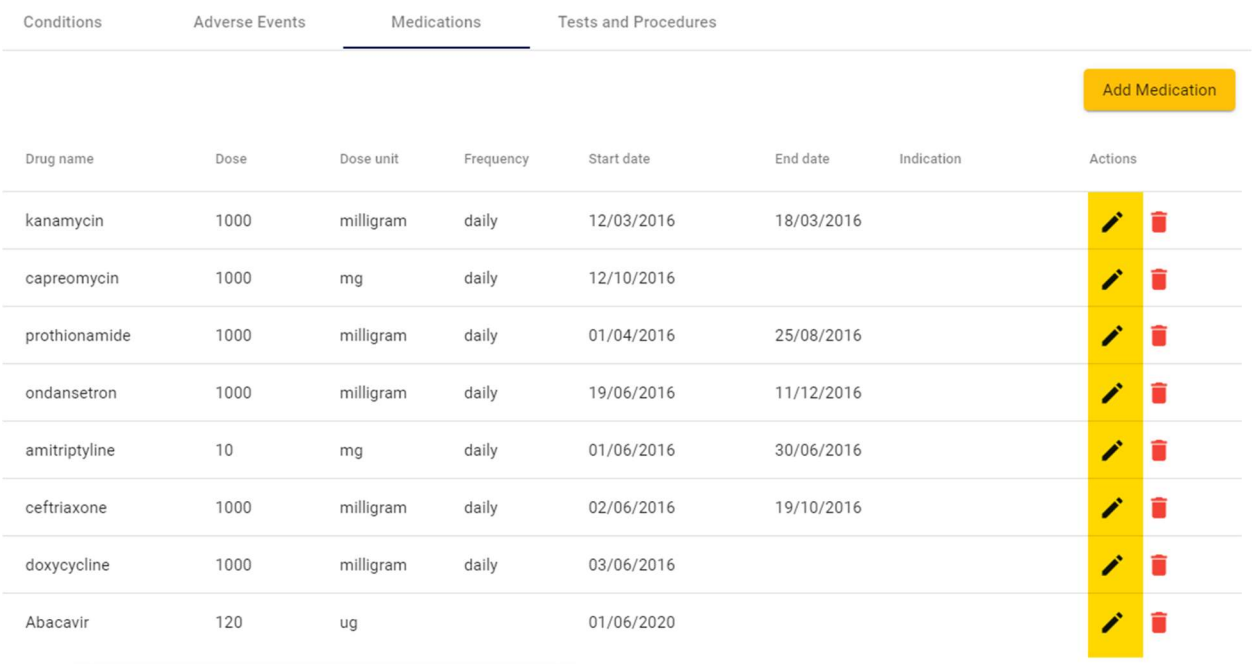


The fields on the **Add Medication** form are described below:

Medication description (As stated by patient or reporter)	Text field; enter medicine term as stated in the medical records
Find by term	Text field; Enter name of medication
Term results	System generated list; Select appropriate medication from list
Start Date	Text field; Enter date patient started taking the medication
End Date	Text field; Enter date patient stopped taking the medication
Dose	Text field; Enter the dose prescribed
Dose Unit	Dropdown list; Select the unit prescribed
Dose Frequency	Text field; Enter the dose frequency prescribed
Route	Dropdown list; Select the route of administration
Frequency in days per week	Dropdown list; Select number of days per week the medicine is administered
Still On Medication	Dropdown list; Select Yes or No
Indication	Text field; Enter the reason the medicine was prescribed
Type of Indication	Dropdown list; Select Primary, Pre-existing Condition, or Treat AE
Reason For Stopping	Dropdown list; Select from the list provided (e.g., Adverse Event, Cost, Course Completed)
Clinician action taken with regard to medicine if related to AE	Dropdown list; Select Dose Not Changed, Dose Reduced, Drug Interrupted, Drug Withdrawn, or Not Applicable
Batch Number	Text field; Enter the medicine Batch Number
Effect OF Dechallenge (D) & Rechallenge (R)	Dropdown list; Select from the list provided (e.g., Not Applicable, D – AE improved/resolved when medicine dose reduced/interrupted/withdrawn, R – AE Recurred on medicine re-admission/dose increase)

4.2.4.6 Edit an Existing Patient Medication

In either the patient or encounter view for the patient, navigate to the **Clinical Information** tab. Within the **Medications** tab, find the medication you would like to edit within the table and click the **Update Medication** icon. The system will display the **Update Medication** pop up form.



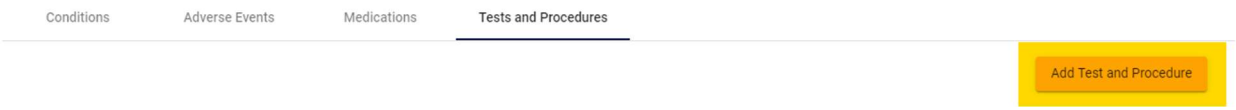
NOTE: The system will not allow you to change the **Medication** name. To change the **Medication,** you will need to delete the record and enter the **Medication** as a new entry.

Add information or make changes to the page as appropriate.

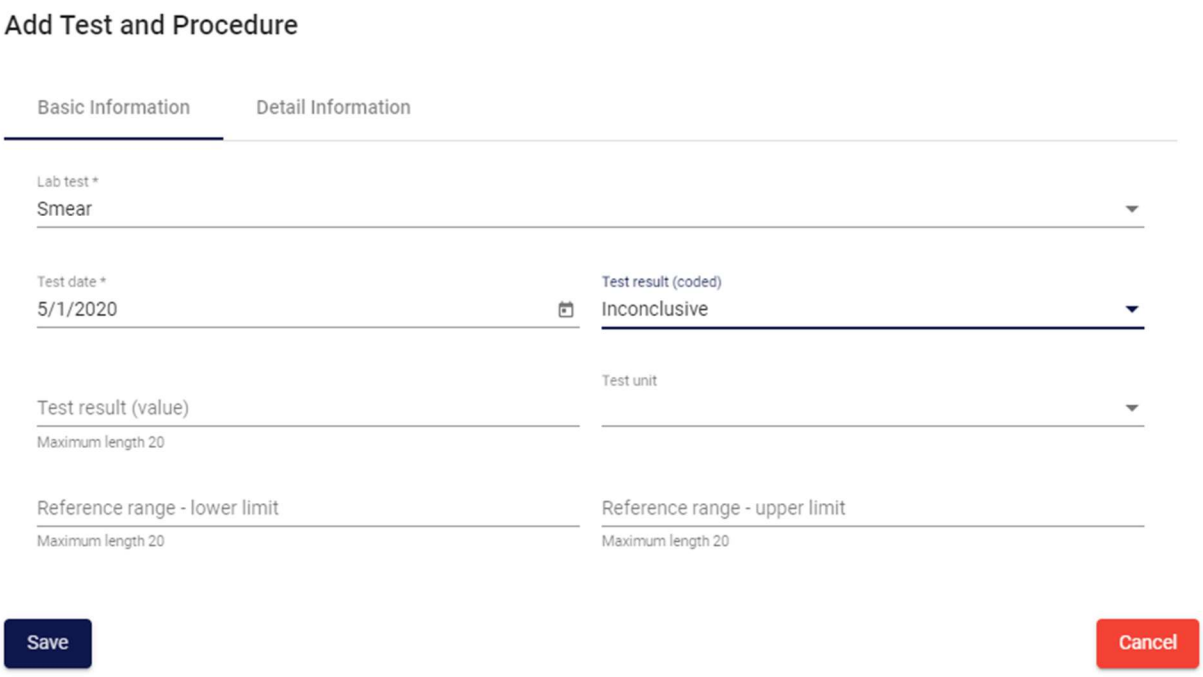
Click the Save button or click the Cancel button to cancel the action and return to the previous page.

4.2.4.7 Add a Test or Procedure

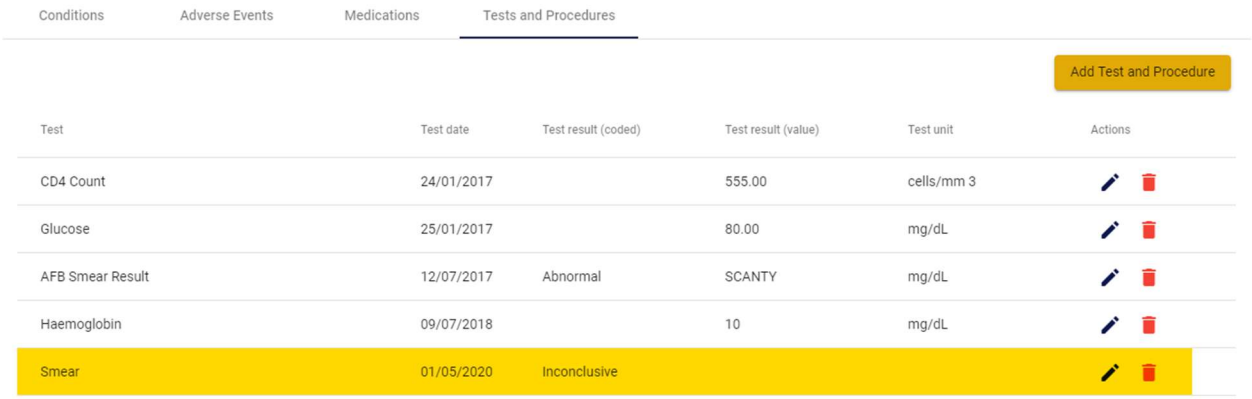
At the **Tests and Procedures** tab click the **Add Test and Procedure** button, after which the system will display an **Add Test and Procedure** pop up form.



- Select the test or procedure.
- Enter the date the patient had the test completed.
- Complete any other fields for which you have data.
- Click the **Save** button to create the test or procedure or click the **Cancel** button to cancel the action and go back to the previous page.



- After clicking on **Save** the system will display the updated the table under the **Tests and Procedures** tab.

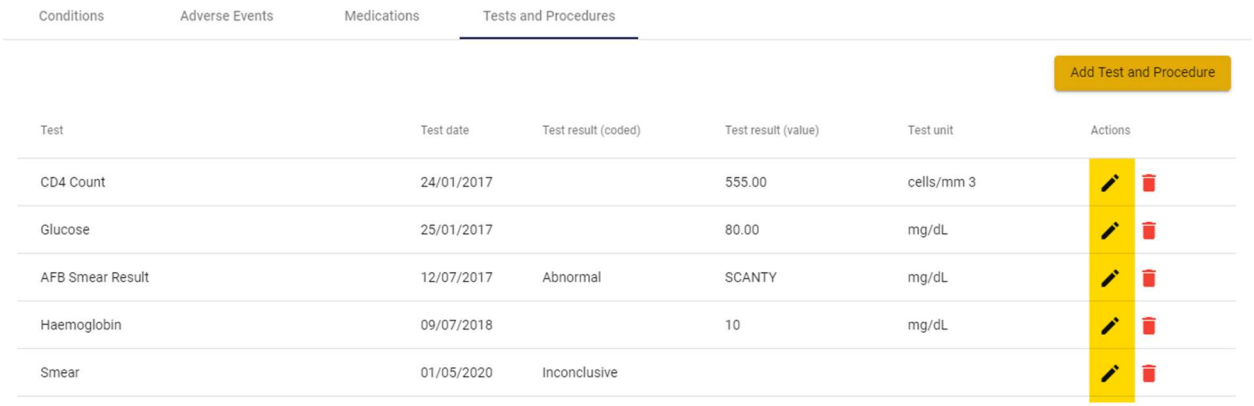


The fields on the **Add Test and Procedure** form are described below:

Lab test	Dropdown list; Select name of the Test or Procedure (e.g., Blood Glucose, Chest X-ray)
Test date	Text field (dates only); Enter date the Test or Procedure was performed
Test result (coded)	Dropdown list; Select qualitative Test or Procedure result (e.g., Positive, Negative, Normal, Abnormal) if appropriate
Test result (value)	Text field (numbers only); Enter the value for the test result
Test Unit	Dropdown list; select corresponding unit for the Test or Procedure Result (e.g., %, mg, millisecond)
Reference Range – Lower Limit	Text field (numbers only); Enter the value for the lower limit of normal defined by the laboratory
Reference Range – Upper Limit	Text field (numbers only); Enter the value for the upper limit of normal defined by the laboratory
Comments	Text field; Enter additional information about the Test or Procedure if needed

4.2.4.8 Edit an Existing Test or Procedure

In either the patient or encounter view for the patient, navigate to the **Clinical Information** tab. Within the **Tests and Procedures** tab, find the test or procedure you would like to edit within the table and click the **Update Test and Procedure** icon. The system will display the **Update Test and Procedure** pop up form.



NOTE: The system will not allow you to change the **Test** or **Procedure** name. To change the **Test** or **Procedure** name you will need to delete the record and enter the **Test** or **Procedure** as a new entry.

Add information or make changes to the page as appropriate.

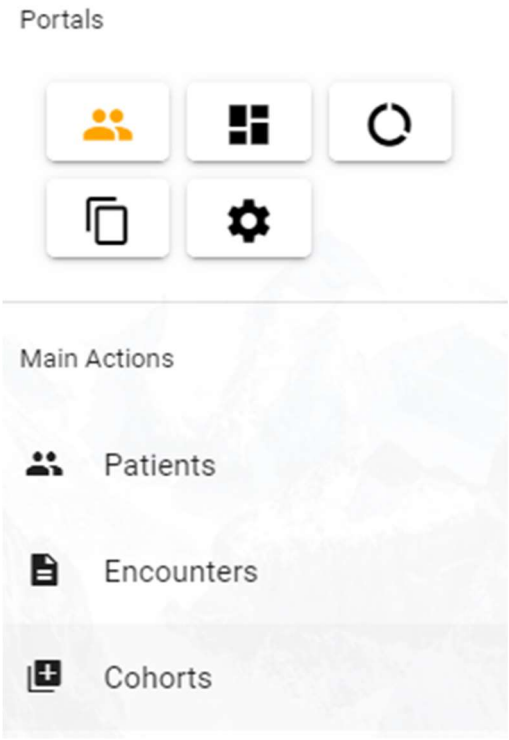
Click the Save button or click the Cancel button to cancel the action and return to the previous page.

4.3 Cohorts

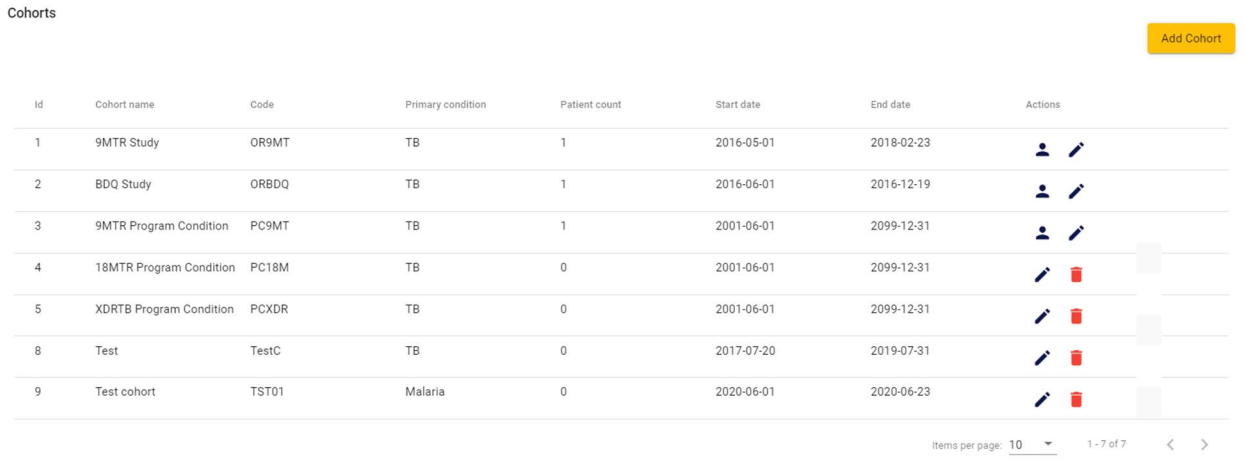
Cohorts can be used to track a sub-group of patients within a Condition Group. Cohorts in the system are determined by the Public Health Program or the System Administrator.

4.3.1 View Cohorts

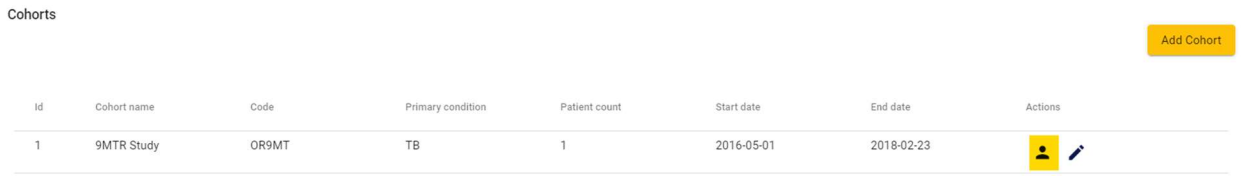
The **Cohort** function can be accessed through the **Cohorts** menu.



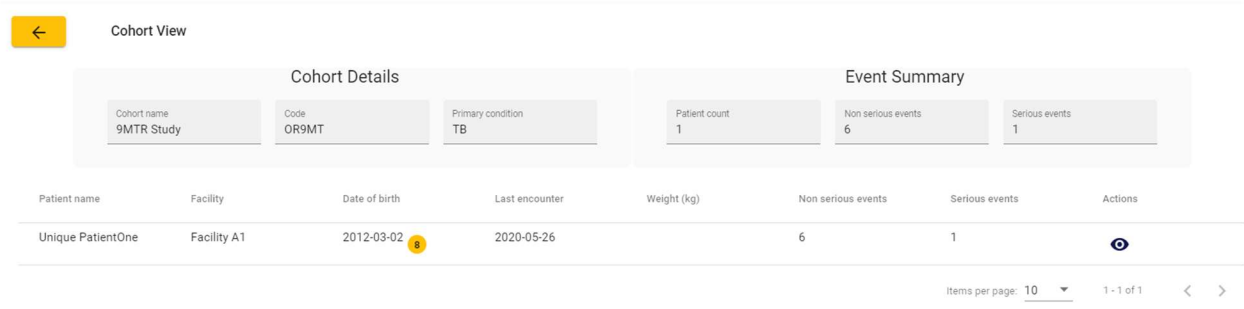
The system will display the **Cohorts** page, which lists all currently registered cohorts in the system.



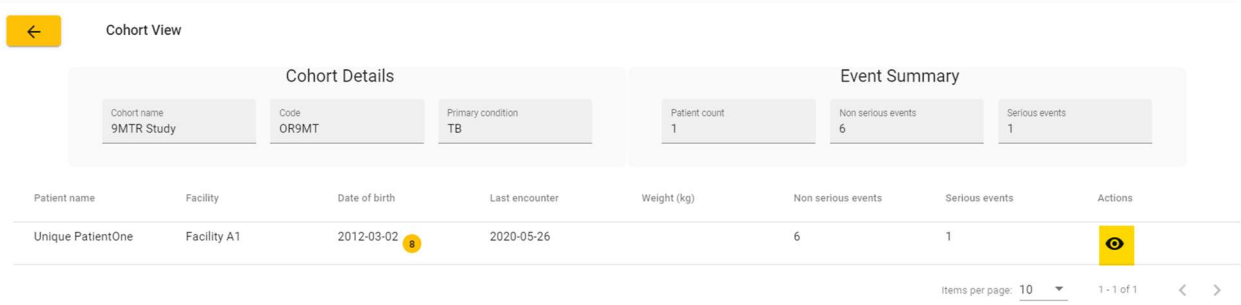
Find the **Cohort** you would like to view in the table. Click the **View Cohort** icon in the Action Column for the cohort to view.



The system will display the **Cohort View** page with a table listing all of the patients enrolled in that cohort.



From the **Cohort View** page, you are able to view a patient enrolled in the cohort. Find the patient to view in the table.



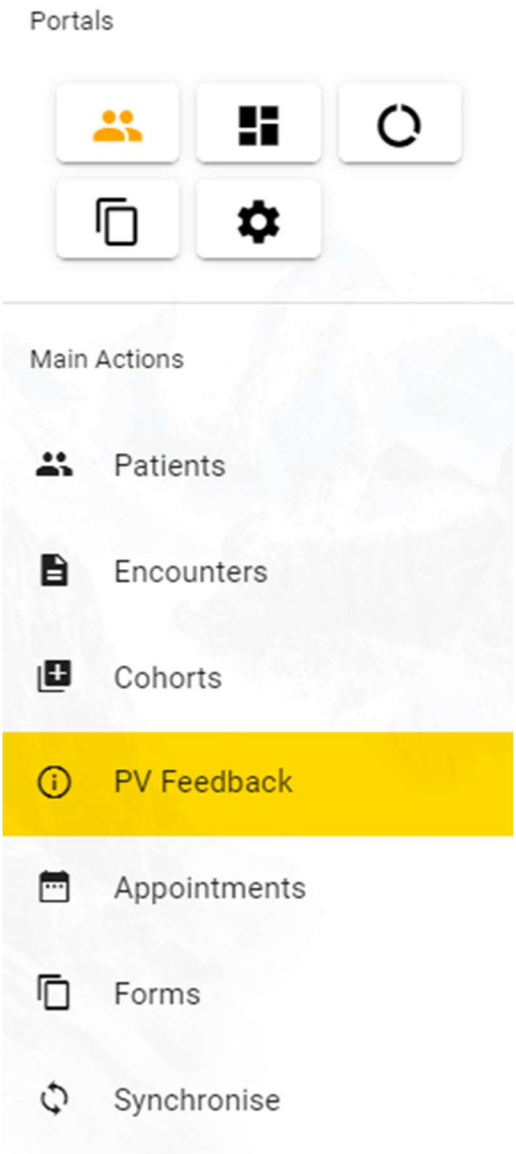
Click the **View Patient** button in the Action column for the patient to view. The system will display the **Patient View** page.

4.4 Feedback

Feedback can be used to track all analysis concluded by the Pharmacovigilance technician for all adverse events logged within the clinical portal. Feedback is generated by the system as causality and terminology has been set by the technician.

4.4.1 View Feedback

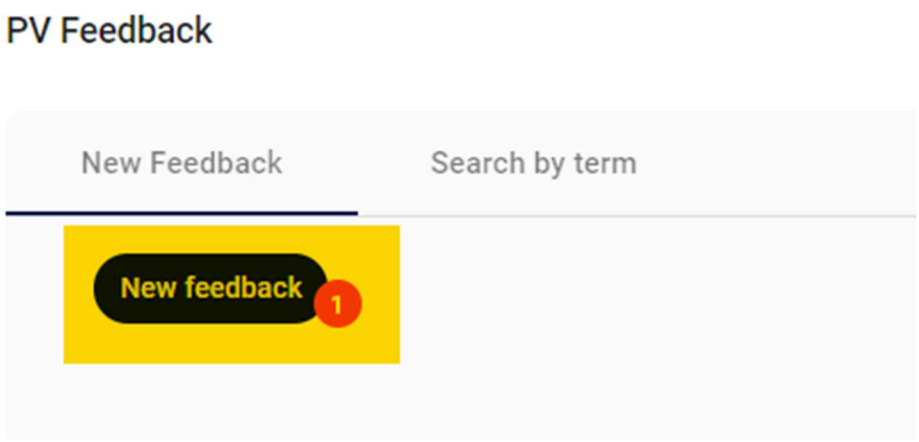
The **Feedback** function can be accessed through the **PV Feedback** menu.



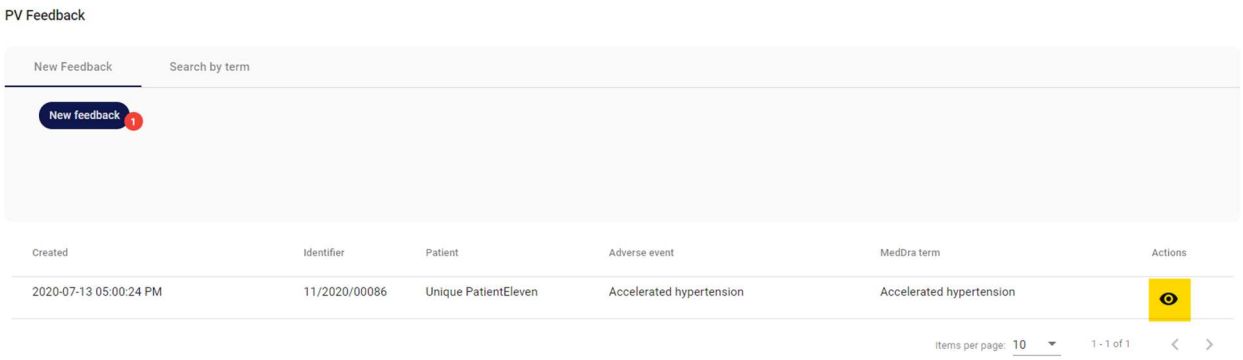
The system will display the **feedback** page, which provides the user with the ability to view feedback via a new and search by term tab.

4.4.2 View New Feedback

The **New feedback** tab represents all new feedback that has been generated by the technician within a recent time period. This period is administrable by your system administrator within the administrator portal. The number in the red circle represents the number of new reports that have recent feedback.

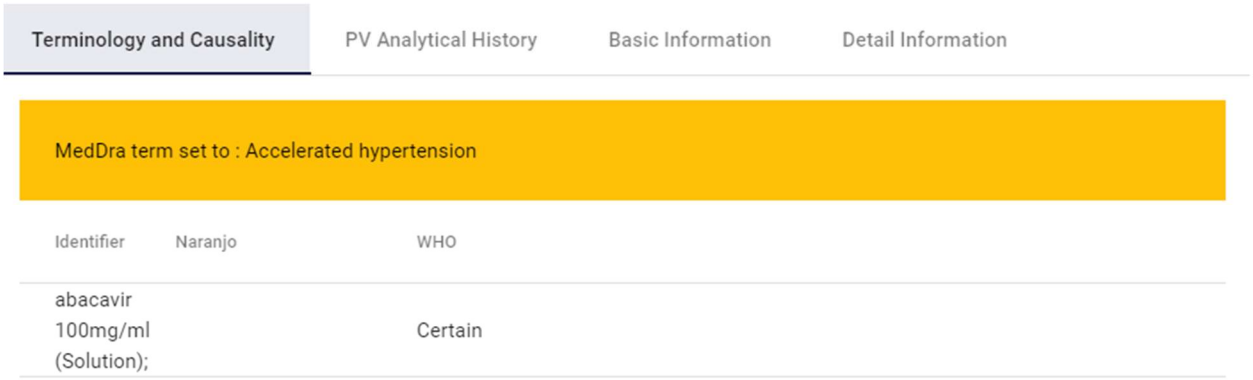


To view feedback for a report, click on the visibility icon for the report you would like to view feedback for.

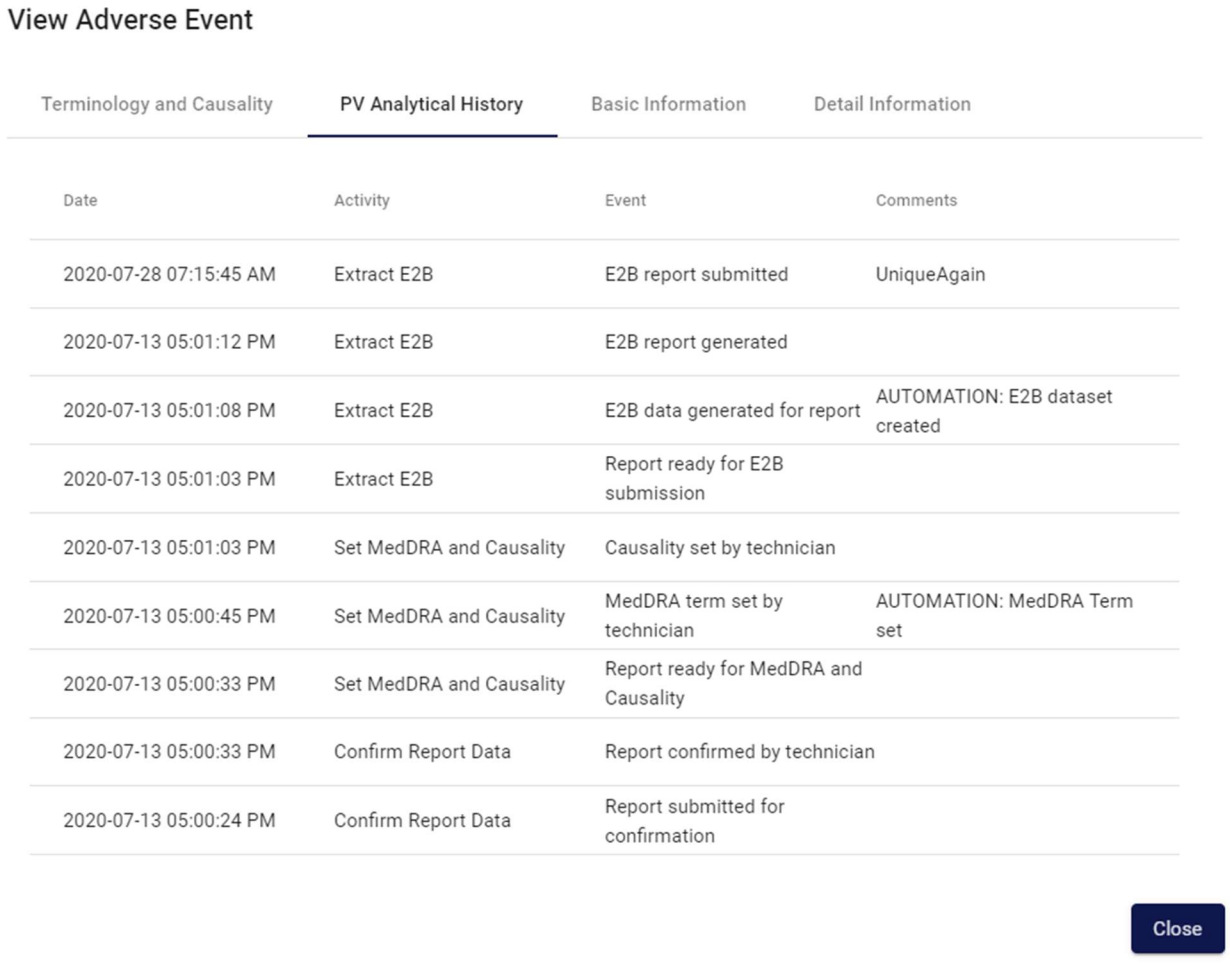


The system will navigate the user to a **View Adverse Event** pop up form where the user will be able to view the following feedback:

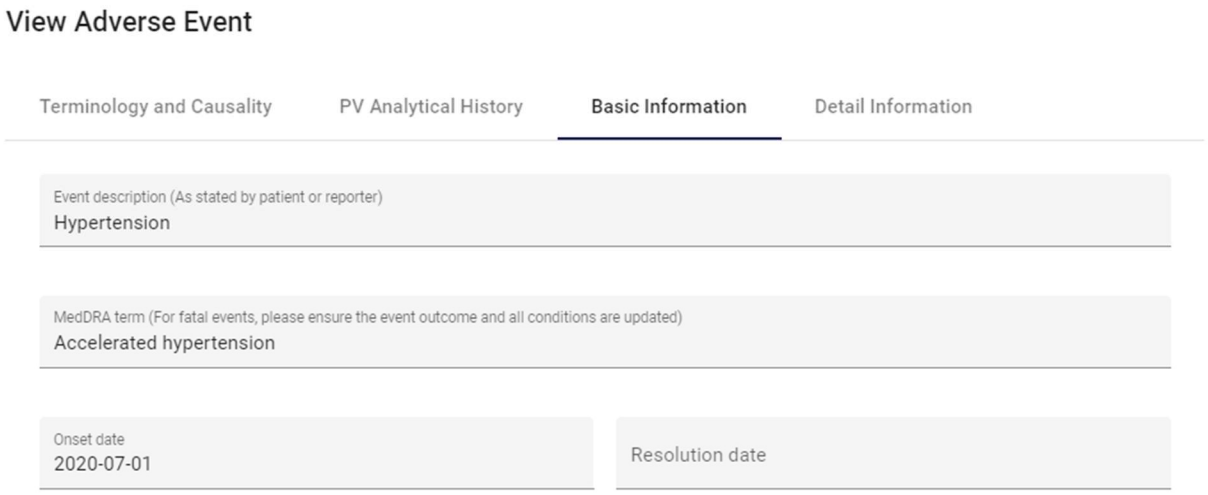
- **Terminology and Causality.** Ability to view MedDRA term and causality per medication as set by the technician



- **PV Analytical History**. Ability to view a full history of all activity by the technician



- **Basic and Detailed Information.** Adverse event information captured in the clinical portal

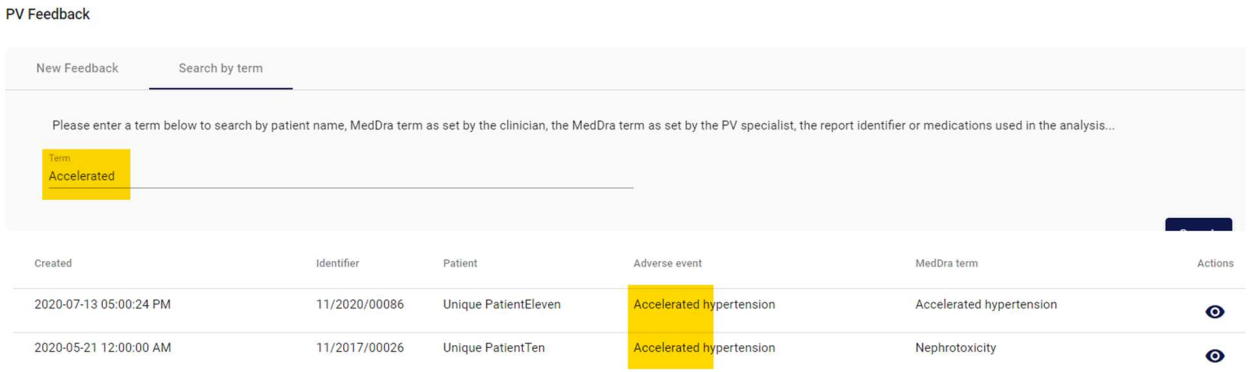


4.4.3 Search for Feedback for Report

The **Search by term** tab allows the user to search for feedback for a report using the following criteria:

- Patient name
- MedDRA term as set by the clinician and technician
- Report identifier
- Medications

The search function uses a partial term to search all reports on the terms noted above. For instance, searching for the term Accelerated returns all reports that have that partial term in the MedDRA term for the report.

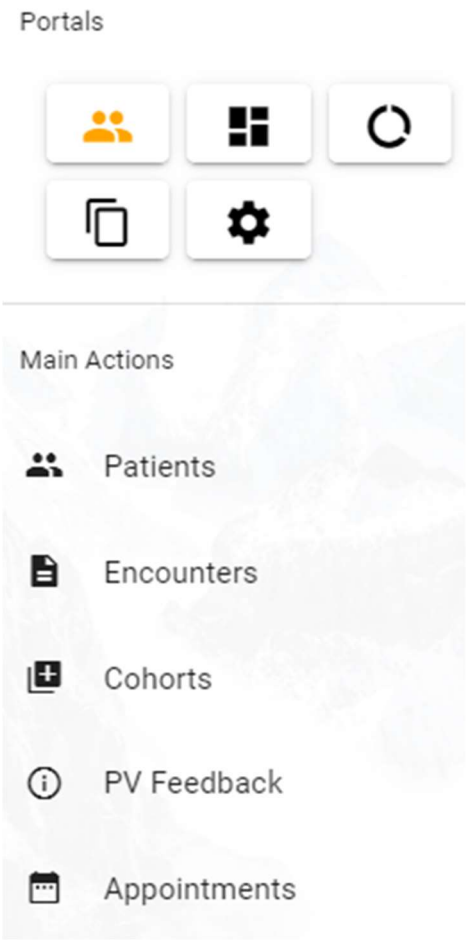


4.5 Appointments

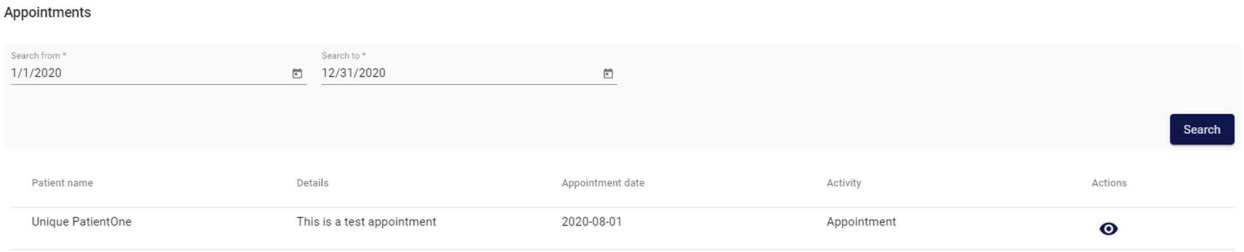
Appointments can be used to monitor an patient return date to the facility. This is particularly effective for patients in a cohort where ongoing monitoring for adverse events is expected.

4.5.1 View Appointments

The **Appointments** function can be accessed through the **Appointments** menu.



The system will display the **Appointments** page, which lists all appointments registered within the system for a given date range.



The columns in the appointments table are described below:

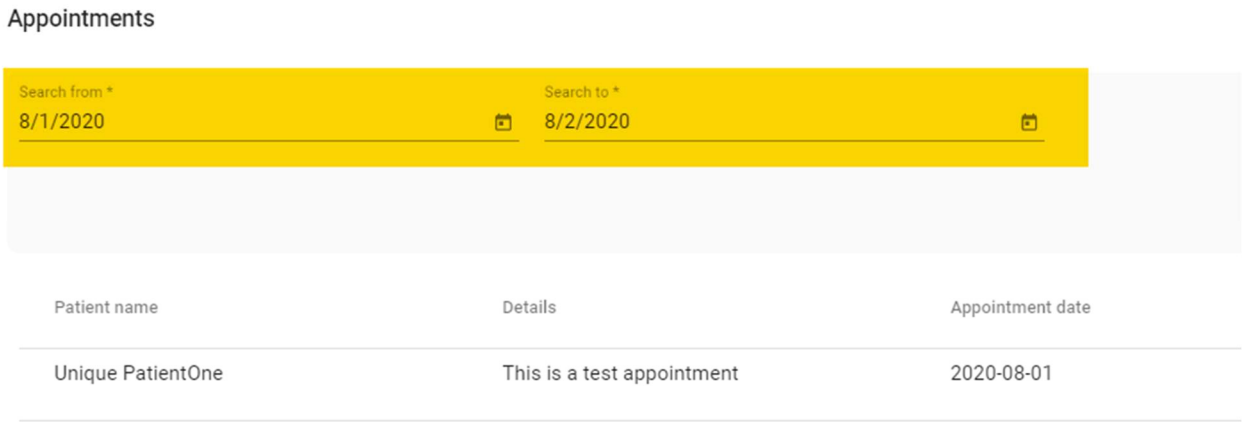
Patient name	Reflecting the patient's name and surname as captured in the system when the appointment was scheduled.
Details	Reason for the appointment
Appointment date	The date of the appointment
Activity	Has the patient arrived for their appointment? Has the patient missed their appointment?
Action	Ability to view patient Ability to view encounter Ability to mark the appointment as Did Not Arrive

PViMS Term - Did Not Arrive

If an appointment is marked as **Did Not Arrive**, this means the patient has been confirmed as missing their appointment. This status serves to confirm this scenario in situations where encounters are retrospectively captured into the system in a delayed data capture mode.

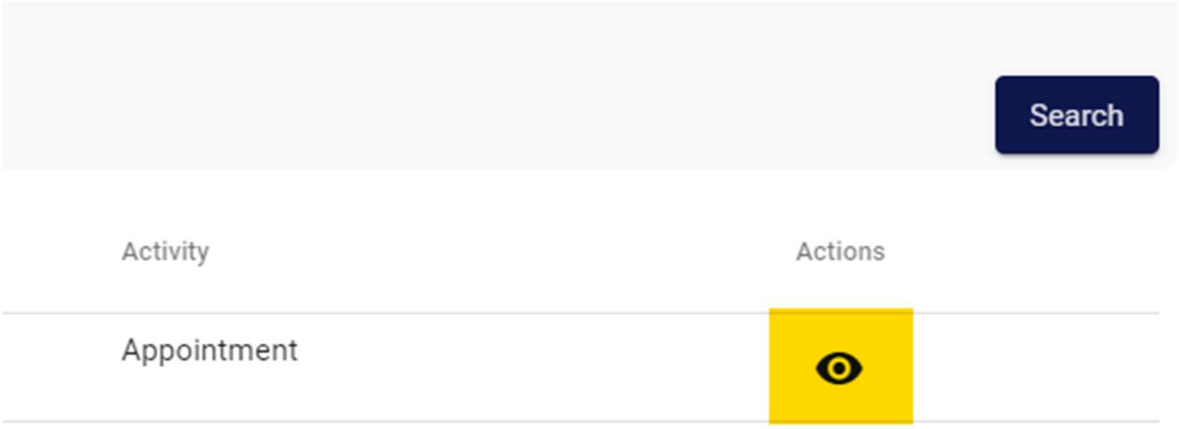
4.5.2 View Appointments for a specified day

Enter the specified day in the **Show Appointments For** field and click the **Search** button. The system will display the **Appointments** page for the specified day.



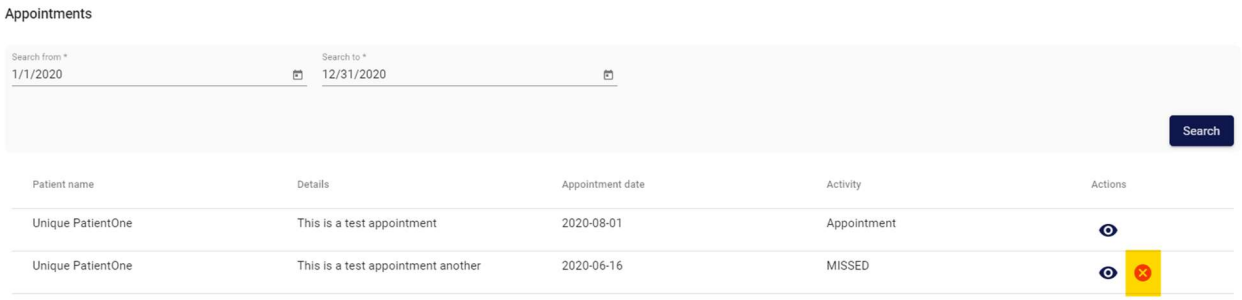
4.5.3 View Patient Record

Select the patient in the table whose record you wish to view and click the **View Patient** icon. The system will display the **Patient View** page for the selected patient.

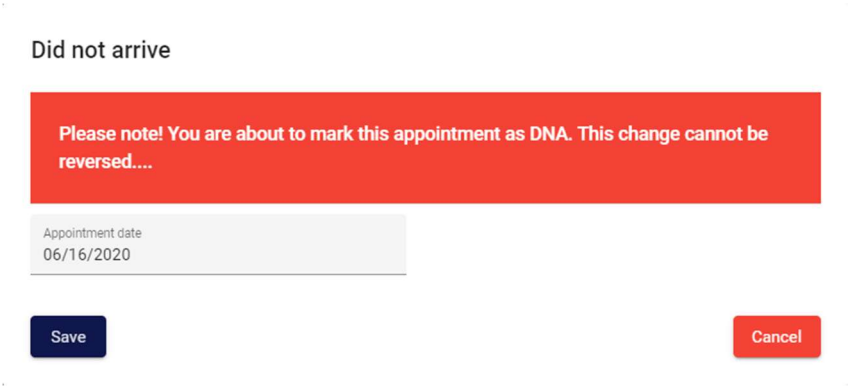


4.5.4 **Mark Appointment as Did Not Arrive**

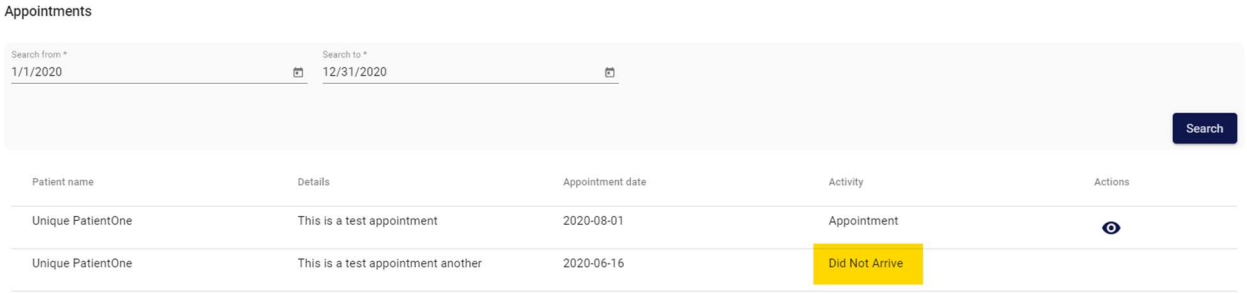
Select the patient in the table whose record you wish to confirm as did not arrive and click the **Did Not Arrive** icon.



Click the **Did Not Arrive** icon to confirm that the patient did not arrive for their visit.



The system will update the appointment with the update reflected in the appointments table.



You will not be able to mark an appointment as DNA until at least 3 days have passed from the original appointment date.

4.6 Deleting Records

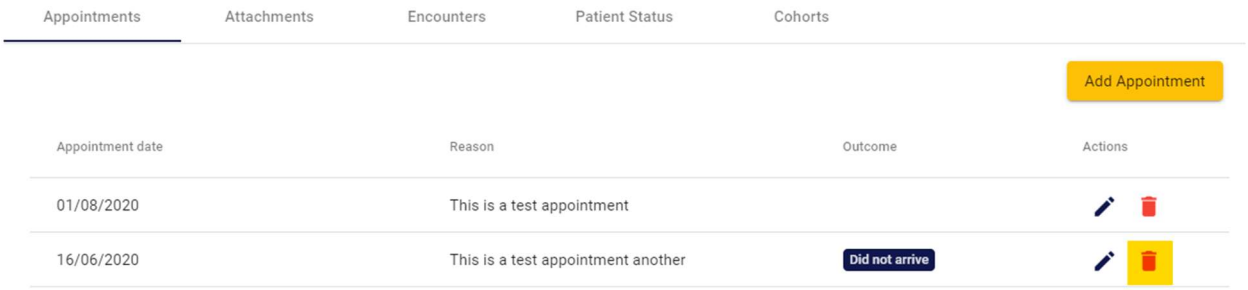
User rights assignment policies determine which Users or User Profile groups are able to delete records from the system. Check with your administrator regarding user right assignments.

After deletion the record is placed in an archive, thus not permanently deleted from the system.

4.6.1 Patient View - Additional Information

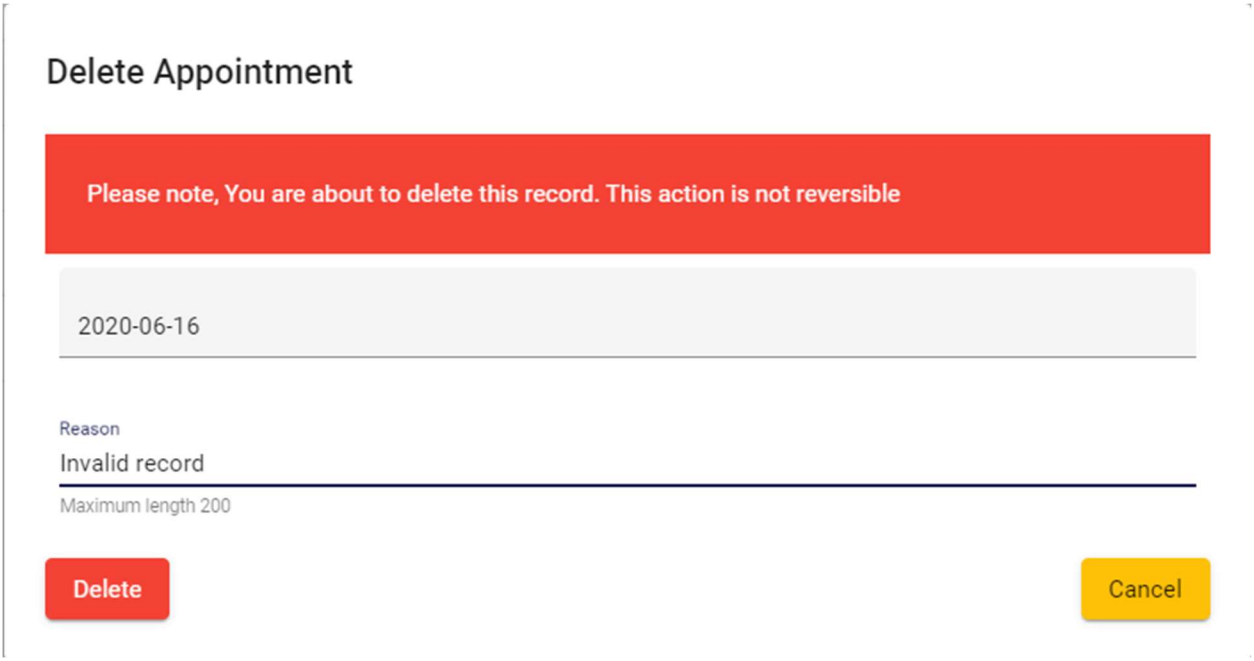
4.6.1.1 Delete an Appointment

Within the **Patient View,** at the **Appointments** tab, find the appointment you would like to delete.



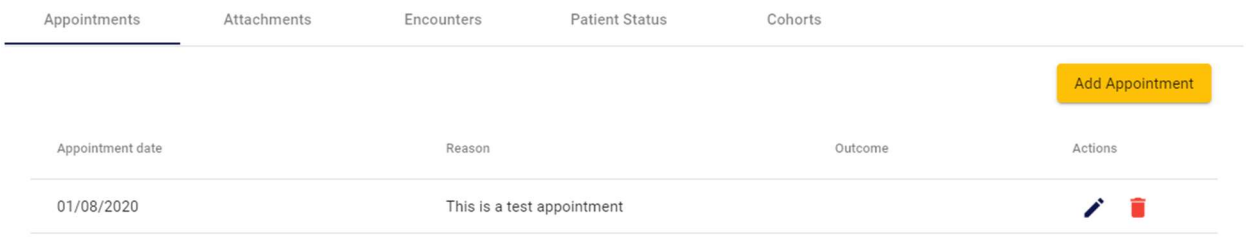
You will only have the opportunity to delete appointments within the future. Existing appointments will need to be cancelled.

Click the **Delete Appointment** icon after which the system will take you to a **Delete Appointment** pop up form.



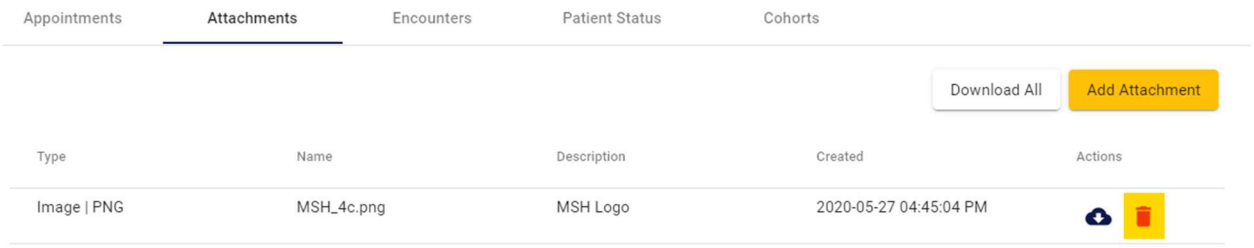
Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the appointment or click the **Cancel** button to undo the action.

After confirming the deletion, the system will update the **Appointment** table.

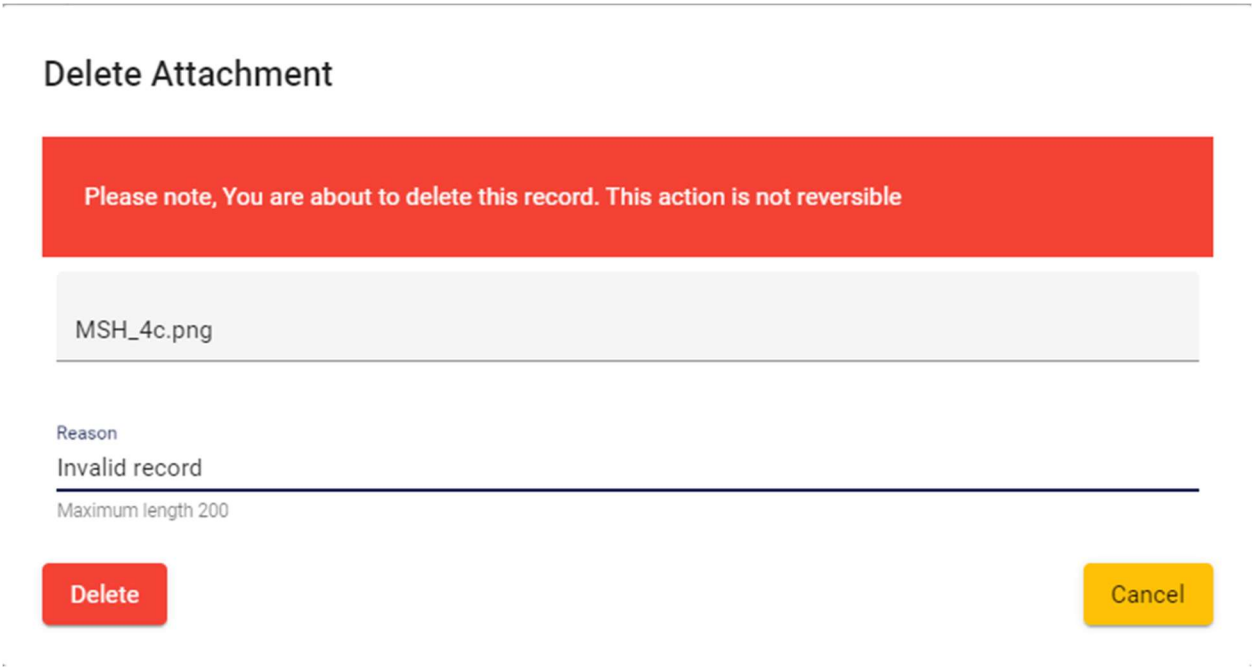


4.6.1.2 Delete an attachment

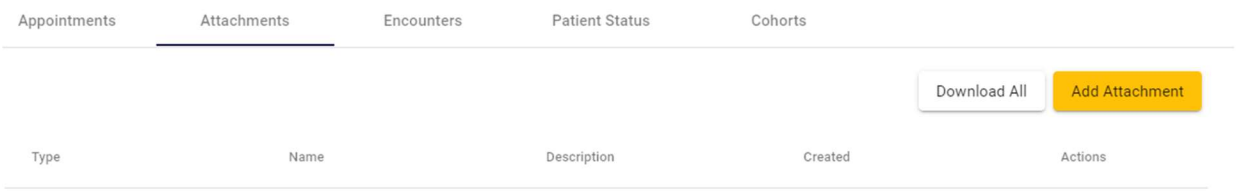
Within the **Patient View,** at the **Attachments** tab, once you have uploaded an attachment, the attachment can easily be removed again.



Click the **Delete Attachment** icon after which the system will take you to a **Delete Attachment** pop up form.

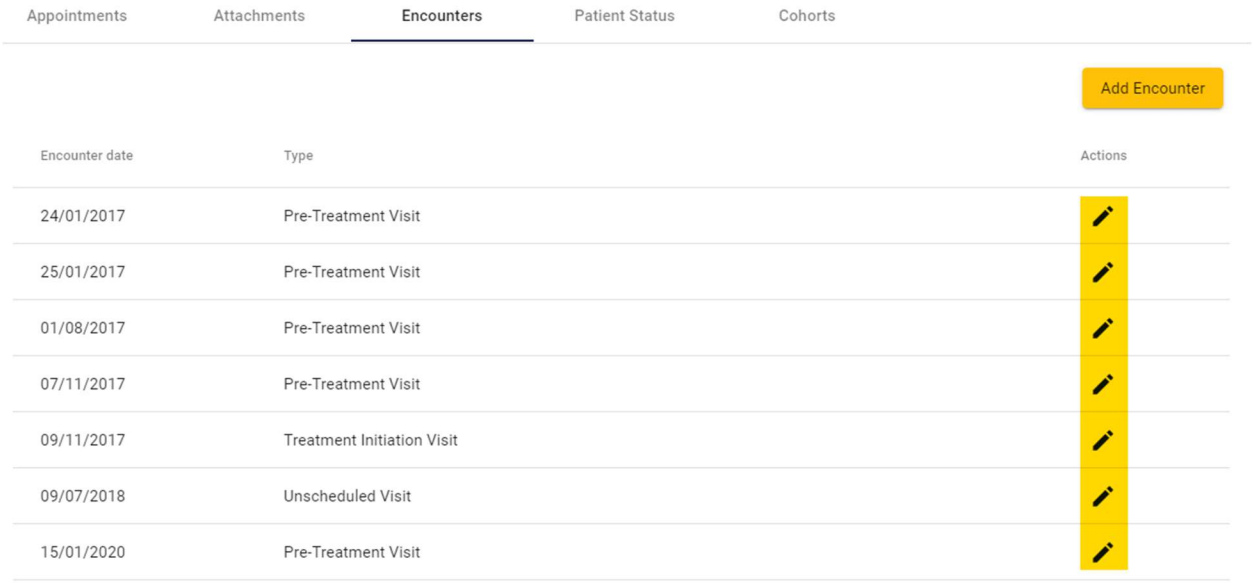


Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the attachment or click the **Cancel** button to undo the action. After confirming the deletion, the system will update the **Attachment** table.

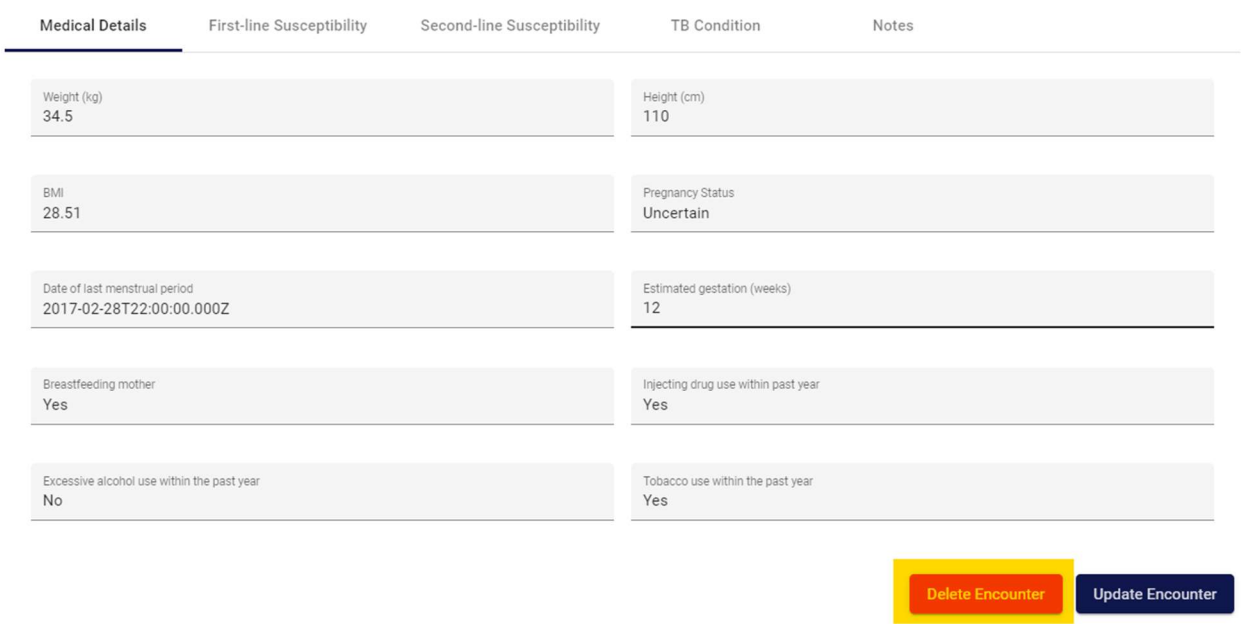


4.6.1.3 **Delete an Encounter**

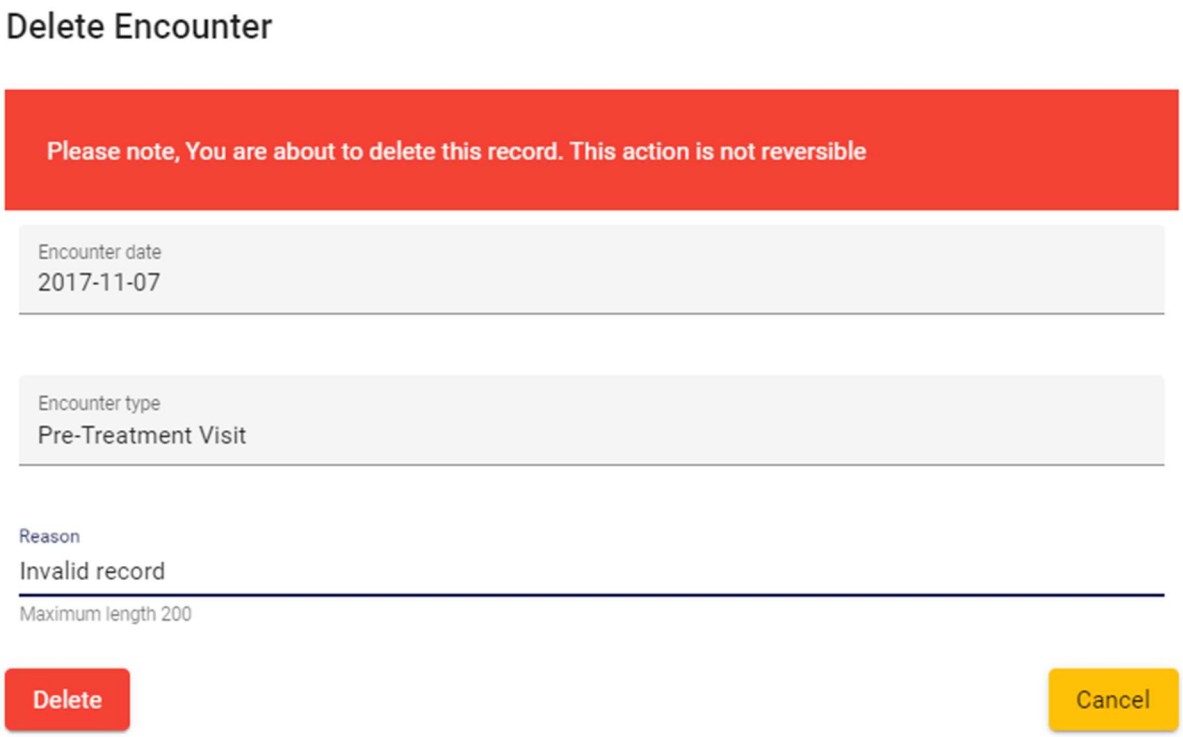
Within the **Patient View,** at the **Encounters** tab, locate the **Encounter** you would like to remove and click the **View Encounter** icon.



The system will navigate you to the **Encounter View.**



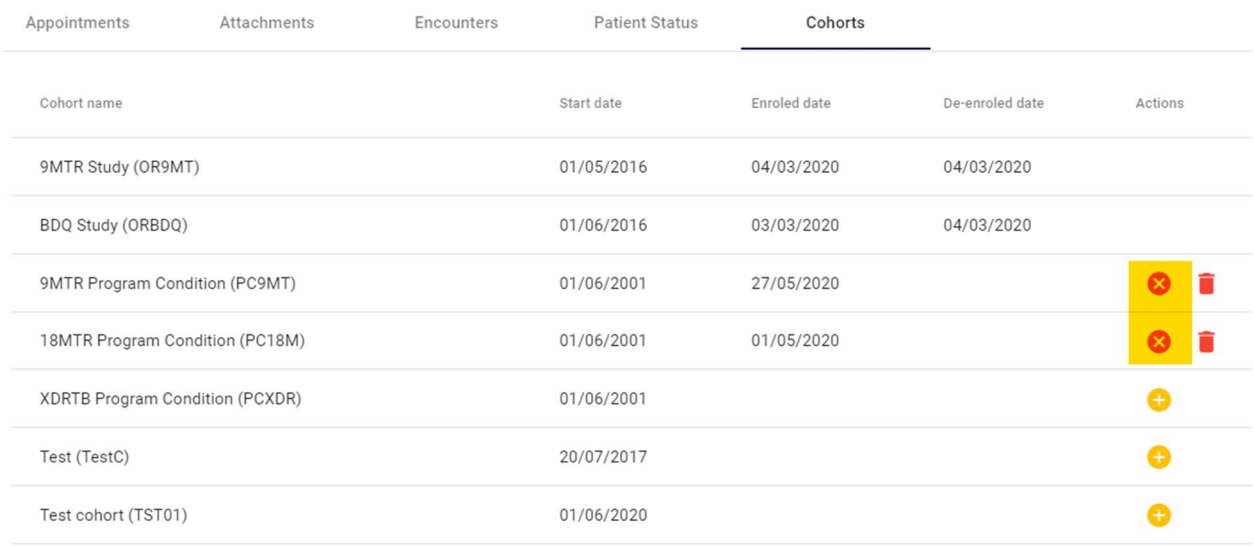
Click the **Delete Encounter** button after which the system will pop up a **Delete Encounter** form.



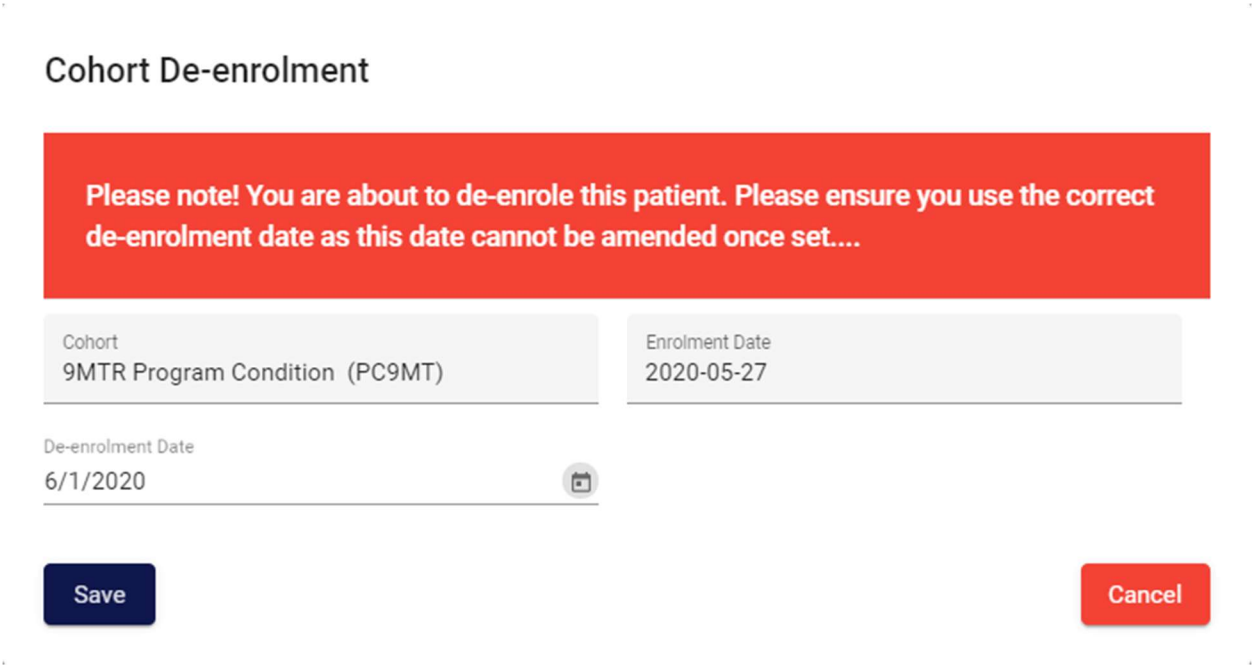
Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the encounter or click the **Cancel** button to undo the action. After confirming the deletion, the system will update the **Encounter** table.

4.6.1.4 De-enroll a Patient from a Cohort

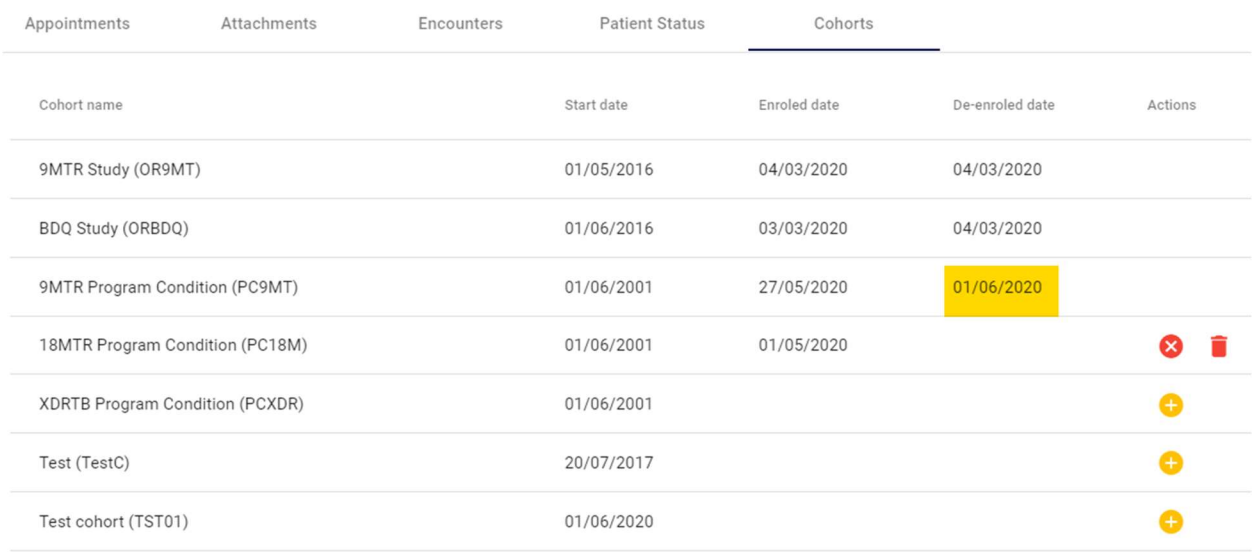
Within the **Patient View,** at the **Cohorts** tab, find the **Cohort** you would like to de-enroll the patient from.



Click the **De-enroll** icon after which the system will display a **Cohort De-enrollment** confirmation pop up form.

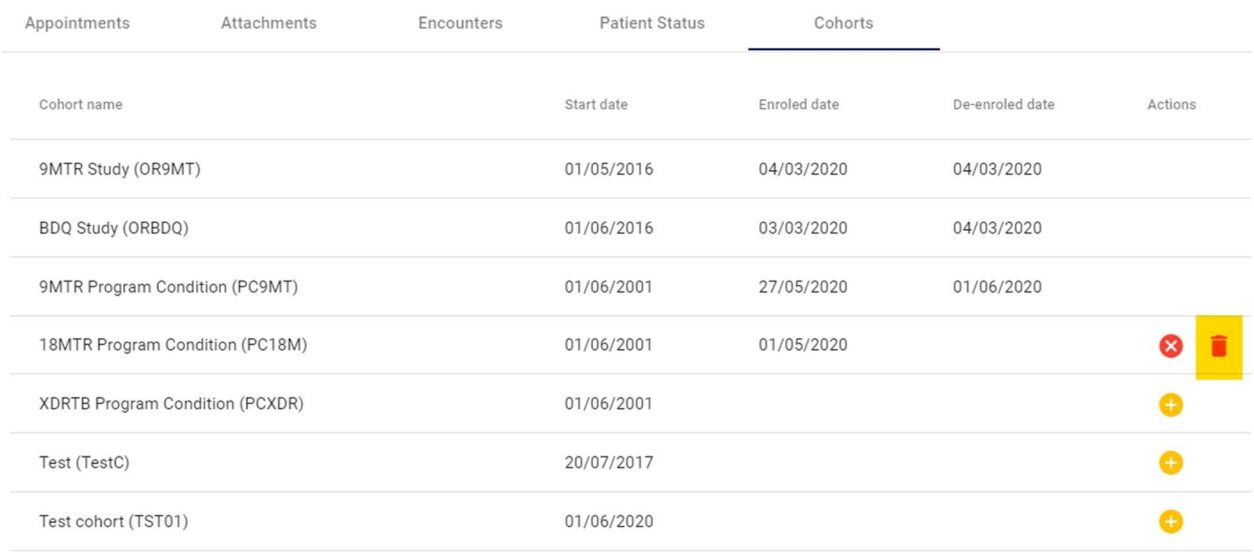


Enter the de-enrollment date and click the **Save** button. The system will display the updated **Cohorts Table**.

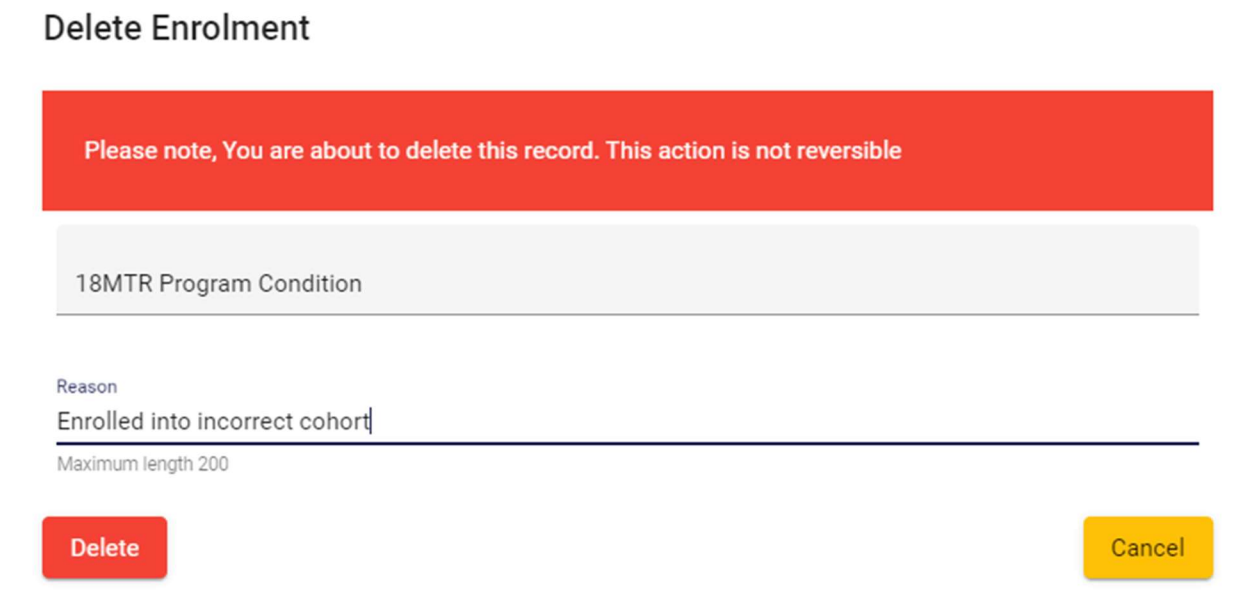


4.6.1.5 Deleting A Cohort Enrolment

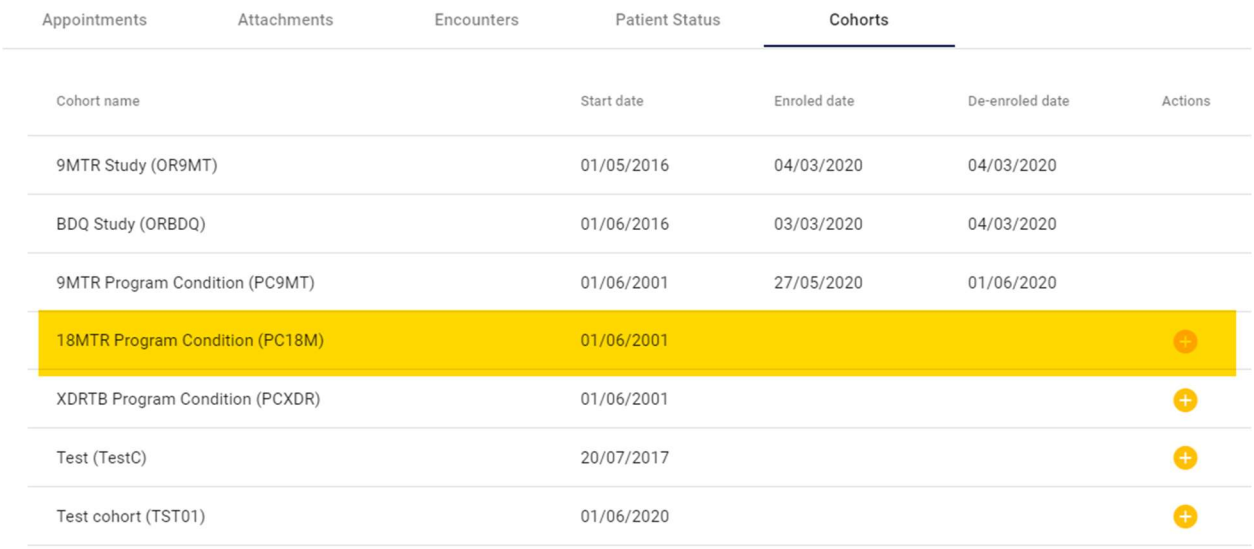
Within the **Patient View,** at the **Cohorts** tab, find the **Cohort** you would like to delete the enrolment for .



Click the **Delete** icon after which the system will display a **Cohort Enrollment Deletion** confirmation pop up form.



Enter the reason for deletion and click the **Delete** button. The system will display the updated **Cohorts Table**.

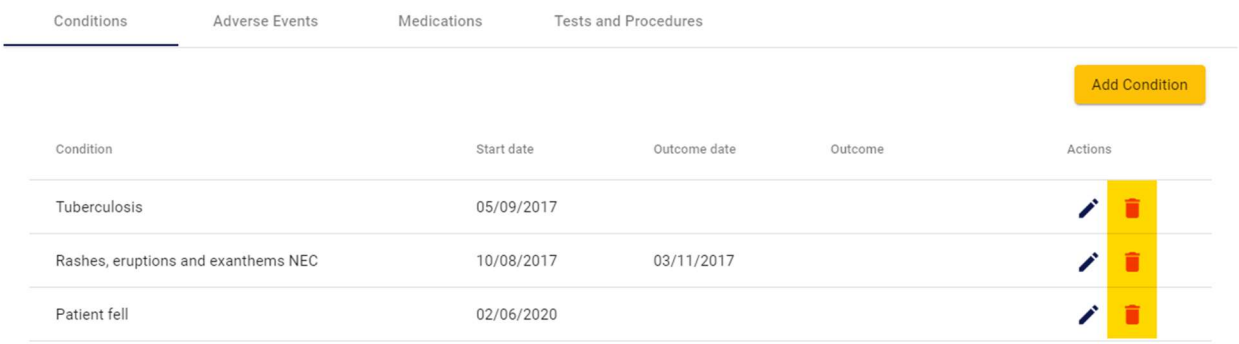


Deleting a cohort enrolment allows the patient to be re-enrolled into that cohort. De-enrollment means the patient cannot be re-enrolled into the same cohort again.

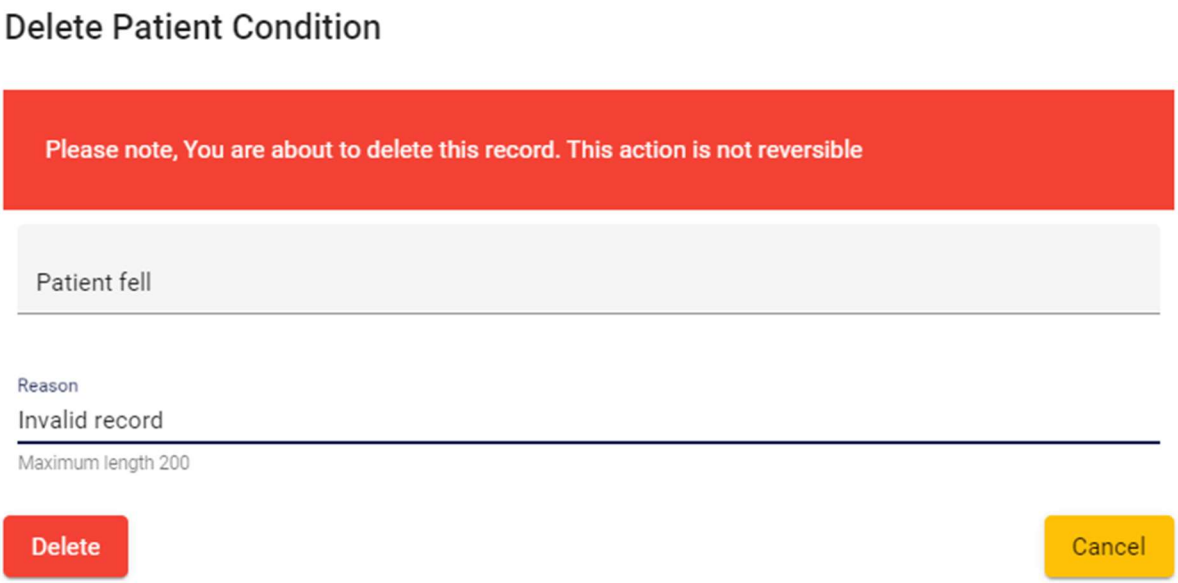
4.6.2 Encounter View – Clinical Information

4.6.2.1 Delete A Patient Condition

Within the **Encounter or Patient View,** at the **Patient Conditions** tab and locate the condition you would like to delete.



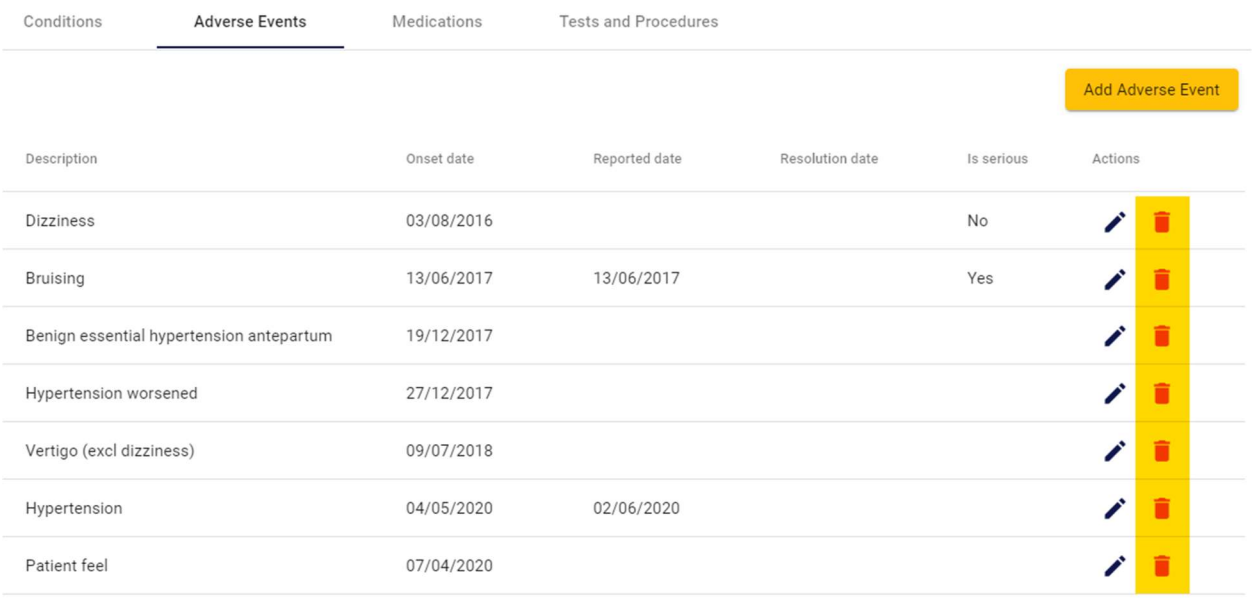
Click the **Delete Condition** icon after which the system will take you to a **Delete Condition** pop up form.



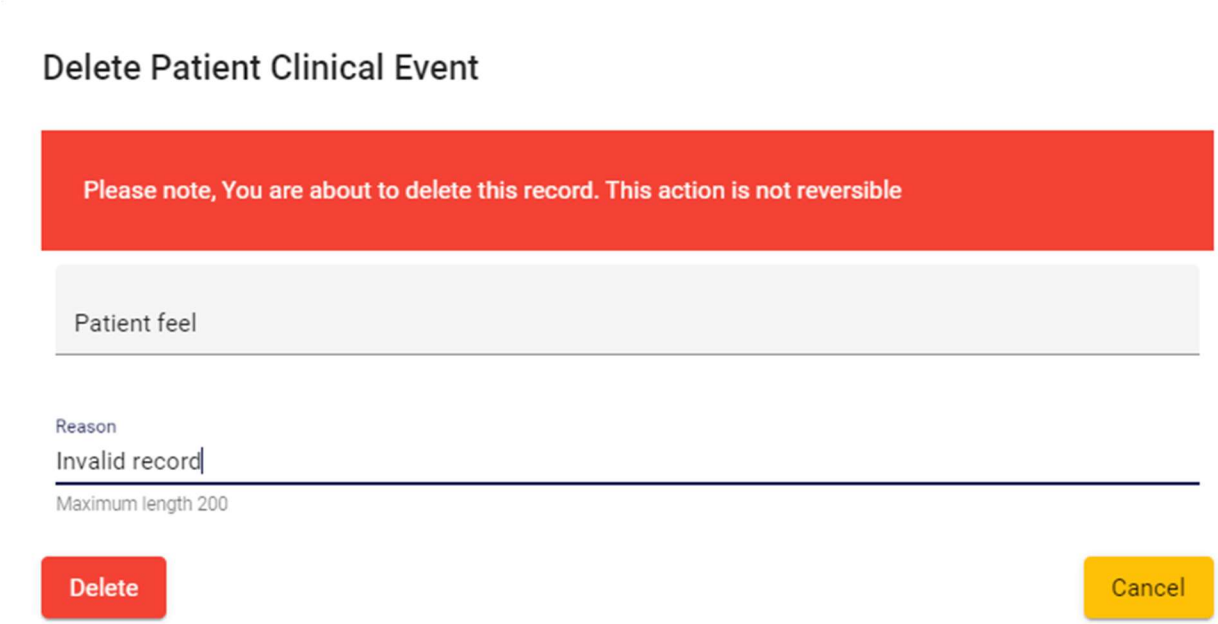
Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the condition or click the **cancel** button to undo the action. After confirming the deletion, the system will update the **Patient Condition** table.

4.6.2.2 **Delete an Adverse Event**

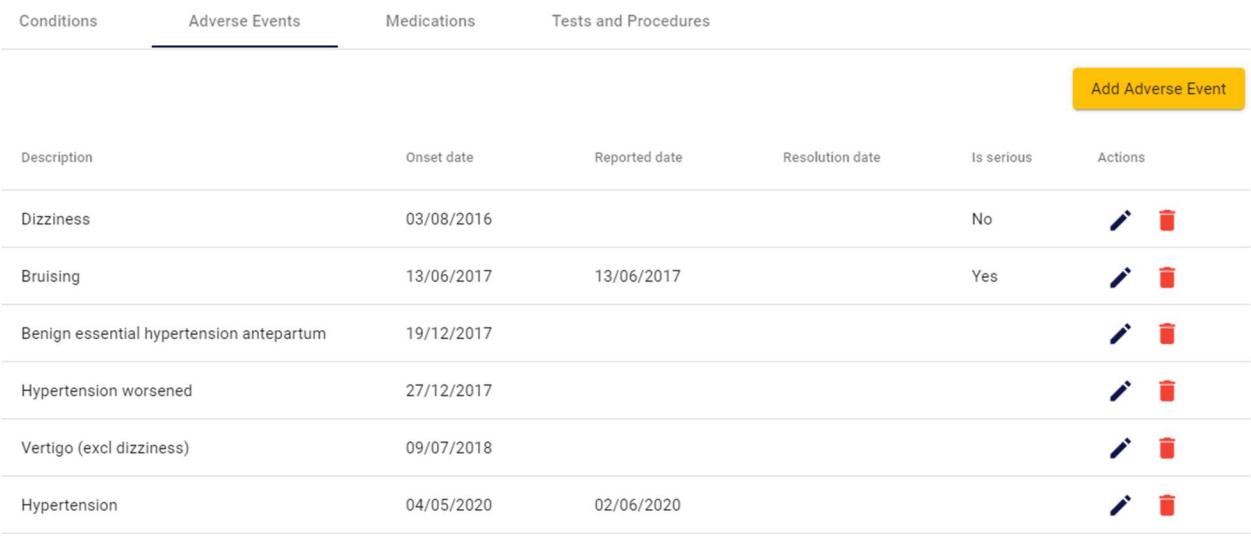
Within the **Encounter or Patient View,** at the **Adverse Events** tab and locate the event you would like to delete.



Click the **Delete Adverse Event** icon after which the system will take you to a **Delete Adverse Event** pop up form.

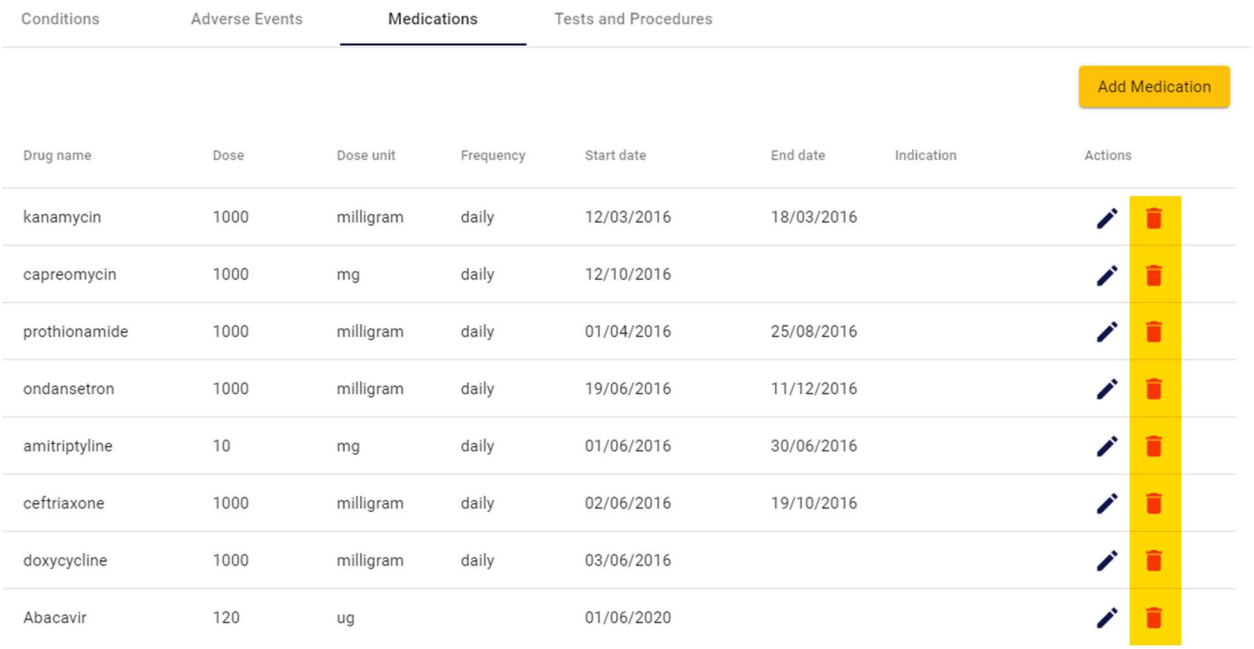


Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the adverse event or click the **Cancel** button to undo the action. After confirming the deletion, the system will update the **Adverse Events** table.

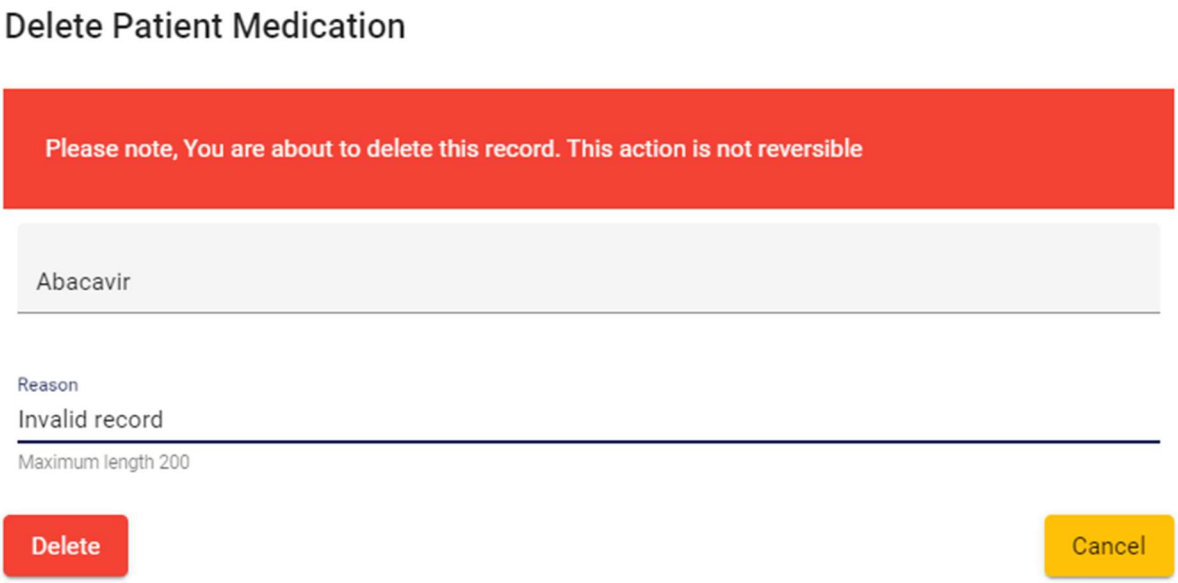


4.6.2.3 Delete A Patient Medication

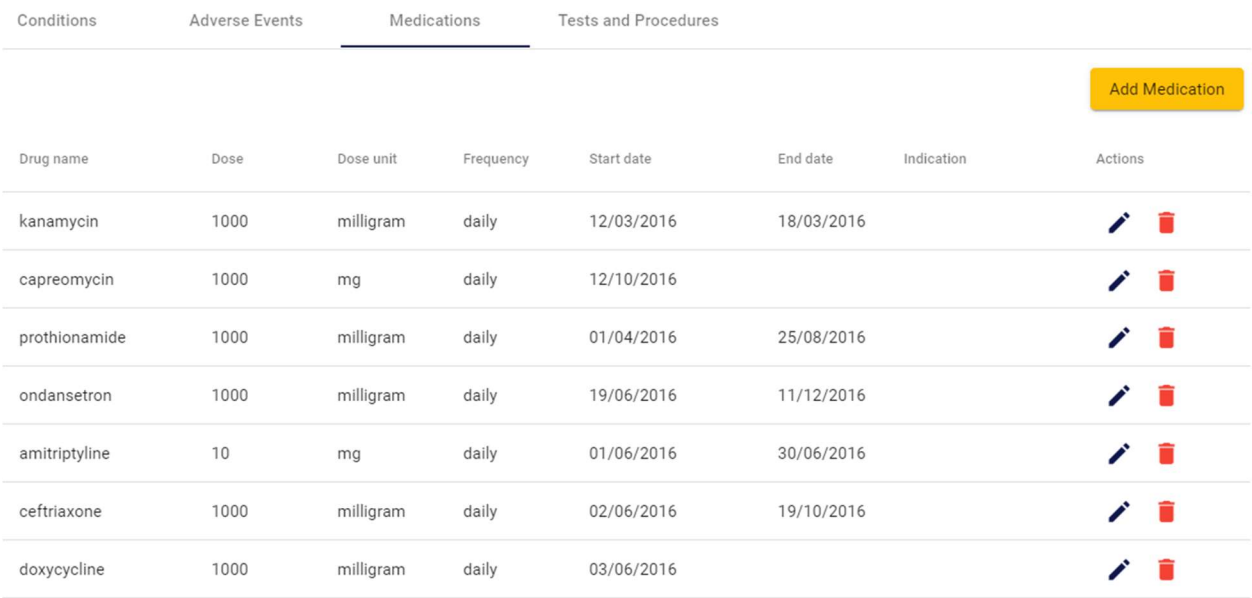
Within the **Encounter or Patient View,** at the **Patient Medications** tab and locate the medication you would like to delete.



Click the **Delete Medication** icon after which the system will take you to a **Delete Patient Medication** pop up form.



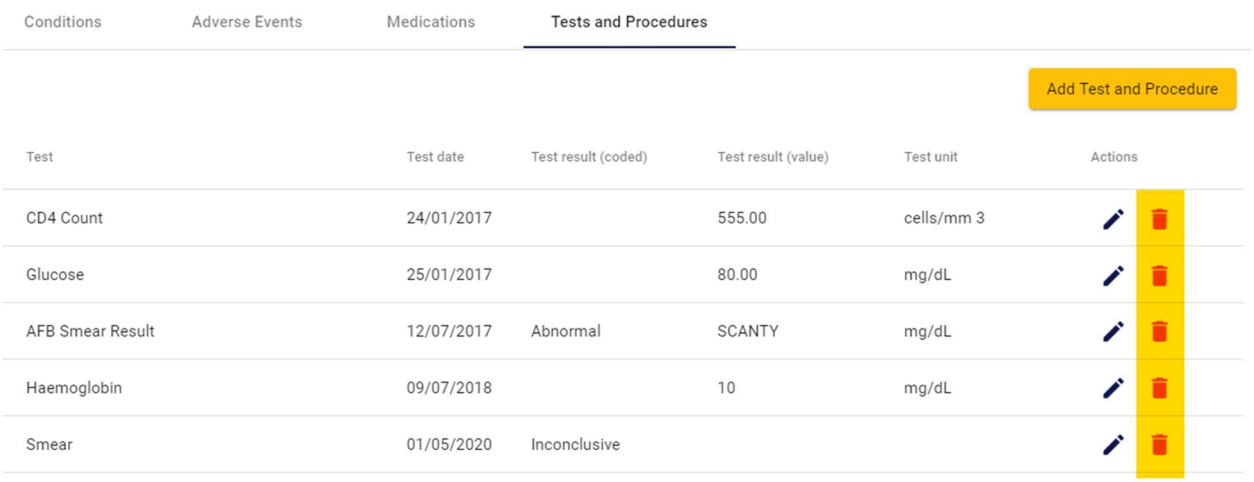
Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the medication or click the **Cancel** button to undo the action. After confirming the deletion, the system will update the **Patient Medications** table.



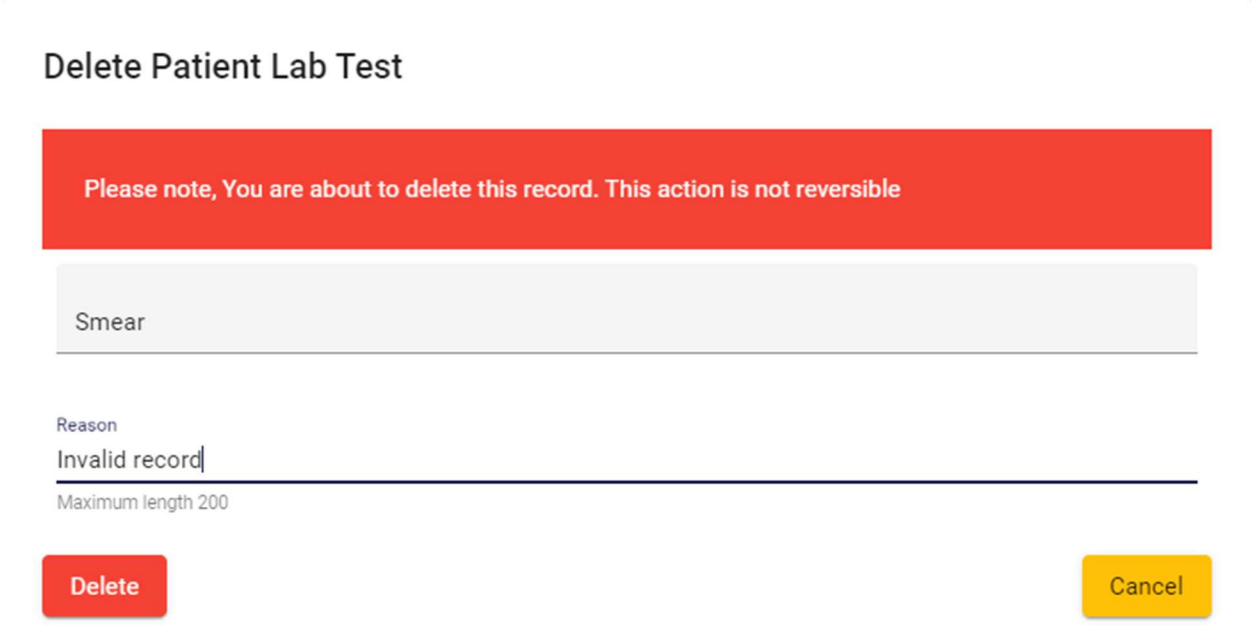
Drug name	Dose	Dose unit	Frequency	Start date	End date	Indication	Actions
kanamycin	1000	milligram	daily	12/03/2016	18/03/2016		
capreomycin	1000	mg	daily	12/10/2016			
prothionamide	1000	milligram	daily	01/04/2016	25/08/2016		
ondansetron	1000	milligram	daily	19/06/2016	11/12/2016		
amitriptyline	10	mg	daily	01/06/2016	30/06/2016		
ceftriaxone	1000	milligram	daily	02/06/2016	19/10/2016		
doxycycline	1000	milligram	daily	03/06/2016			

4.6.2.4 Delete A Test and Procedure

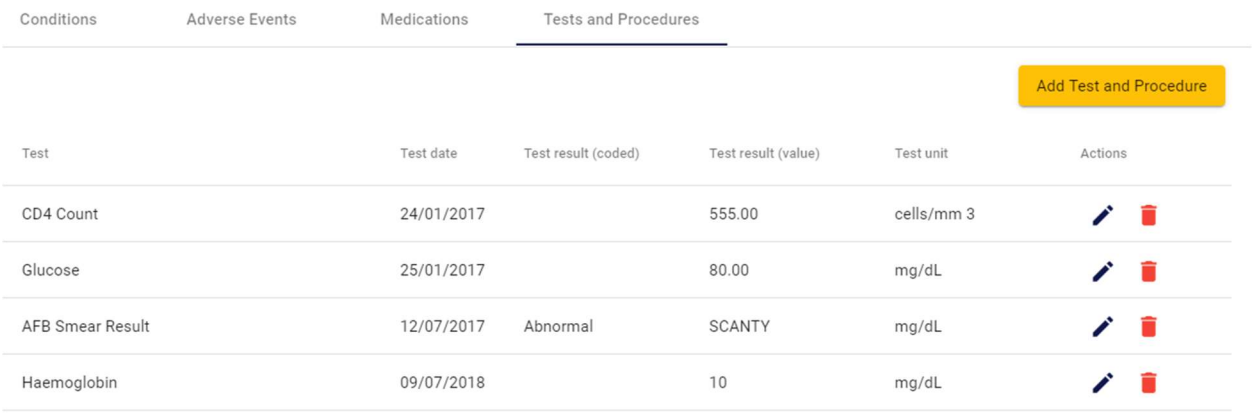
Within the **Encounter or Patient View,** at the **Tests and Procedures** tab and locate the test and procedure you would like to delete.



Click the **Delete Test and Procedure** icon after which the system will take you to a **Delete Test and Procedure** pop up form.

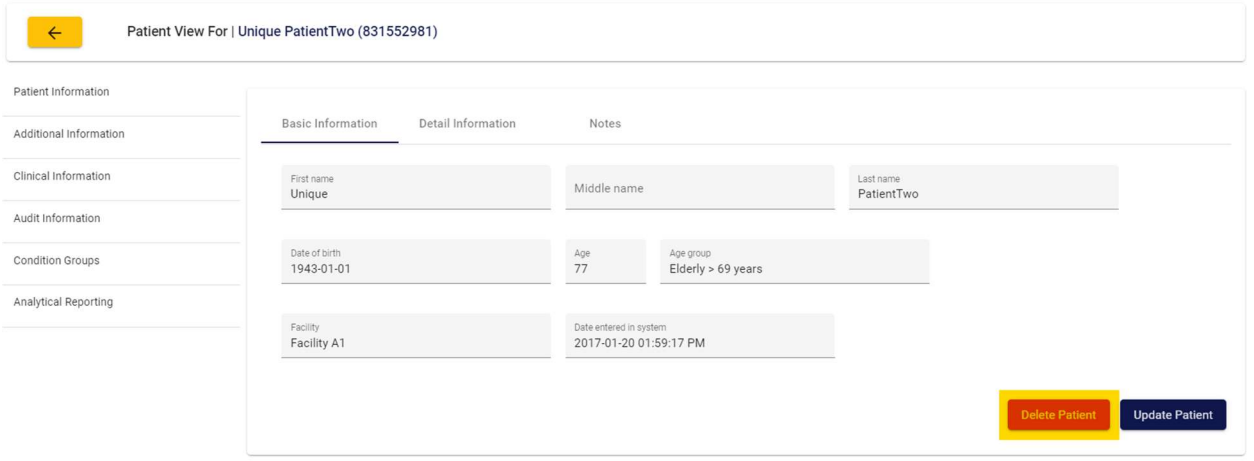


Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the test and procedure or click the **Cancel** button to undo the action. After confirming the deletion, the system will update the **Tests and Procedures** table.

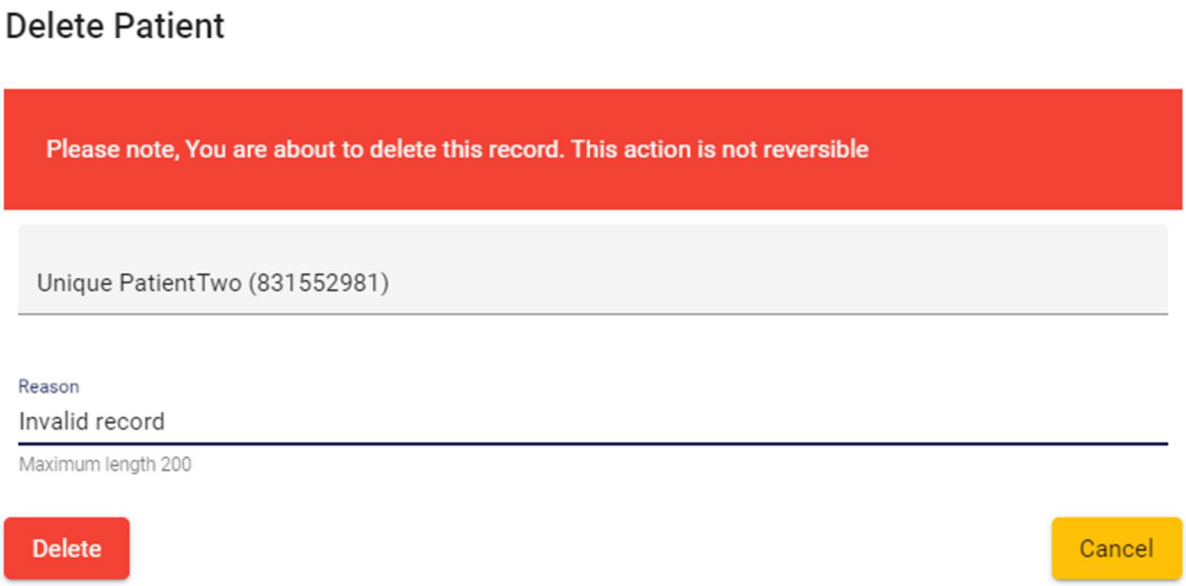


4.6.3 **Delete an Entire Patient Record**

Within the **Patient View, click** the **Delete** button to delete an entire patient record.



The system will take you to a **Delete Patient** pop up form.



Enter a reason for the deletion (this is compulsory) and click on the **Delete** button to delete the patient or click the **Cancel** button to undo the action. After confirming the deletion, the return you to the patient search form.

Deleting a patient will archive all patient information and will remove the patient from any analysis they may have been part of previously.

5 Analytical Portal

At the **Home Page** of the Analytical Portal**,** you will be presented with the following options:

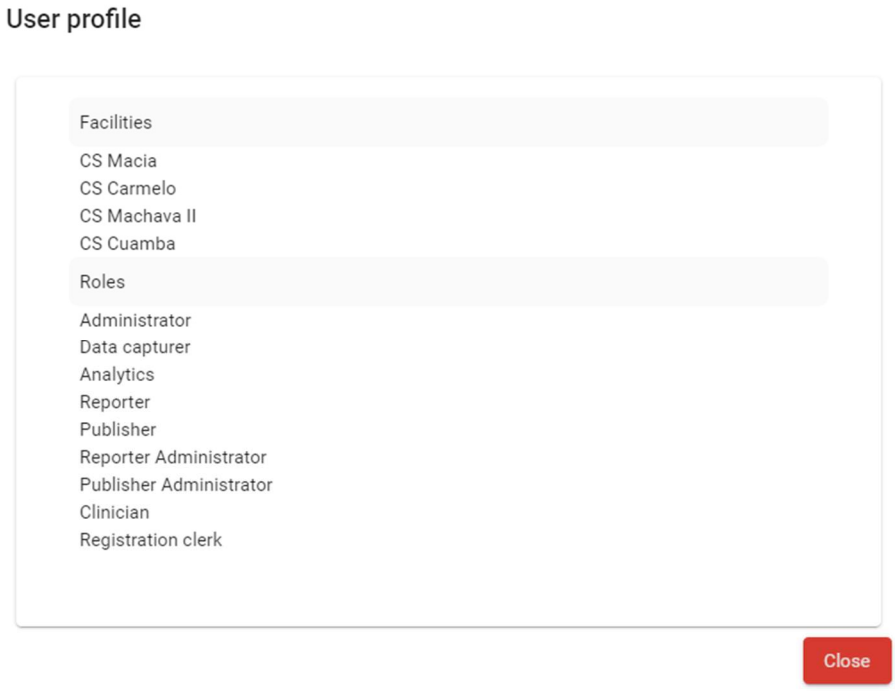
- Spontaneous and Active Reports
- Spontaneous and Active Analyzer

The analytical portal is the centralized hub for causative drug assessment using traditional recognized rating scales, standardized terminology, and risk detection with exposed versus non-exposed risk ratios.

Note: the following roles have access to the analytical portal:

- **Administrator.** The administrator has FULL permissions to the analytical portal.
- **Analyst.** An analytics user is able to assign terminology, set causality and run analysis on collected data.

Click on the user profile menu option which appears when clicking on the profile icon to view roles and facilities you currently have access to.



5.1 Spontaneous Reporting

Spontaneous reports are termed spontaneous as they take place during the clinician's normal diagnostic appraisal of a patient, when the clinician is drawing the conclusion that the drug may be implicated in the causality of the event.

Spontaneous reporting system relies on vigilant physicians, other healthcare professionals, and patients who not only generate a suspicion of an ADR, but also report it.

It is an important source of regulatory actions such as taking a drug off the market or a label change due to safety problems. Spontaneous reporting is the core data-generating system of international pharmacovigilance, relying on healthcare professionals (and, in some countries, consumers) to identify and report any adverse events to their national pharmacovigilance center.

Spontaneous reports are, by definition, submitted voluntarily.

PViMS facilitates a public facing interface that allows clinicians or the public itself to spontaneously report on adverse event related data. See chapter 8 for more information.

5.2 Active Reporting

Active surveillance for monitoring the safety and effectiveness of medical products is increasingly recognized as a complement to spontaneous reporting commonly used by pharmacovigilance systems. Integrated mechanisms and processes for monitoring the safety of medicines are essential to a well-functioning pharmaceutical sector. A positive benefit-to-risk balance should precede access to market; however, most regulatory decisions take place early in the product lifecycle and are based on limited data from clinical trials that may be of relatively short duration with limited numbers and types of subjects.

It is critical, therefore, that medicines continue to be monitored for safety and effectiveness once they enter the market under real-life conditions. For some medicines, issues will only emerge under real-world conditions as a result of prolonged use, use in specific subpopulations or in patients with multiple comorbidities, or use in combination with other medicines. In some cases, rare adverse effects only emerge after a product is used for many years, by large numbers of patients, or both.

Active surveillance is particularly important to support the introduction of new essential medicines in LMICs whose regulatory systems are being developed and are in need of support. In resource-limited settings, active surveillance can help determine the real-life frequency, risk factors, and impact of clinically significant adverse medicine events on treatment outcomes.

5.3 Pharmacovigilance Activities

This section describes the common processes adopted by the pharmacovigilance unit in responding to both spontaneous and active reports submitted through PViMS.

PLEASE NOTE: In relation to spontaneous reports, the patient identified in the report does **NOT** form part of the active reporting patient dataset. Therefore, spontaneous reports do not form part of any analysis conducted through the analyzer.

The following activities are facilitated in the Analytical portal:

- Verify quality of report data
- Set terminology for adverse drug reaction (MedDRA)
- Set causality per drug (Naranjo or WHO)
- Create and update E2B files (Export to XML)
- Generate a patient summary

5.3.1 Terminology

Causality Assessment Scale or Terms. The assessment scales or terms were developed to help standardize assessment of causality for all adverse drug reactions. The result is determined by an algorithm designed by Naranjo or WHO for determining the likelihood of whether an adverse drug event is actually due to the drug rather than the result of other factors. Probability is assigned via a score termed definite, probable, possible or doubtful (Naranjo) or certain, probable/likely, possible, unlikely, conditional, unassessable or unclassified (WHO). Values obtained from this algorithm are sometimes used in peer reviews to verify the validity of author's conclusions regarding adverse drug reactions.

MedDRA Terminology. MedDRA or Medical Dictionary for Regulatory Activities is a clinically validated international medical terminology dictionary by regulatory authorities in the pharmaceutical industry during the regulatory process, from pre-marketing to post-marketing activities, and for data entry, retrieval, evaluation, and presentation. In addition, it is the adverse event classification dictionary endorsed by the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH).

E2B. The international standard for transmitting medicine adverse event reports specified by the ICH.

5.3.2 Process Flow

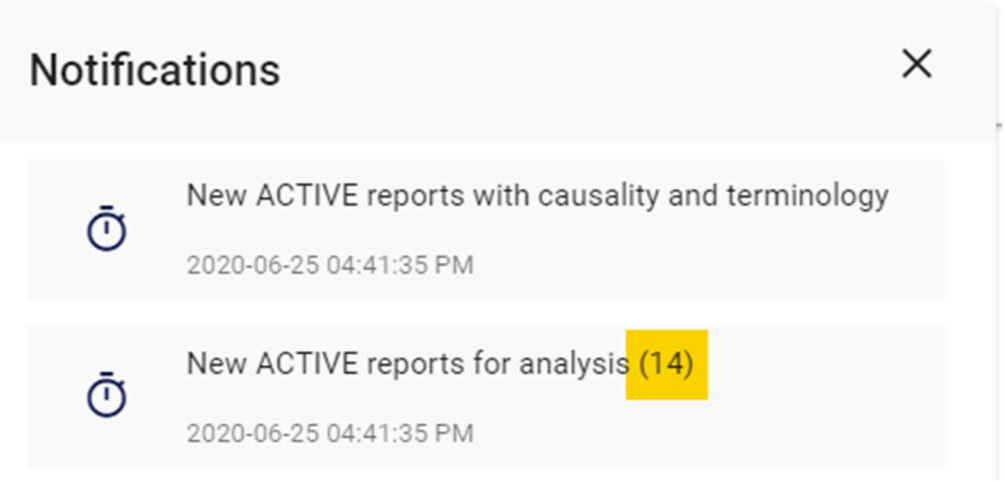
Activity: Start of Process Clinician adds a new adverse event to the patient record. Reporter adds a new spontaneous report. System generates a new report for the PV unit to action: • Report has a unique identifier • Report has a default activity of **Confirm Report Data** • Report has a sub status of **UNCONFIRMED**	**Confirm Report Data** The purpose of this activity is to facilitate the checking, vetting, and updating of the clinical data collected for the adverse event logged. The PV unit will be unable to perform any causality or terminology configurations until the data is confirmed as accurate and comprehensive. The PV specialist has the following options: • Delete Report • Confirm Report
Activity: Confirm Report Data **Step 1:** PV specialist and clinician/reporter update clinical data until there is sufficient information to perform causality and terminology configurations. For active reporting the patient record is modified through the clinical portal. For spontaneous reporting the report is modified directly in the analytical portal. **Step 2:** Once data has been confirmed, the PV specialist will confirm the report is ready for assessment by selecting the Confirm report option. **Step 3:** The report is moved into a new activity: Set MedDRA and Causality.	

Activity: Set MedDRA and Causality **Step 1:** PV specialist sets the terminology for the adverse event. This will be the terminology used for analysis purposes. **Step 2:** PV specialist then sets the causality per medication that was in use at the onset date of the adverse event. *Please note: the specialist is unable to set causality until the MedDRA term has been selected.* **Step 3:** On completion, the PV specialist confirms causality has been set. **Step 4:** The report is moved into a new activity: Extract E2B.	**Set MedDRA and Causality** The purpose of this activity is to facilitate the setting of the MedDRA terminology for the event and to set causality per medication using either the Naranjo or WHO causality scales. The PV specialist has the following options: • Set Terminology • WHO Causality • Naranjo Causality • Confirm Causality Set
Activity: Extract E2B **Step 1:** PV technician creates an E2B XML file for submission. *Please note: a patient summary is stored at the time of generating the E2B XML to allow for data verification.* **Step 2:** PV technician then confirms that the E2B XML file has been submitted to WHO. • A receipt code and time can be noted here *Please note: submission to WHO UMC is a manual process.*	**Extract E2B** The purpose of this activity is to facilitate sharing of E2B adverse drug reactions with the WHO Uppsala Monitoring Centre using the ICH ICSR E2B specification. The PV specialist has the following options: • Create E2B • Confirm E2B submission
PROCESS COMPLETED	

5.3.3 Identifying New Reports

It is possible to identify new active or spontaneous reports using the notification panel in the main system header. A number next to the notification icon indicates you either have a new adverse event report requiring analysis or new terminology has been defined for an existing report.

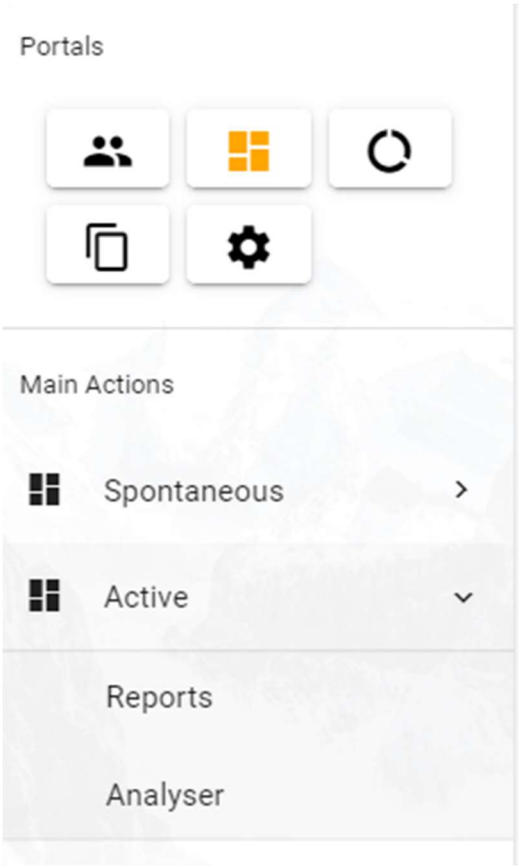
The number of new reports registered within a pre-defined period of time is displayed as a value in the notification list of menu items.



This alert will vary in count depending on the **ReportInstanceNewAlertCount** configuration. Please speak to your system administrator to confirm what this value is set to in days.

5.3.4 Search for a Report

In the **Reports** function for Spontaneous and Active Reporting, you can **Search** for new reports that have been created. The **Reports** function can be accessed either through the Spontaneous or Active menu.



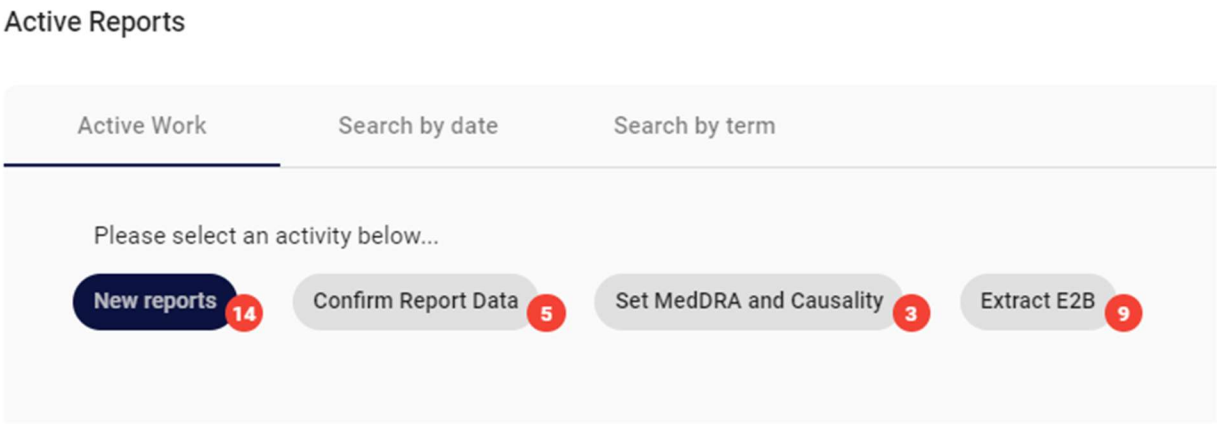
There are three ways to search for a report. You can search by:

- Activity (what stage of analysis the report is currently in)
- Report date range
- By a search term

5.3.4.1 **Search by Active Work**

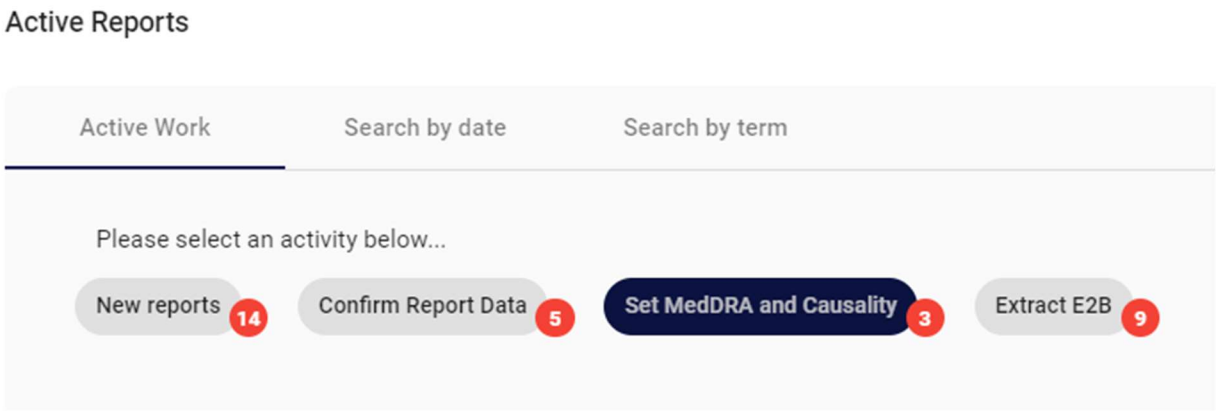
Searching by activity allows you to search for all incomplete reports by the current stage a report may be in or by listing all new reports that have been identified as per the **ReportInstanceNewAlertCount** configuration**.**

- Select the **New Reports** button to view a list of new reports within the system

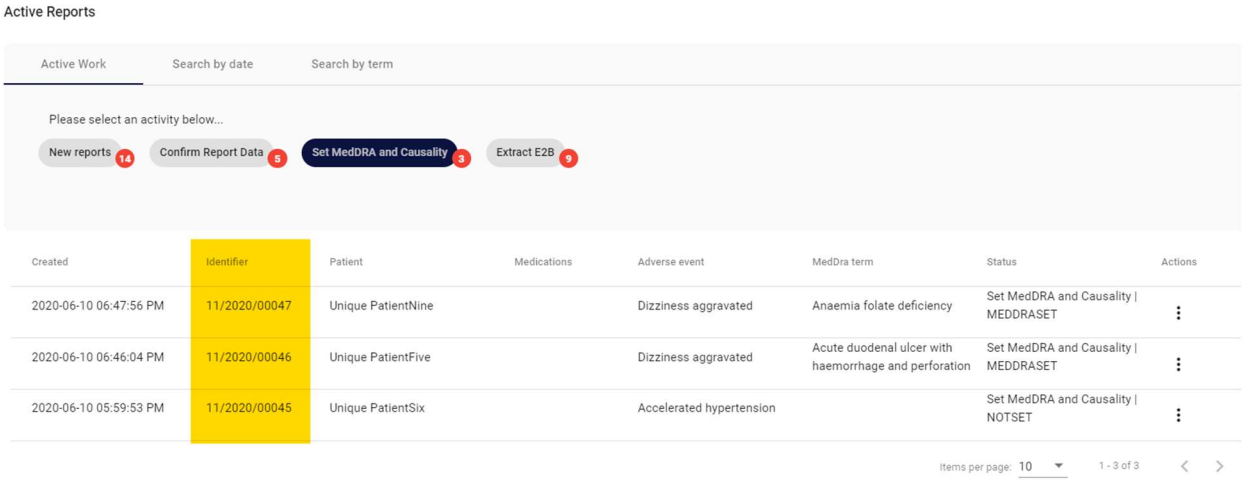


- Or select the current activity stage if you would like to filter reports based on the stage they are currently in

Confirm Report Data Stage: Search on all reports that are newly submitted and are not yet VERIFIED.
Set MedDRA and Causality: Search on all reports that have been VERIFIED and are in the process of being defined through terminology.
Extract E2B: Search on all reports that have all terminology defined and are ready to extract information for submission to the WHO Uppsala Monitoring Centre.



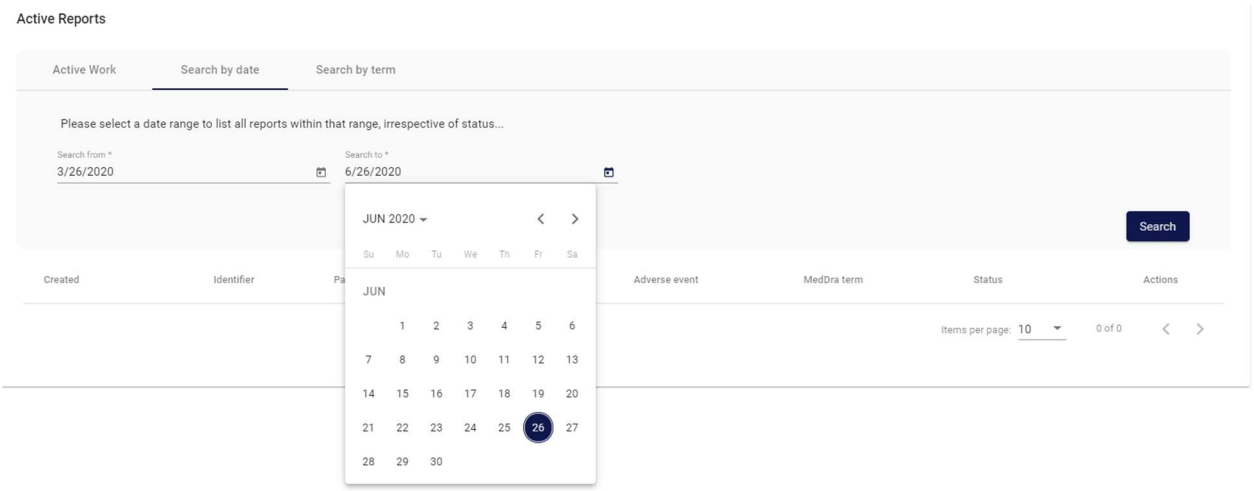
The system will display a list of reports according to the filter selected, please note the Unique Identifier of the report in column 2.



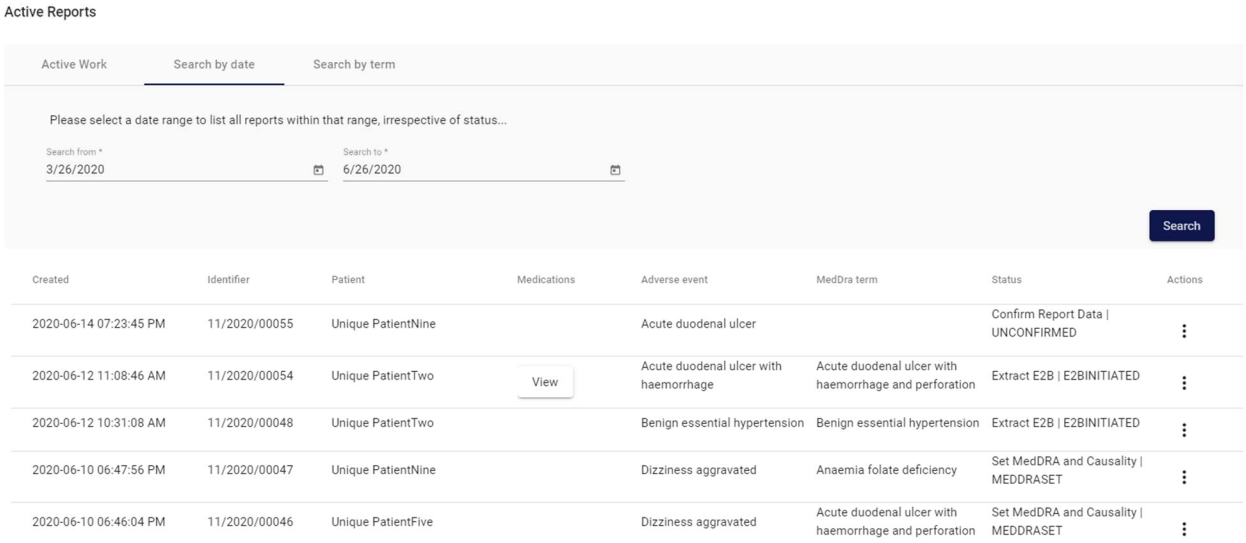
5.3.4.2 Search for all Reports within a Date Range

Search by date range searches for all complete and incomplete reports within a given date range.

- Enter the **Search From** and **Search To date fields** for the date range to search.
- Click the **Search** button.



The system will display a list of reports according to the filter selected, please note the Unique Identifier of the report in column 2.

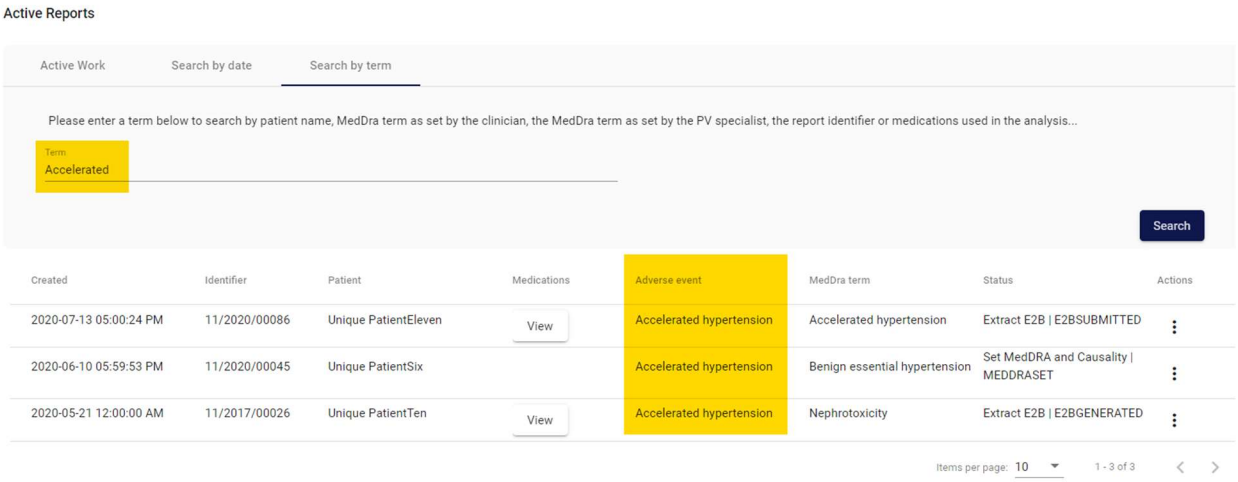


5.3.4.3 Search Reports by Term

The **Search by term** tab allows the user to search for all reports using the following criteria:

- Patient name
- MedDRA term as set by the clinician and technician
- Report identifier
- Medications

The search function uses a partial term to search all reports on the terms noted above. For instance, searching for the term Accelerated returns all reports that have that partial term in the MedDRA term for the report.



The columns in the report table are described below:

Created	The date and time the report was registered in the analytical portal
Identifier	Unique identifier for the report (system generated)
Patient	Patient name
Medication Summary	Overview of all medications associated with the patient at the time of the event, including the WHO or Naranjo causality assessment outcome once that has been set
Adverse Event	The adverse event experienced by the patient
MedDRA Terminology	The unique and internationally recognized MedDRA term defined for the event
Status	Current status of the report
Actions	**Confirm Report Data Stage** View Activity History Confirm or Delete Report View Patient (Active Only) View SAE Report or Patient Summary Update Report (Spontaneous Only) **Set MedDRA and Causality Stage** View Activity History Set Terminology (MedDRA), Naranjo or WHO Causality Term

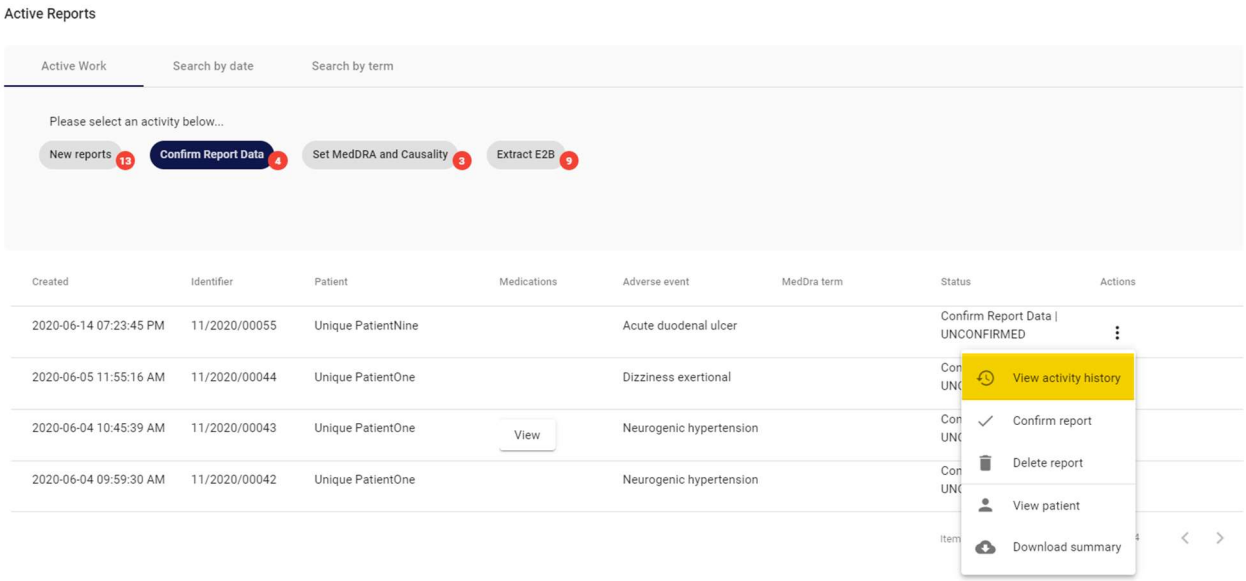
	Confirm Causality Set View Patient (Active Only) View SAE Report or Patient Summary Update Report (Spontaneous Only) **Extract E2B Stage** View Activity History Create or Update an E2B file Prepare or Confirm E2B file for submission Download XML file View Patient (Active Only) View SAE Report or Patient Summary Update Report (Spontaneous Only)

5.4 Pharmacovigilance Activities - General

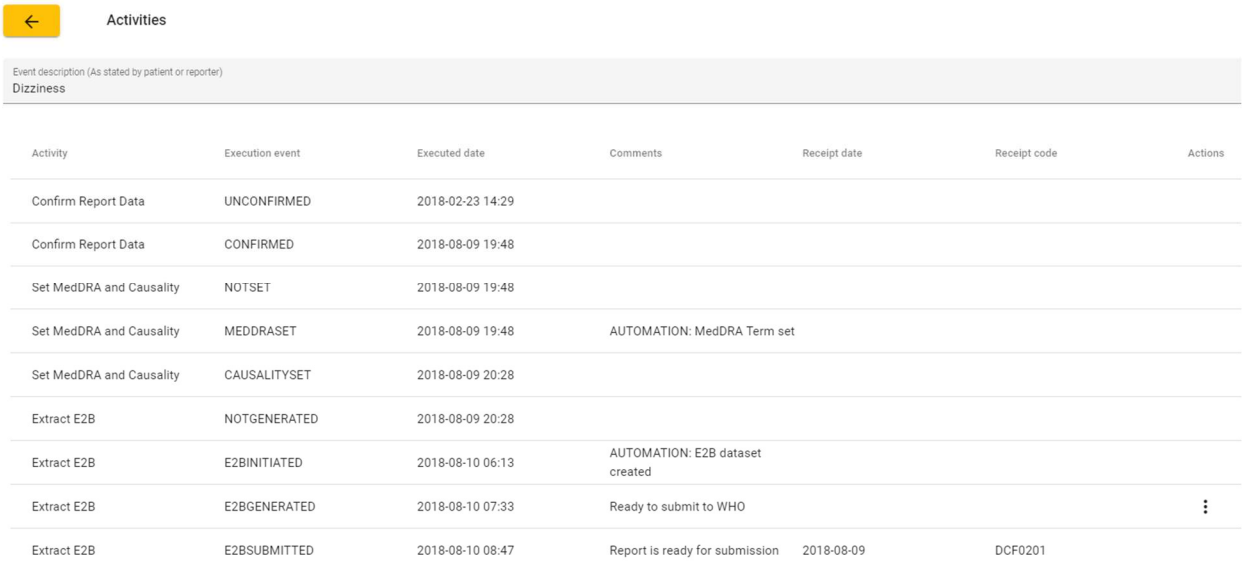
5.4.1 Viewing Activity History for Report

Irrespective of the stage the report is currently in, you will be able to view a comprehensive history for the report detailing all actions that have occurred, who effected the action, and any additional comments.

Once you have searched for a report, click on the **View activity history** menu for the associated report you would like to view.



The system will navigate you to the **View activity history** page.



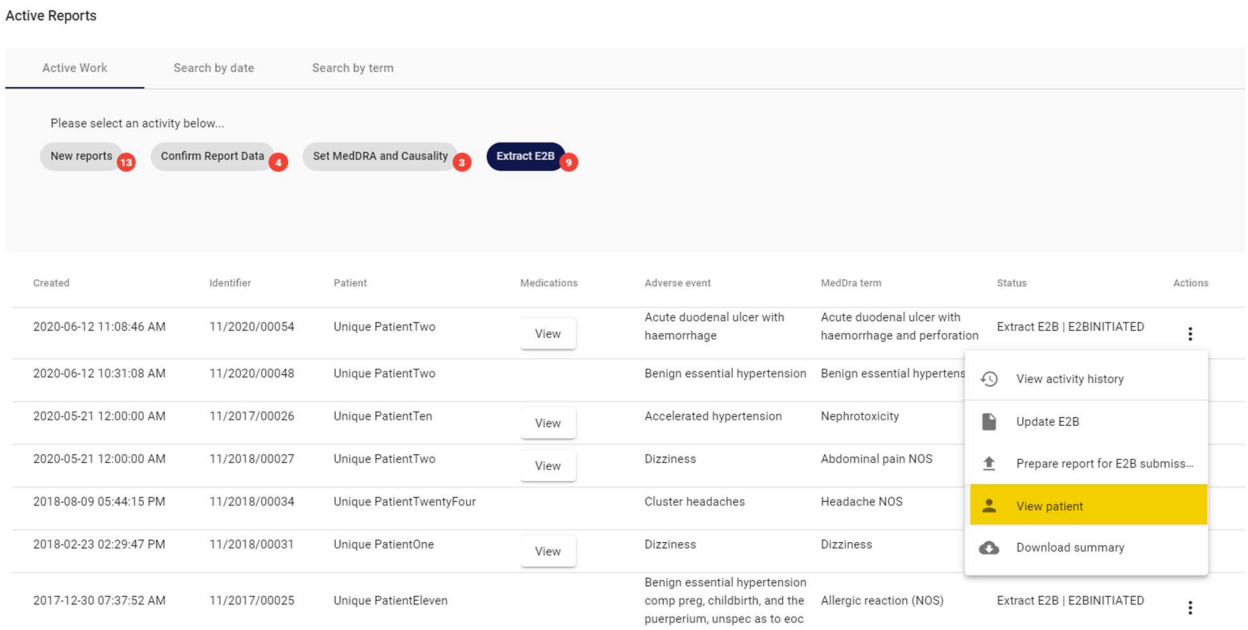
The columns in the activity history table are described below:

Activity	The primary activity stage performed by the analyst
Execution event	The primary activity performed by the analyst. Activities are dependent on the stage the report is in.
Execution date	The date and time the user executed the activity
Comments	Any comments noted by the user at the point of completing the activity
Receipt date	Particular to the E2B Extract Stage. Confirmation an E2B extract has been received by UMC and this is the associated receipt date
Receipt code	Particular to the E2B Extract Stage. Confirmation an E2B extract has been received by UMC and this is the associated receipt code that may be supplied by UMC
Action	Download summary Download extract Download E2B file

5.4.2 Viewing a Patient Record

The **View patient** menu allows the user to navigate to the Patient View for the patient so that the analyst may gather additional information to the adverse event.

Once you have searched for a report, click on the **View patient** menu for the associated report you would like to view.



The system will navigate you to the **Patient View** page in the Clinical Portal.

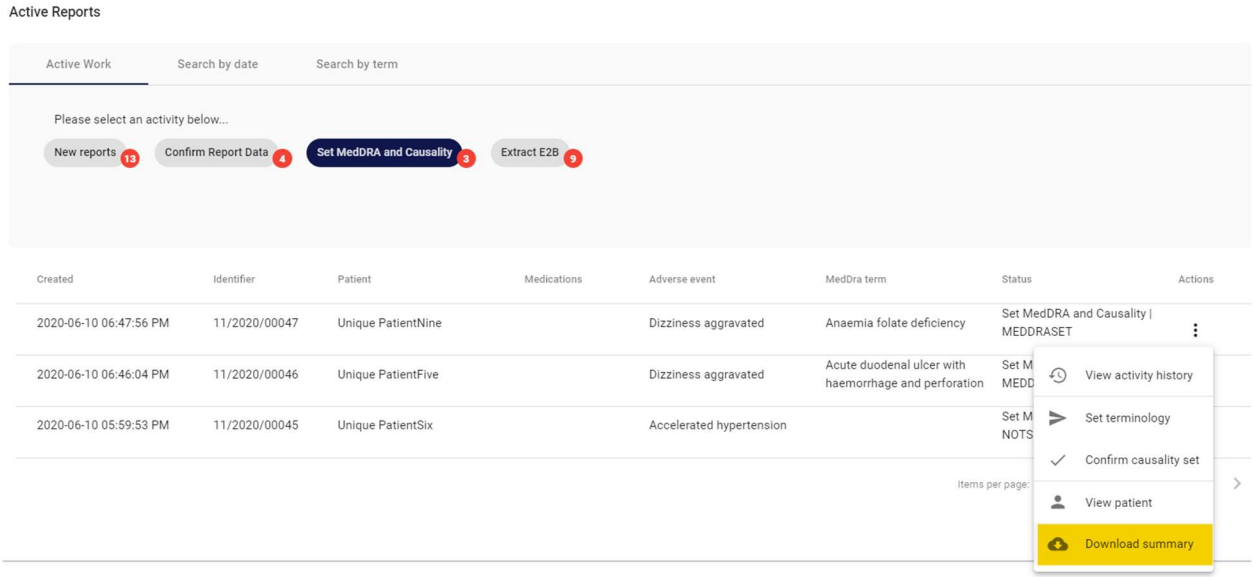
This menu item will only be made available under the following circumstances:

- Active reports only
- If the user has the Registration Clerk, Data Capture or Clinician role assigned to their user profile

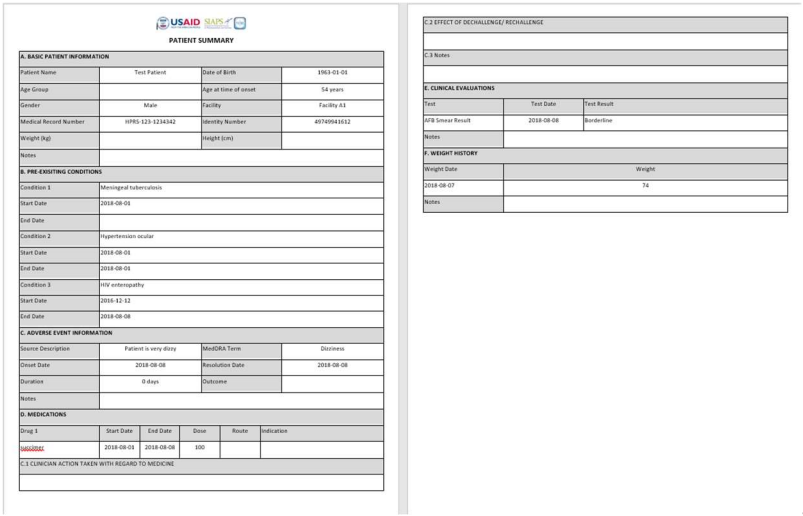
5.4.3 Extracting a Patient Summary

The **Download summary** menu allows the user to generate an overall summary of the patient clinical record in MS Word format. This is the preferred method of granting access to the clinical record for the analyst.

Once you have searched for a report, click on the **Download Summary** menu for the associated report you would like to view.



The system generates an extract of the patient summary:

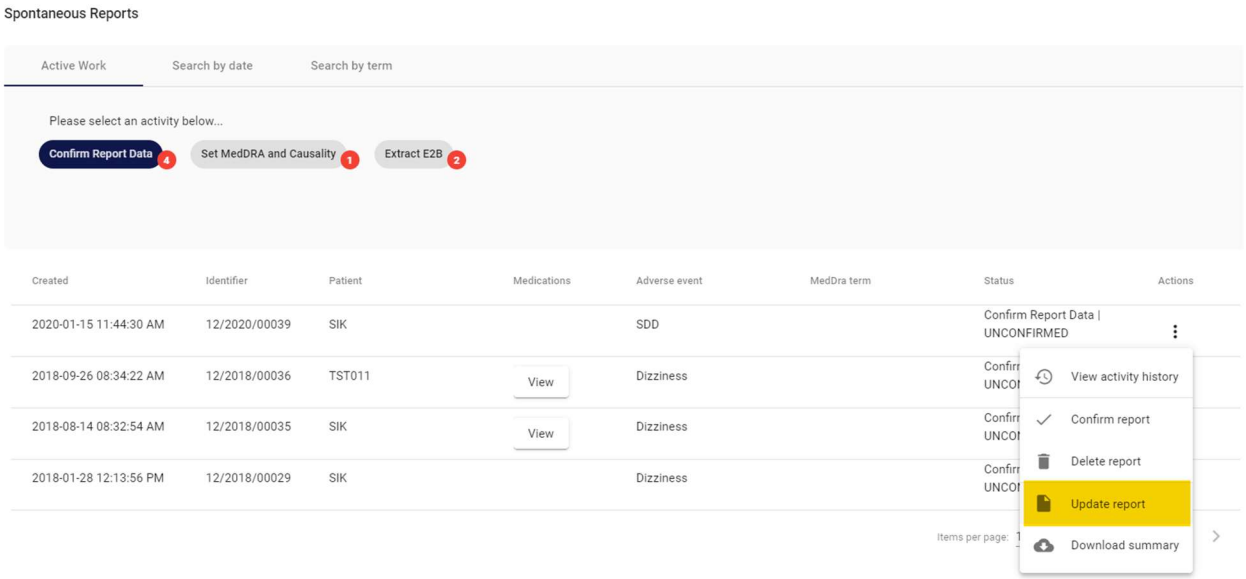


If the event is defined as serious, the system will include additional information that explains the seriousness of the report. The heading of the extract will be changed to **Serious Adverse Event.**

5.4.4 Updating a Spontaneous Report

The **Update report** menu allows the user to navigate to the spontaneous report for the patient and allow the analyst the ability to amend the report.

Once you have searched for a report, click on the **Update report** menu for the associated report you would like to amend.



The system will navigate you to the **Spontaneous View** pop up form for the report.

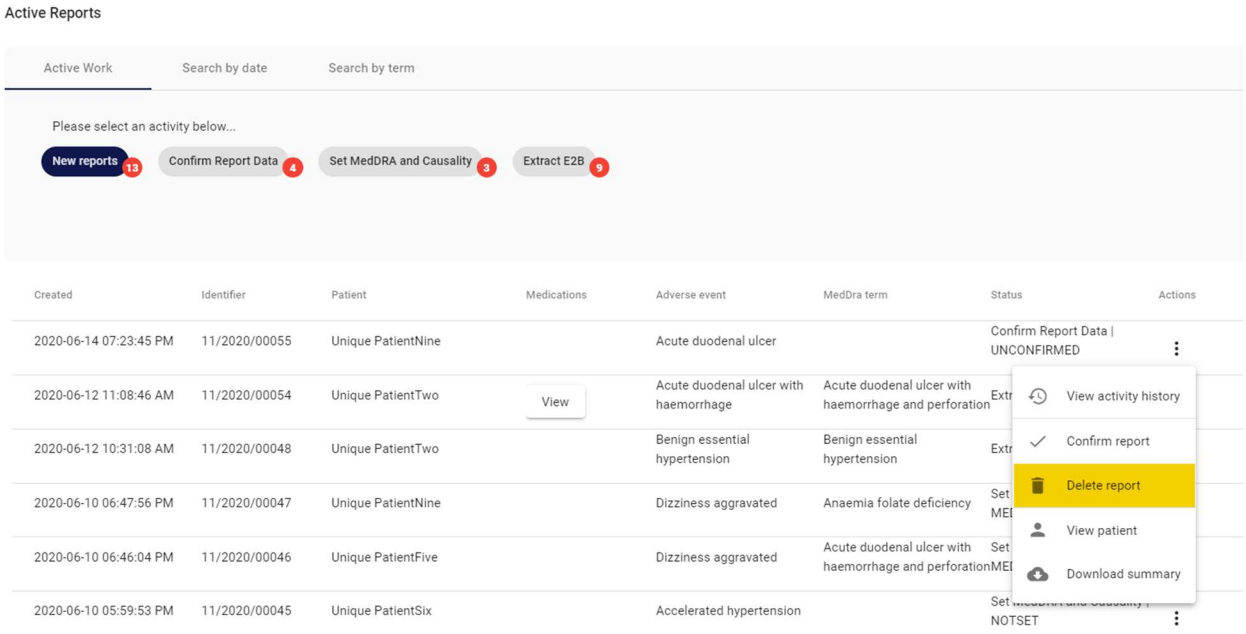
This menu item will only be available for spontaneous reports.

5.5 Pharmacovigilance Activities – Confirm Report Data

5.5.1 Deleting a Report

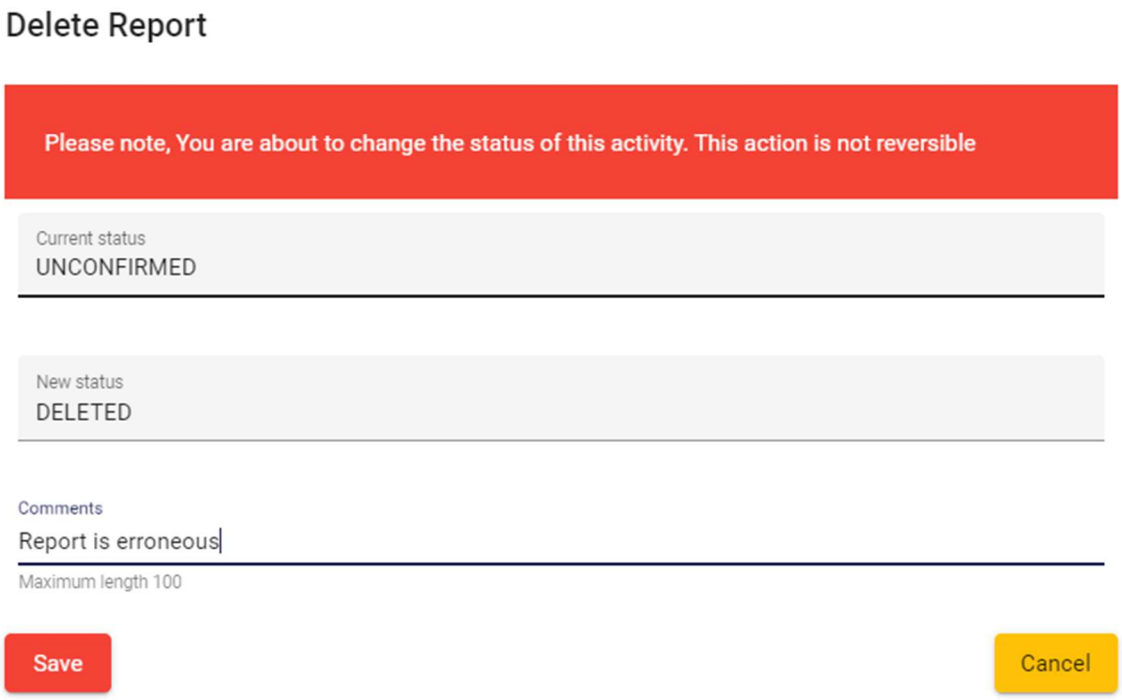
The **Confirm Report Data** stage enforces a process whereby the analyst ensures the necessary clinical data is of sufficient quality to allow terminology and causality to be completed. In the event a report is deemed to be insufficient, inaccurate, or erroneous, a report may be deleted, which will effectively remove it from analysis.

Once you have searched for a report, click on the **Delete report** menu for the associated report you would like to delete.

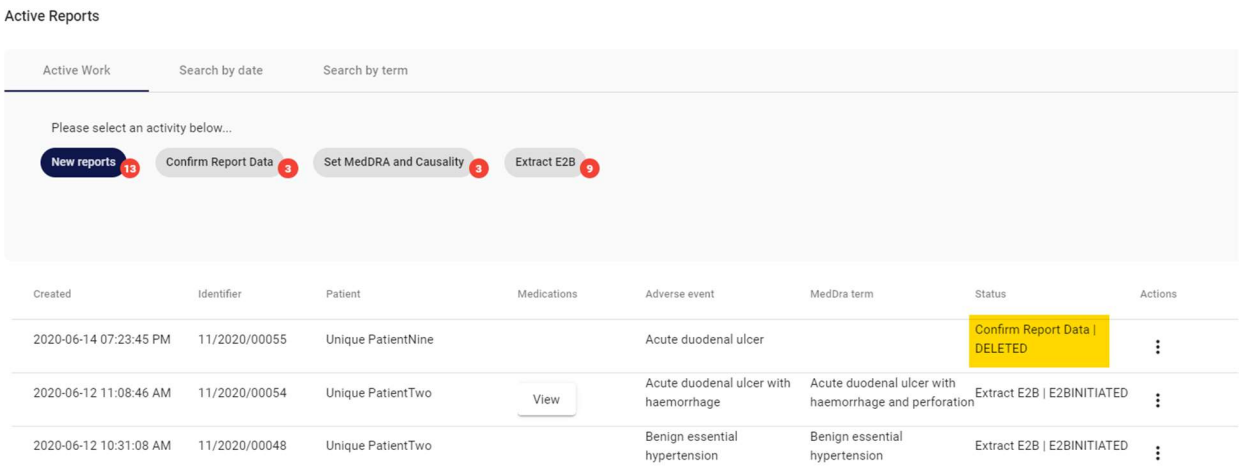


The system will navigate you to the **Add Activity** page.

Specify the reason for deleting the record and click **Save** to confirm the deletion or **Cancel** to cancel the action and return to the previous page.



The system will update the status of the report accordingly.

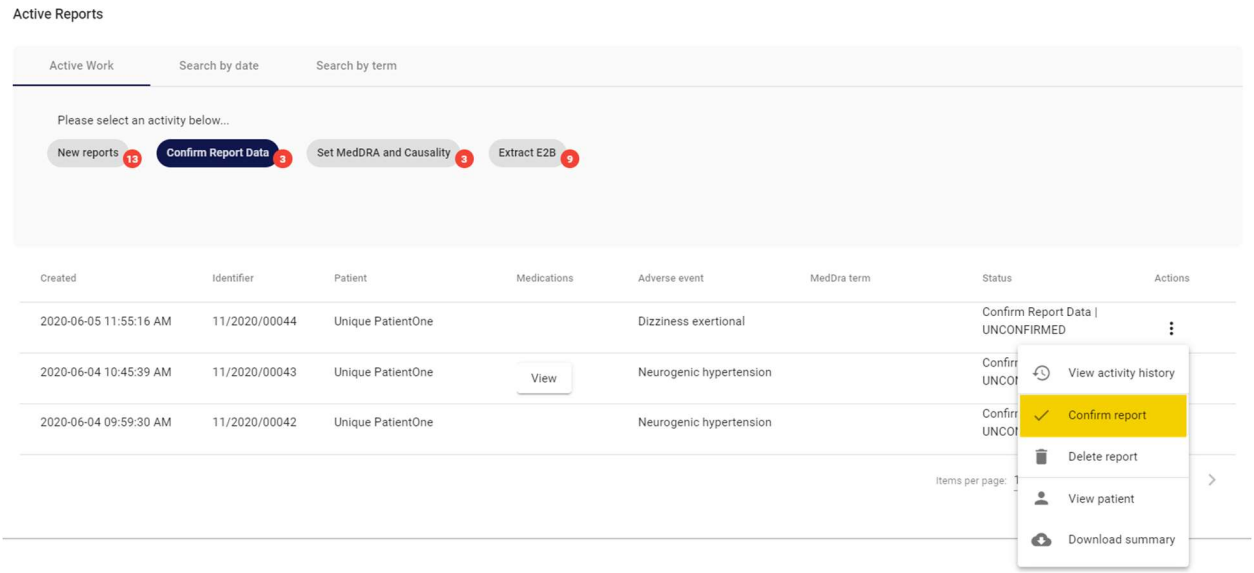


Once a report is deleted, it is not possible to re-include the report for analysis. Please ensure you are correct in deleting the report before committing the action.

5.5.2 Confirming a Report

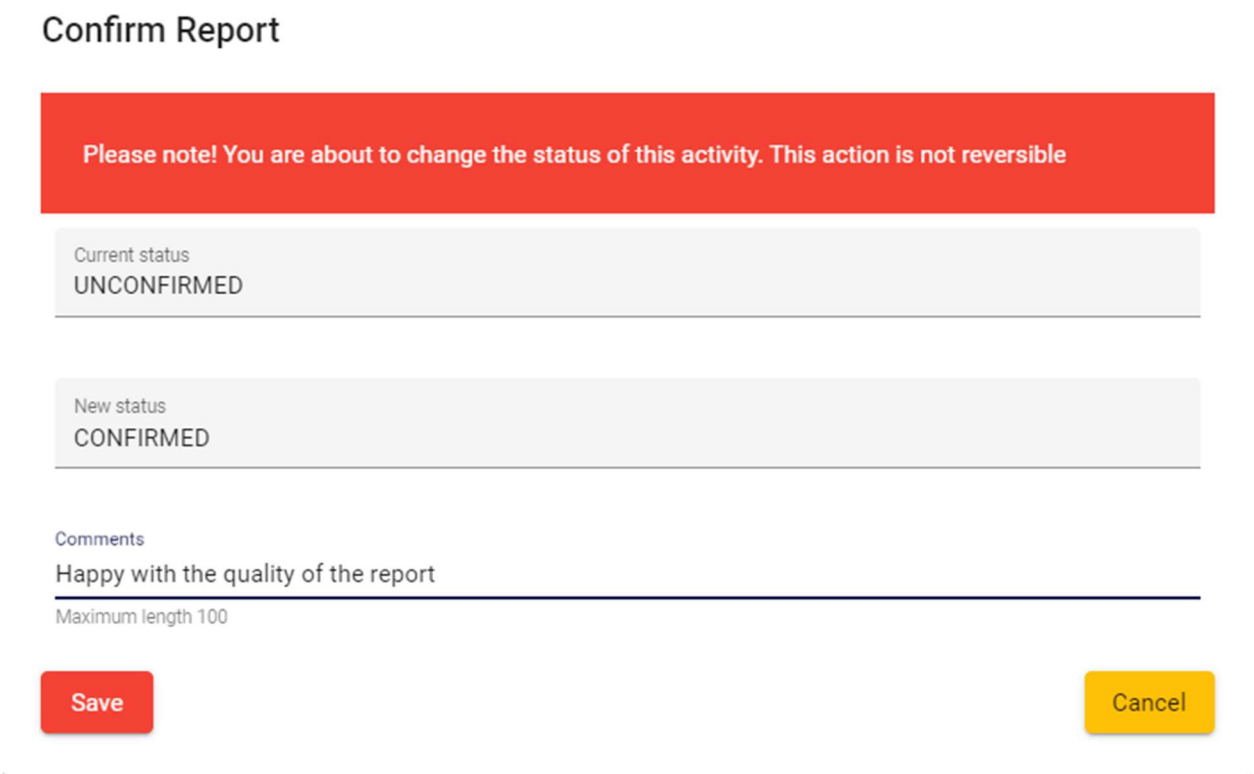
The **Confirm Report Data** stage enforces a process whereby the analyst ensures the necessary clinical data is of sufficient quality to allow terminology and causality to be completed. When a report is deemed to be sufficient and accurate, the report should be confirmed, which will effectively allow terminology definition to commence.

Once you have searched for a report, click on the **Confirm Report** menu for the associated report you would like to confirm.

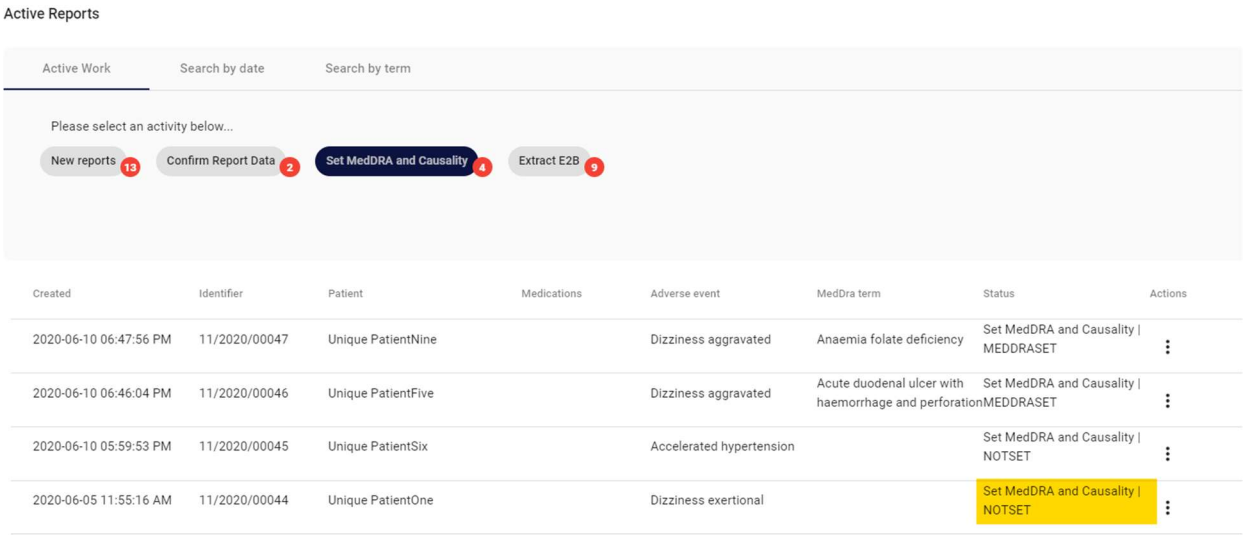


The system will navigate you to the **Add Activity** page.

Specify any additional comments for confirming the record and click **Save** to confirm the deletion or **Cancel** to cancel the action and return to the previous page.



The system will update the status of the report accordingly.



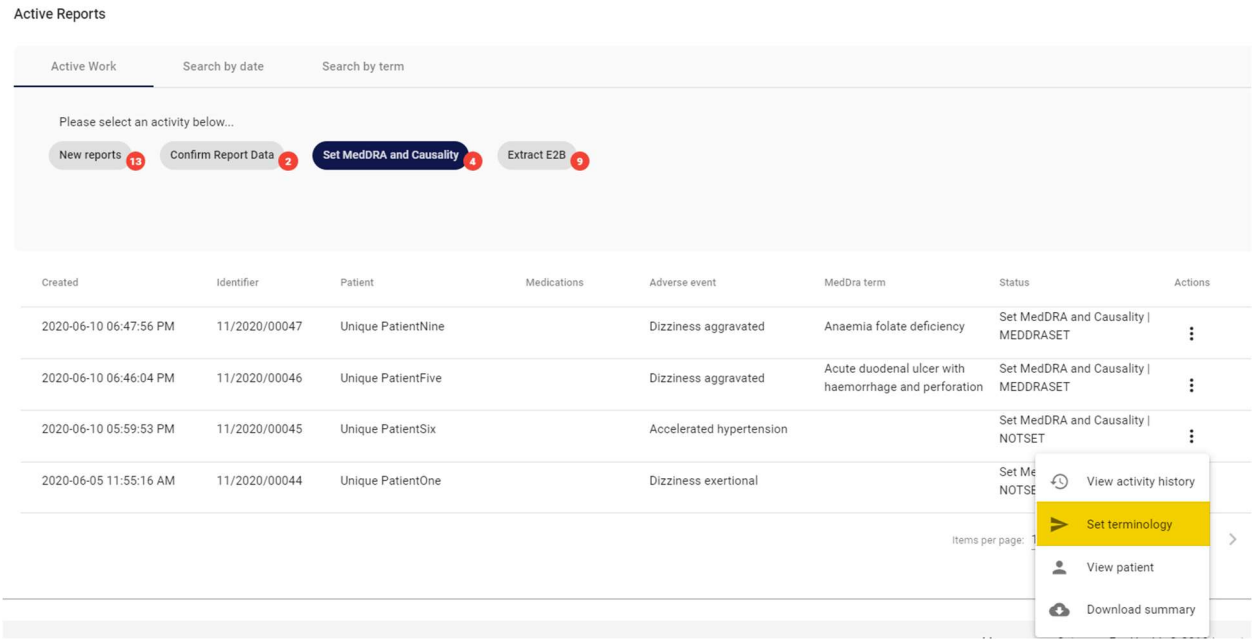
Confirming a report will move the report into the next stage: **Set MedDRA and Causality.**

5.6 Pharmacovigilance Activities – Set MedDRA and Causality

5.6.1 Set MedDRA Terminology

The **Set MedDRA and Causality** stage facilitates the process of confirming the final MedDRA term for the adverse event and allows assignment of causality per relevant medication using either the World Health Organization or the Naranjo Causality Scale.

Once you have searched for a report, click on the **Set Terminology** menu for the associated report you would like to set terminology for.



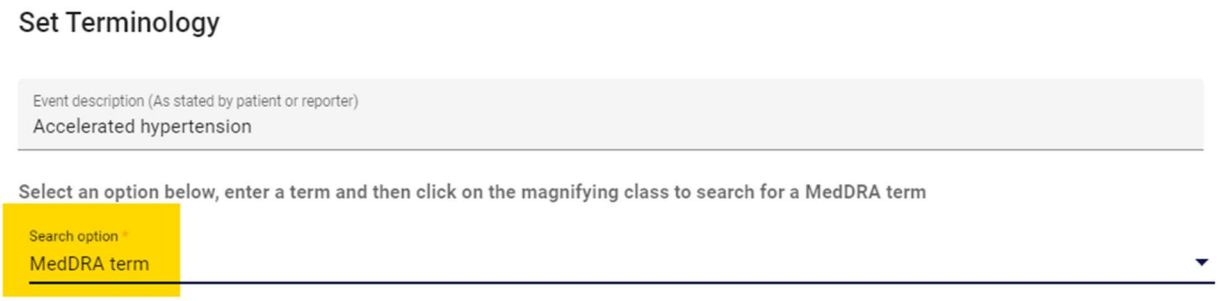
The system will pop up a **MedDRA Terminology** form.

The **MedDRA Terminology** form displays the adverse event as reported by the clinician and/or reporter.

MedDRA terms can be searched in the following ways:

- Search by MedDRA Term
- Search by MedDRA Code
- Search by MedDRA List

5.6.1.1 Search by MedDRA Term

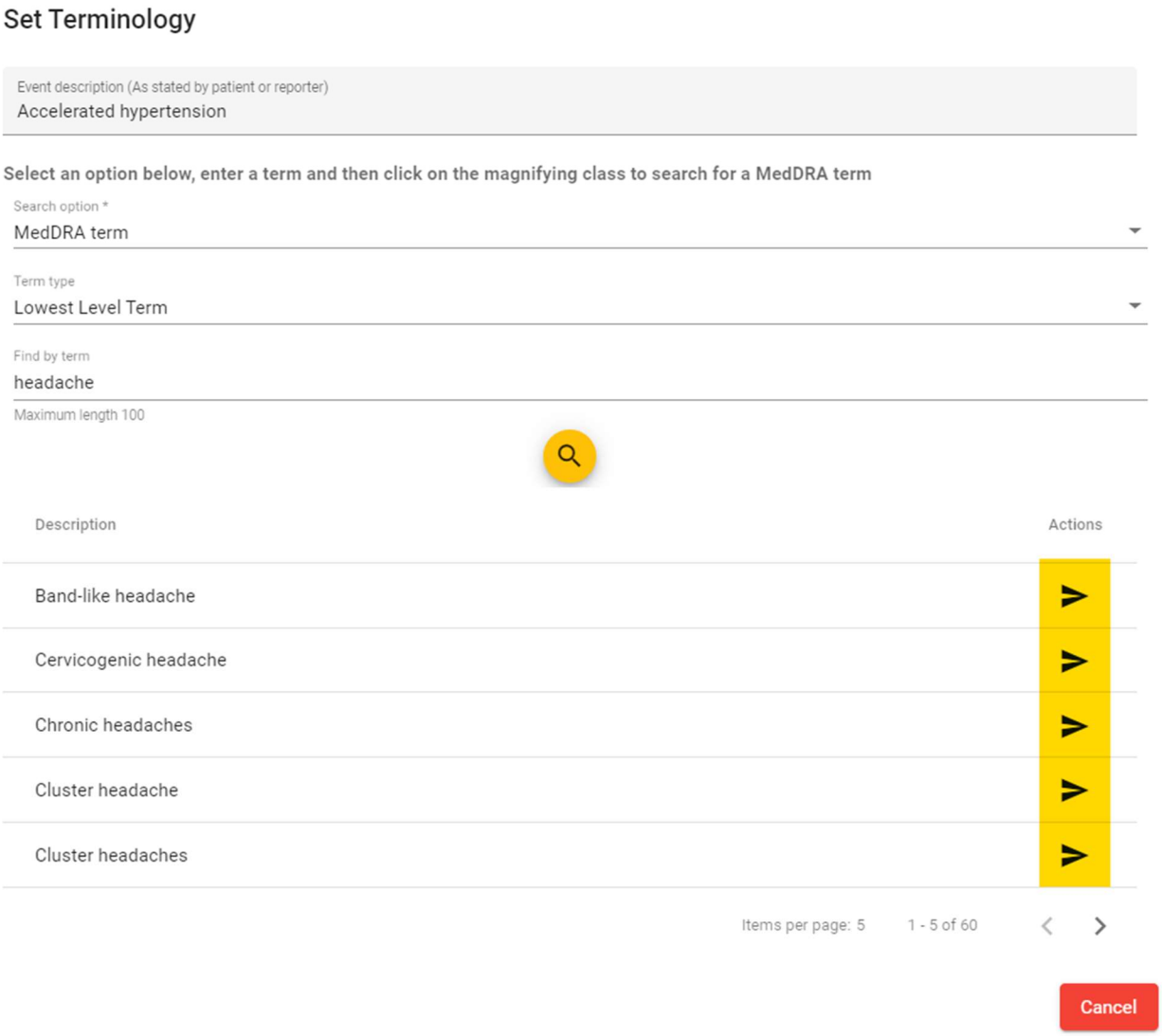


Searching by a MedDRA term allows you the ability to filter through MedDRA using a partial or fully defined term.

In the **Term type** field, select the level of specificity for the search (SOC, HLTG, HLT, PT, or LLT).

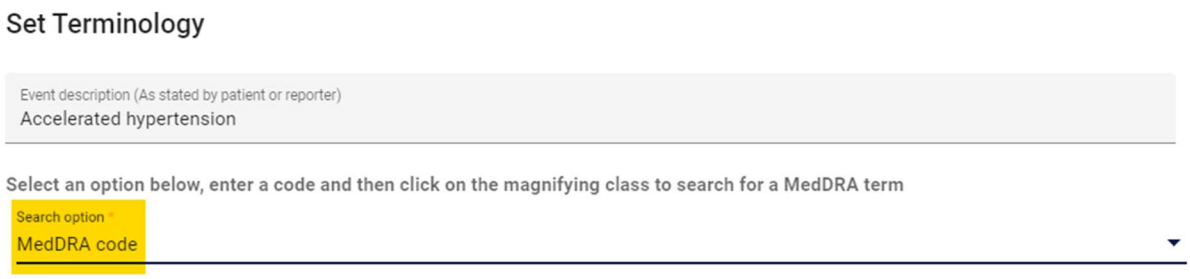
In the **Find by term** field, enter the term you are searching for.

It is possible to complete a partial search by entering as few as 3 characters that form part of the overall term.



Click the **Select** icon next to the term you would like to assign or click the **Cancel** button to cancel the action and return to the previous page. The report will be updated with the new term accordingly.

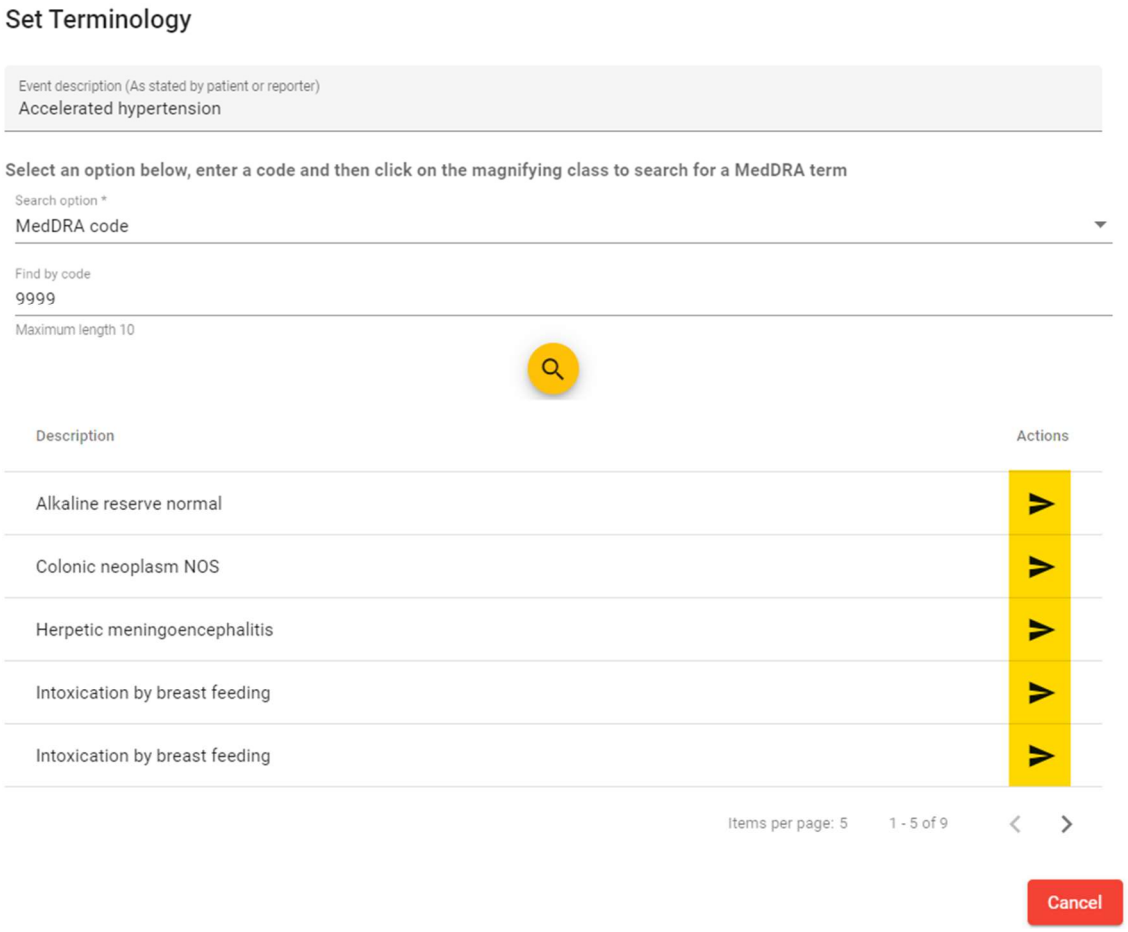
5.6.1.2 Search by MedDRA Code



Searching by a MedDRA code allows you the ability to filter through MedDRA using an associated code which can be fully defined or partial in nature.

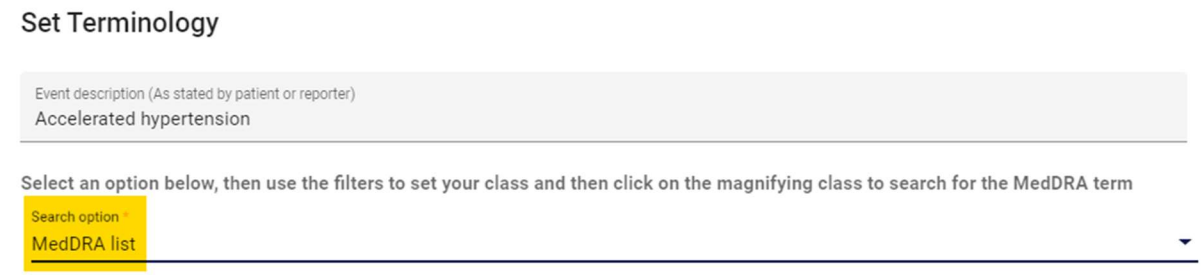
In the **Find by code** field, enter the code you are searching for.

It is possible to complete a partial search by entering as few as 4 numerical characters that form part of the overall code.

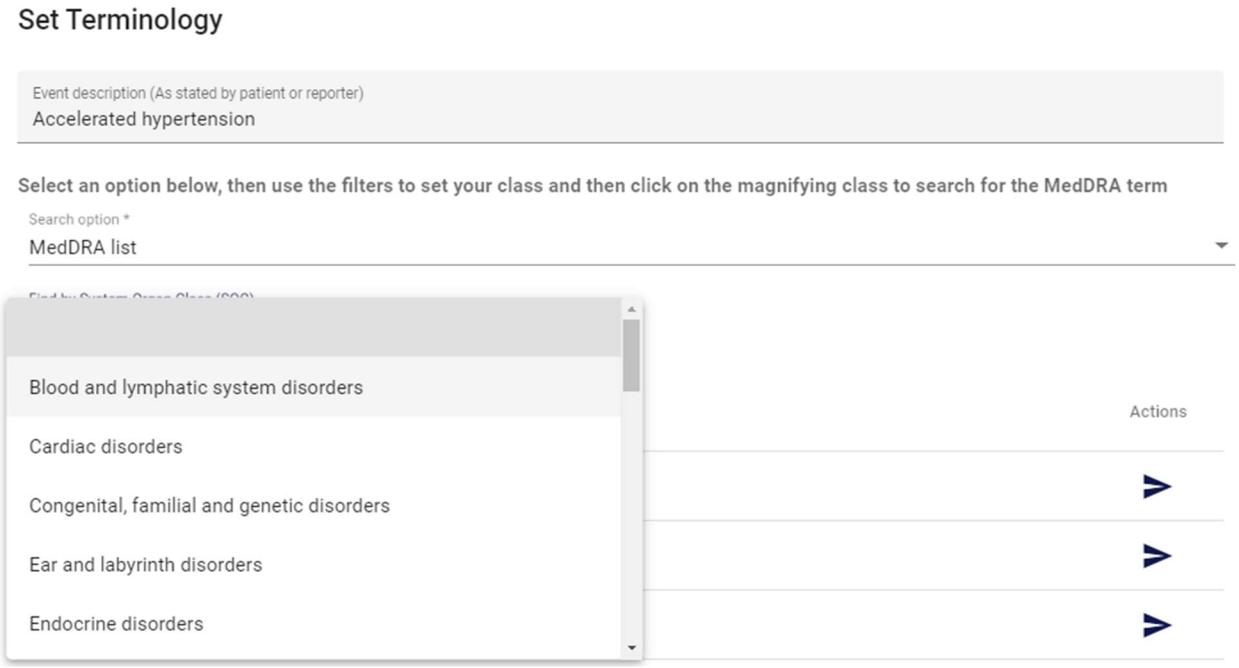


Click the **Select** icon next to the term you would like to assign or click the **Cancel** button to cancel the action and return to the previous page. The report will be updated with the new term accordingly.

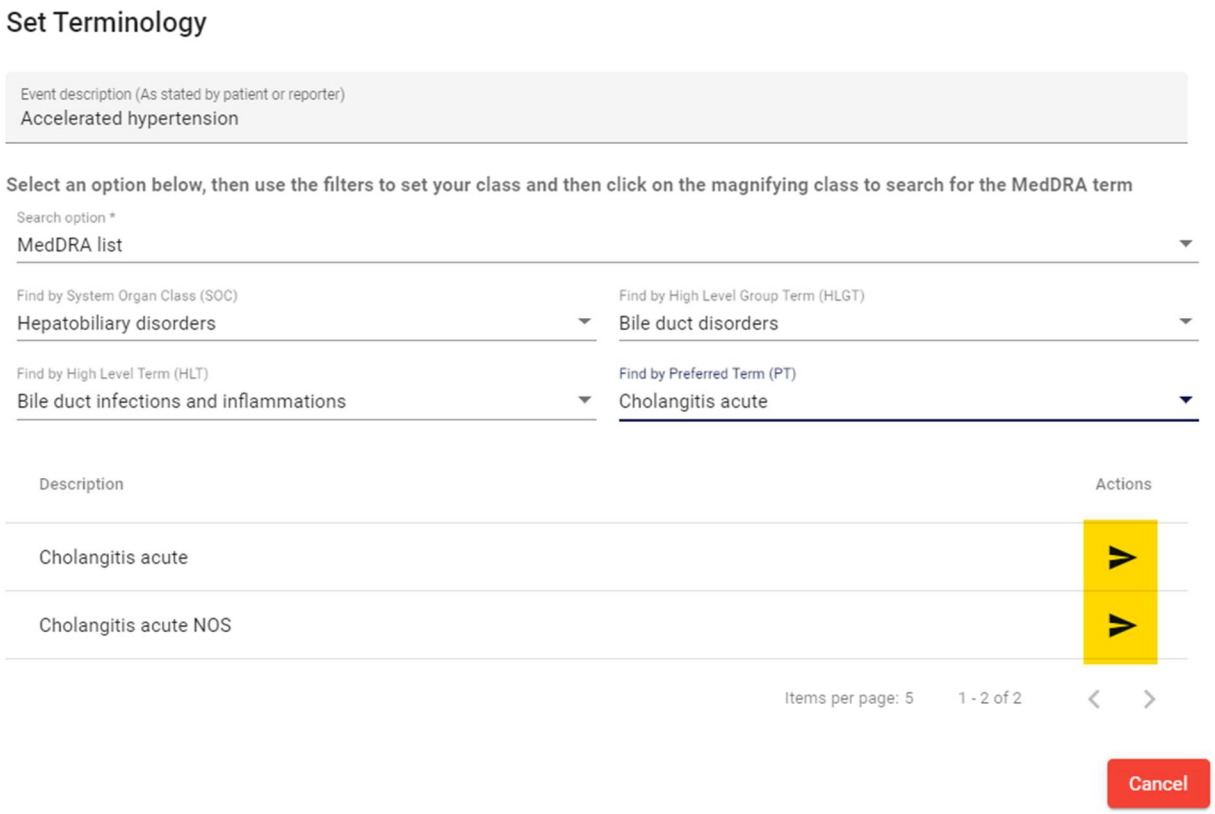
5.6.1.3 Search for MedDRA Term by List



The list search function allows the user to navigate the MedDRA dictionary using the hierarchical structure of the dictionary. Select the System Organ Class field to select a SOC.



Continue selecting through the hierarchy (HLGT, HLT, PT) until you have selected the Low-Level Term that matches the adverse event.

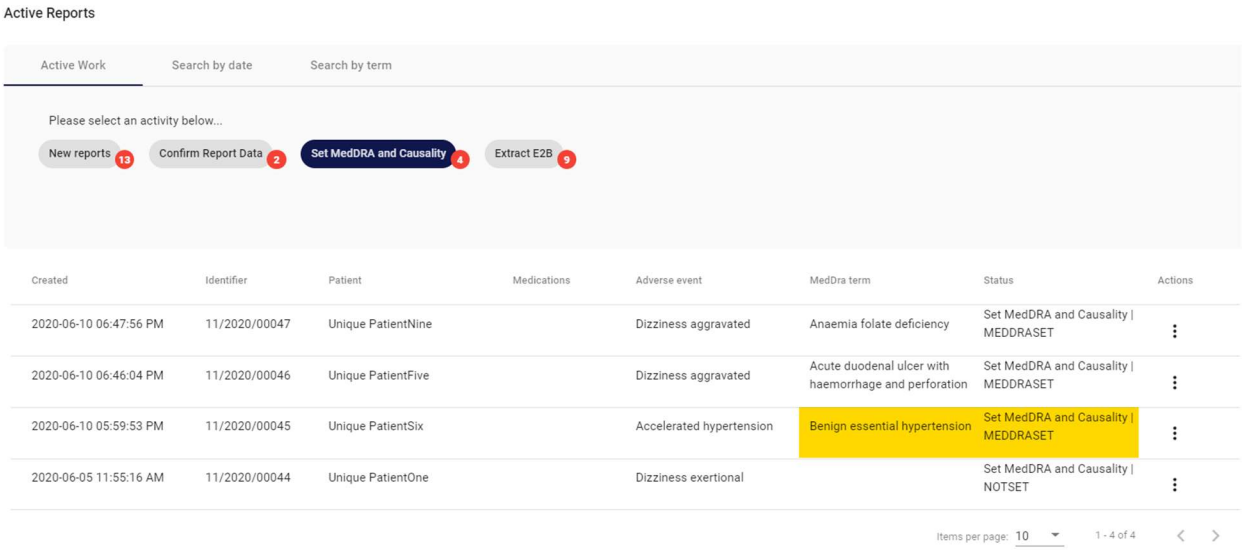


Click the **Select** icon next to the term you would like to assign or click the **Cancel** button to cancel the action and return to the previous page. The report will be updated with the new term accordingly.

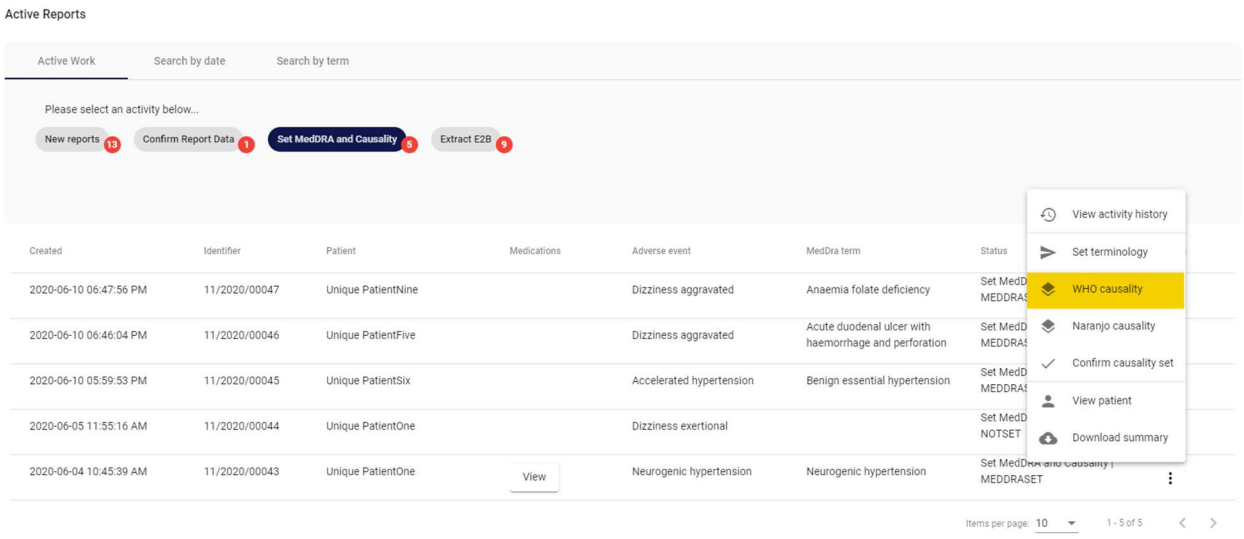
5.6.2 Causality Assessment using the WHO Scale

The **Set MedDRA and Causality** stage facilitates the process of confirming the final MedDRA term for the adverse event and allows assignment of causality per relevant medication using either the World Health Organization or the Naranjo Causality Scale.

Once the MedDRA terminology for the event has been set, the report will be updated accordingly.

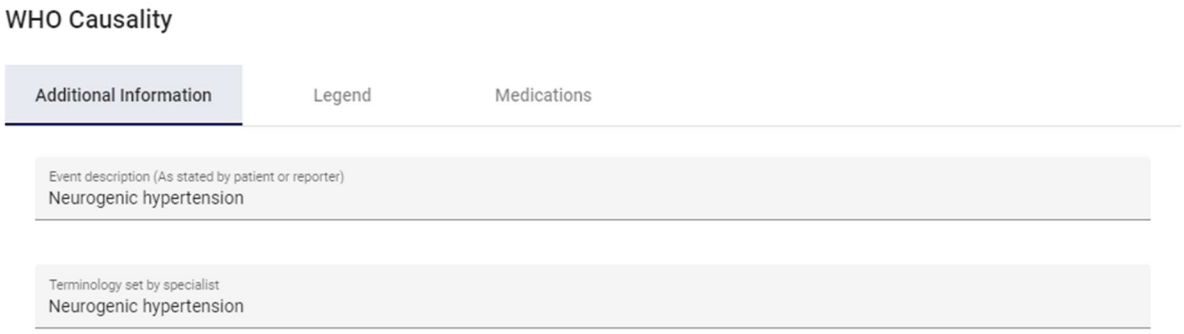


Setting the terminology will further allow the setting of causality per medication. Click on the **WHO Causality** menu option for the associated report you would like to set causality for.



The system will open up the **WHO Causality Assessment** pop up form.

The **WHO Causality Assessment** form displays the adverse event as reported by the clinician and/or reporter and the MedDRA Terminology that has been set for the event.

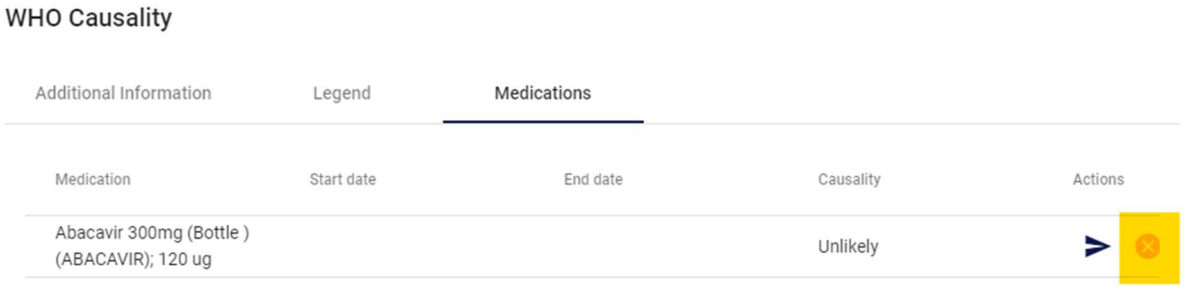


The **WHO Causality Assessment** form also displays the list of medications that the patient was exposed to at the onset of the adverse event.

The analyst can perform the following actions against each medication the patient was exposed to:

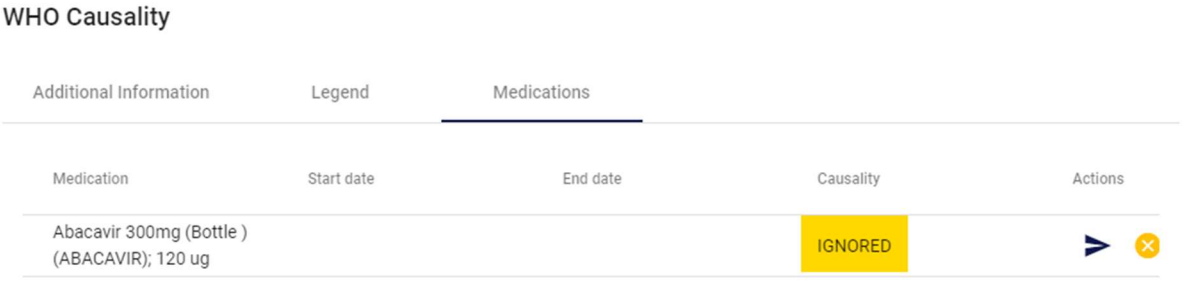
- Ignore the medication (the medication is definitely not responsible for the adverse event)
- Set causality for the medication

5.6.2.1 Ignoring the Medication

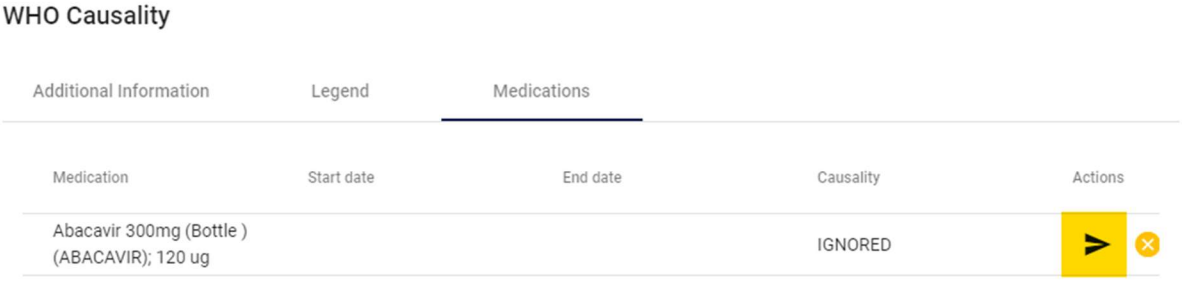


Ignoring the medication means that a causality assessment need not be set for that medication.

To ignore assigning a WHO causality assessment for this medication, click the **Ignore Medication** icon. The system will assign **Ignored** as the causality term.



5.6.2.2 Set Causality

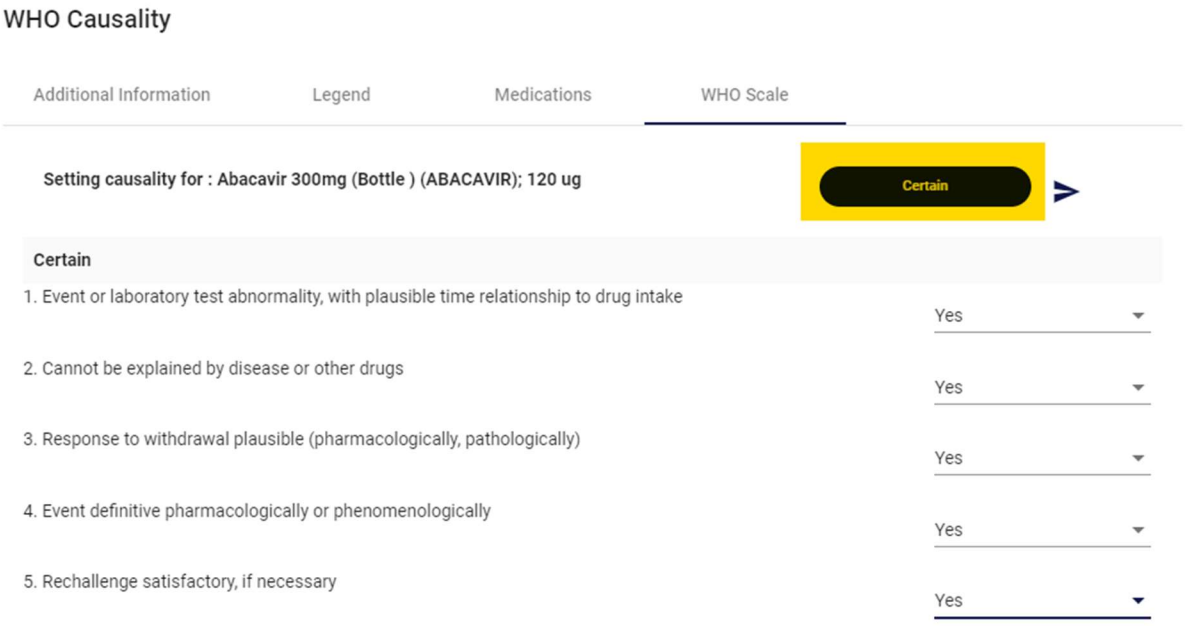


By clicking the **Set Causality** icon, the system will enable you to set causality for this medication.

The system displays a series of questions for assigning the WHO causality term.

Click the appropriate response (Yes or No) for question 1 and continue to respond to the first five questions.

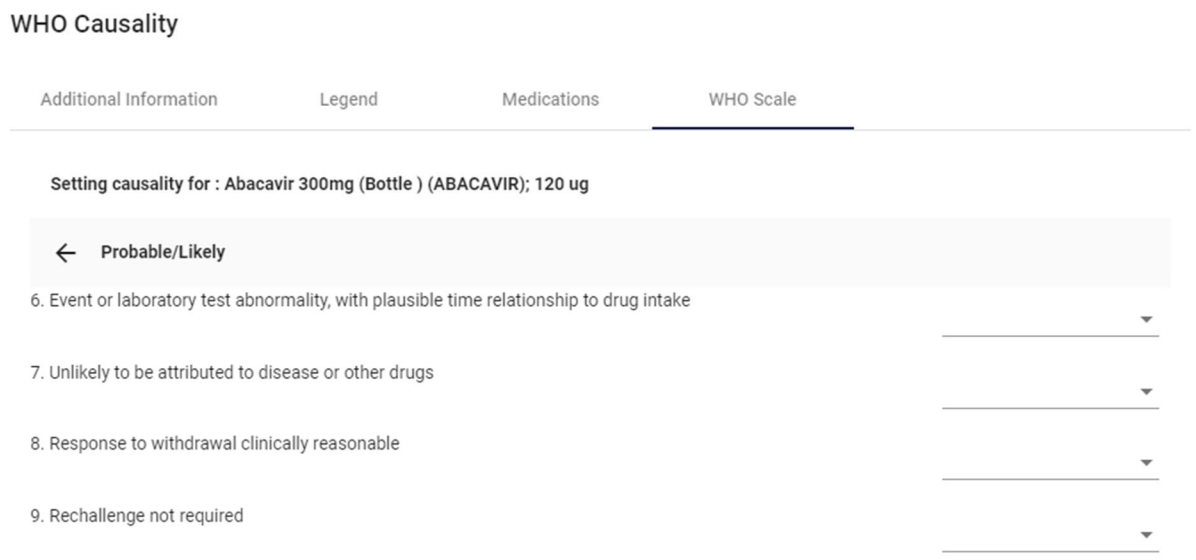
If responses to all five questions are **Yes,** the system will assign **Certain** as the causality term.



Click the **Set causality icon** next to the causality assessment and the system will assign the term as Certain in the system.

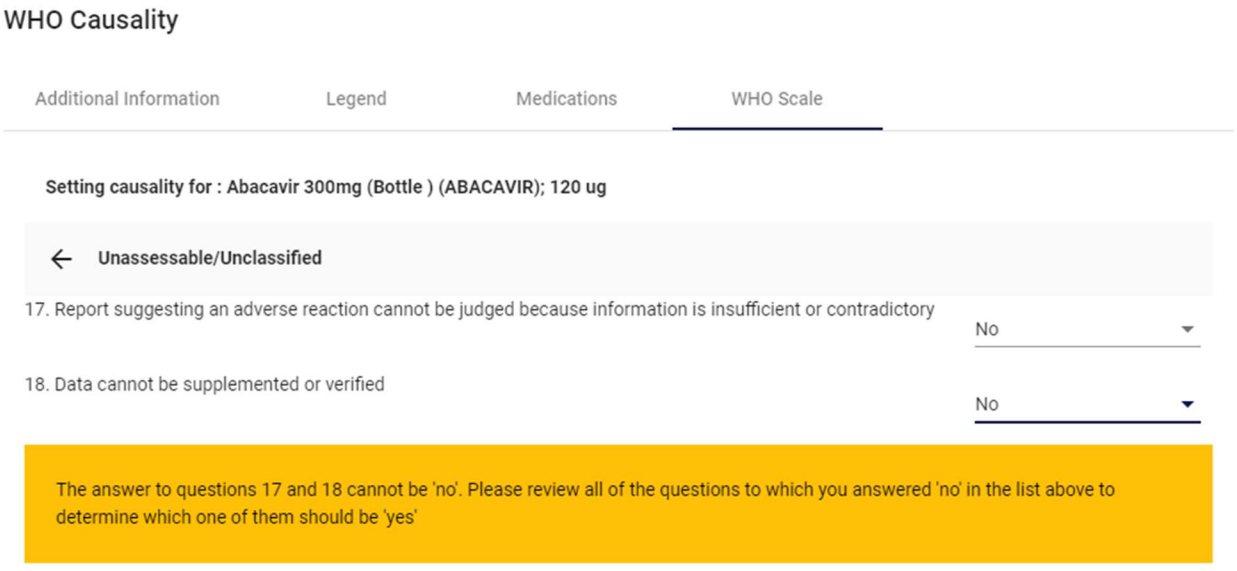
Click the **Close** button at any time to undo the action and return to the list reports.

If any of the questions in the Certain category have a response of **No** the system will display questions for the Probable/Likely category.

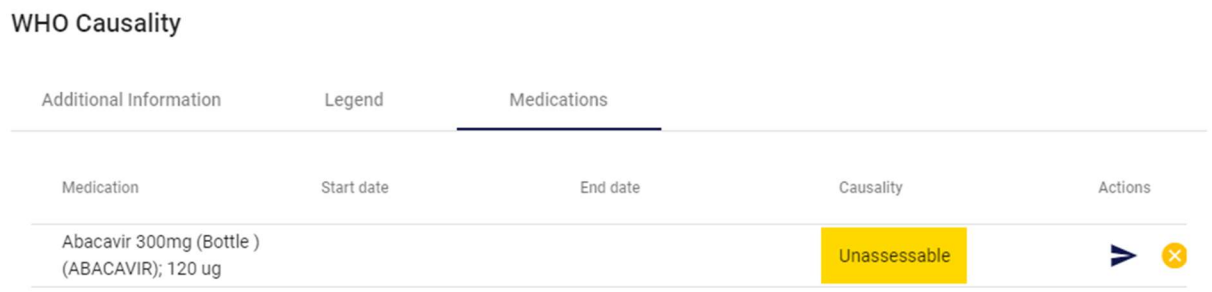


Continue this process until you have answered **Yes** to all questions in a specific category. This will result in the classification for that medication being set to that category.

In the event you reach the **Unassessable/Unclassified** category, if one of the two questions asked in that category is answered as **No,** the system will trigger an alert.



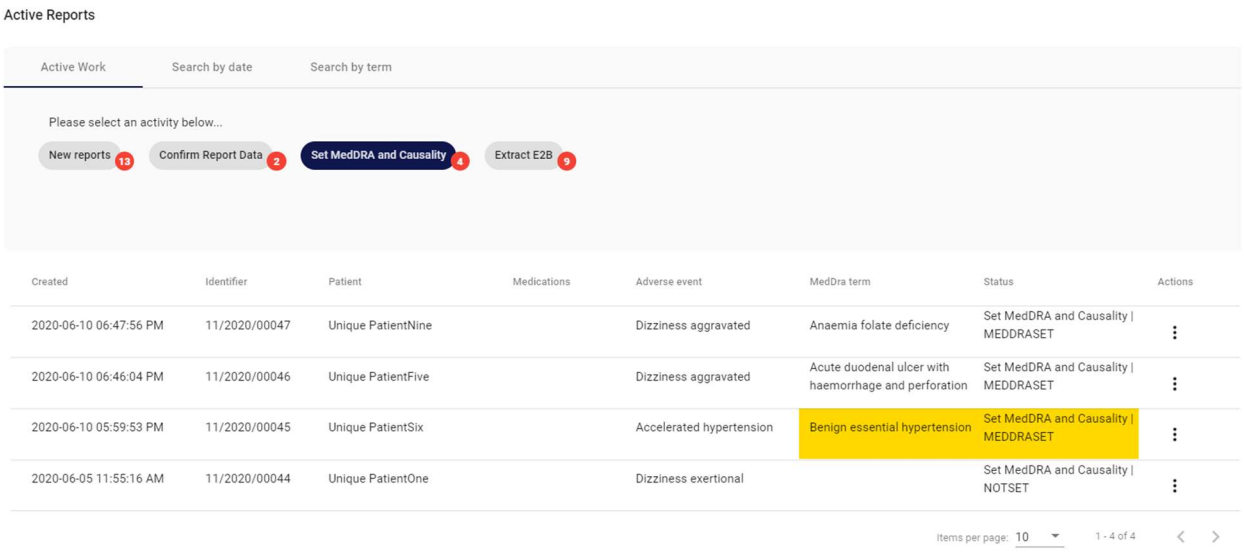
Once the term has been assigned click the **Close** button to continue.



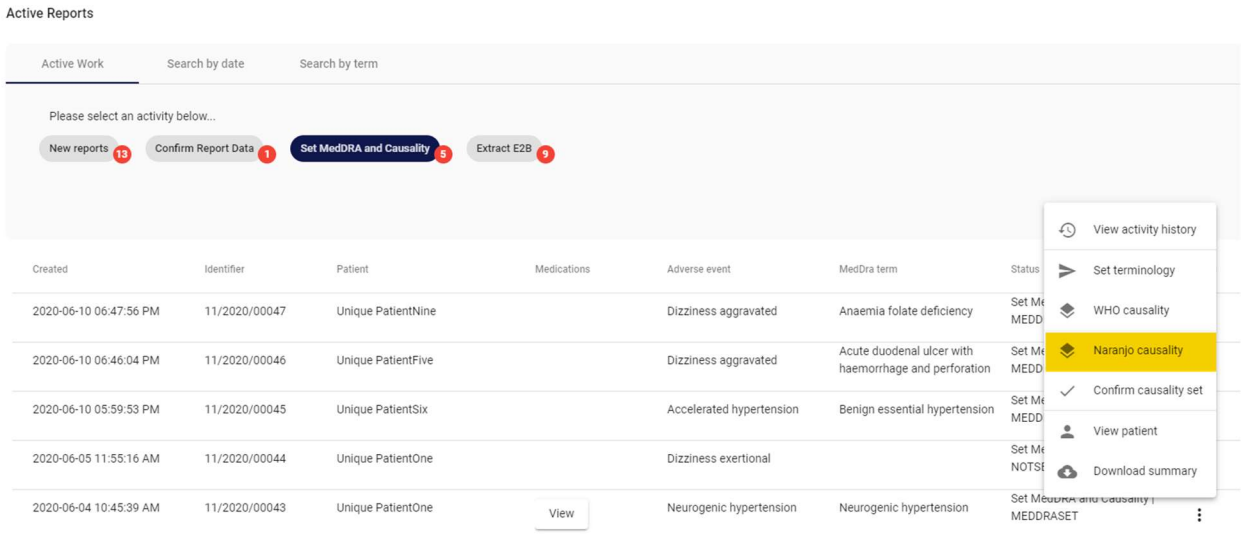
5.6.3 Causality Assessment using the Naranjo Scale

The **Set MedDRA and Causality** stage facilitates the process of confirming the final MedDRA term for the adverse event and allows assignment of causality per relevant medication using either the World Health Organization or the Naranjo Causality Scale.

Once the MedDRA terminology for the event has been set, the report will be updated accordingly.

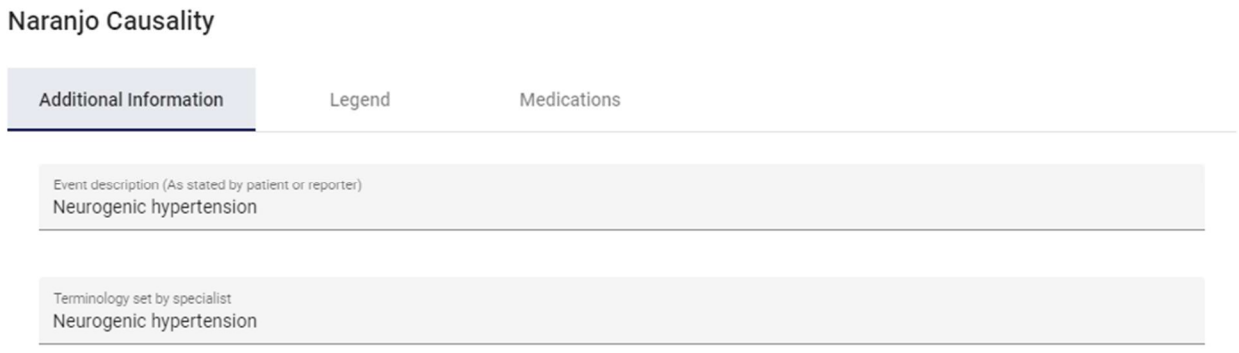


Setting the terminology will further allow the setting of causality per medication. Click on the **Naranjo Causality** menu for the associated report you would like to set causality for.



The system will open up the **Naranjo Causality Assessment** pop up form.

The **Naranjo Causality Assessment** form displays the adverse event as reported by the clinician and/or reporter and the MedDRA Terminology set.

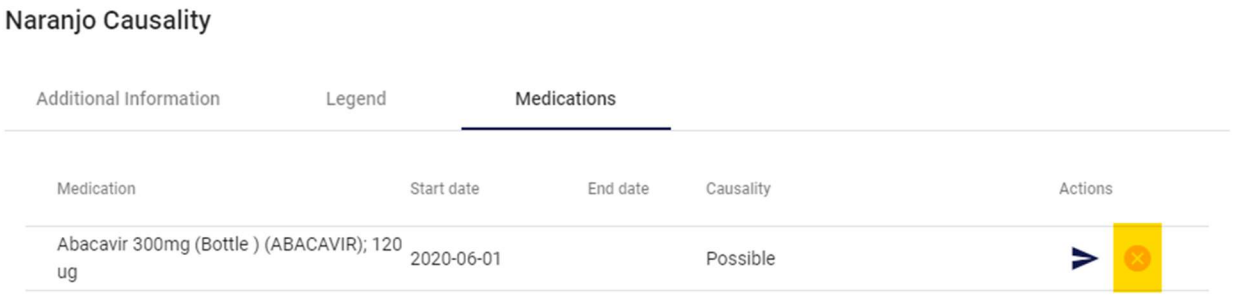


The **Naranjo Causality Assessment** form also displays the list of medications that the patient was exposed to at the onset of the adverse event.

The analyst can perform the following actions against each medication the patient was exposed to:

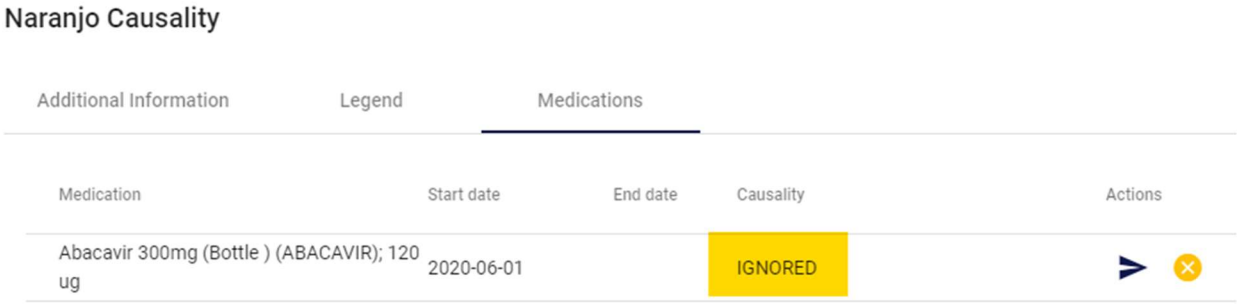
- Ignore the medication (the medication is definitely not responsible for the adverse event)
- Set causality for the medication

5.6.3.1 Ignoring the Medication

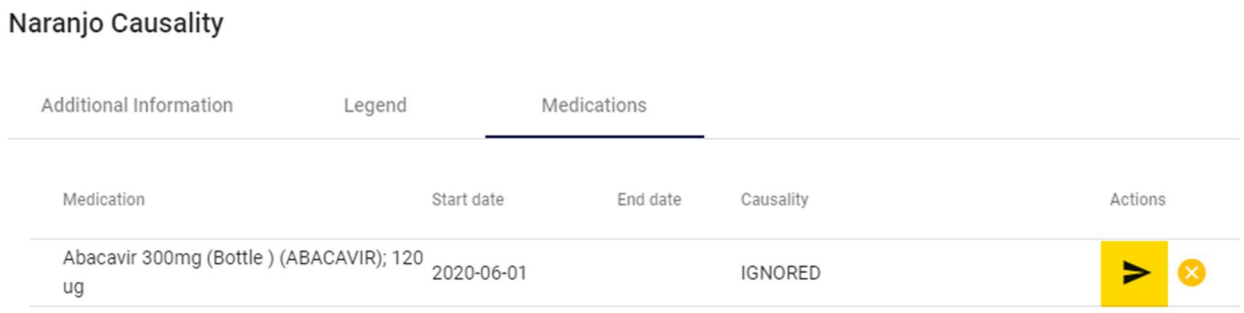


Ignoring the medication means that a causality assessment need not be set for that medication.

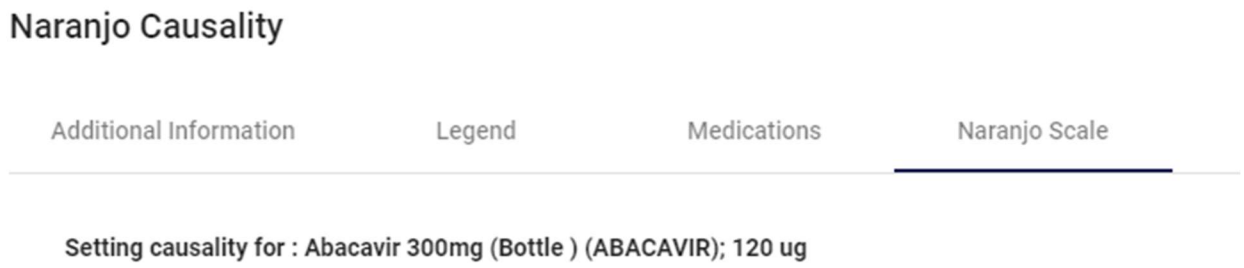
To ignore assigning a Naranjo causality assessment for this medication, click the **Ignore Medication** icon. The system will assign **Ignored** as the causality term.



5.6.3.2 **Set Causality**

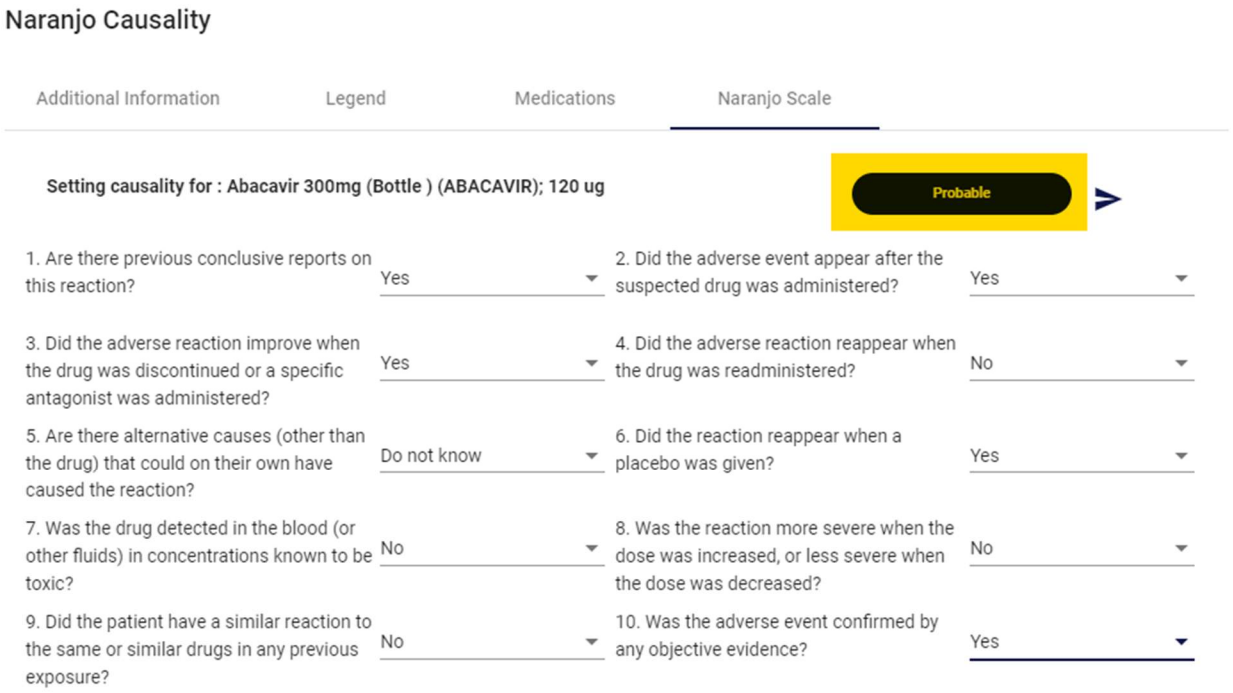


By clicking the **Set Causality** icon, the system will enable you to set causality for this medication.



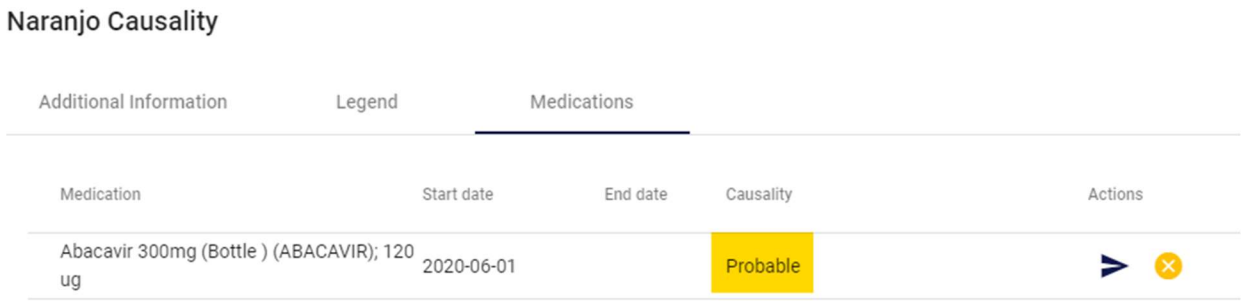
The system displays a series of questions for calculating the Naranjo causality score.

Click the appropriate response (Yes or No) for question 1 and continue to respond to all remaining questions. Once all responses have been selected, the system will automatically calculate the causality assessment.



Click the **Set causality icon** next to the causality assessment and the system will assign the term as per the calculation in the system.

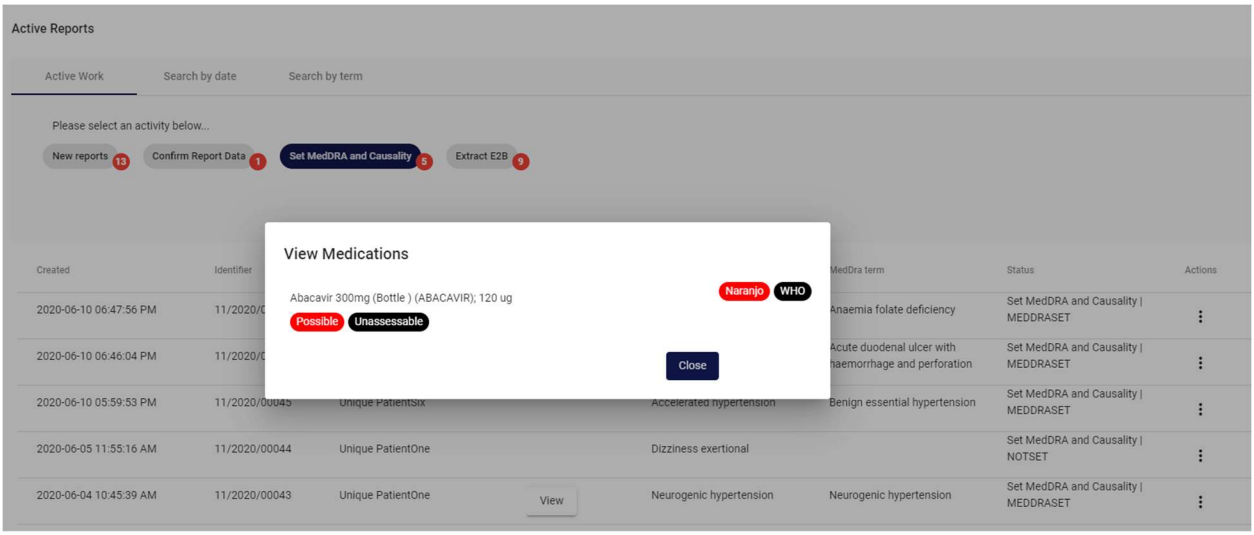
Click the **Close** button at any time to undo the action and return to the list reports.



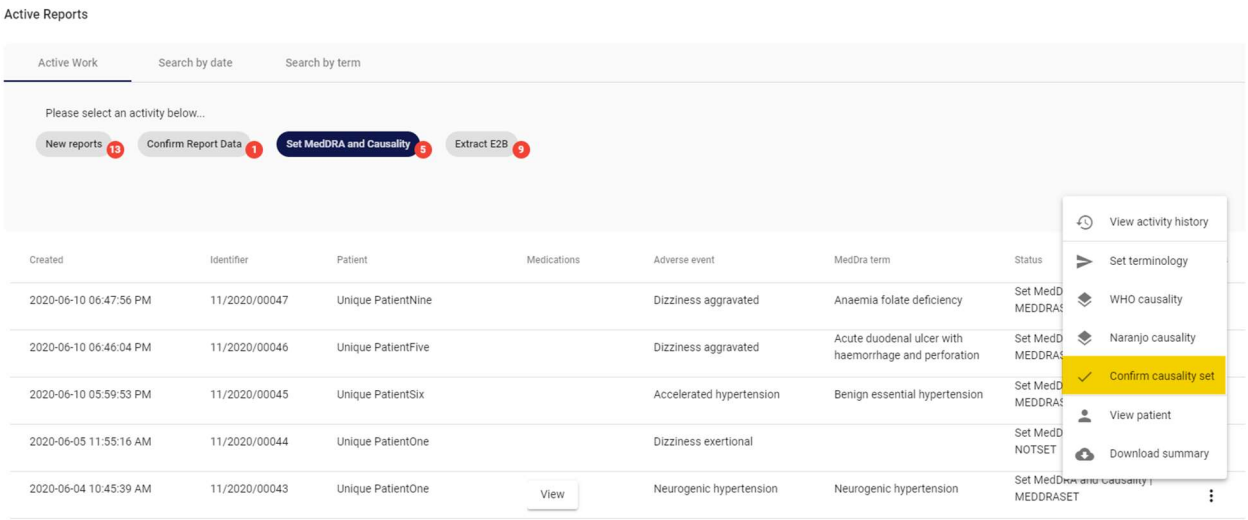
5.6.4 Confirming Causality Set

The **Set MedDRA and Causality** stage facilitates the process of confirming the final MedDRA term for the adverse event and allows assignment of causality per relevant medication using either the World Health Organization or the Naranjo Causality Scale.

Once causality for all medications has been set, the report will be updated accordingly.

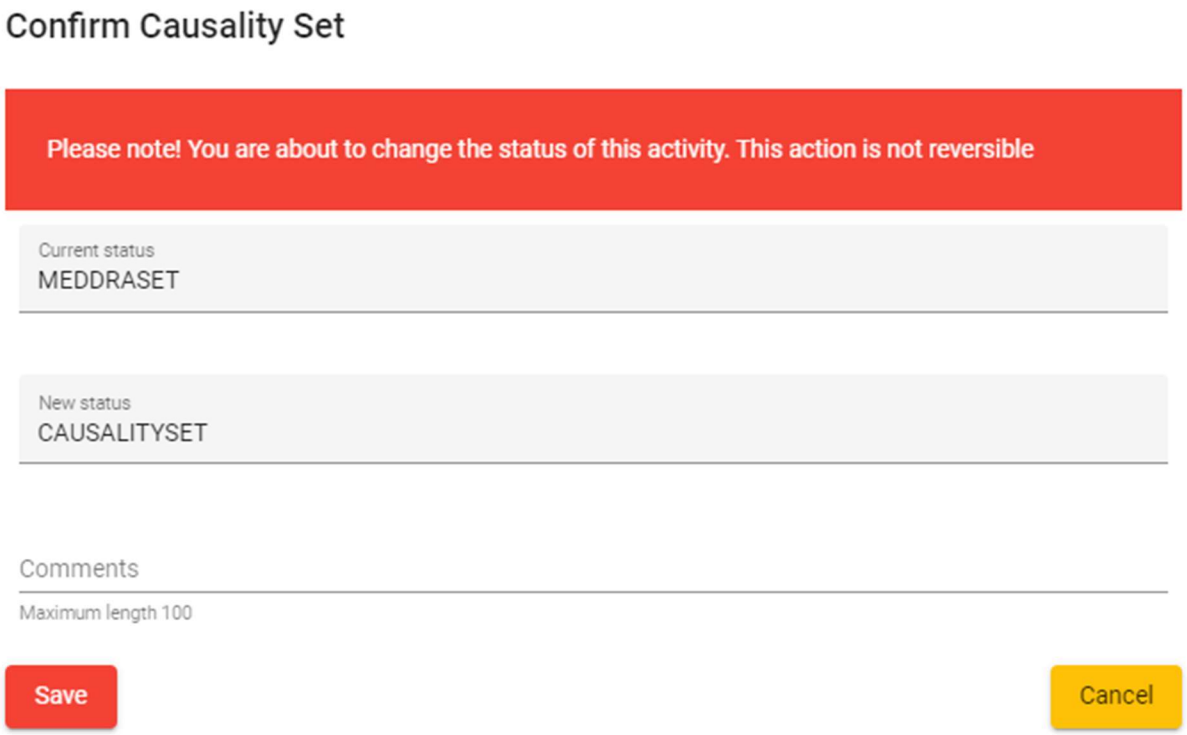


Click on the **Confirm Causality Set** menu for the associated report you would like to confirm causality set for.

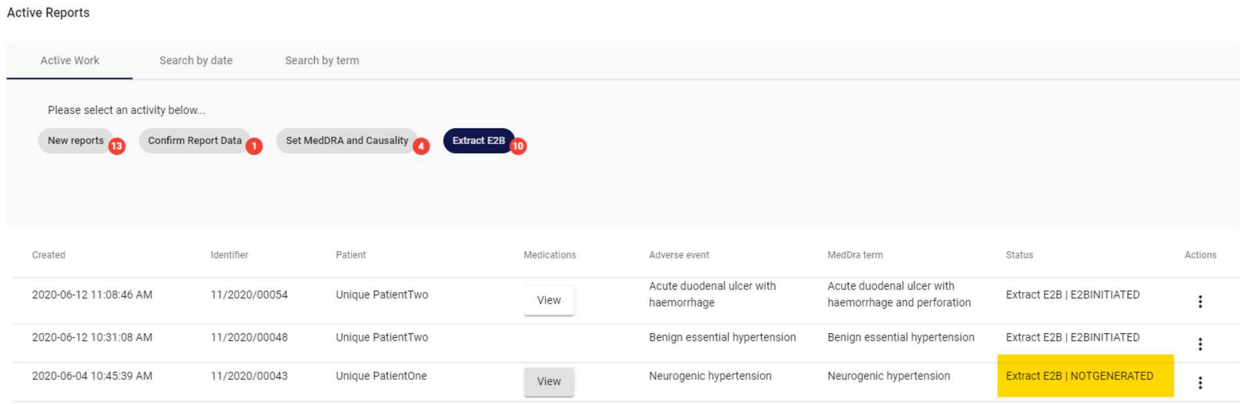


The system will navigate you to the **Confirm Causality Set** form.

Specify any comment if necessary and click **Save** to confirm the setting of the causality or **Cancel** to cancel the action and return to the previous page.



The system will update the status of the report accordingly. Please note the report will now be located in the Extract E2B tab.



5.7 Pharmacovigilance Activities – Extract E2B

5.7.1 Create E2B

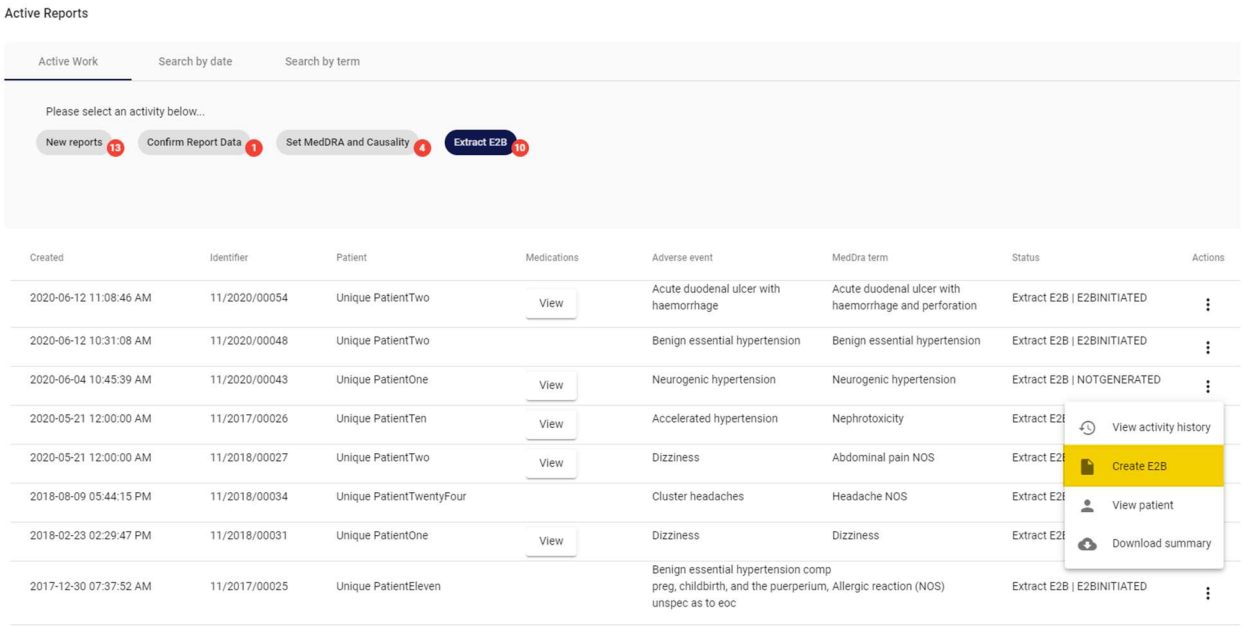
BUZZWORDS

ICH E2B: An E2B dataset facilitates the Electronic Transmission of Individual Case Safety Reports (ICSRs) and can be used to submit such reports to WHO. The E2B dataset within PViMS is implemented using the standard adopted by the ICH1 for electronic transmission of ICSRs according to the ICH E2B(R3) message standard.

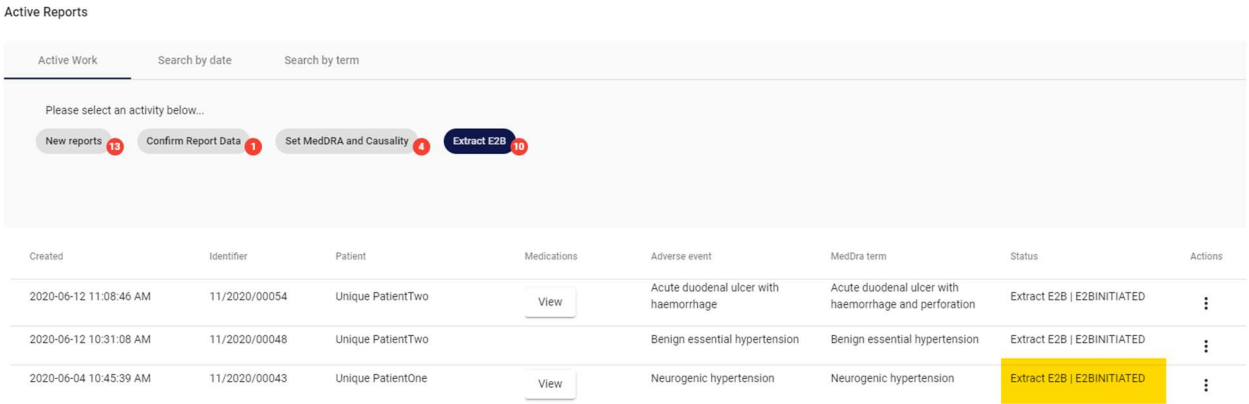
The **Extract E2B** stage generates an E2B extract for submission to the World Health Organization Uppsala Monitoring Centre.

By clicking this menu item, the system generates an E2B dataset that is populated with clinical information from the source form. This E2B dataset can be amended by you to reflect additional clinical information pertinent to WHO. This menu item is only available for forms that are in the Extract E2B | NOTGENERATED stage.

Once you have searched for a report, click on the **Create E2B** menu for the associated report you would like to create E2B for.



Once the system has created the E2B dataset, the status of the report will be updated to E2BINITIATED to indicate the dataset has been created.



The E2B dataset is constructed with the following sections (these sections are based on R2 of the E2B ICH definition):

- Message Header
- Safety Report
- Primary Source
- Sender
- Receiver
- Patient
- Medical History Episode
- Past Drug Therapy
- Patient Death
- Reaction
- Test
- Drug
- Summary

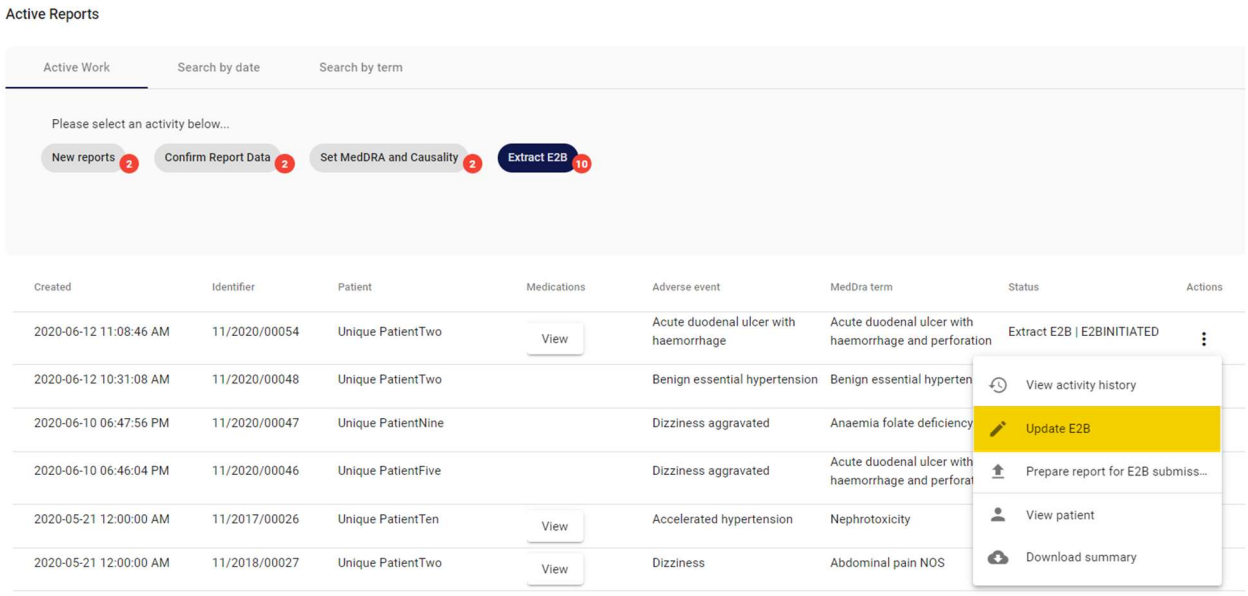
ICH E2B: Please see the electronic Transmission of Individual Case Safety Reports Message Specification for detail specifications on the above categories. This specification can be found on

https://admin.ich.org/sites/default/files/inline-files/ICH_ICSR_Specification_V2-3.pdf

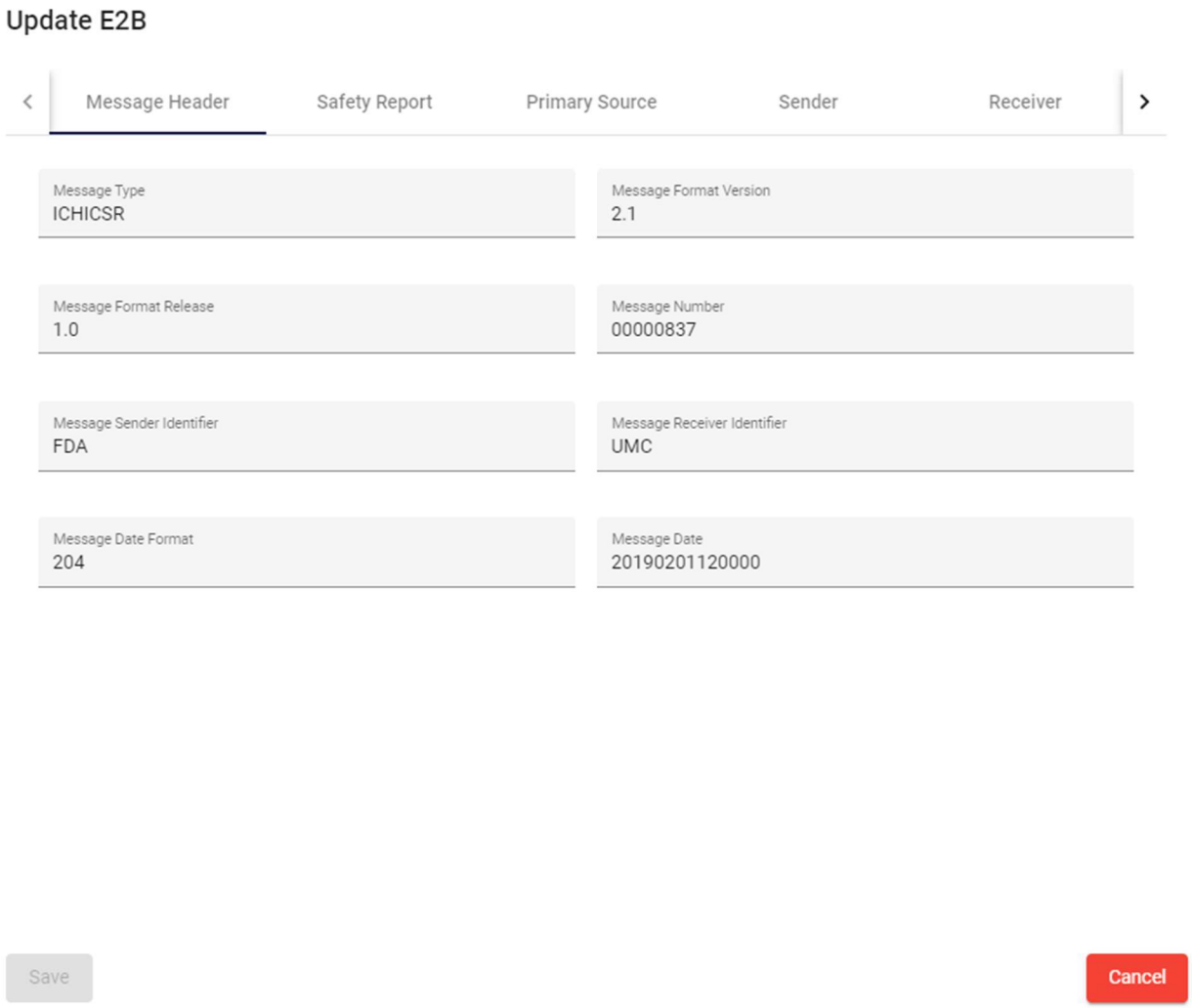
5.7.2 Adding Information to and Updating an E2B File

Once an E2B file is created, the **Update E2B** menu allows the user to add information to or updating existing information populated in an E2B file.

Once you have searched for a report, click on the **Update E2B** menu for the associated file you would like to amend.



The system will pop up a **E2B ICH Report** form for the file.

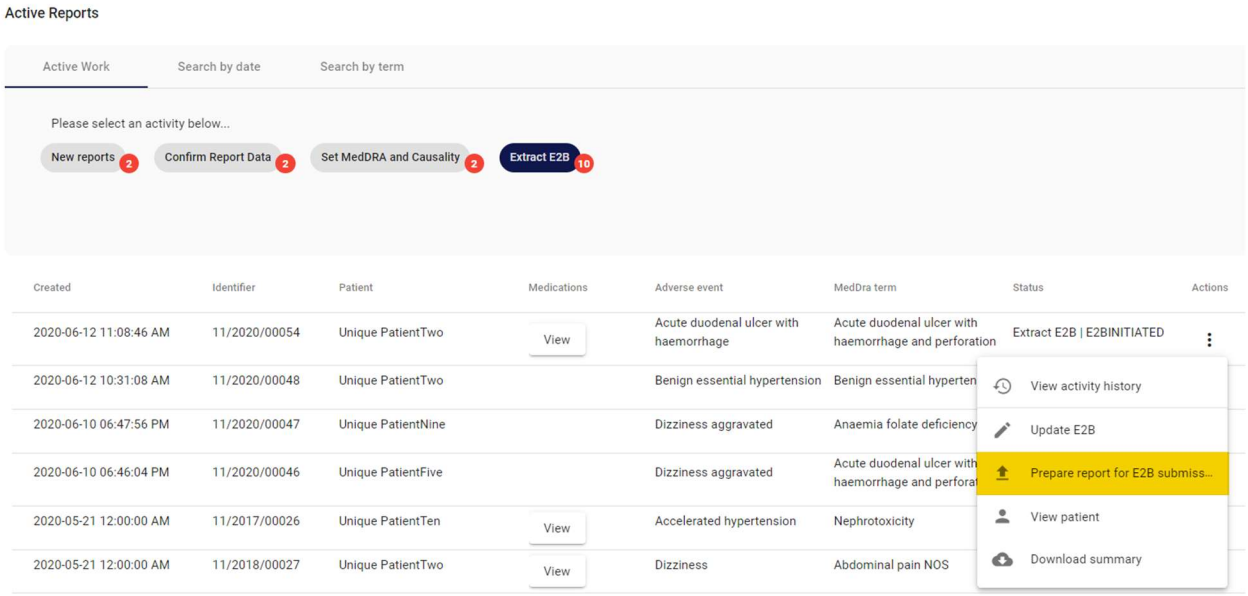


Once you completed making amendments, click on the **Save** button to save the changes or the **Cancel** button to cancel the action and return to the previous page.

5.7.3 **Preparing a Report for E2B Submission**

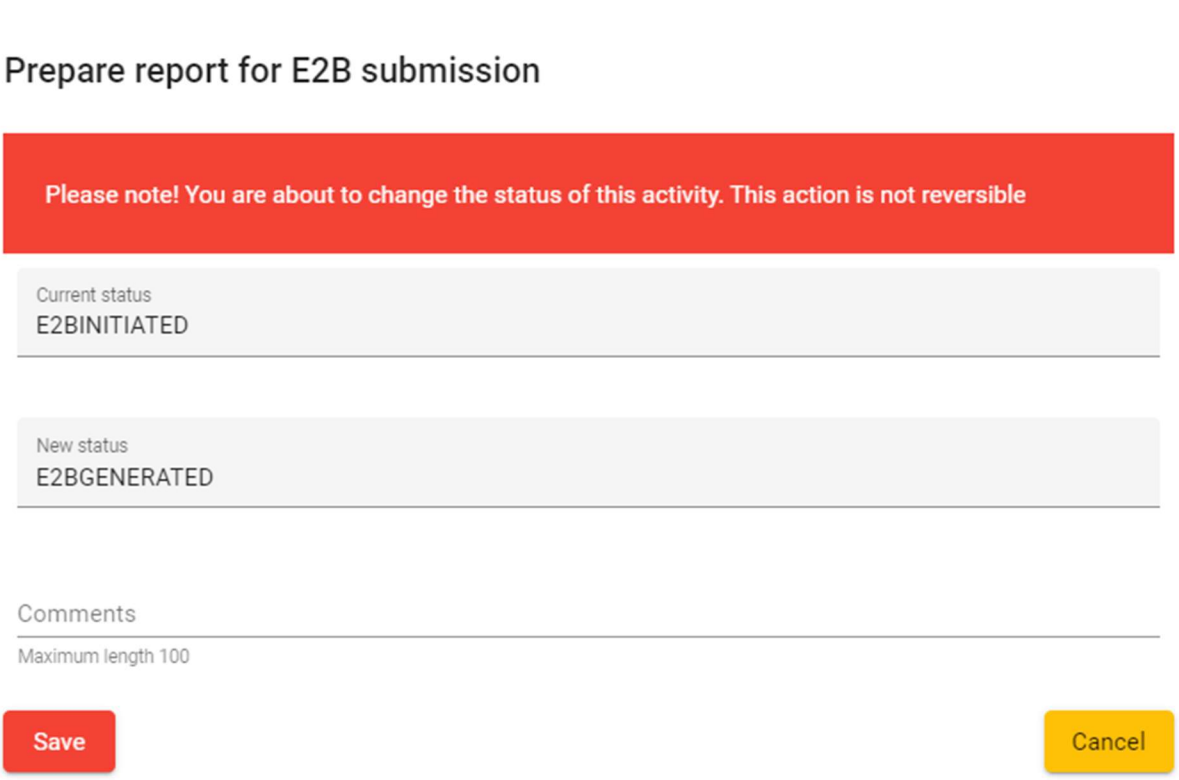
The **Extract E2B** stage facilitates the process of generating an E2B extract for submission to the World Health Organization Uppsala Monitoring Centre.

Once you have created an E2B file and searched for the report, click on the **Prepare Report for E2B Submission** menu for the associated report you would like to prepare.

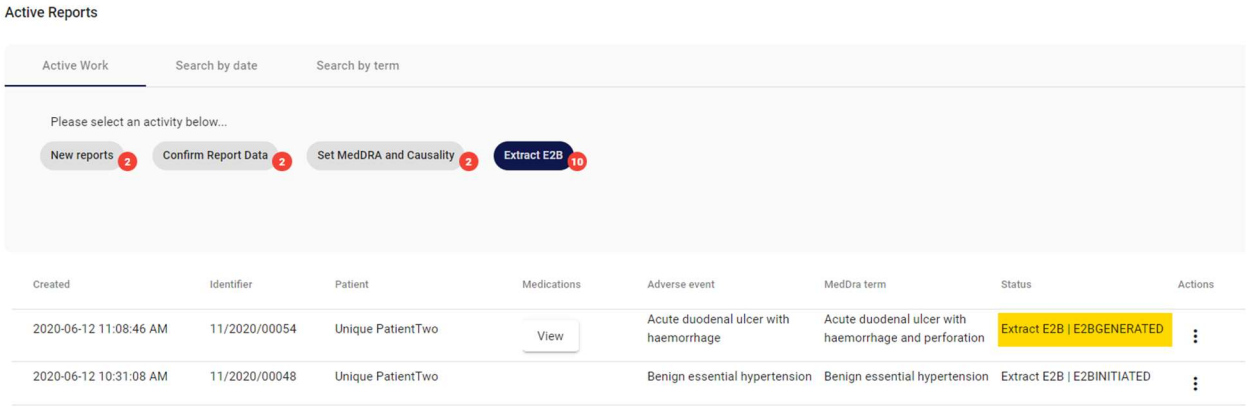


The system will navigate you to the **Prepare report for E2B submission** pop up form.

Specify any additional comments for preparing the record and click **Save** to confirm the preparation or **Cancel** to cancel the action and return to the previous page.



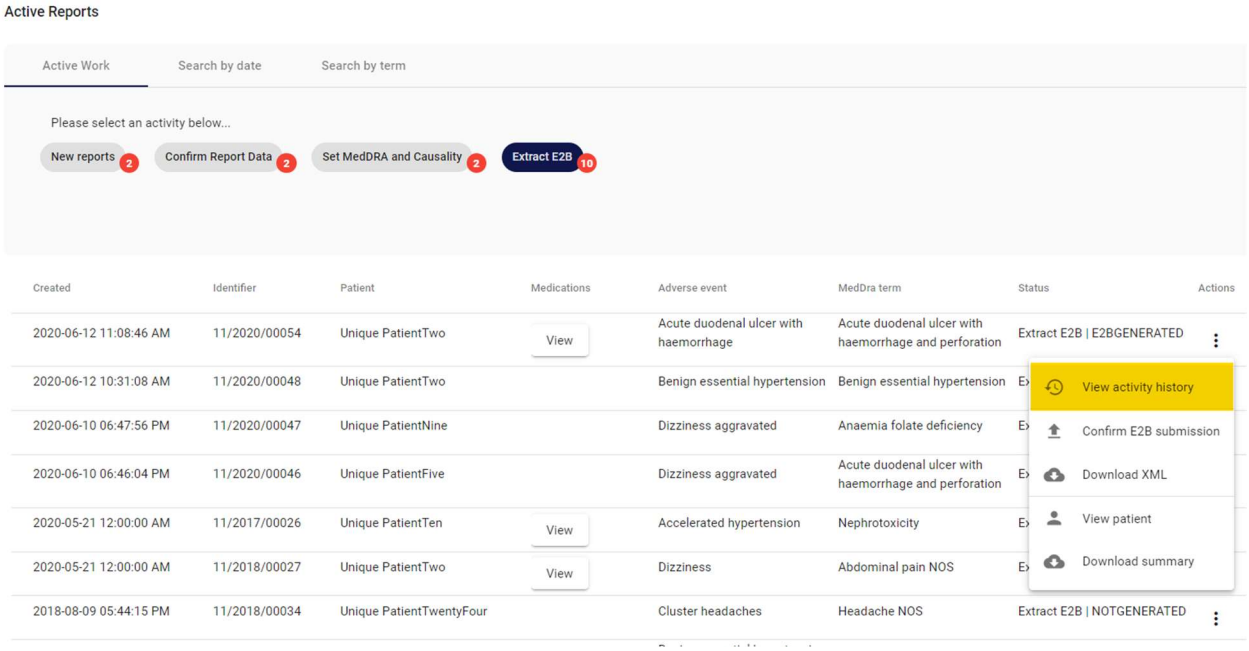
The system will update the status of the report accordingly.



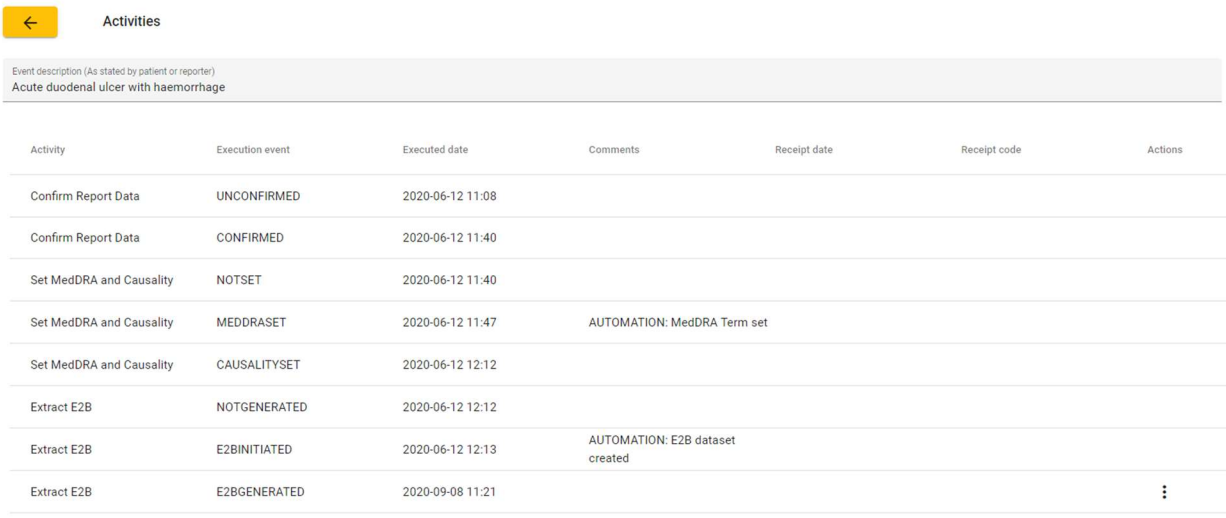
It is during this stage that PViMS prepares the XML file for submission. It also prepares an associated patient summary and patient extract that correspond to the clinical data at the point of generating the submission.

5.7.4 Viewing the E2B XML File

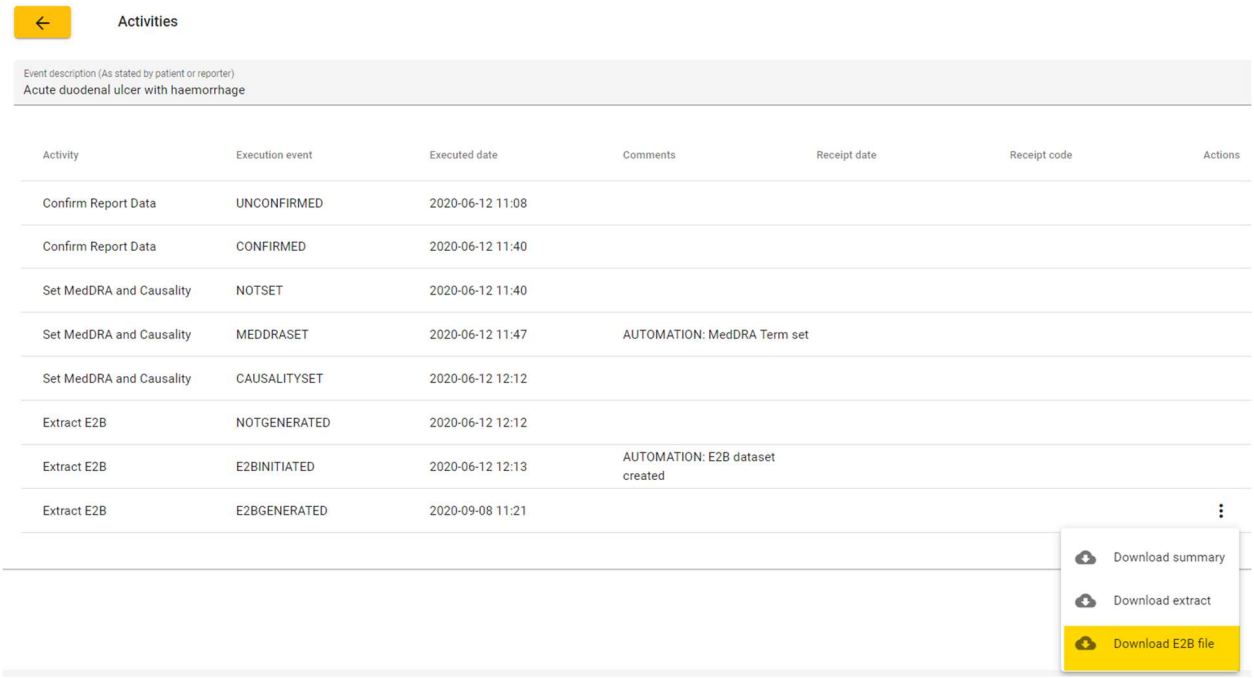
Once you have prepared the E2B file for submission and searched for the report, click on the **View Activity History** menu for the associated report you would like to view the XML file for.



The system will navigate you to the **Activities** page where you will be able to view a comprehensive history of activities by the analyst against this adverse event.



Locate the **E2BGENERATED** execution event for the report, click the menu for that event, and select the **Download E2B File** menu.



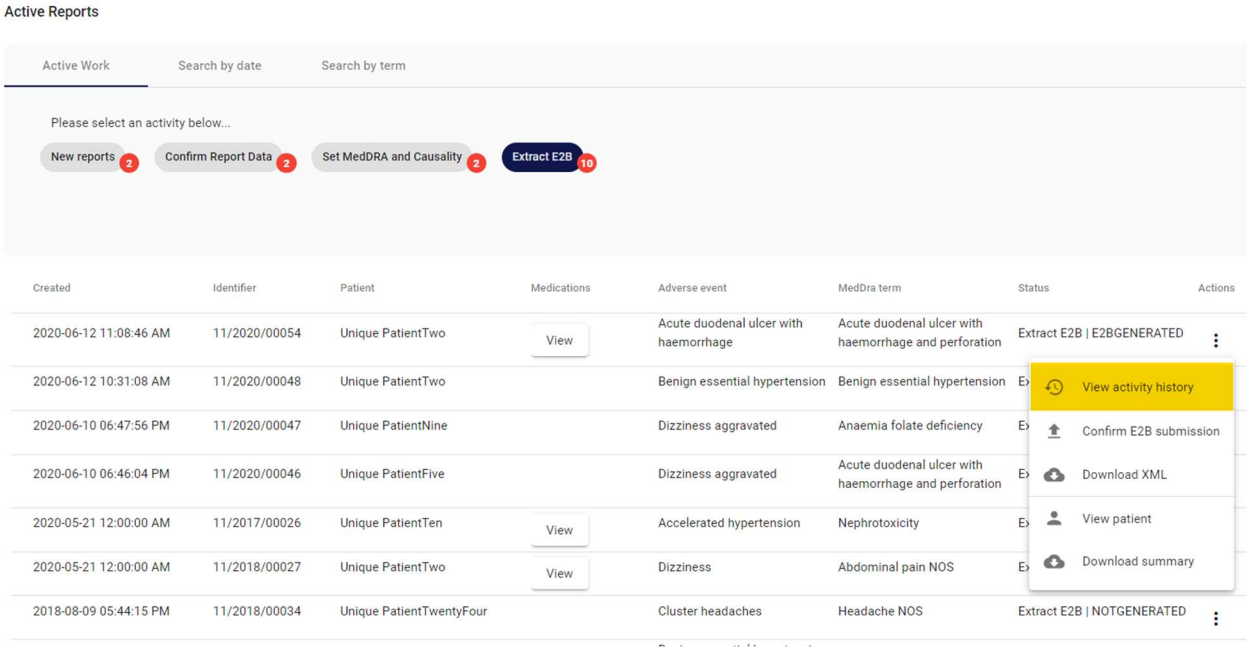
The system will automatically generate the XML file for the E2B submission to WHO. Save the file on your local computer for referral when sending to WHO.

The extract below is the XML generated for the Message Header section of the XML file.

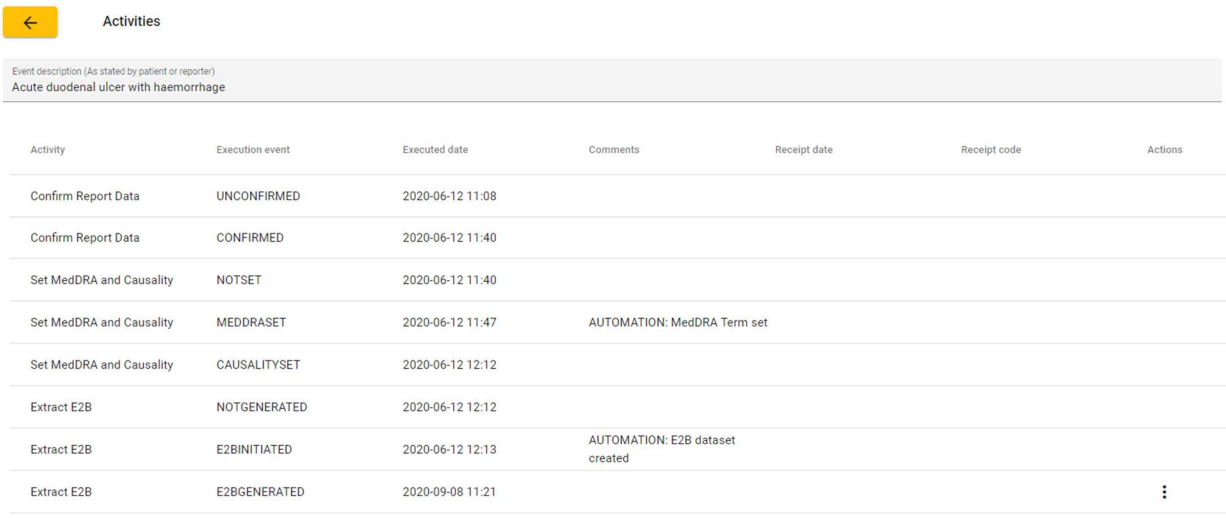
```
<?xml version="1.0" encoding="utf-16"?>
<ichicsr lang="en">
  <ichicsrmessageheader>
    <messagetype>ICHICSR</messagetype>
    <messageformatversion>2.1</messageformatversion>
    <messageformatrelease>1.0</messageformatrelease>
    <messagenumb>00000770</messagenumb>
    <messagesenderidentifier>FDA</messagesenderidentifier>
    <messagereceiveridentifier>UMC</messagereceiveridentifier>
    <messagedateformat>204</messagedateformat>
    <messagedate>20180810120000</messagedate>
  </ichicsrmessageheader>
```

5.7.5 Viewing the Clinical Data Associated to the E2B XML File

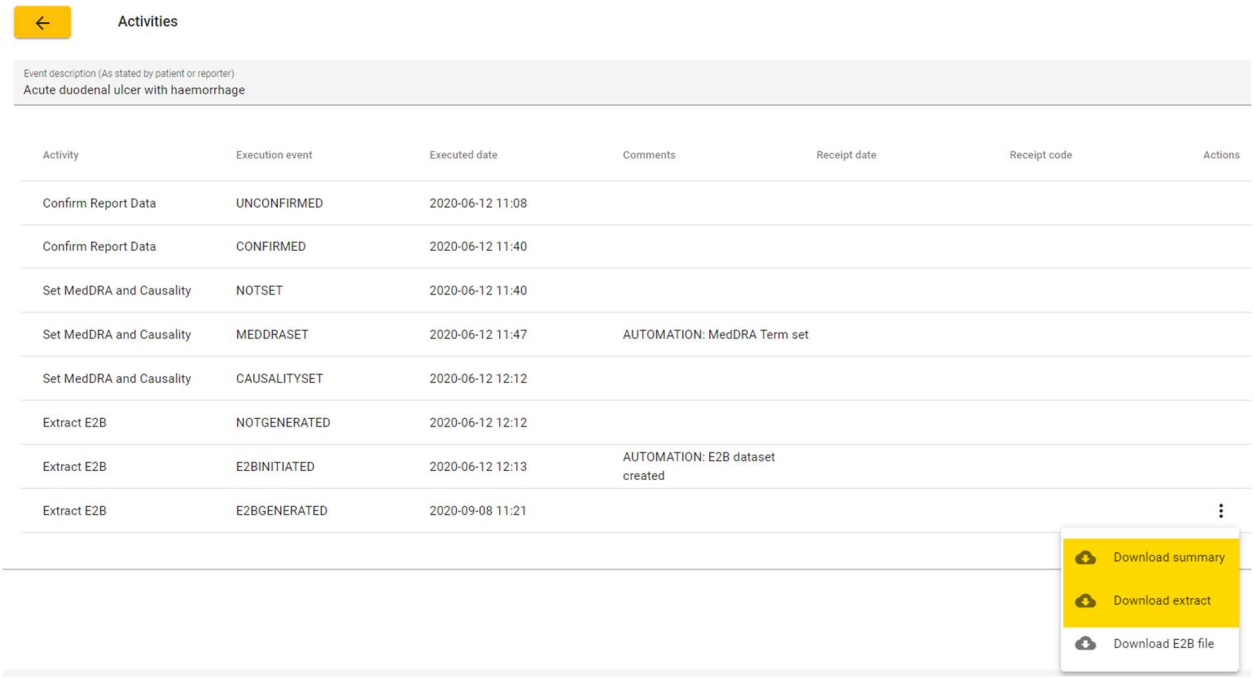
Once you have prepared the E2B file for submission and searched for the report, click on the **View Activity History** menu for the associated report you would like to view the XML file for.



The system will navigate you to the **Activities** page where you will be able to view a comprehensive history of activities by the analyst against this adverse event.

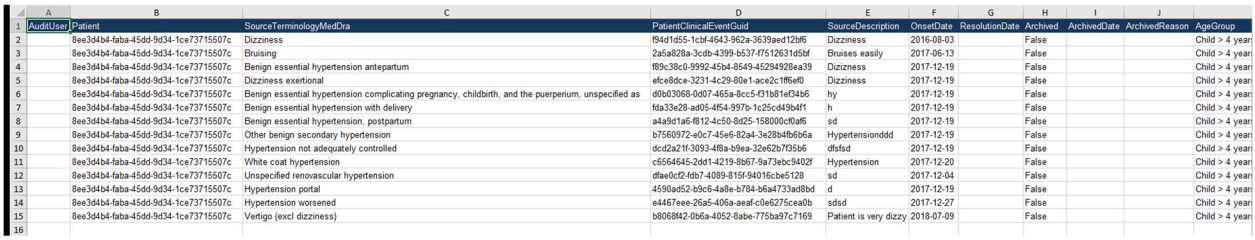


Locate the **E2BGENERATED** execution event for the report, click the action menu, and select the **Download summary or Download extract** menu.



If you select the **Download summary** menu, the system will generate a MS Word extract of the associated clinical data. See Extracting a Patient Summary.

If you select the **Download extract** menu, the system will generate an MS Excel extract of the associated clinical data.

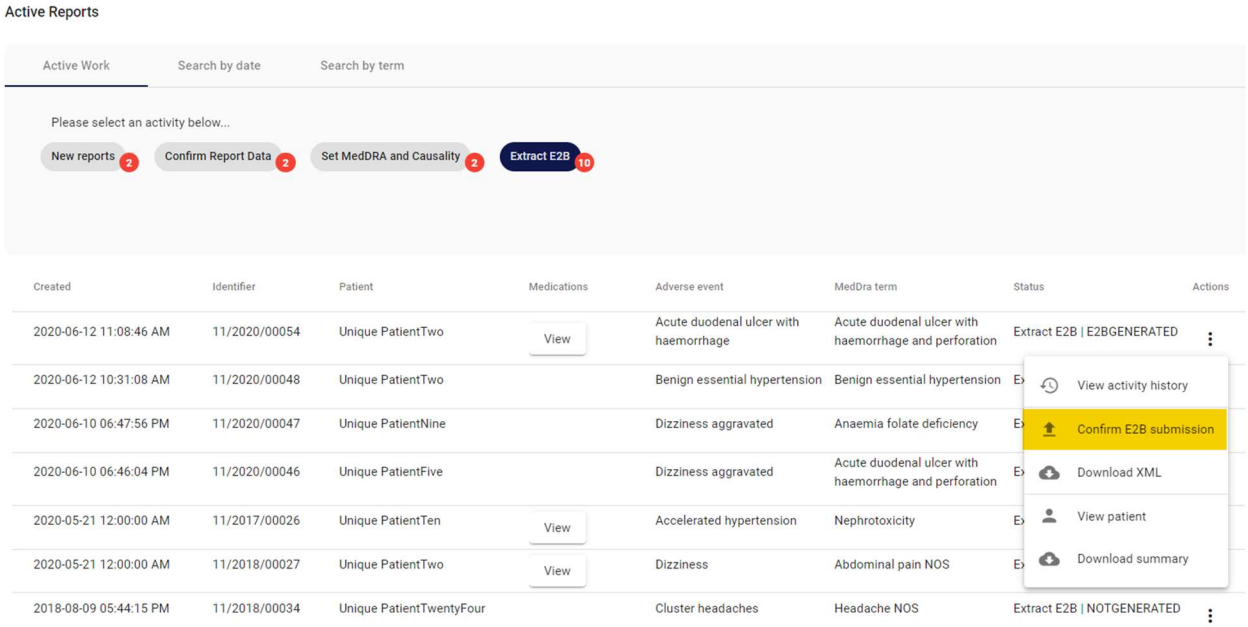


AuditUser	Patient	SourceTerminologyMedDra	PatientClinicalEventGuid	SourceDescription	OnsetDate	ResolutionDate	Archived	ArchivedDate	ArchivedReason	AgeGroup
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Dizziness	f94d1d55-1cbf-4643-962a-3639aed12bf6	Dizziness	2016-08-03		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Bruising	2a6a828a-3cdb-4399-b537-f7512631d5bf	Bruises easily	2017-06-13		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Benign essential hypertension antepartum	f89c38c0-9992-45b4-8549-45294928ea39	Dizizness	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Dizziness exertional	efce8dce-3231-4c29-80e1-ace2c1ff6ef0	Dizziness	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Benign essential hypertension complicating pregnancy, childbirth, and the puerperium, unspecified as	d0b03068-0d07-465a-8cc5-f31b81ef34b6	hy	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Benign essential hypertension with delivery	fda33e28-ad05-4f54-997b-1c25cd49b4f1	h	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Benign essential hypertension, postpartum	a4a9d1a6-f812-4c50-8d25-158000cf0af6	sd	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Other benign secondary hypertension	b7560972-e0c7-45e6-82a4-3e28b4fb6b6a	Hypertensionddd	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Hypertension not adequately controlled	dcd2a21f-3093-4f8a-b9ea-32e62b7f35b6	dfsfsd	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	White coat hypertension	c6564645-2dd1-4219-8b67-9a73ebc9402f	Hypertension	2017-12-20		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Unspecified renovascular hypertension	dfae0cf2-fdb7-4089-815f-94016cbe5128	sd	2017-12-04		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Hypertension portal	4590ad52-b9c6-4a8e-b784-b6a4733ad8bd	d	2017-12-19		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Hypertension worsened	e4467eee-26a5-406a-aeaf-c0e6275cea0b	sdsd	2017-12-27		False			Child > 4 years
	8ee3d4b4-faba-45dd-9d34-1ce73715507c	Vertigo (excl dizziness)	b8068f42-0b6a-4052-8abe-775ba97c7169	Patient is very dizzy	2018-07-09		False			Child > 4 years

In both the Patient Extract and Summary above, these files are stored along with the XML file as a reference to the clinical data associated with the XML file at the point the XML file was generated.

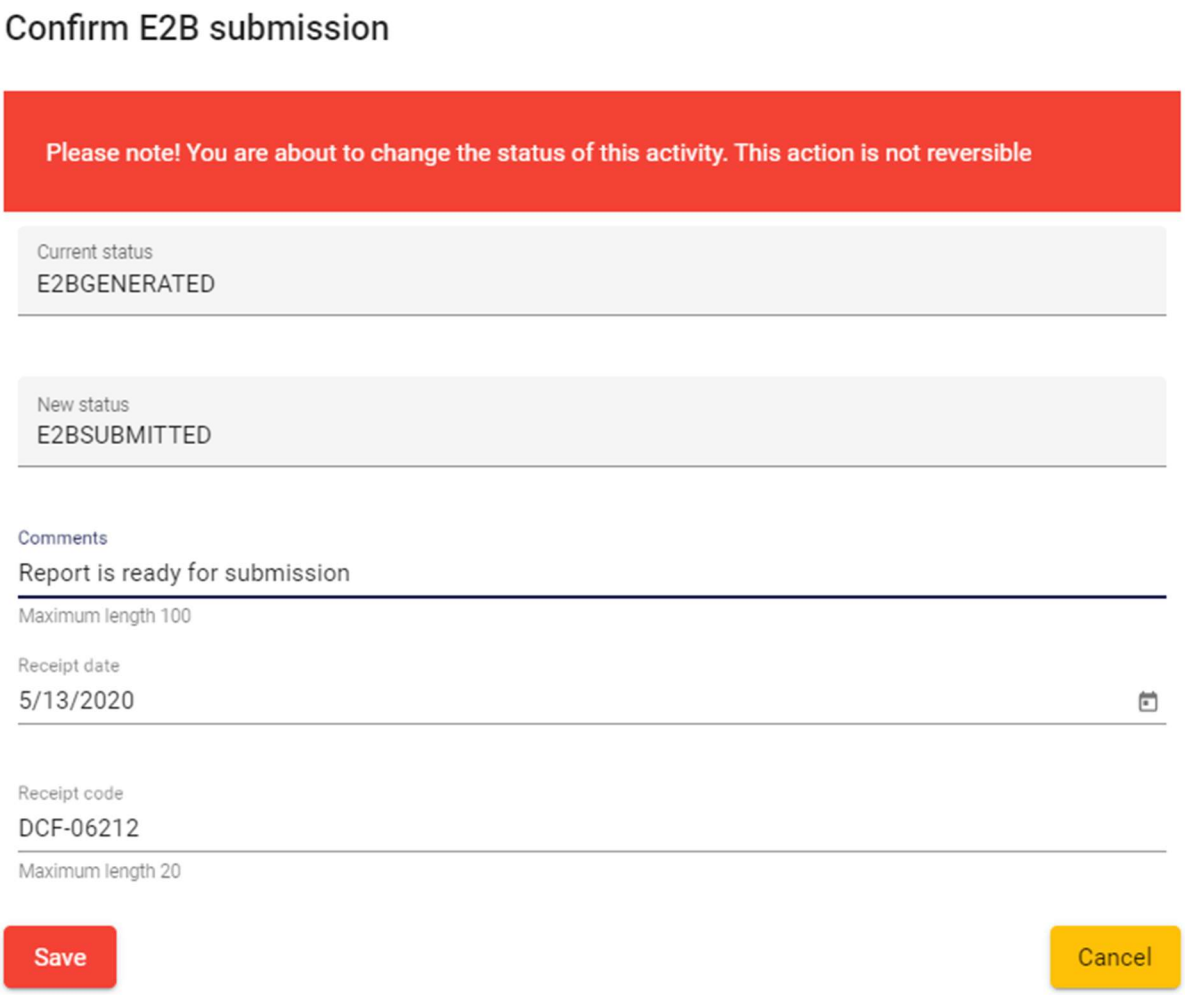
5.7.6 Confirming a Report for E2B Submission

Once you have prepared the E2B file for submission and searched for the report, click on the **Confirm E2B Submission** menu for the associated report you would like to submit.



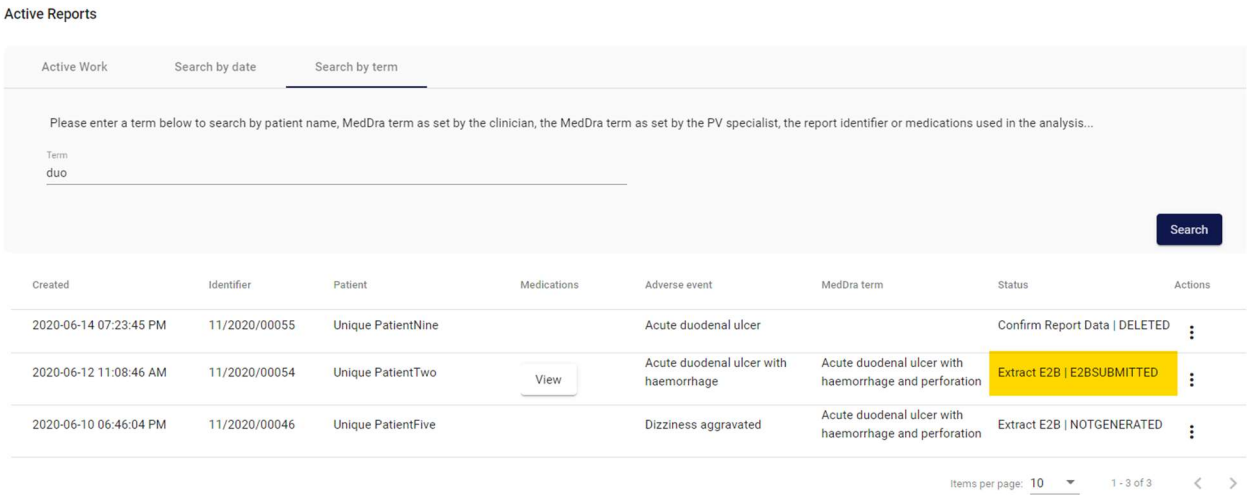
The system will navigate you to the **Confirm E2B Submission** pop up form.

Specify any additional comments for confirming the submission and click **Save** to confirm the submission or **Cancel** to cancel the action and return to the previous page.



The receipt date and code can be used to note correspondence with WHO on the receipt of the E2B XML submission file.

The system will update the status of the report accordingly. Please note that the report is removed from the Extract E2B tab as work for this report is now classified as finished. Use the **Search by date** or **Search by term** tabs to locate this report again.



5.8 **Analyser**

This section is used to generate the relative risk for a specified adverse drug reaction based on an exposed and non-exposed population set over a defined period of time.

The analyser user can access the following functionality within the analytical portal:

- Define population set (cohort or condition group)
- Define reporting period
- Specify additional risk factors
- View risk ratios per exposed drug
- Download dataset for further analysis

6.8.1 **Methodology**

The following formulas, calculations and definitions are used in the calculation of the relative risk for a specific medication and adverse reaction.

5.8.1.1 Incidence Rate

The incidence rate is the number of new cases per population in a given time period

IR = (ADR / Population) * 1000

Where
IR = Incidence Rate
ADR = Number of adverse drug reactions *
Population = Total patient years in reporting period **

Note
* Where causality is set to possible, probably/likely or Certain for WHO assessments and where causality is set to possible, probable or definite for Naranjo assessments

** Population is represented in patient years. For example, if the reporting period is 30 days, and 10 patients were on treatment for all 30 days, the total patient years is 300 /365.25 which is 0.82.

Example
Cases = 11
Non-Cases = 170
Population = 181 (11 + 170)
Incidence Rate = 11/181 * 1000 = **60.77**

5.8.1.2 Relative Risk

Relative risk is defined as the incidence in the exposed over incidence in the non-exposed.

RR= IR1 / IR2

Where
RR = Relative Risk
IR1 = Incidence Rate for exposed population. The exposed population is defined as the patient population that have been exposed to a medication in the reporting period.
IR2 = Incidence Rate for non-exposed population. The exposed population is defined as the patient population that have NOT been exposed to a medication in the reporting period.

Example
Incidence Rate Exposed = 60.77
Incidence Rate Non-Exposed = 45.12
RR = **1.35**

5.8.1.3 Confidence Interval

A confidence interval is a range of values so defined that there is a specified probability that the value of a parameter lies within it. Most commonly, the 95% confidence interval is used.

$$CI = \log(RR) \pm \mathrm{SE} \times z_\alpha$$

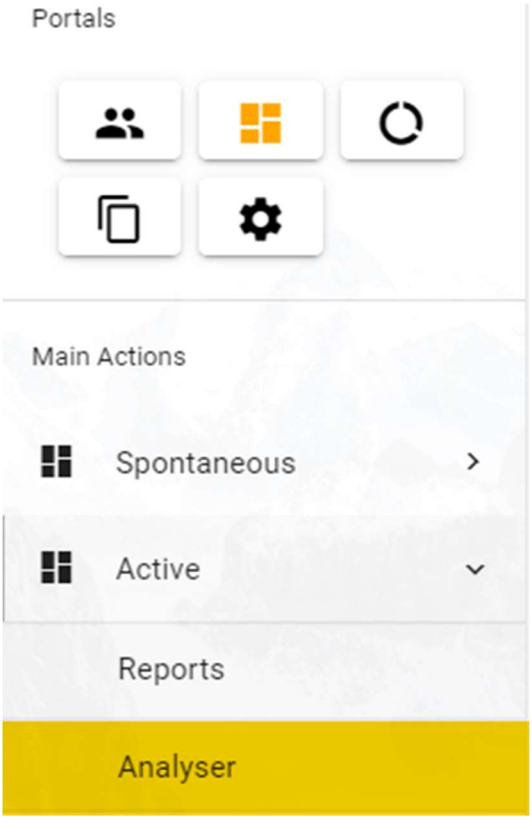
Where
z_α is the standard score for the chosen level of significance and SE the standard error.

5.8.2 Generating Unadjusted Relative Risk Ratios

To implement the methodology for generating an **Unadjusted Relative Risk Ratio,** the following parameters will need to be specified:

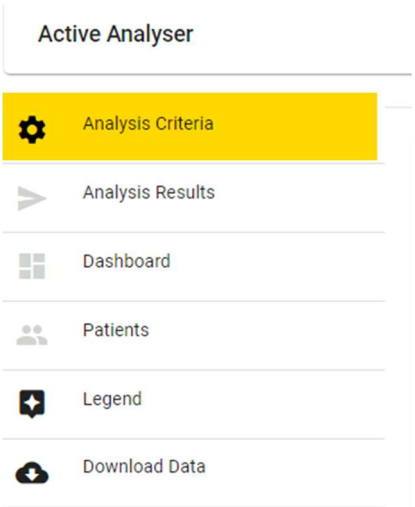
- The population target (condition group or Cohort)
- The date range for the analysis

The **Analyser** function can be accessed through the main menu.

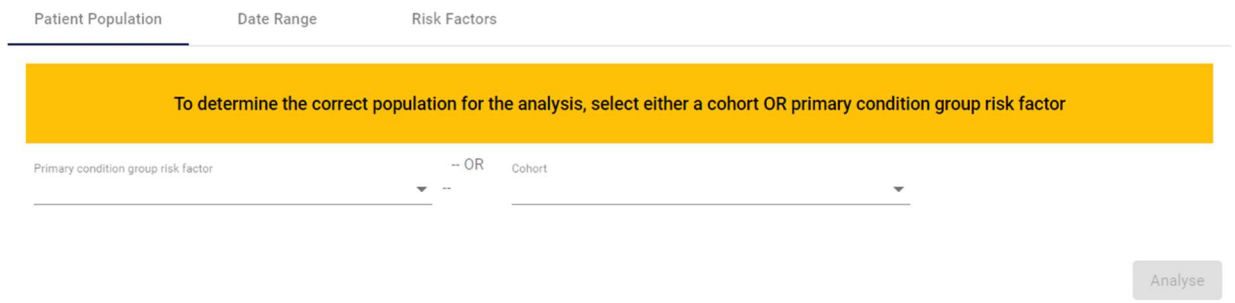


5.8.2.1 Specifying the Population Group

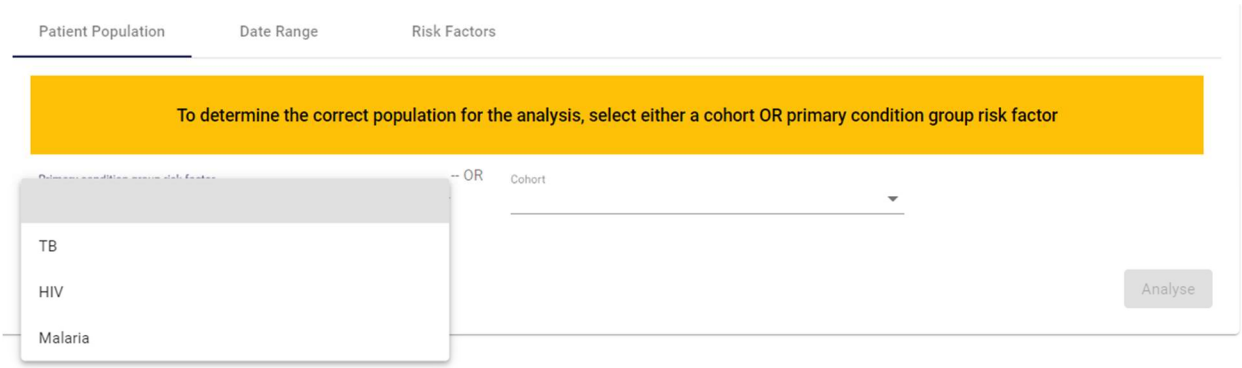
Ensure the **Analysis Criteria** tab is selected to allow the selection of the patient population.



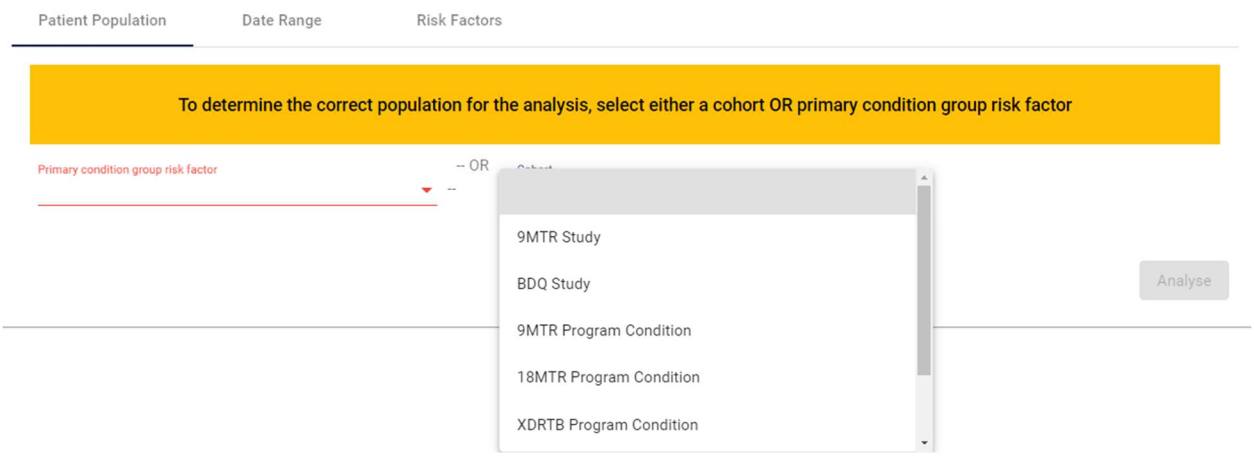
By selecting a **Condition Group** or **Cohort**, you are effectively able to target a specific set of patients for analysis.



To specify a **Condition Group,** click the Primary Condition Group Risk Factor field and select the primary condition you would like to run analysis against. If this option is selected, all patients that belong to the corresponding Condition Group will be included in the analysis.



To specify a **Cohort,** click the Cohort field and select the primary cohort you would like to run analysis against. If this option is selected, all patients that belong to the corresponding Cohort will be included in the analysis.

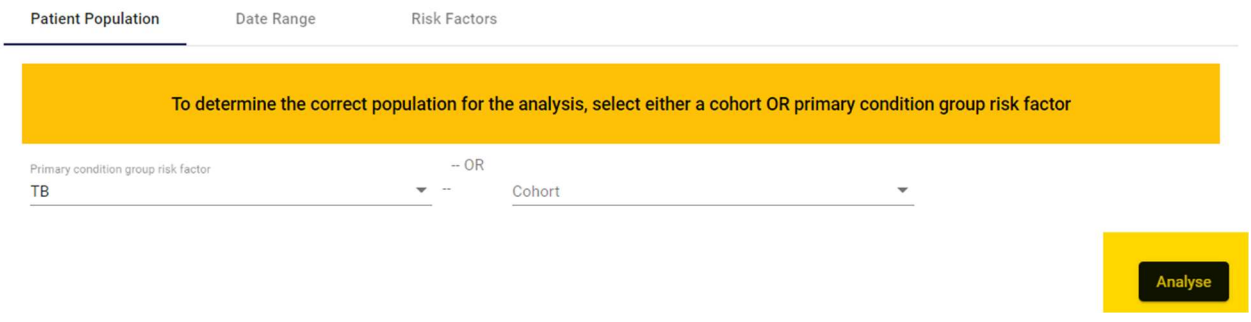


5.8.2.2 Specifying the Date Range for the Analysis

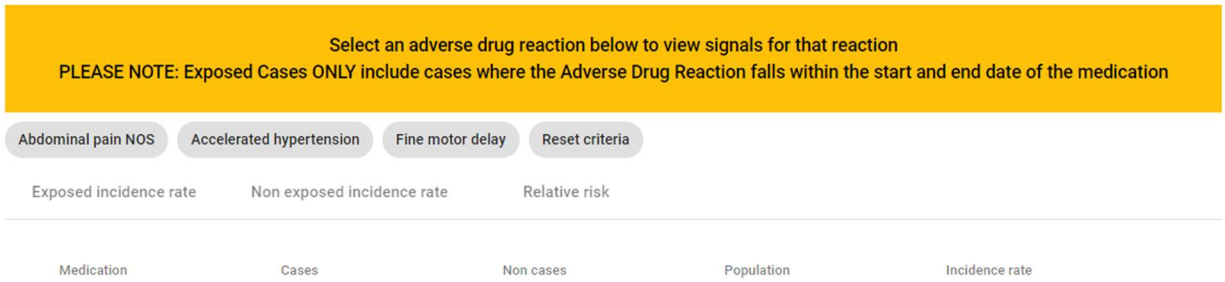
By selecting a **Date Range**, the system will determine which patients should be included into the analysis from the Patient Population specified in the previous step. Patients that have been actively exposed to medication within that range will be included.

5.8.2.3 Running the Analysis

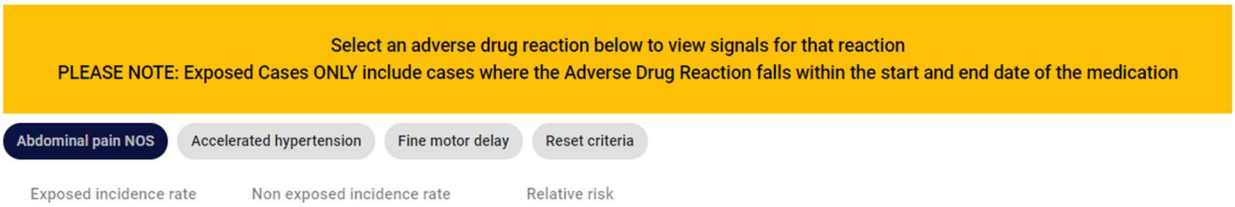
Once the **Patient Population** and **Date Range** parameters have been selected, click on the **Analyse** button to execute the analysis.



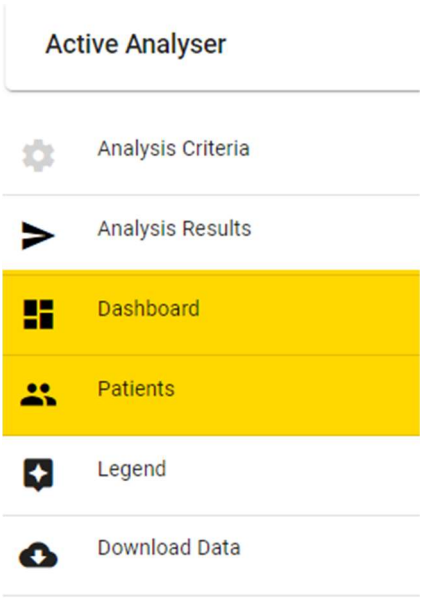
The system will conduct an initial analysis that will identify what Adverse Drug Reactions have been identified over the reporting period. All results are included in the **Analysis Results** tab. Please note that only adverse drug reactions where a MedDRA term has been selected by the PV technician will be selected.



Select the **Adverse Drug Reaction** that you would like to detect signals for by clicking on the tab for that reaction. The selected tab will turn a dark blue to signify that it has been selected.



Please note that selecting the Adverse Drug Reaction will enable the Dashboard and Patients tab.

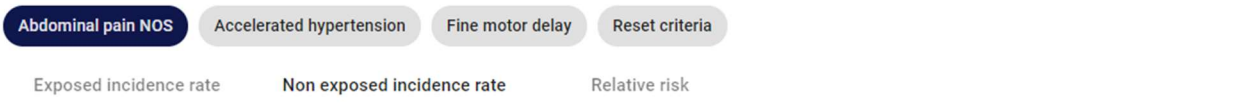


You are able to now view a 2 by 2 table illustrating relative risk and 95% Confidence Interval for the selected reaction.

- Incidence Rate for exposed group

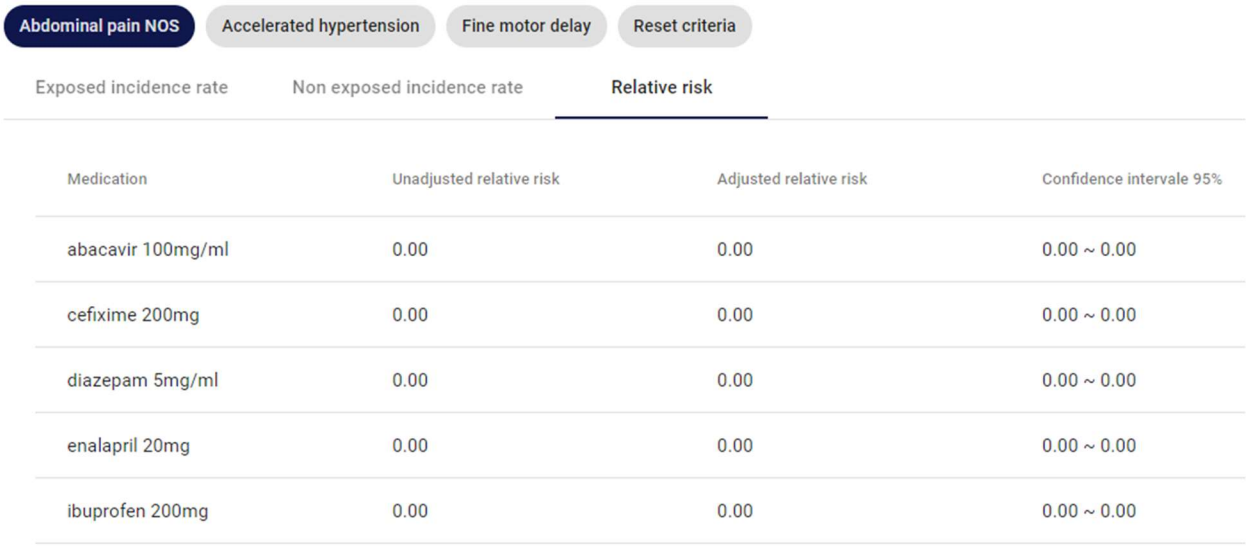
Medication	Cases	Non cases	Population	Incidence rate
abacavir 100mg/ml	0	1	0.21	0.00
cefixime 200mg	0	4	-3.04	0.00
diazepam 5mg/ml	0	4	-3.04	0.00
enalapril 20mg	0	1	-3.07	0.00
ibuprofen 200mg	0	3	-4.04	0.00

- Incidence Rate for non-exposed group



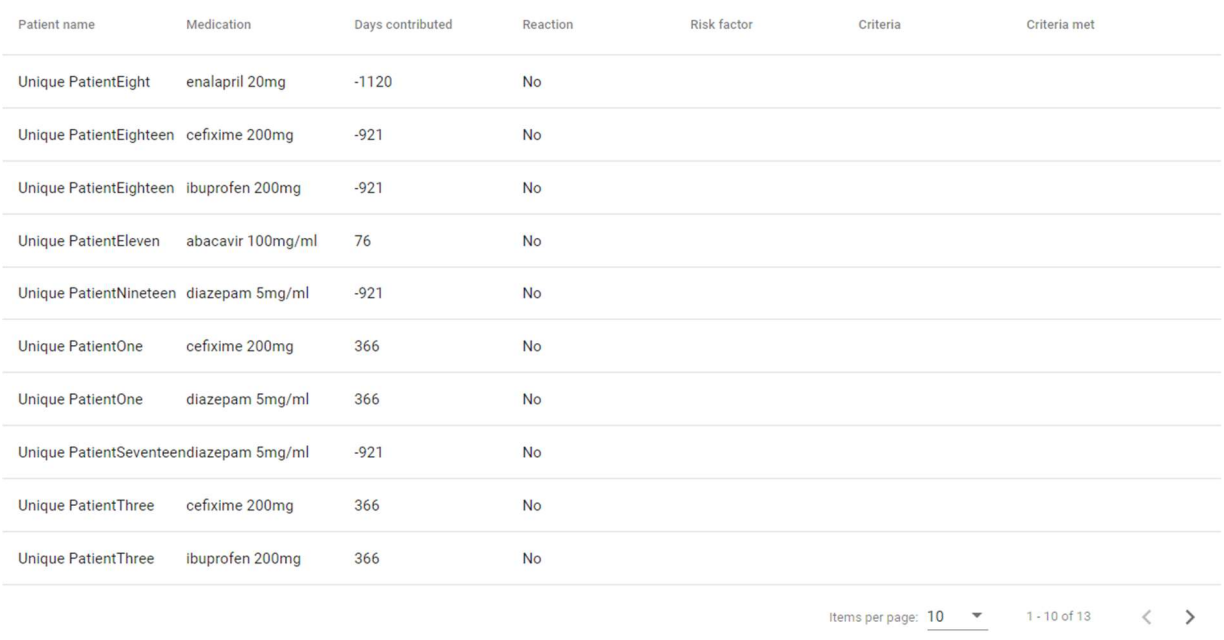
Medication	Cases	Non cases	Population	Incidence rate
abacavir 100mg/ml	0	8	-13.19	0.00
cefixime 200mg	0	5	-9.94	0.00
diazepam 5mg/ml	0	5	-9.94	0.00
enalapril 20mg	0	8	-9.91	0.00
ibuprofen 200mg	0	6	-8.94	0.00

- Relative Risk for the associated medication
- 95% Confidence Interval for the associated medication



5.8.2.4 Viewing the Contributing Patient List

Once analysis has been executed, it is possible to view the list of patients that have contributed to the analysis population set through the **Patients** tab.

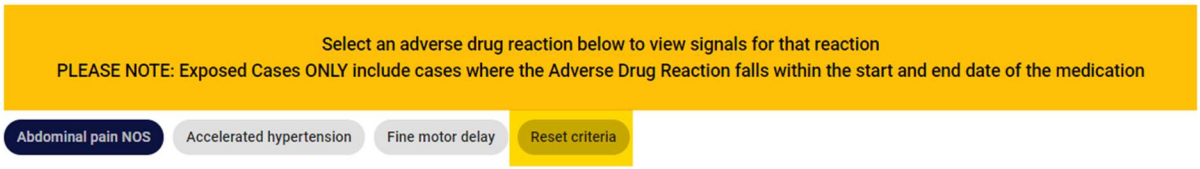


The columns in the patient list table are described below:

Patient name	The name of the patient contributing to the patient population
Medication	The drug the patient was exposed to during the period of analysis
Days contributed	The number of days the patient contributed to the patient population
Reaction	Did the patient suffer a reaction during the period of analysis
Risk Factor	Which risk factor does the patient match

5.8.2.5 **Resetting Criteria**

If at any point you would like to alter the patient population or date range, click the **Reset criteria** tab alongside the list of Adverse Drug Reactions to reset analysis. Selecting this tab will remove existing analysis and reopen the Analysis Criteria tab for modification.

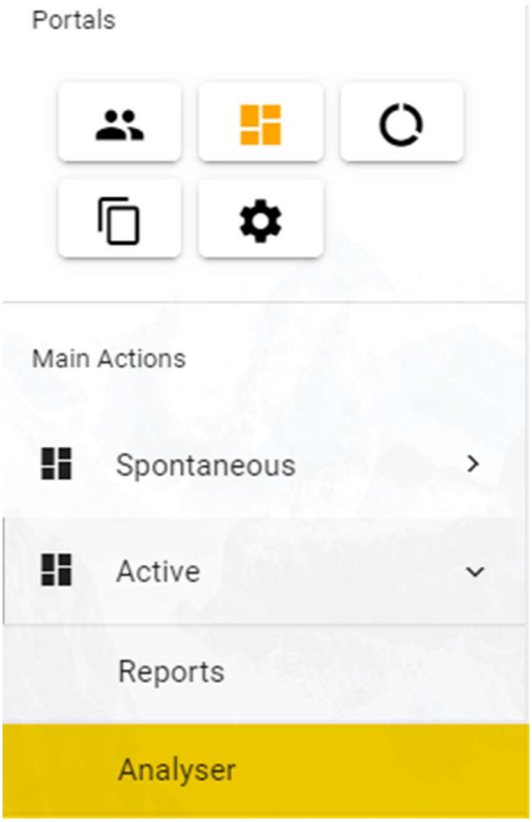


5.8.3 **Generating Adjusted Relative Risk Ratios**

To implement the methodology for generating an **Adjusted Relative Risk Ratio,** the following parameters will need to be specified:

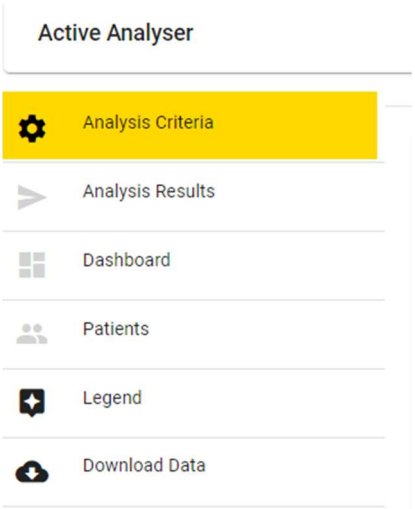
- The population target (condition group or Cohort)
- The date range for the analysis
- The risk factors to be applied

The **Analyser** function can be accessed through the main menu.

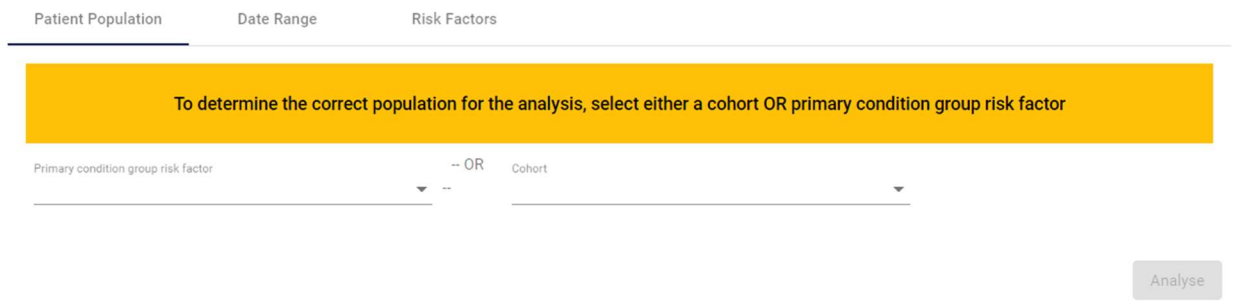


5.8.3.1 Specifying the Population Group

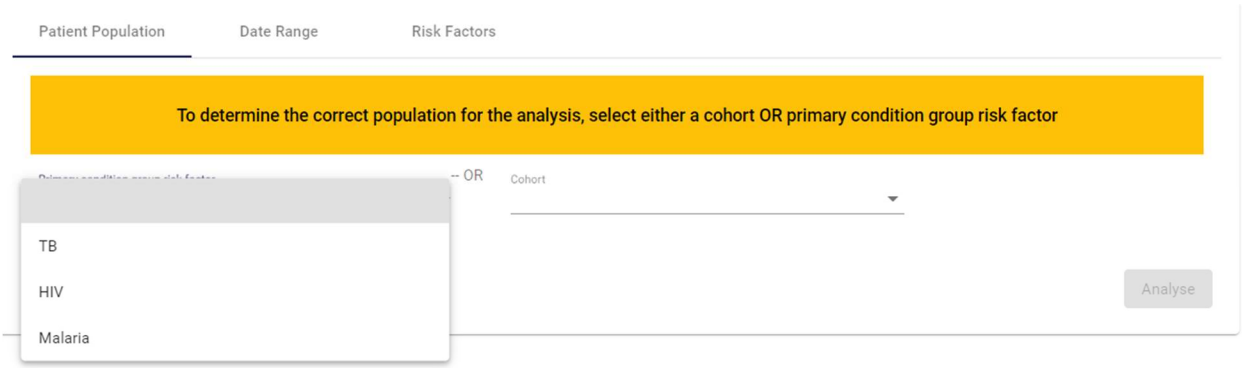
Ensure the **Analysis Criteria** tab is selected to allow the selection of the patient population.



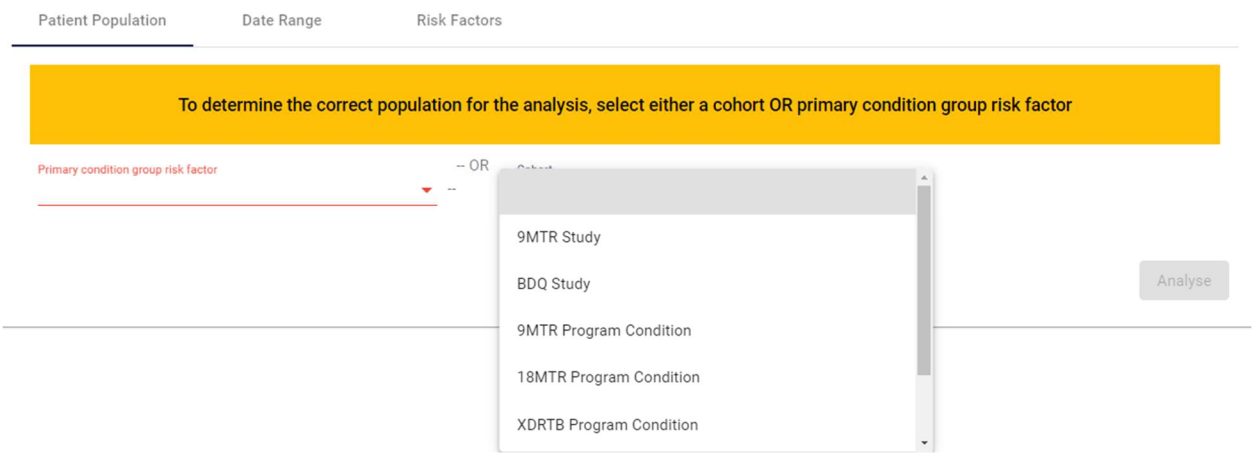
By selecting a **Condition Group** or **Cohort**, you are effectively able to target a specific set of patients for analysis.



To specify a **Condition Group,** click the Primary Condition Group Risk Factor field and select the primary condition you would like to run analysis against. If this option is selected, all patients that belong to the corresponding Condition Group will be included in the analysis.



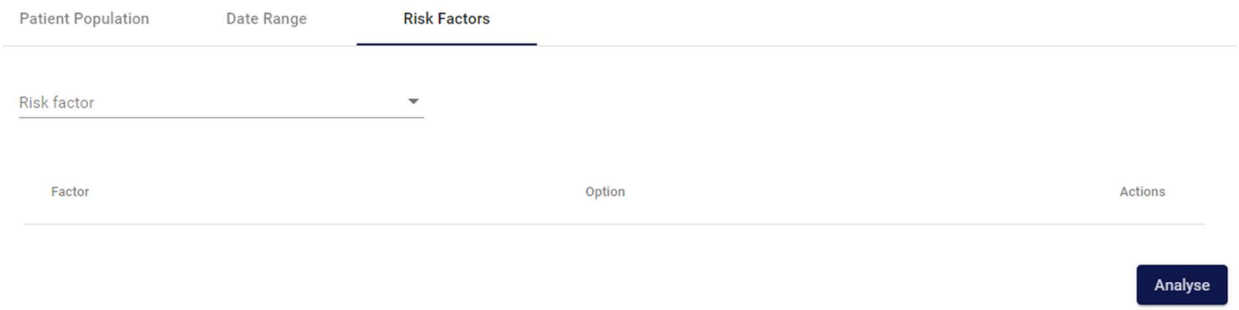
To specify a **Cohort,** click the Cohort field and select the primary cohort you would like to run analysis against. If this option is selected, all patients that belong to the corresponding Cohort will be included in the analysis.



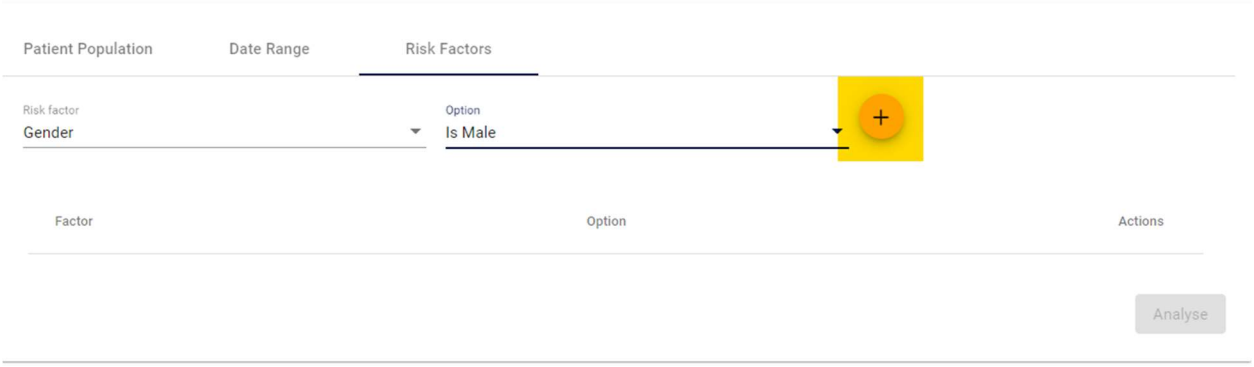
5.8.3.2 Specifying the Date Range for the Analysis

By selecting a **Date Range**, the system will determine which patients should be included into the analysis from the Patient Population specified in the previous step. Patients that have been actively exposed to medication within that range will be included.

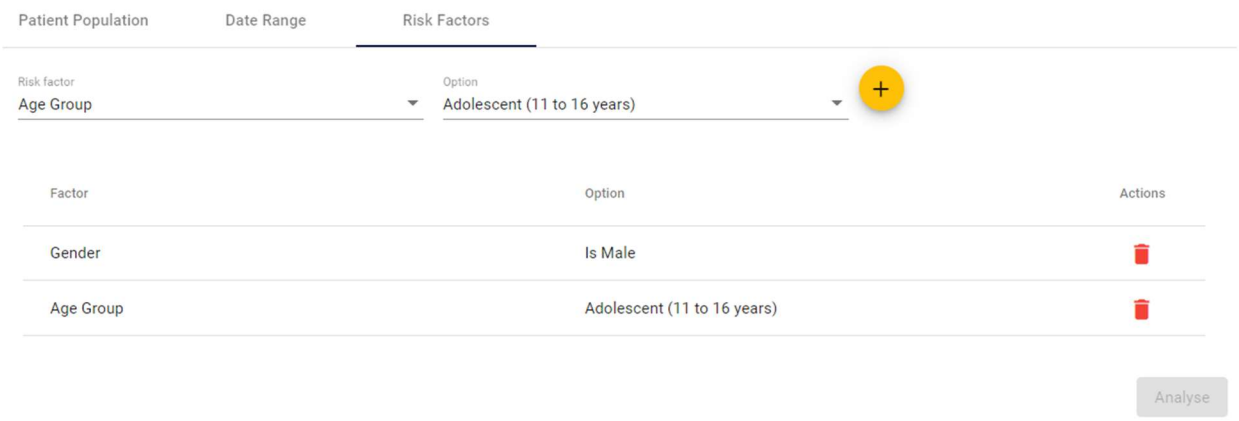
5.8.3.3 Specifying Risk Factors for the Analysis



To specify a **Risk Factor,** click the Risk factor field and select the risk factor you would like to include into the analysis. Once you have selected the risk factor, select the appropriate option associated to that risk factor and click the **add** icon to add the risk factor to the selected list.



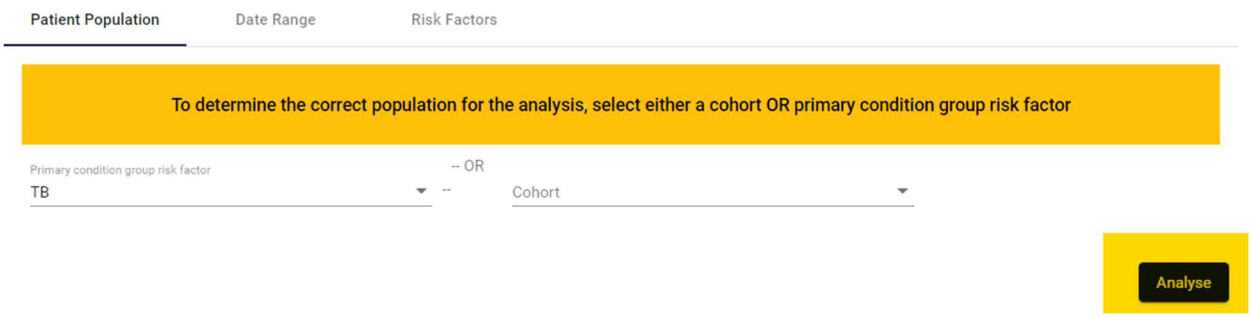
You are able to add as many risk factors as you would like to include into the analysis by repeating the above process.



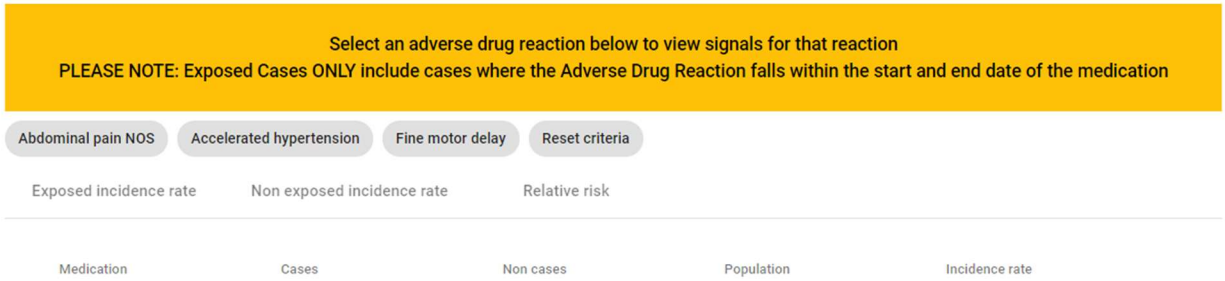
By including **Risk Factors** into the analysis, the system will determine which patients match the criteria stipulated by the set of risk factors and the corresponding **Relative Risk Ratio** will be adjusted based on the new population set.

5.8.3.4 Running the Analysis

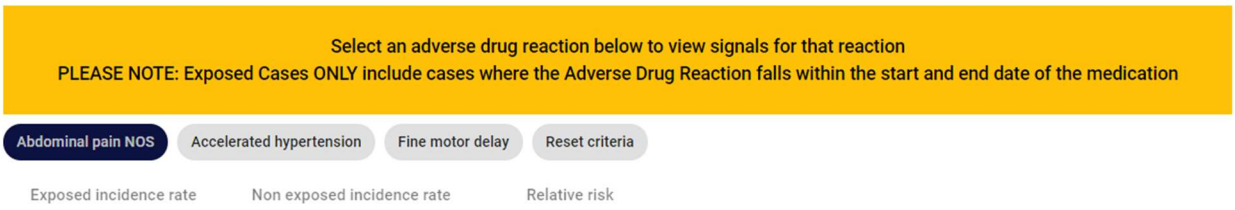
Once the **Patient Population, Date Range, and Risk Factor** parameters have been selected, click on the **Analyse** button to execute the analysis.



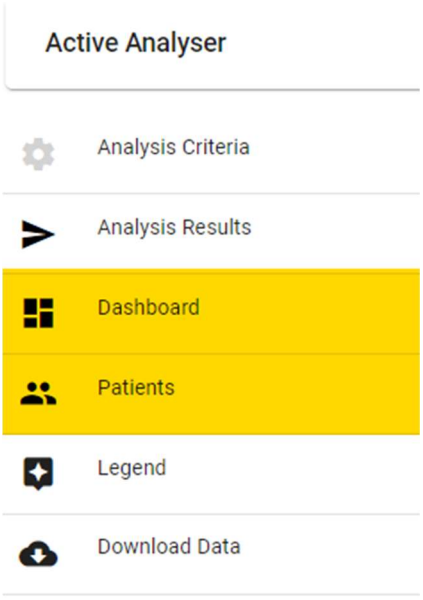
The system will conduct an initial analysis that will identify what Adverse Drug Reactions have been identified over the reporting period. All results are included in the **Analysis Results** tab. Please note that only adverse drug reactions where a MedDRA term has been selected by the PV technician will be selected.



Select the **Adverse Drug Reaction** that you would like to detect signals for by clicking on the tab for that reaction. The selected tab will turn a dark blue to signify that it has been selected.

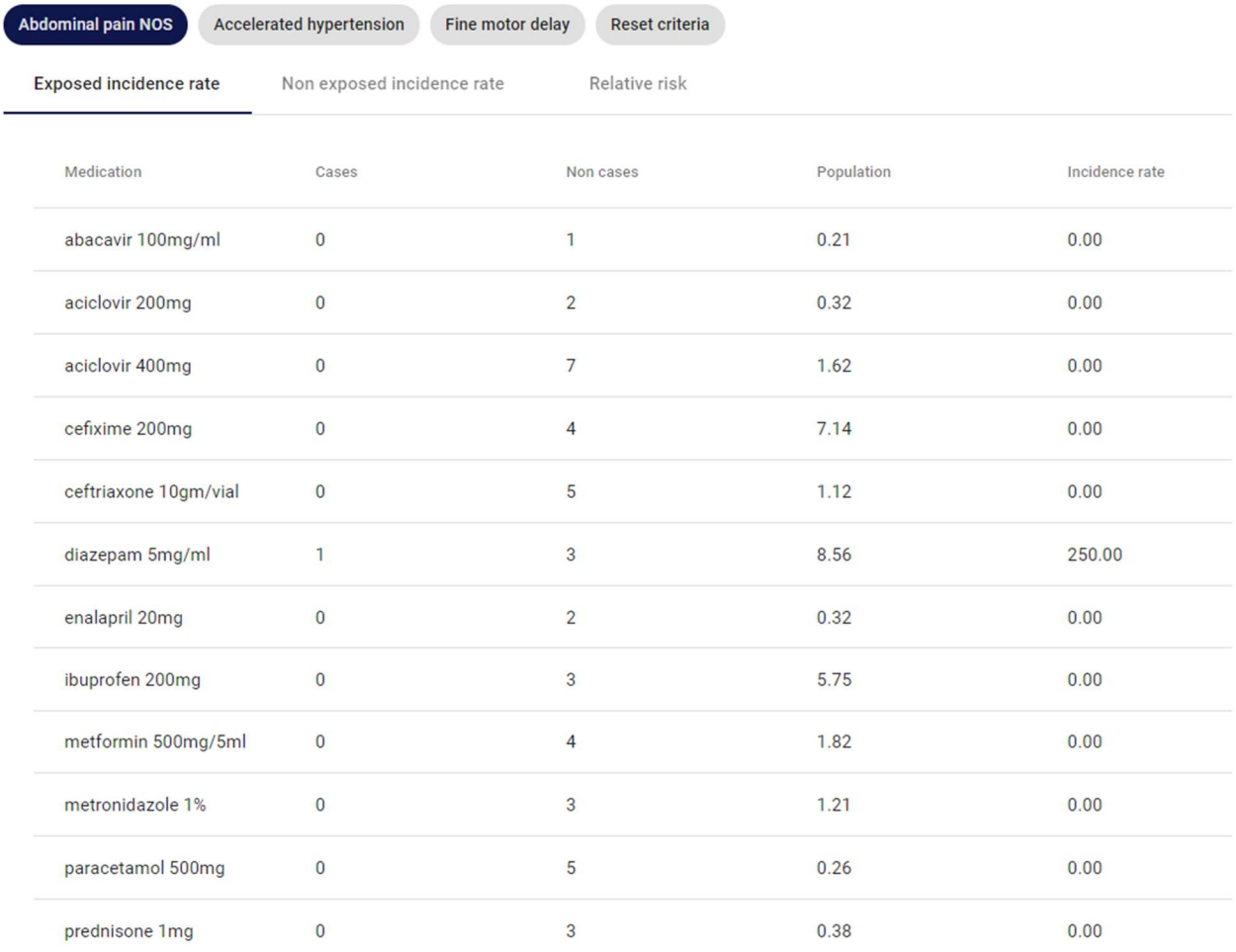


Please note that selecting the Adverse Drug Reaction will enable the Dashboard and Patients tab.



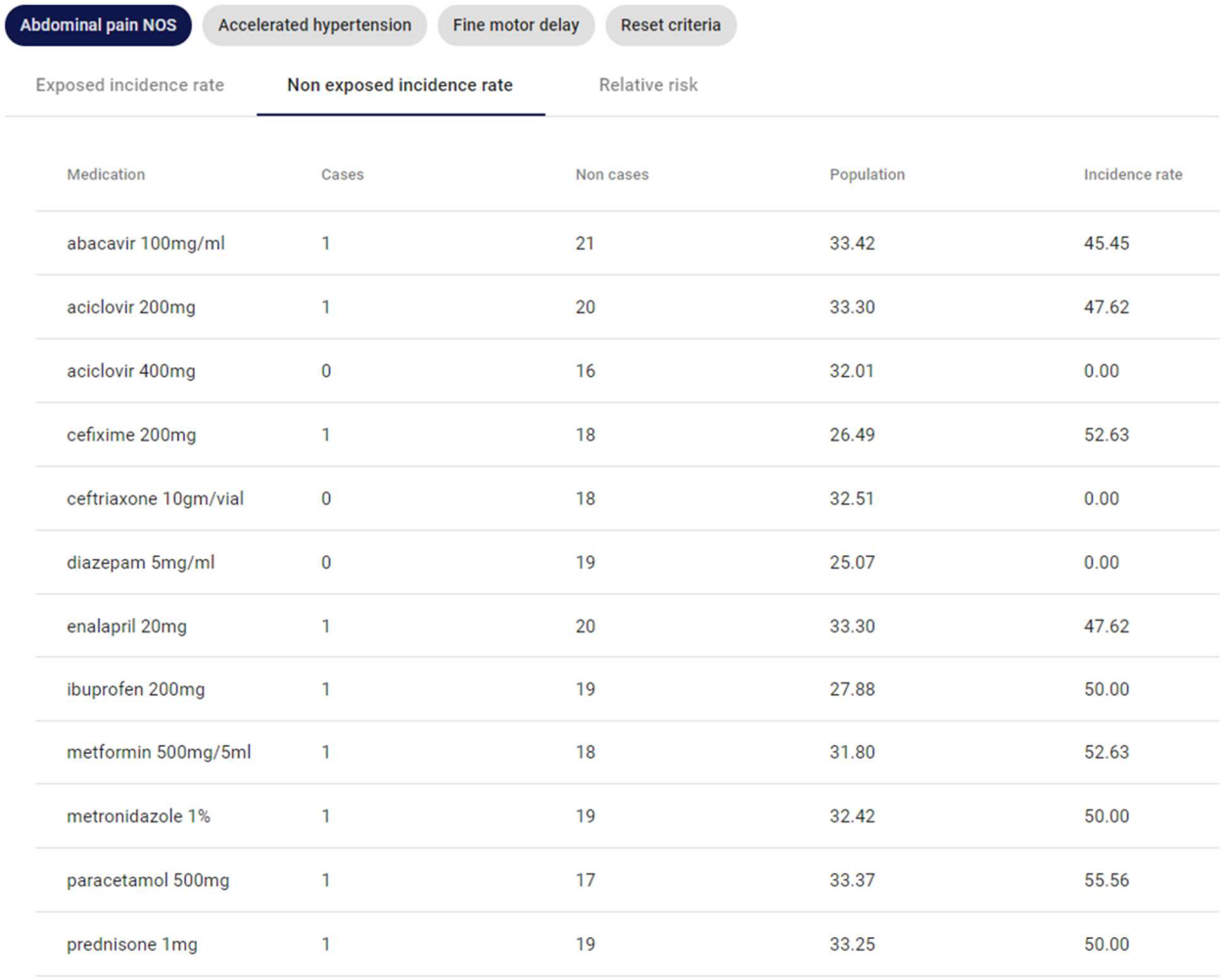
You are able to now view a 2 by 2 table illustrating relative risk and 95% Confidence Interval for the selected reaction.

- Incidence Rate for exposed group



Medication	Cases	Non cases	Population	Incidence rate
abacavir 100mg/ml	0	1	0.21	0.00
aciclovir 200mg	0	2	0.32	0.00
aciclovir 400mg	0	7	1.62	0.00
cefixime 200mg	0	4	7.14	0.00
ceftriaxone 10gm/vial	0	5	1.12	0.00
diazepam 5mg/ml	1	3	8.56	250.00
enalapril 20mg	0	2	0.32	0.00
ibuprofen 200mg	0	3	5.75	0.00
metformin 500mg/5ml	0	4	1.82	0.00
metronidazole 1%	0	3	1.21	0.00
paracetamol 500mg	0	5	0.26	0.00
prednisone 1mg	0	3	0.38	0.00

- Incidence Rate for non-exposed group



Medication	Cases	Non cases	Population	Incidence rate
abacavir 100mg/ml	1	21	33.42	45.45
aciclovir 200mg	1	20	33.30	47.62
aciclovir 400mg	0	16	32.01	0.00
cefixime 200mg	1	18	26.49	52.63
ceftriaxone 10gm/vial	0	18	32.51	0.00
diazepam 5mg/ml	0	19	25.07	0.00
enalapril 20mg	1	20	33.30	47.62
ibuprofen 200mg	1	19	27.88	50.00
metformin 500mg/5ml	1	18	31.80	52.63
metronidazole 1%	1	19	32.42	50.00
paracetamol 500mg	1	17	33.37	55.56
prednisone 1mg	1	19	33.25	50.00

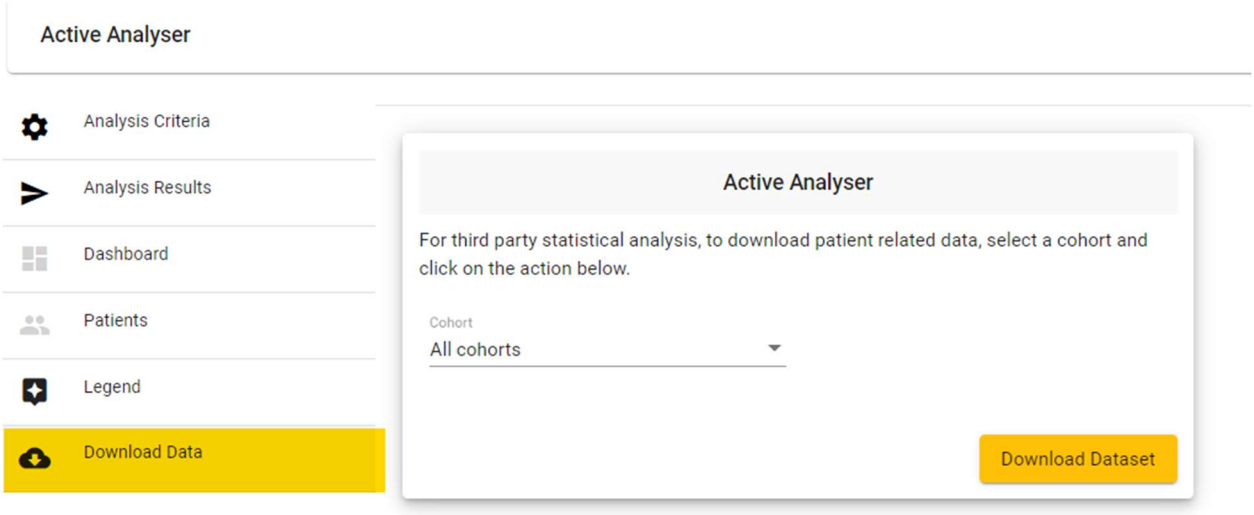
- Relative Risk for the associated medication
- 95% Confidence Interval for the associated medication

Exposed incidence rate | Non exposed incidence rate | Relative risk

Medication	Unadjusted relative risk	Adjusted relative risk	Confidence intervale 95%
abacavir 100mg/ml	0.00	0.00	0.00 ~ 0.00
aciclovir 200mg	0.00	0.00	0.00 ~ 0.00
aciclovir 400mg	0.00	0.00	0.00 ~ 0.00
cefixime 200mg	0.00	0.00	0.00 ~ 0.00
ceftriaxone 10gm/vial	0.00	0.00	0.00 ~ 0.00
diazepam 5mg/ml	0.00	0.00	0.00 ~ 0.00
enalapril 20mg	0.00	0.00	0.00 ~ 0.00
ibuprofen 200mg	0.00	0.00	0.00 ~ 0.00
metformin 500mg/5ml	0.00	0.00	0.00 ~ 0.00
metronidazole 1%	0.00	0.00	0.00 ~ 0.00
paracetamol 500mg	0.00	0.00	0.00 ~ 0.00
prednisone 1mg	0.00	0.00	0.00 ~ 0.00

5.8.4 **Downloading a Dataset for Further Analysis**

Select the **Download Data** tab and then click the **Download Dataset** button to able to download a comprehensive dataset of patient clinical data in XLSX format for importation into a third-party statistical tool.



If you are unable to locate this function, please liaise with your system administrator as the ability to download a dataset for external consumption will need to be assigned to your user profile.

6 Reporting Portal

The reports portal is the centralized hub for system reporting.

Note: the following roles have access to the report's portal:

- **All users.** All users have view access to pages defined within the portal.
- **Administrator.** The administrator has FULL permissions to the information portal.
- **Reporter Administrator.** The reports administrator has the ability to add and customize reports.

6.1 List of Standard System Reports

The reports portal includes several reports as part of the base configuration of the system. These reports are listed below:

Patients on Treatment	Aggregated number of patients per facility that have a serious event, non-serious event, and the percentage that have events
Adverse Events	Number of patients with an adverse event by age group, facility and drug
Quarterly Adverse Events	Number of patients with an adverse event by MedDRA system organ class per quarter and grade
Annual Adverse Events	Number of patients with an adverse event by MedDRA system organ class per year and grade
Causality	List adverse events where causality has been set and not set
Patients by Drug	Number of patients on a specific medication
Outstanding Visits	Patients who did not attend an appointment

6.2 Report Customization

The Reports Portal gives PViMS report publishers the ability to add new and modify existing reports within this portal. This provides PViMS analysts with an integrated platform to customize what reports are available to end users.

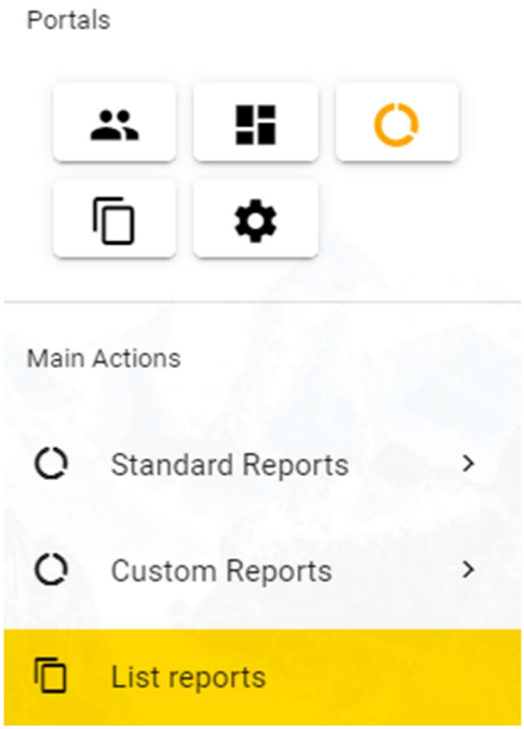
6.2.1 Types of Reports

When adding a new report, you first need to specify the type of report to be incorporated. There are currently two report types that can be customized within PViMS: A summary report that provides aggregated reporting based on the stratification criteria specified, and a list report that allows for a line by line rendering of the report in a non-aggregated manner.

6.2.2 Adding a New Report

In order to add a new report to the Reports Portal, you need to have the **Reporter Administrator** role assigned to your user profile.

To add a new report, click on the **List Reports** menu.



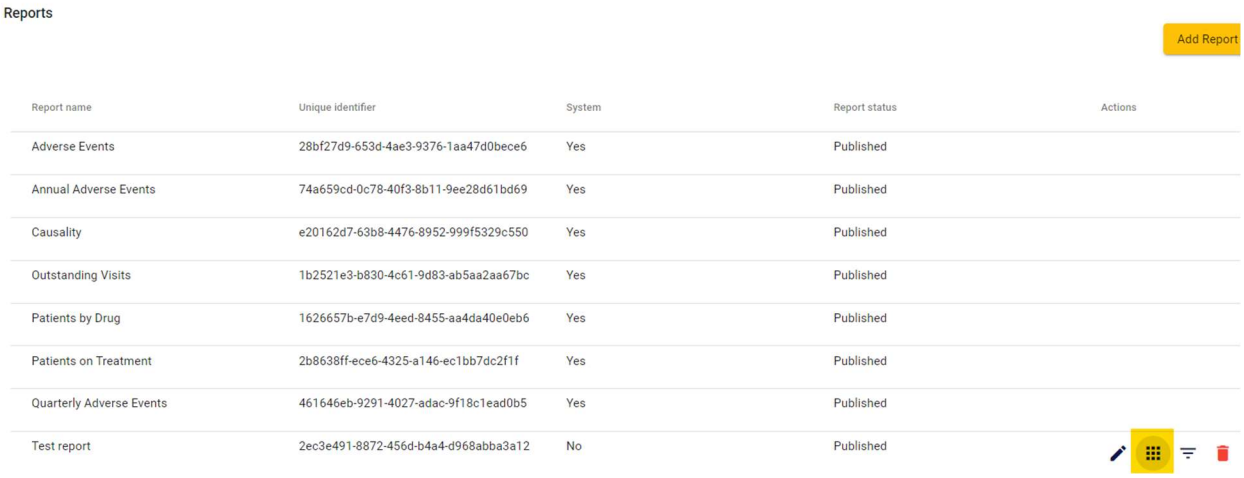
Then click on the **Add Report** button so you can define the characteristics of the new report.

The following information must be entered when publishing a new report:

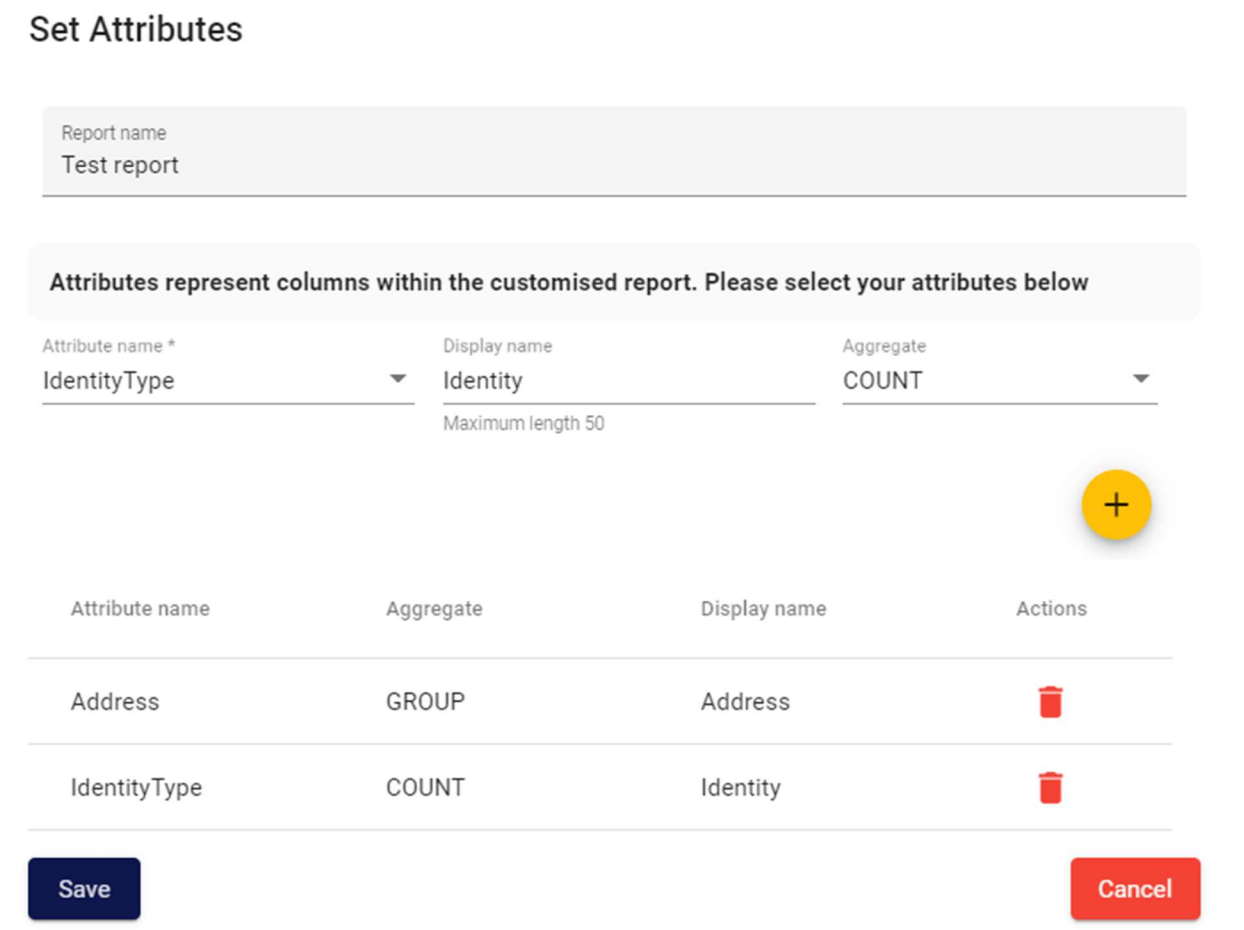
Report name	The unique name for the report
Definition	Provide additional information that describes the report
Report status	Is the report to be published or unpublished. Published reports appear in the custom reports list in the reporting portal
Report type	Is this a summary or a list report?
Core Entity	The primary entity that should be reported on: • Patient, report on patient specific criteria • PatientClinicalEvent, report on adverse event information • PatientCondition, report on concomitant conditions • PatientFacility, report on facility and patient • PatientLabTest, report on lab test information • PatientMedication, report on medication history • Encounter, report on clinical data collected per encounter • CohortGroupEnrolment, report of cohort enrolment data

6.2.2.1 **Summary Report**

Once the base report is configured, stratification-related information may now be specified. All attributes specified as part of the stratification list will be aggregated based on these attributes. To add a new stratification item, first select the attributes menu option for the report you would like to customize.

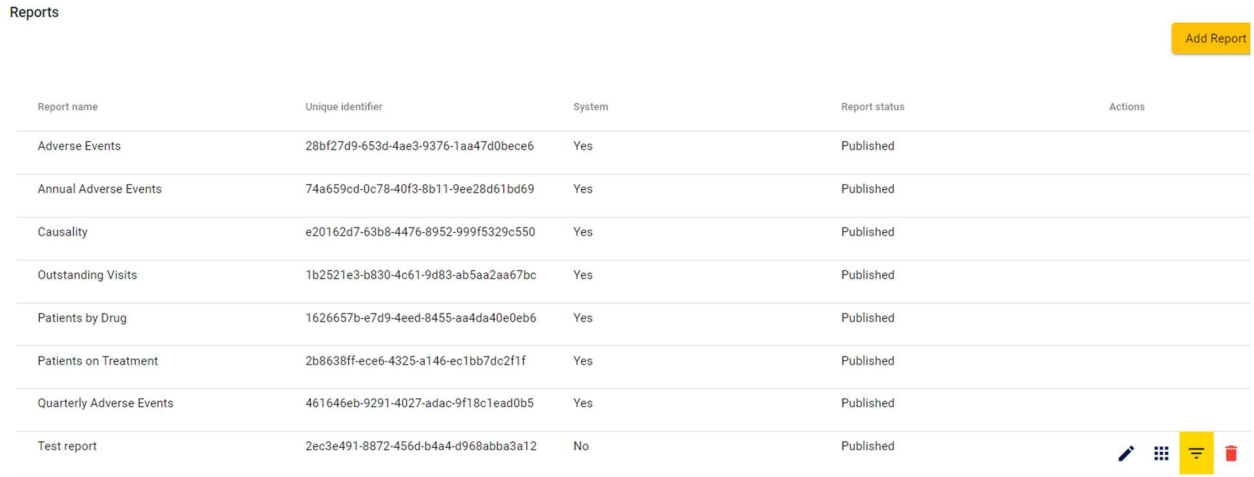


Select the attribute from the list, specify the name of column in the display field, the type of aggregation and click the **add** icon. The attribute will be added to the stratification list.



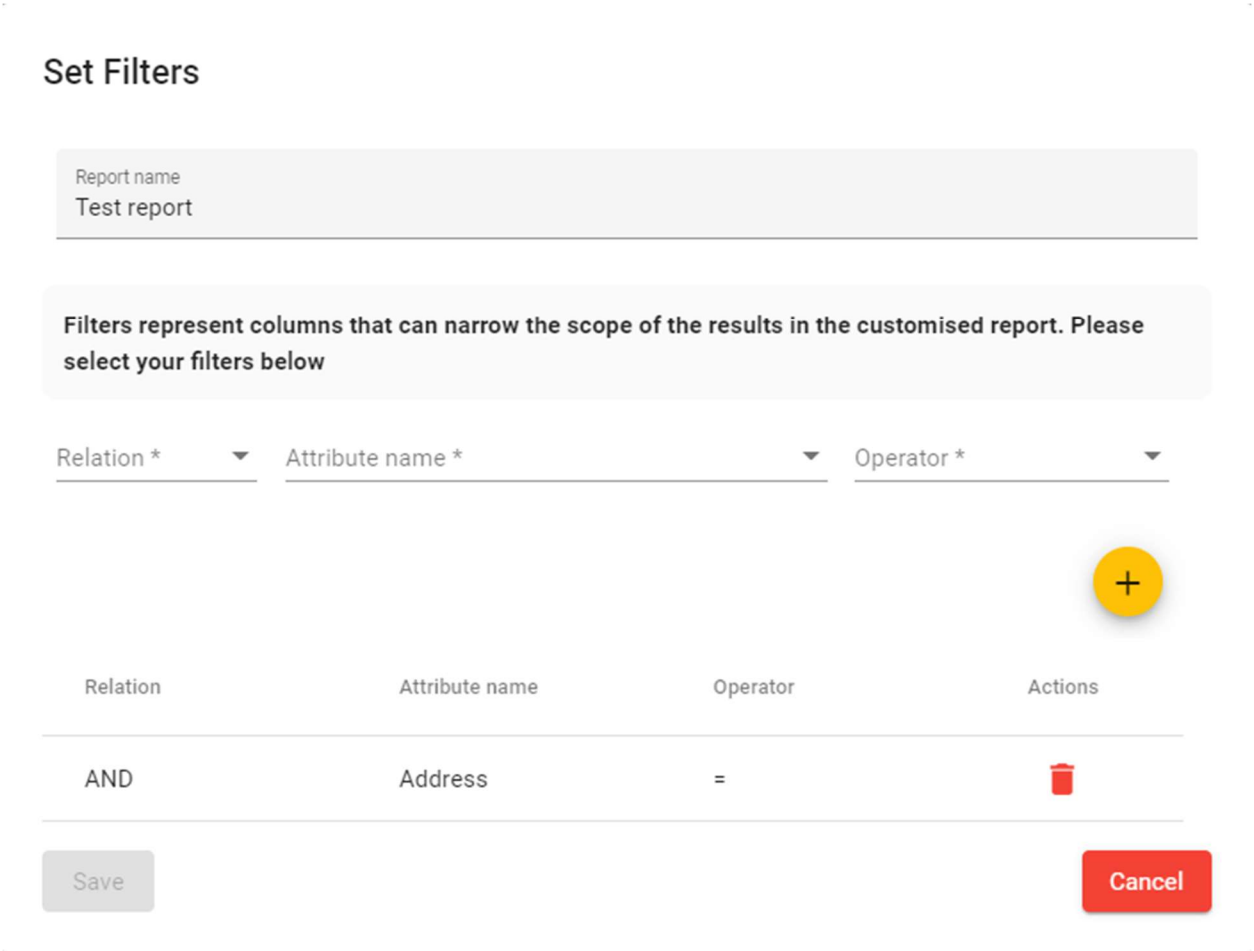
Please note that at least one attribute must be specified as a group field and at least one attribute must be specified as an aggregate. A group field is the field that will be used for stratification. Once all attributes have been added, click the **Save** button to add each attribute to the report or the **Cancel** button to cancel this action and return to the previous page.

Once the stratification list is specified, filter-related information may now be specified. All attributes specified as part of the filter list will be used to filter the result set by the end user. To add a new filter item, first select the filters menu option for the report you would like to customize.



Add the filter item as per the field description below and click the **add** icon:

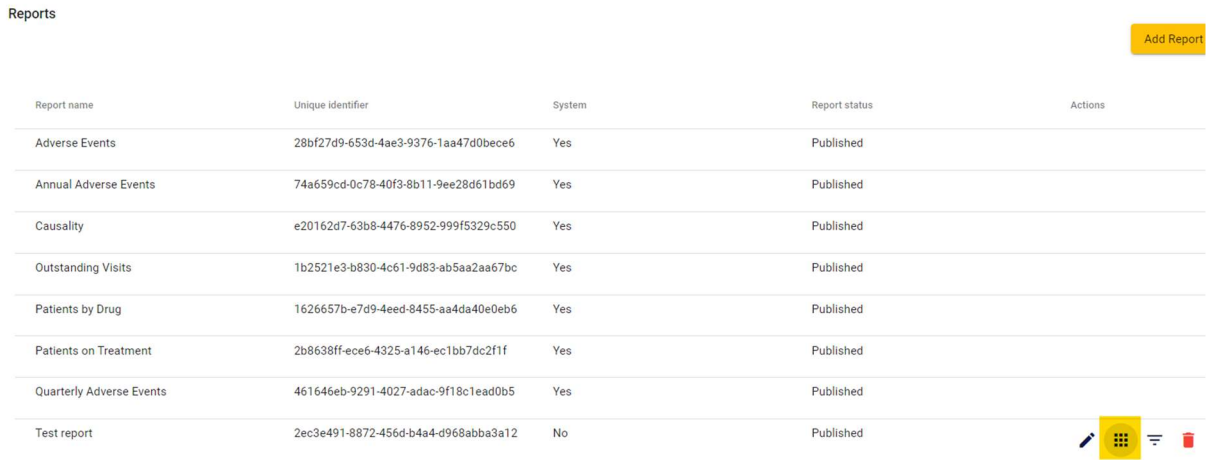
Relationship	Specify **AND** if this filter criteria must be true in conjunction with other attributes Specify **OR** if this filter criteria or other criteria must be true
Attribute	The attribute that is being filtered on
Operator	The operator that will be applied to the filter: • Dates and numerics allow the following operators: Equals, Not Equals, Greater Than, Less Than, GreaterEqual Than, LessEqual Than, Between • Text fields allow the following operators: Equals, Not Equals • DropDown Lists allow the following operators: Equals, Not Equals, In



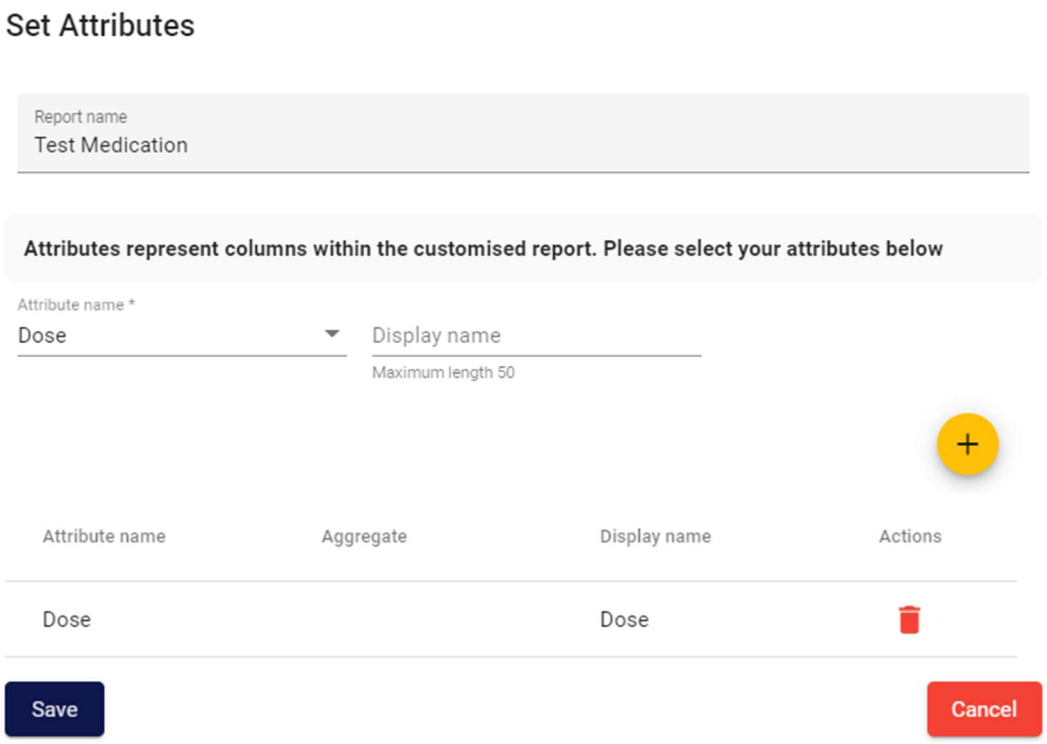
Once you are satisfied with the report configuration, change the status of the report to published. The new report will appear in the custom report menu once it is published. Please note, you may need to log out and log back in to view the new report..

6.2.2.2 List Report

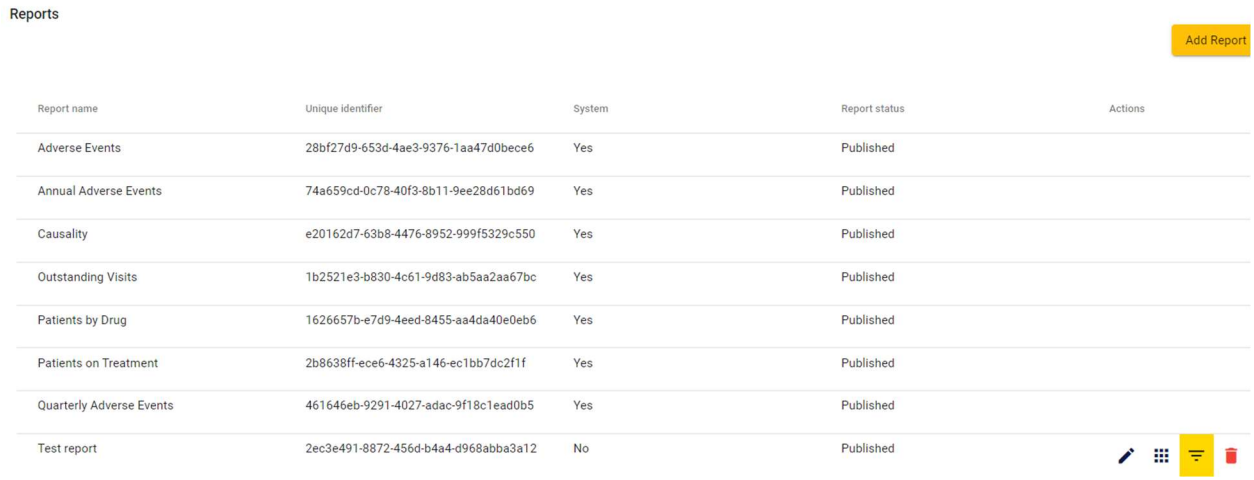
Once the base report is configured, list-related information may now be specified. All attributes specified as part of the list will be included as separate columns in the report. To add a new list item, first select the attributes menu option for the report you would like to customize.



Then select the attribute from the list, specify the name of column in the display field, and click the **add** icon. The attribute will be added to the list.

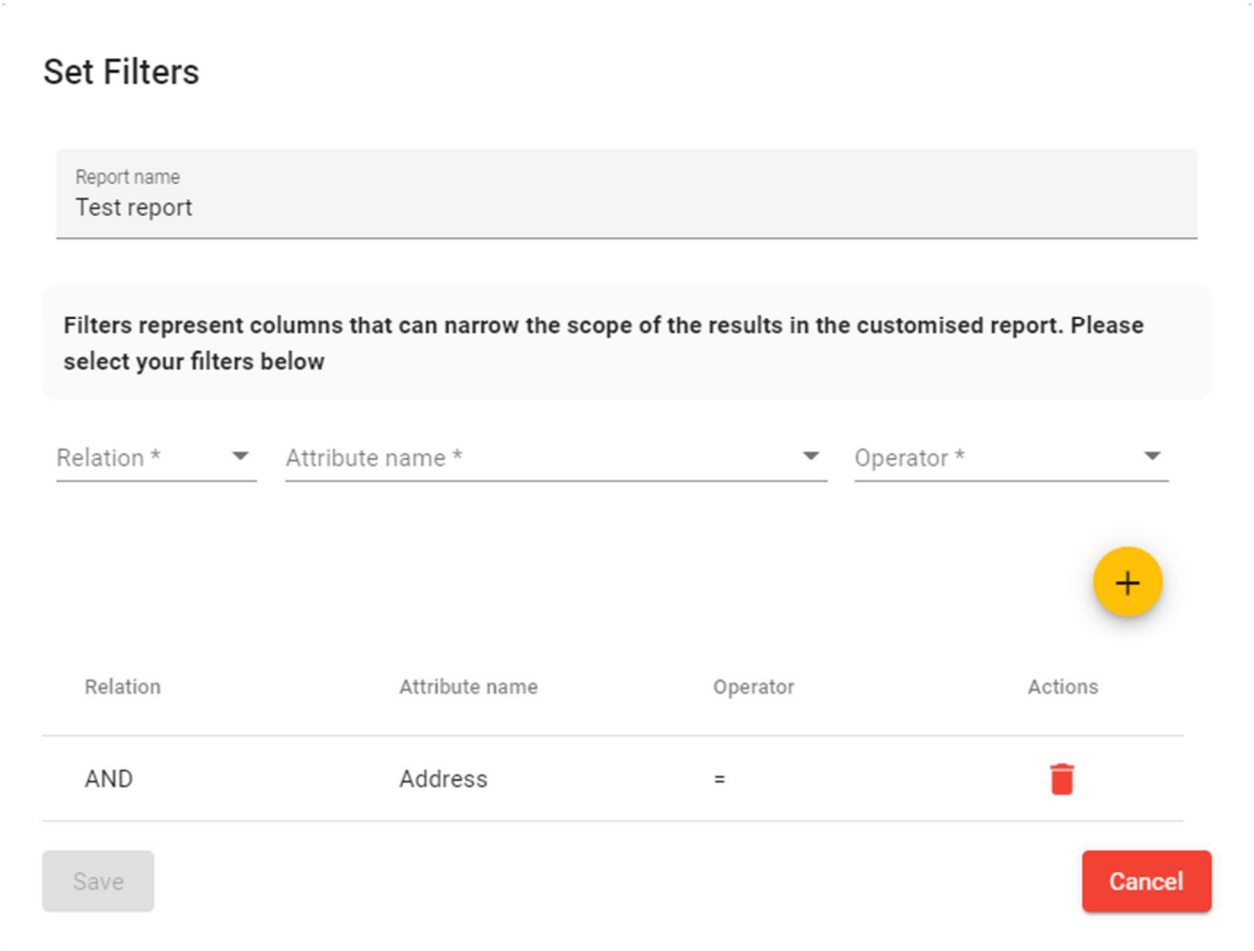


Once the list is specified, filter-related information may now be specified. All attributes specified as part of the filter list will be used to filter the result set by the end user. To add a new filter item, first select the filters menu option for the report you would like to customize.



Add the filter item as per the field description below and click the **add** icon:

Relationship	Specify **AND** if this filter criteria must be true in conjunction with other attributes Specify **OR** if this filter criteria or other criteria must be true
Attribute	The attribute that is being filtered on
Operator	The operator that will be applied to the filter: • Dates and numerics allow the following operators: Equals, Not Equals, Greater Than, Less Than, GreaterEqual Than, LessEqual Than, Between • Text fields allow the following operators: Equals, Not Equals • DropDown Lists allow the following operators: Equals, Not Equals, In

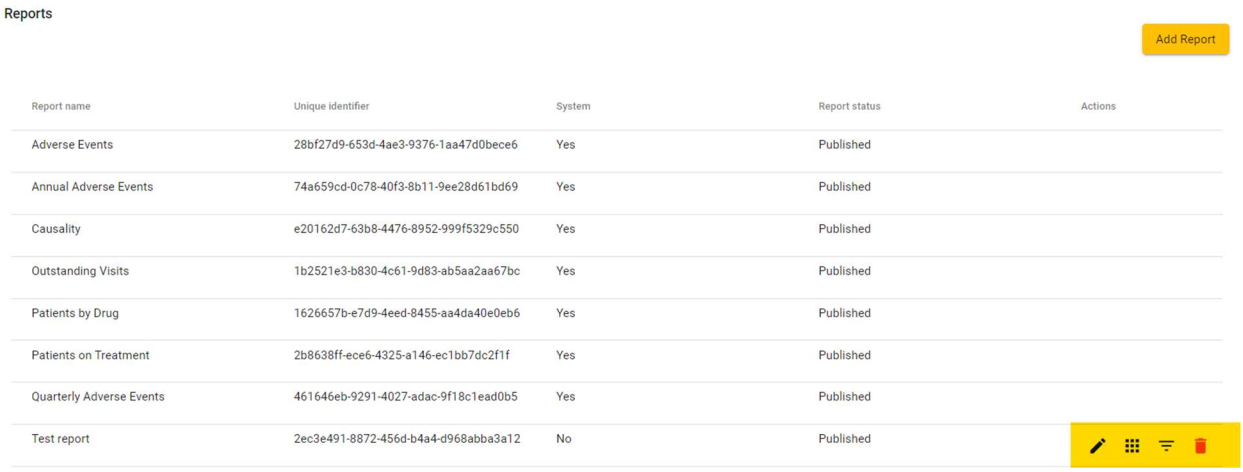


Once you are satisfied with the report configuration, change the status of the report to published. The new report will appear in the custom report menu once it is published. Please note, you may need to log out and log back in to view the new report.

6.2.3 Modifying and Deleting an Existing Report

Browse to the report using the custom report menu in the Reports Portal. To modify or delete a report, you need to have the **Reporter Administrator** role assigned to your user profile.

To modify a report, click on the **Update report, Set attributes or Set filters** icons. To delete a report, click on the **Delete report** icon.



7 Information Portal

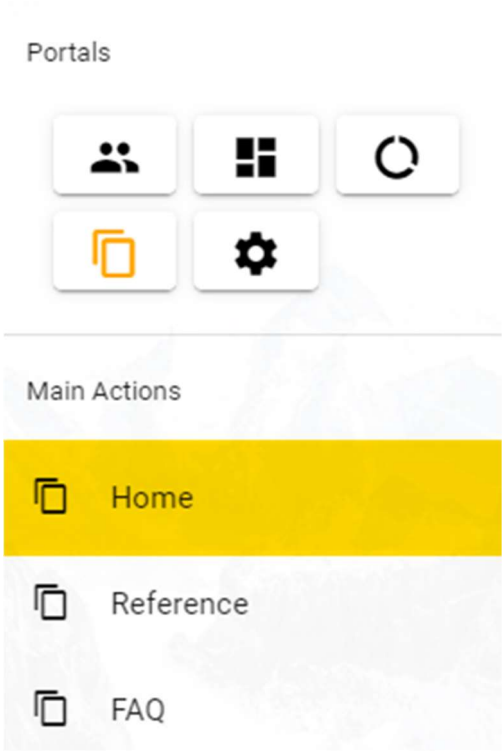
The information portal is the centralized hub for publication and presentation. PVIMS information publishers have the ability to share trends, analysis, graphs, and important information about pharmacovigilance activities.

Note: the following roles have access to the information portal:

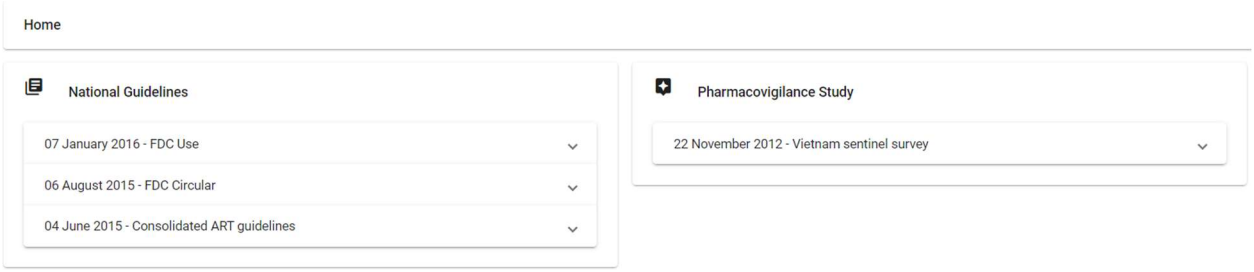
- **All users.** All users have view access to pages defined within the portal.
- **Administrator.** The administrator has FULL permissions to the information portal.
- **Publisher.** The publisher has the ability to add new pages and update content of existing pages

7.1 Viewing the home page

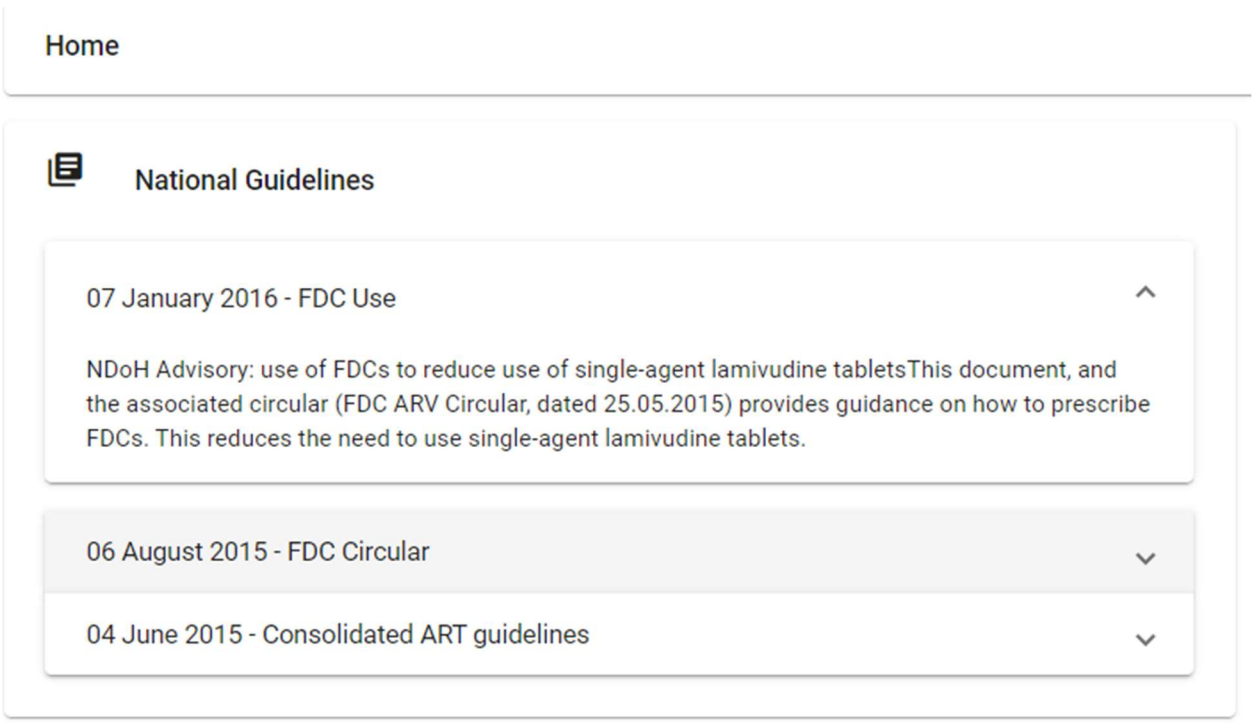
The Information Portal **Home Page** can be used to show information about upcoming pharmacovigilance activities as well as outcomes from existing and previous activities and reports. To access the **Home Page,** click on the **Home** menu in the Information Portal.



The system will navigate you to the **Home Page** where you are able to view information such as national guidelines or other information you post that would be relevant to most PViMS users.

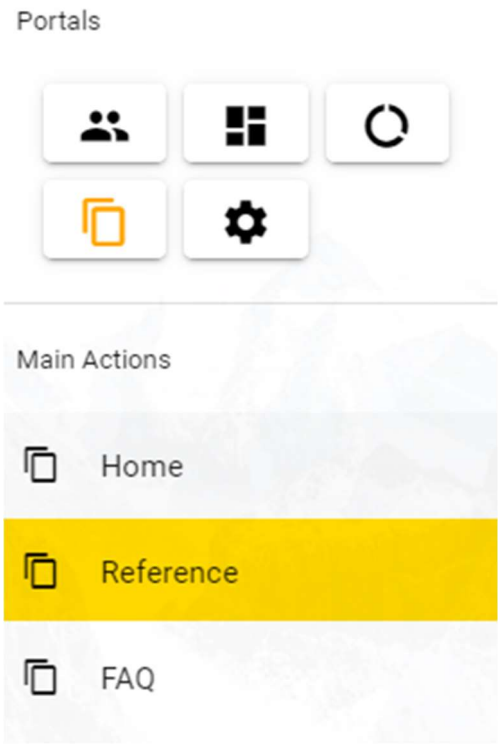


To view information for an existing date, click anywhere on the row expand or compress that guideline as required.

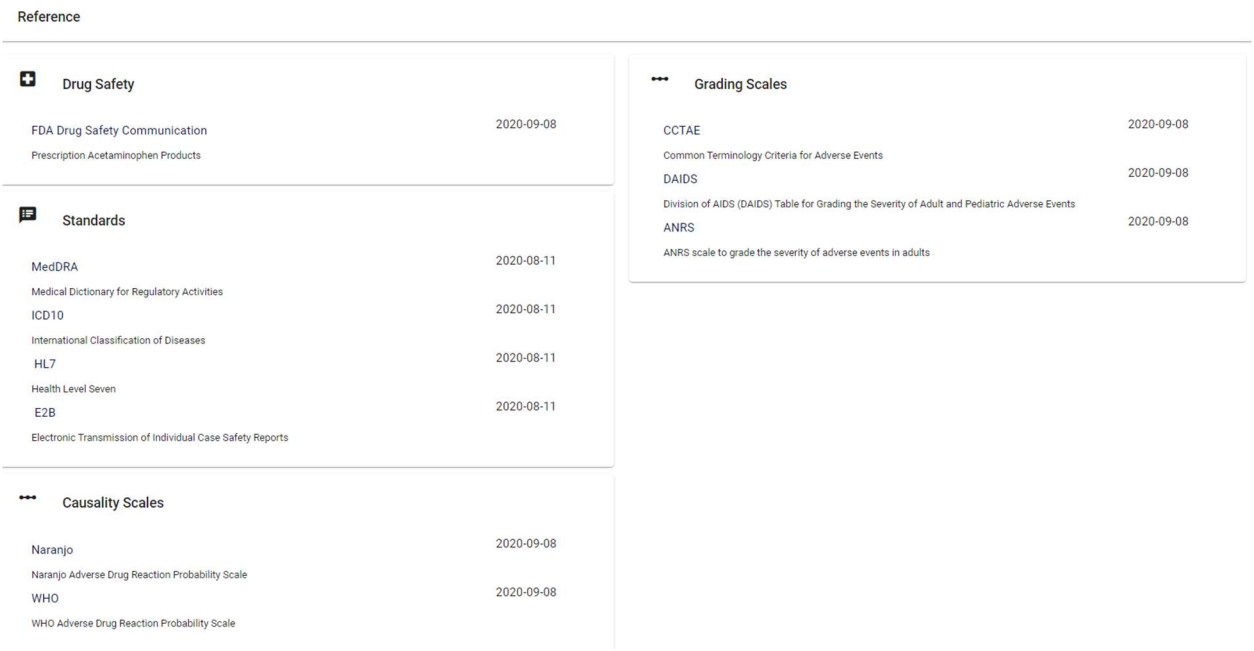


7.2 Viewing the Reference page

The Information Portal **Reference Page** contains a set of reference data particular to the implementation of PViMS for Pharmacovigilance activities. To access the **Reference Page,** click on the **Reference** menu in the Information Portal.



The system will navigate you to the **Reference** page where you are able to view this information.



The following reference data is available for viewing: -

MedDRA	Medical Dictionary for Regulatory Activities
ICD10	International Classification of Diseases
HL7	Health-Level 7
E2B	Electronic Transmission of Individual Case Safety Reports
Naranjo	Naranjo Adverse Drug Reaction Probability Scale
WHO	WHO Adverse Drug Reaction Probability Scale
CCTAE	Common Terminology Criteria for Adverse Events
DAIDS	Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events
ANRS	ANRS scale to grade the severity of adverse events in adults

To view information about the reference, click on the header for that specific reference.

The system will navigate you to a reference pop up which contains additional information to that reference item.

Reference Page 2

PViMS Standards Used

Medical Dictionary For Regulatory Activities

In the late 1990s, the International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) developed MedDRA, a rich and highly specific standardised medical terminology to facilitate sharing of regulatory information internationally for medical products used by humans. ICH?s powerful tool, MedDRA is available to all for use in the registration, documentation and safety monitoring of medical products both before and after a product has been authorised for sale. Products covered by the scope of MedDRA include pharmaceuticals, biologics, vaccines and drug-device combination products. Today, its growing use worldwide by regulatory authorities, pharmaceutical companies, clinical research organisations and health care professionals allows better global protection of patient health.Go to the MedDRA website...

International Classification of Diseases

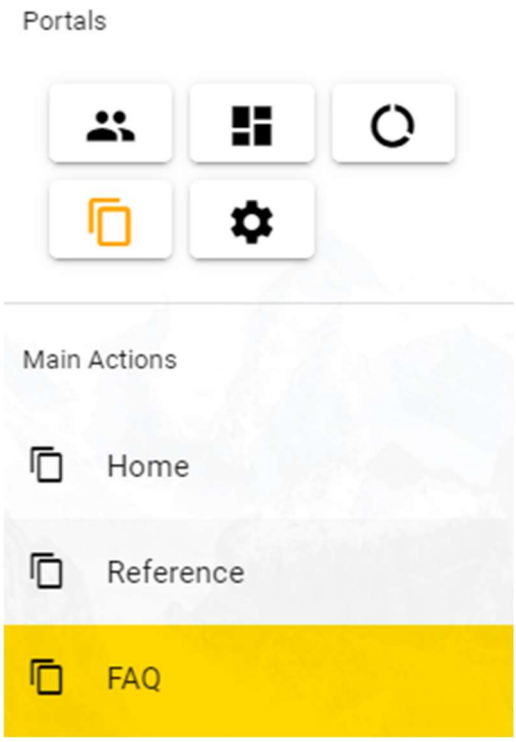
Health Level Seven

Electronic Transmission of Individual Case Safety Reports

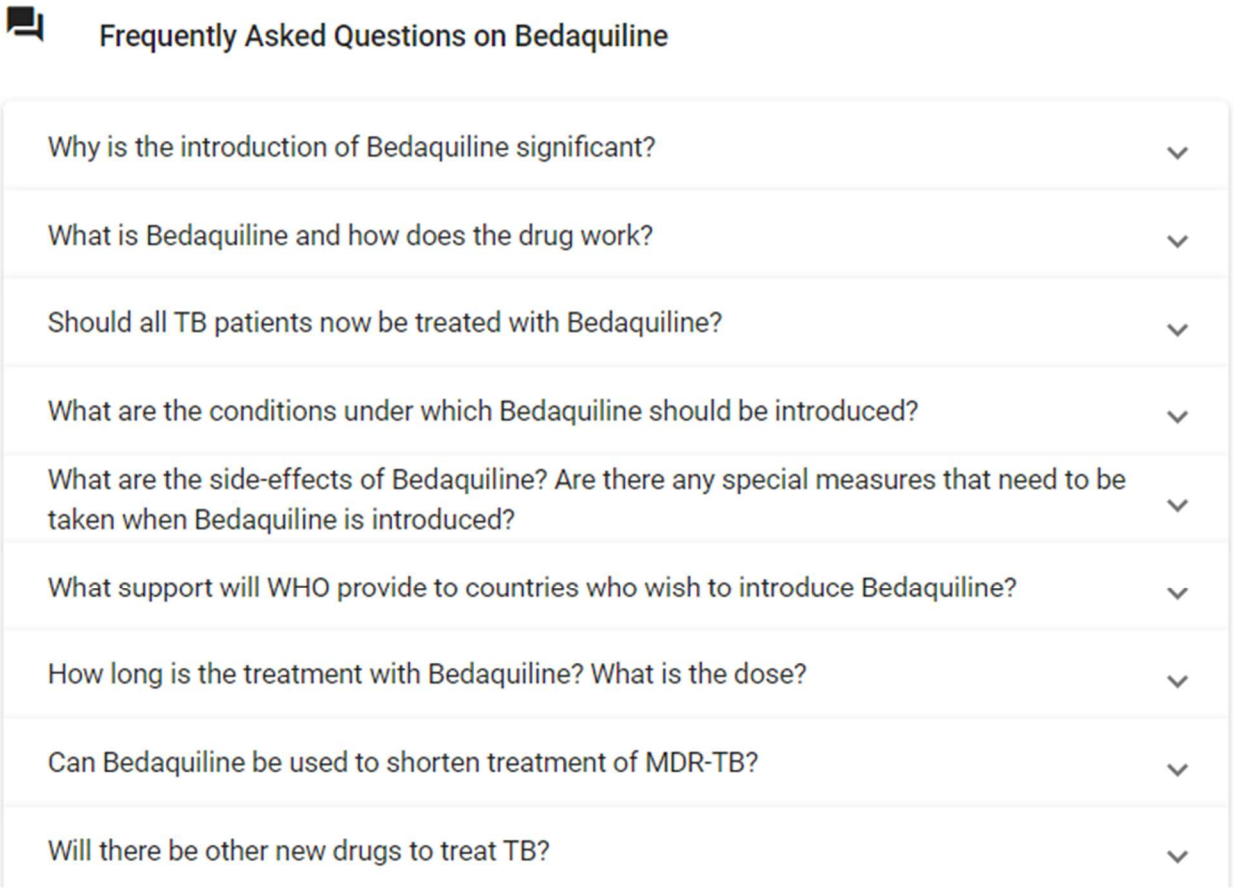
Close

7.3 Viewing the Frequently Asked Questions page

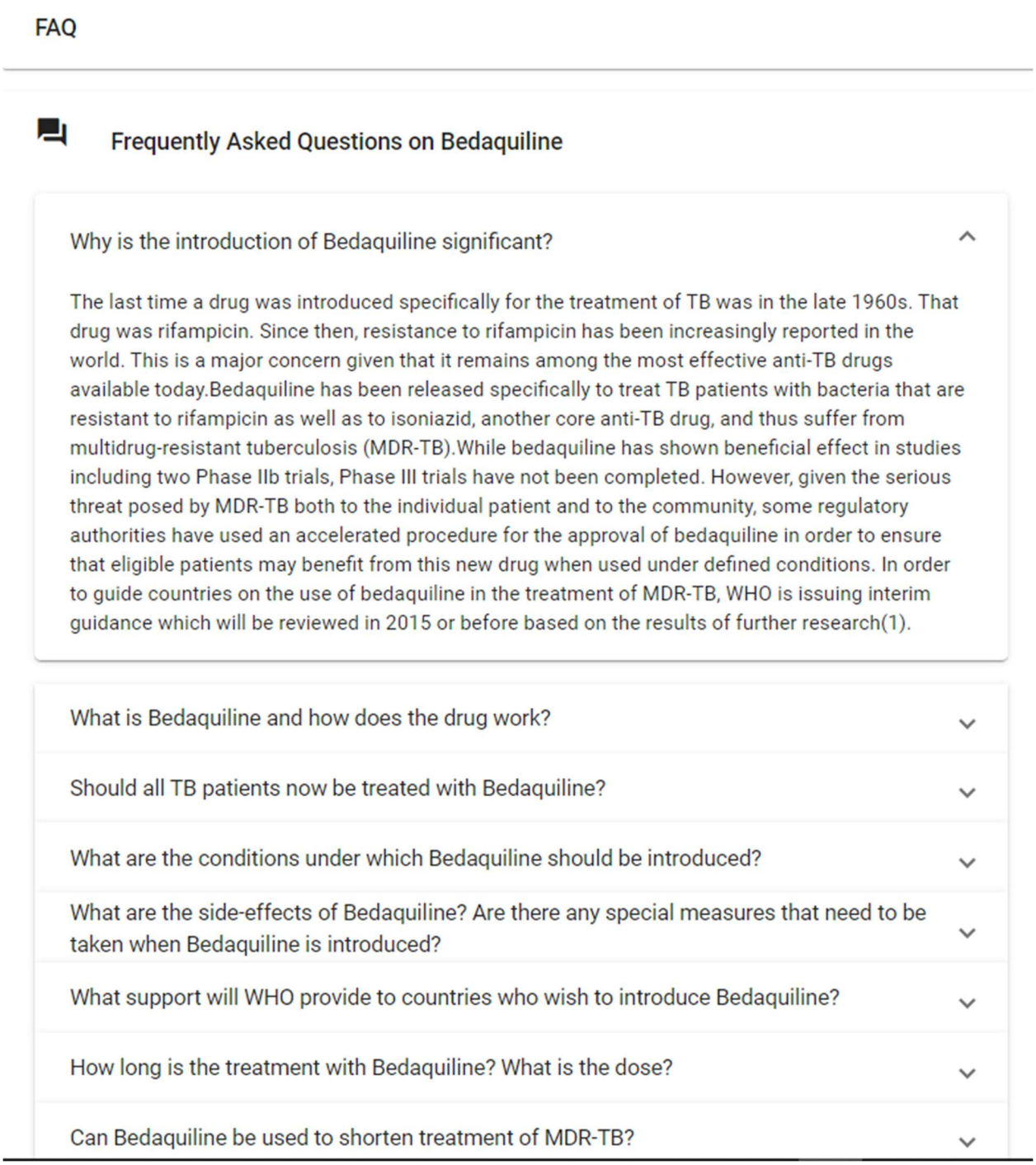
The Information Portal **FAQ Page** contains a list of frequently asked questions particular to the implementation of PViMS for Pharmacovigilance activities in relation to the use of Bedaquiline. To access the **FAQ Page,** click on the **FAQ** menu in the Information Portal.



The system will navigate you to the **FAQ** page where you are able to view these questions.



To view the answer for a specific question, click anywhere on the row to expand or compress the answer for that question.



7.4 Modifying Content in the Information Portal

The information Portal is based on a content management platform whereby PViMS publishers have the ability to add new and modify existing content within this portal in a collaborative manner. This provides PViMS analysts with an integrated platform to share details of their ongoing analysis to the health community at large.

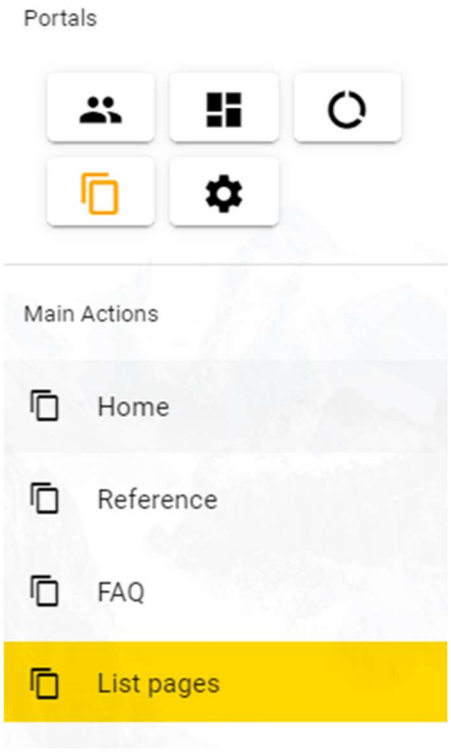
7.4.1 Adding a New Page

BUZZWORDS

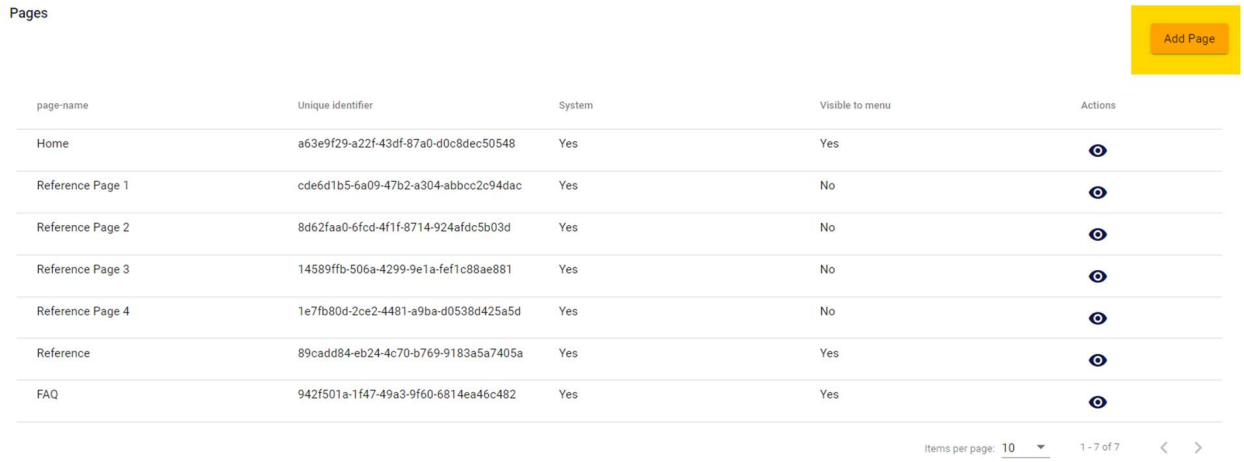
Page: Consists of 1 to 6 widgets, with each widget containing content that is rendered as part of the overall page. Each page is accessible either from the main information portal menu or from within an existing widget.

When adding a new page, you are creating a platform to add content that is specific to the overall theme of that page. To add a new page to the Information Portal, you need to have the **Publisher Administrator** role assigned to your user profile.

To add a new page, click on the **List Pages** menu.



Then click on the Add Page button to configure the core characteristics of the new page.



The following information must be entered when saving a new page:

Page name	The unique name of the page. This is also the name of page header when viewing the page.
Definition	Provide additional information that describes the page
Breadcrumb	The name of the menu option that you can access to view the page
Visible to Menu	Should this page appear on the main list of menu items in the Information Portal?

Enter the relevant information for the page and either click the **Save** button to save your new page or the **Cancel** button to cancel your action and return to the previous page of the portal.

The new page will appear in the main Information Portal menu if the visible to menu field is set to yes. See the next section for information about how to add content to a page using widgets.

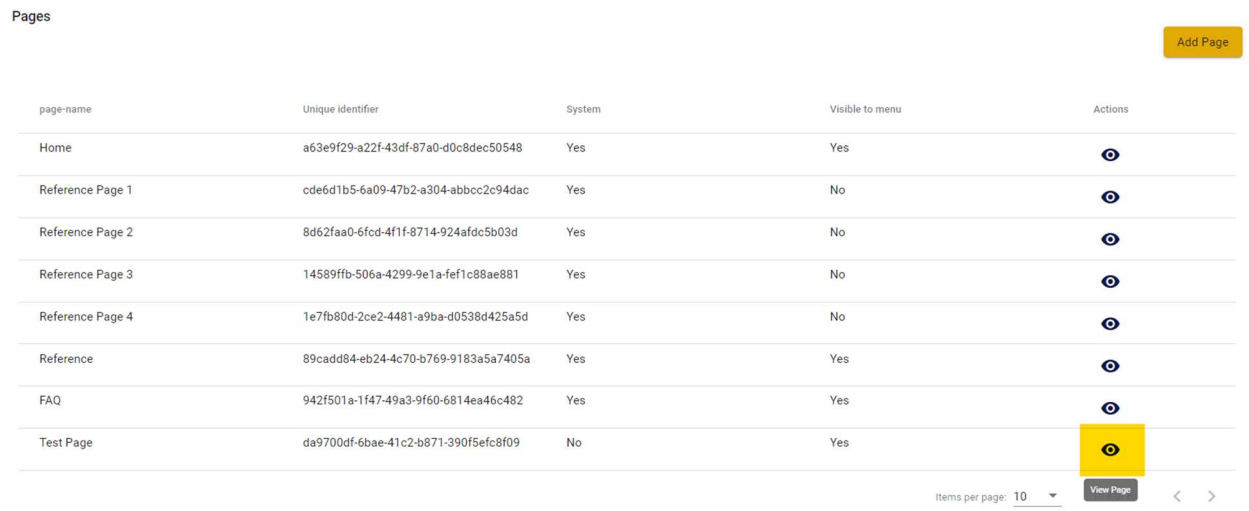
7.4.2 Adding a Widget to a Page

BUZZWORDS

Widget: A widget is an individual panel within a page that provides a container for rendering dynamic content defined by the PViMS publisher. A page within the Information Portal can store up to 6 widgets per page.

When adding a new widget, you are adding a container for new content that is specific to the overall theme of that page. To add a new widget to a page, you need to have the **Publisher Administrator** role assigned to your user profile.

To add a new widget, navigate to the page that you would like to add the widget to by selecting the visibility icon for the appropriate report in the list reports menu.

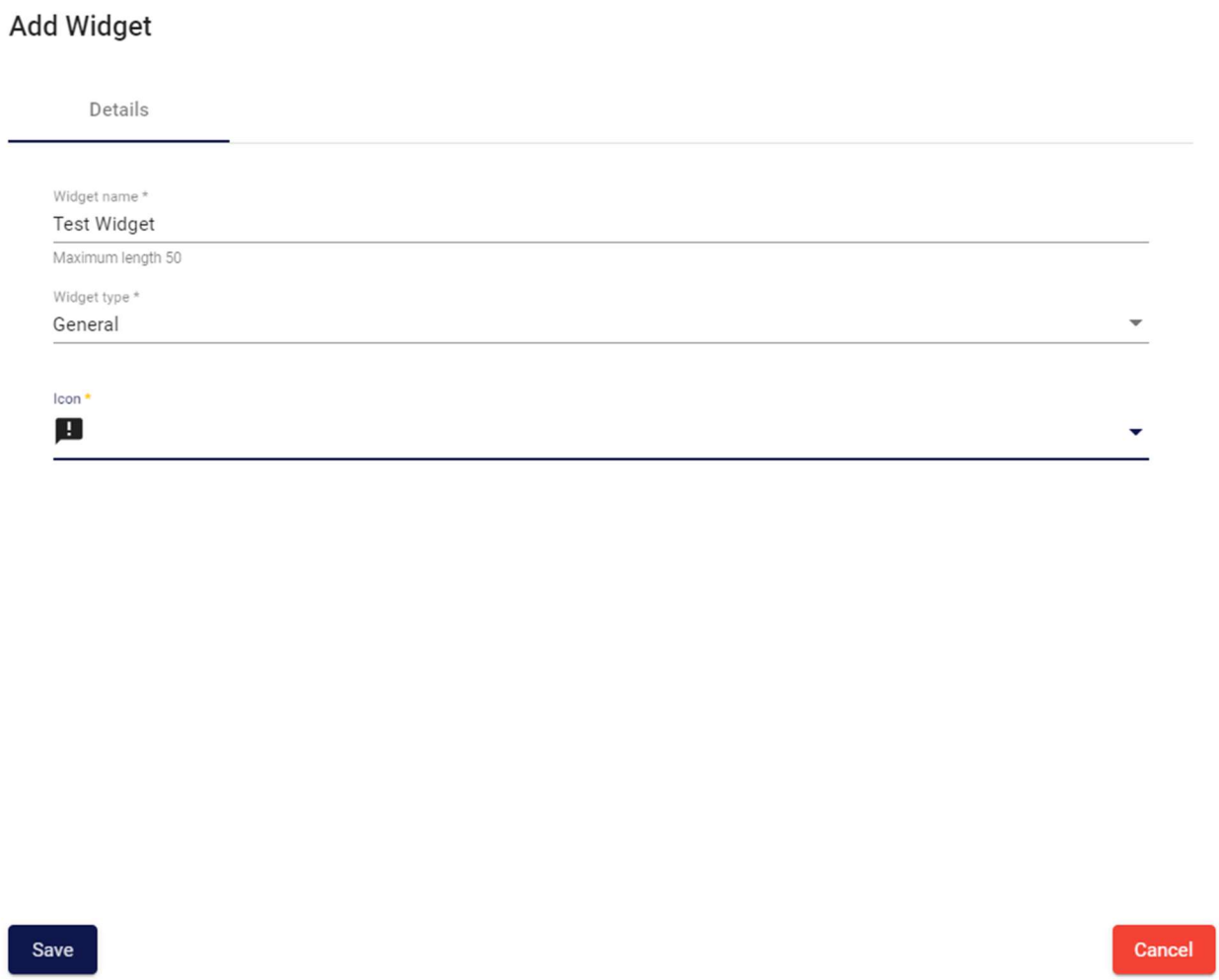


Once the system has navigated you to the newly created page, click on the **Add Widget** icon that appears in the page header.

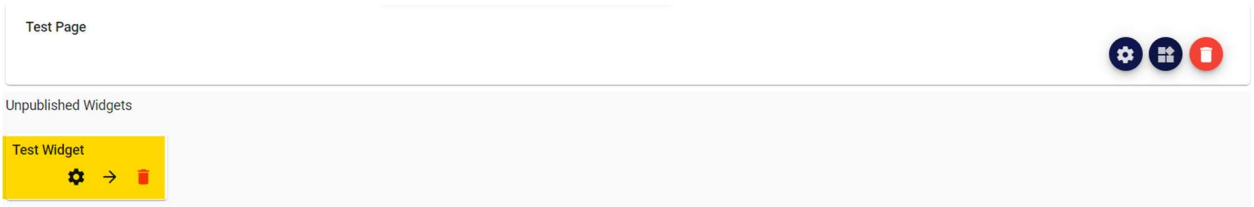
The following information must be entered when adding a new widget:

Widget name	The unique name for the widget. This name will be displayed as the title for the content.
Widget type	The type of widget to be added (General, ItemList, SubItems)
Widget Status	The status of the widget (Published, Unpublished) **Please note:** new widgets have to be added in an unpublished status
Icon	The icon that should accompany the title of the widget

Enter the relevant information for the widget and either click the **Save** button to add your new widget or the **Cancel** button to cancel your action and return to the page.



The system will add the new widget to the page in the unpublished widgets section and will now allow the user the ability to add content to the widget. Please note that a unique ID has now been allocated to the widget.



Please see the section on adding content to a widget for further information on modifying content for the new widget.

Please note: the new widget is allocated to the page as an unpublished widget. This widget therefore cannot be viewed by users who do not have the **Published Administrator** role assigned to their user profile.

7.4.3 Adding or Changing a Widget's Content

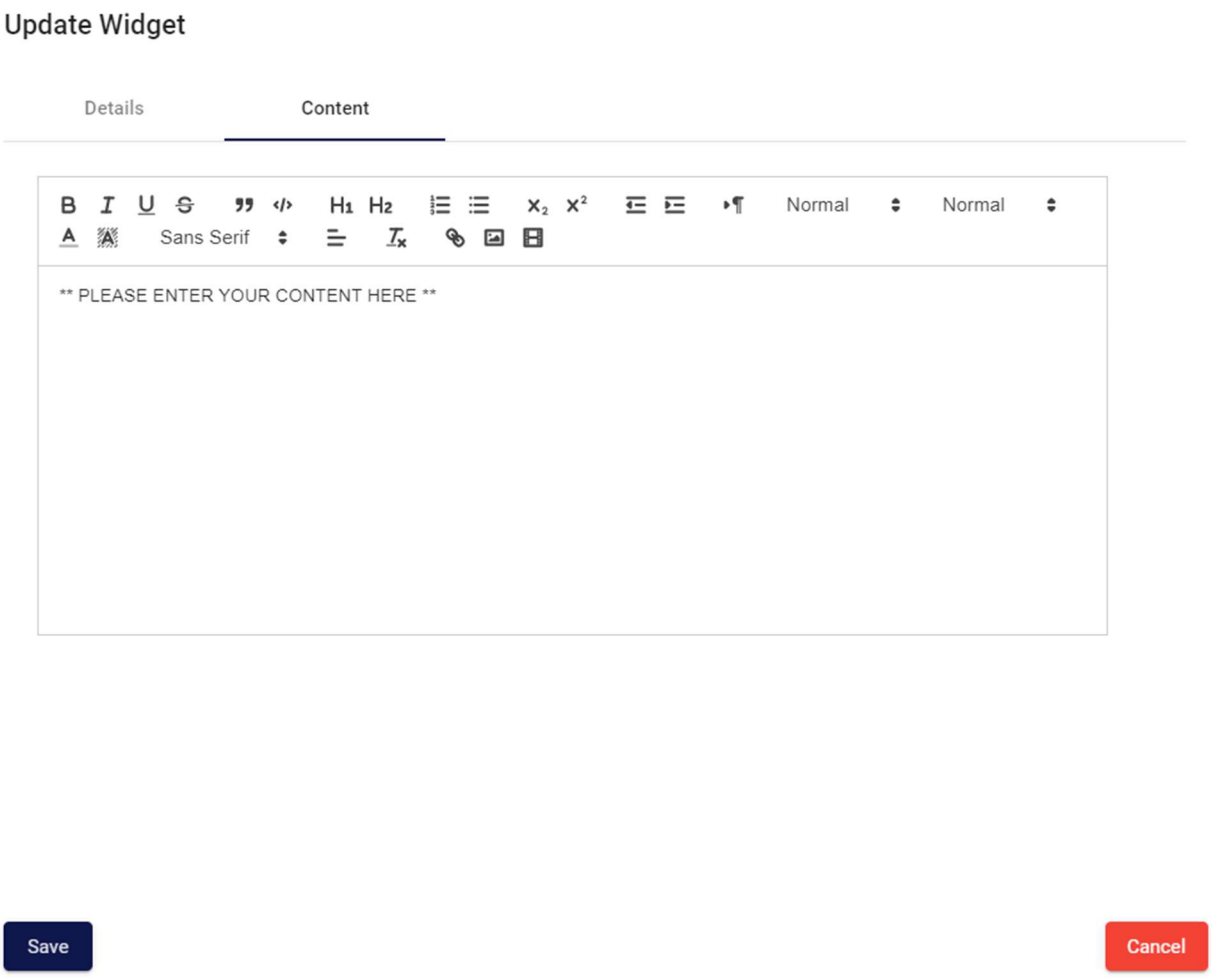
To add content to a new widget or edit content within an existing widget, you need to have the **Publisher Administrator** role assigned to your user profile.

To add or edit content within a widget, navigate to the page that you would like to modify the content, locate the widget you would like to modify and click the **Update widget** icon for this widget.

The system will navigate you to the **Edit Widget** pop up form where you will have the ability to edit your content.

7.4.3.1 General Widget Content

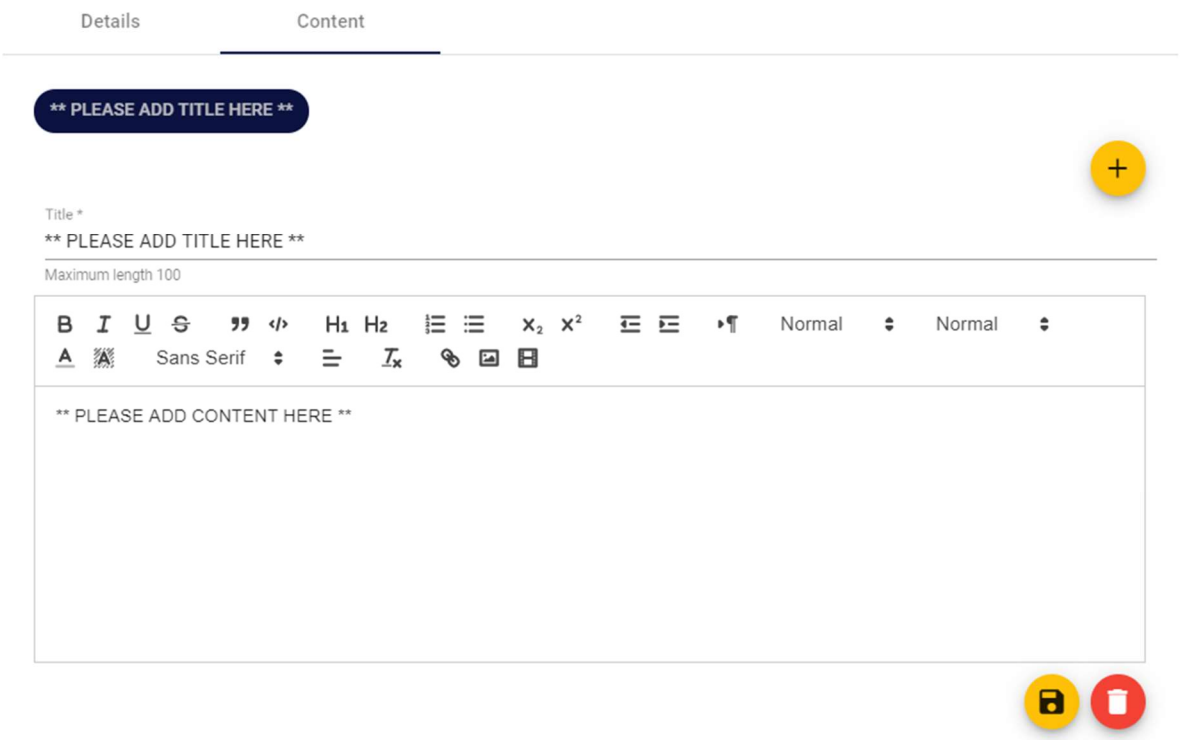
To change content for the general widget, locate the editor field in the General Content section for the widget and edit your content as necessary. Click the **Save** button to save your changes or the **Cancel** button to cancel this action and return to the page view.



7.4.3.2 ItemList Widget Content

To change content for the ItemList widget, locate the Item Content section for the widget. To add a new item to the item list, click the **add** icon.

Select the tab you would like to edit content for and then enter the title for the new item as well as the content associated to this item.



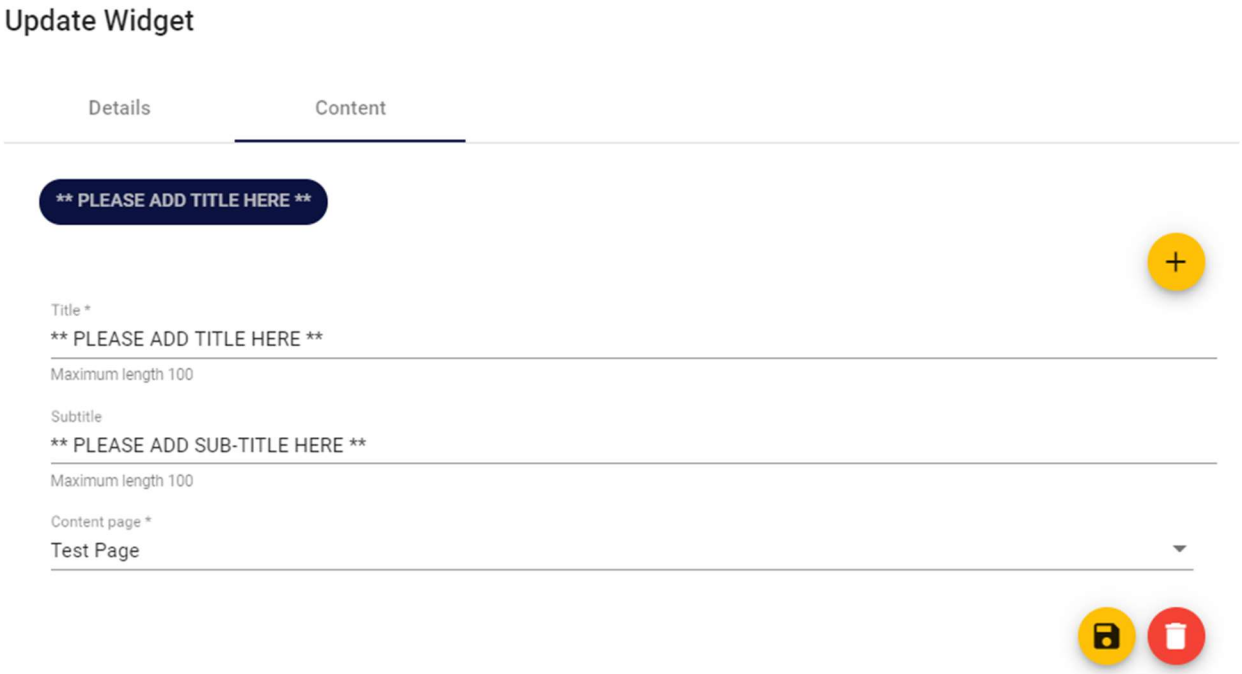
Click the **Save** icon to save the new content or the **Delete** icon to delete the tabbed item.

Once you are finished, click the **Save** button to save your changes or the **Cancel** button to cancel this action and return to the page view.

7.4.3.3 Wiki Widget Content

To change content for the Wiki widget, locate the Wiki Content section for the widget. To add a new item to the item list, click the **Add** icon.

Select the tab you would like to edit content for and then enter the title for the new item, as well as the sub-title and the page that the Wiki item should be routed to when clicked.



Click the **Save** icon to save the new content or the **Delete** icon to delete the tabbed item.

Once you are finished, click the **Save** button to save your changes or the **Cancel** button to cancel this action and return to the page view.

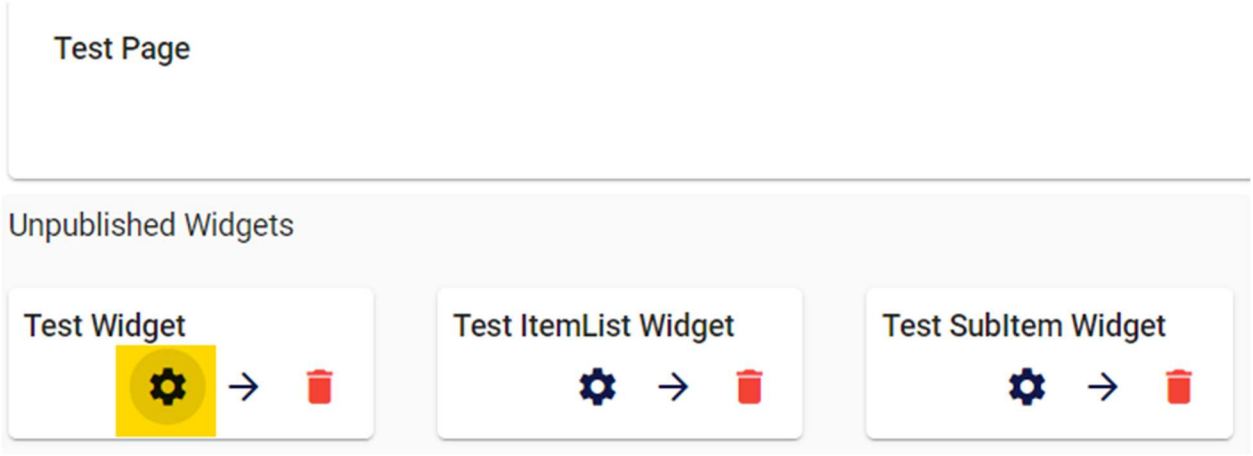
7.4.4 **Publishing a Widget**

BUZZWORDS

Publish: Widgets in an unpublished status are not viewable by standard users of the Information Portal (those who do not have the Publisher Administrator role assigned to their user profile). Conversely, widgets in a published status are viewable by all users who have access to this portal.

Widgets are only to be published once the publisher is happy with the content submitted.

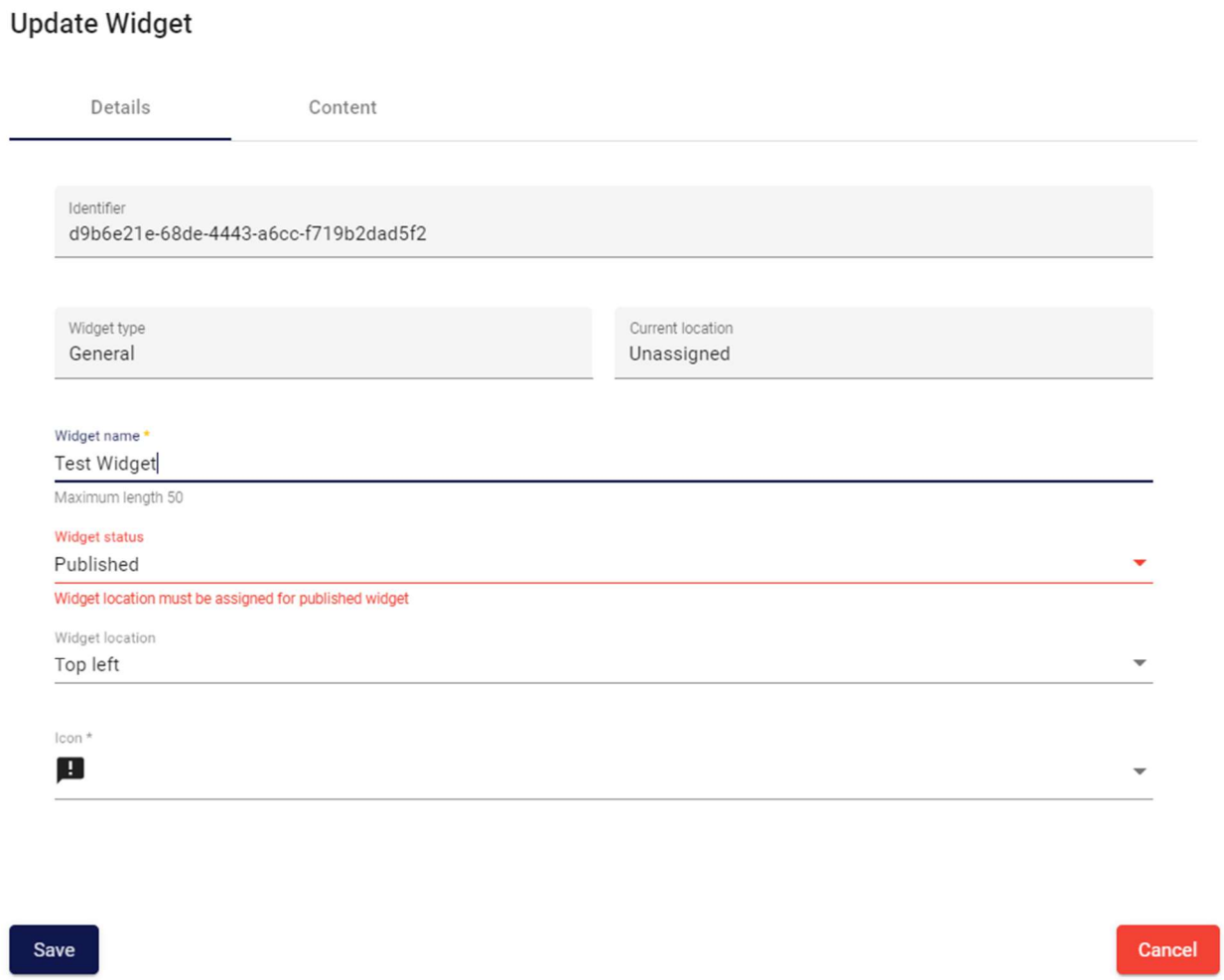
To publish a widget, navigate to the page that you would like to modify, locate the widget you would like to publish and click the **Update widget** icon for this widget.



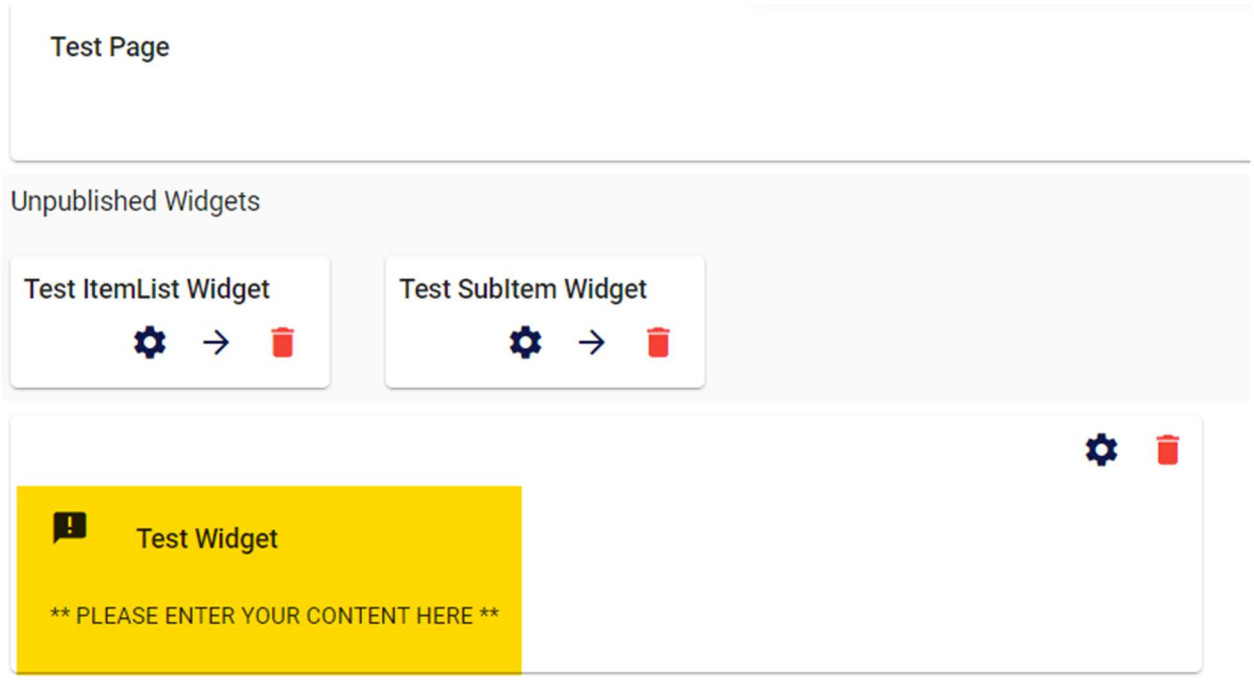
The system will navigate you to the **Update Widget** pop up form where you will have the ability to configure the widget.

Change the status of the widget to **Published** and confirm the location of the widget. Click the **Save** button to save your changes or the **Cancel** button to cancel this action and return to the page view.

The location determines where on the page the widget will appear. Only one widget can occur in each of the 6 allocated spaces on a page.



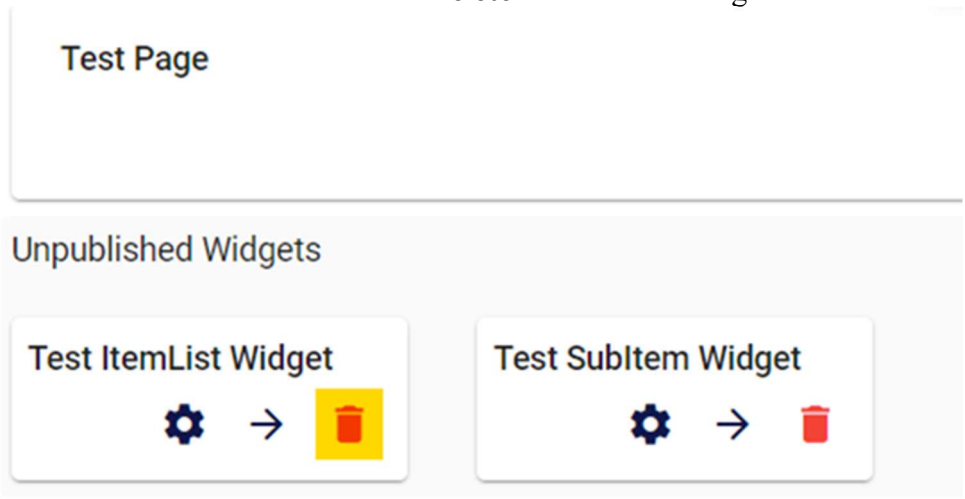
The system will navigate you to the page that you have published the widget on. Please note that the widgets will be removed from the Unpublished Widgets section and the widget will now be viewable on the page itself.



Saving a widget as unpublished will remove the widget from the page and move it to the Unpublished Widgets section. Widgets in this section are **not** viewable by standard users of the Information Portal.

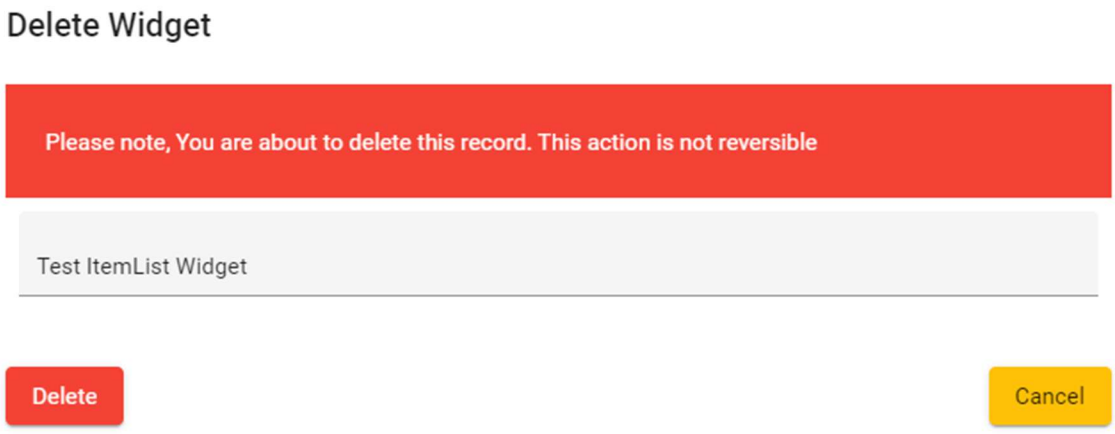
7.4.5 **Deleting a Widget**

To delete a widget, navigate to the page that you would like to modify, locate the widget you would like to delete and click the **Delete** icon for this widget.



The system will navigate you to a **Delete Widget** pop up form where you will have the ability to delete the widget.

Click the **Delete** button to confirm the deletion or the **Cancel** button to cancel this action and return to the page view.

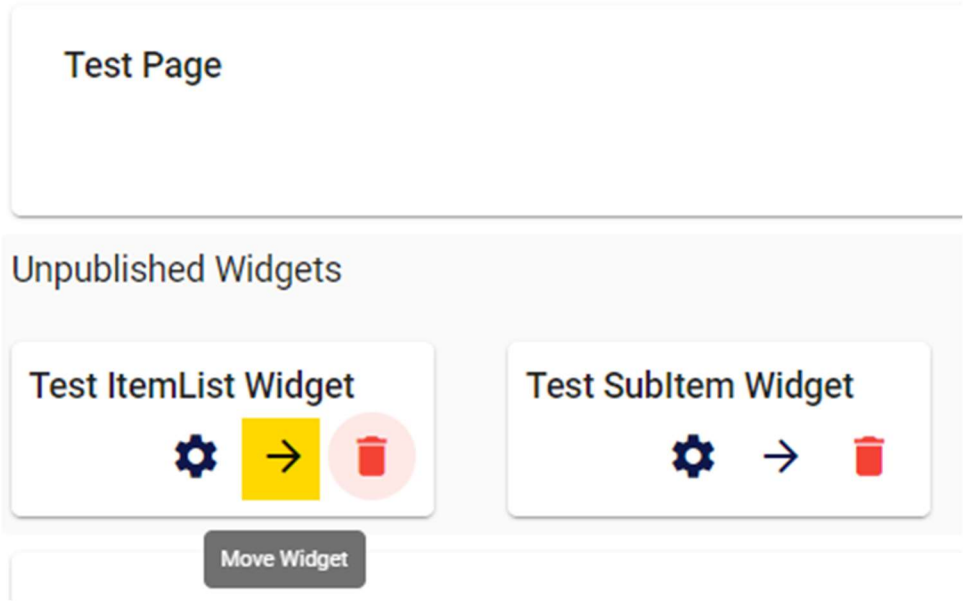


The widget and all related content will be deleted when clicking the **Delete** button.

7.4.6 Moving a Widget to a New Page

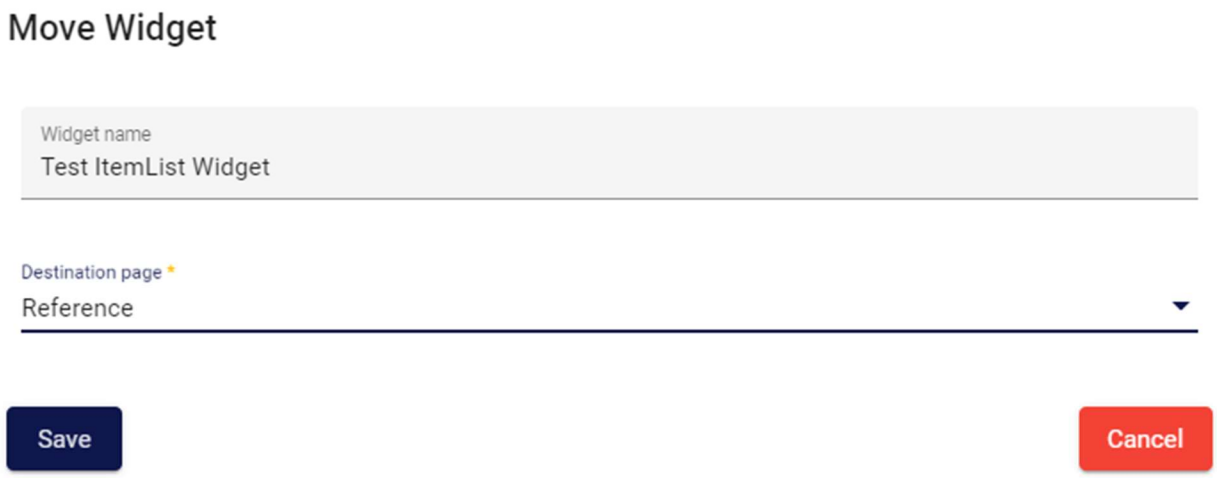
To move a widget, navigate to the page that you would like to modify, locate the widget you would like to move and click the **Move** icon for this widget.

You are only able to move a widget if it is currently in an **Unpublished** status on the page it resides on.



The system will navigate you to a **Move Widget** pop up form where you will have the ability to select the destination page for the widget.

Select the destination page and click the **Save** button to confirm the move or the **Cancel** button to cancel this action and return to the page view.



The widget and all related content will be moved to the new page and will exist in an **Unpublished** status on the new page.

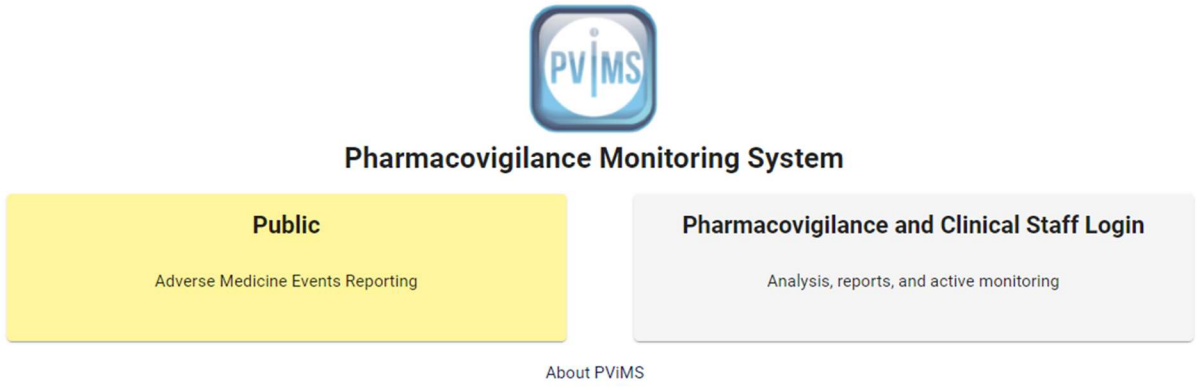
8 Spontaneous Reporting

PViMS provides the mechanism to register spontaneous reports by the public. While these reports form part of the overall PViMS adverse event repository where Pharmacovigilance activities can be performed against the report, they do not form part of the analysis.

Note: spontaneous reporting is available to the public and no login to PViMS is required.

8.1 Accessing Spontaneous Reporting

When you enter the correct URL to access PViMS, the system navigates you the login page. This page contains the primary link to register a spontaneous report.



To register a spontaneous report, click on the **Public** button. The system will navigate you to a page where you can enter the spontaneous report.

8.1.1 **Add a New Report**

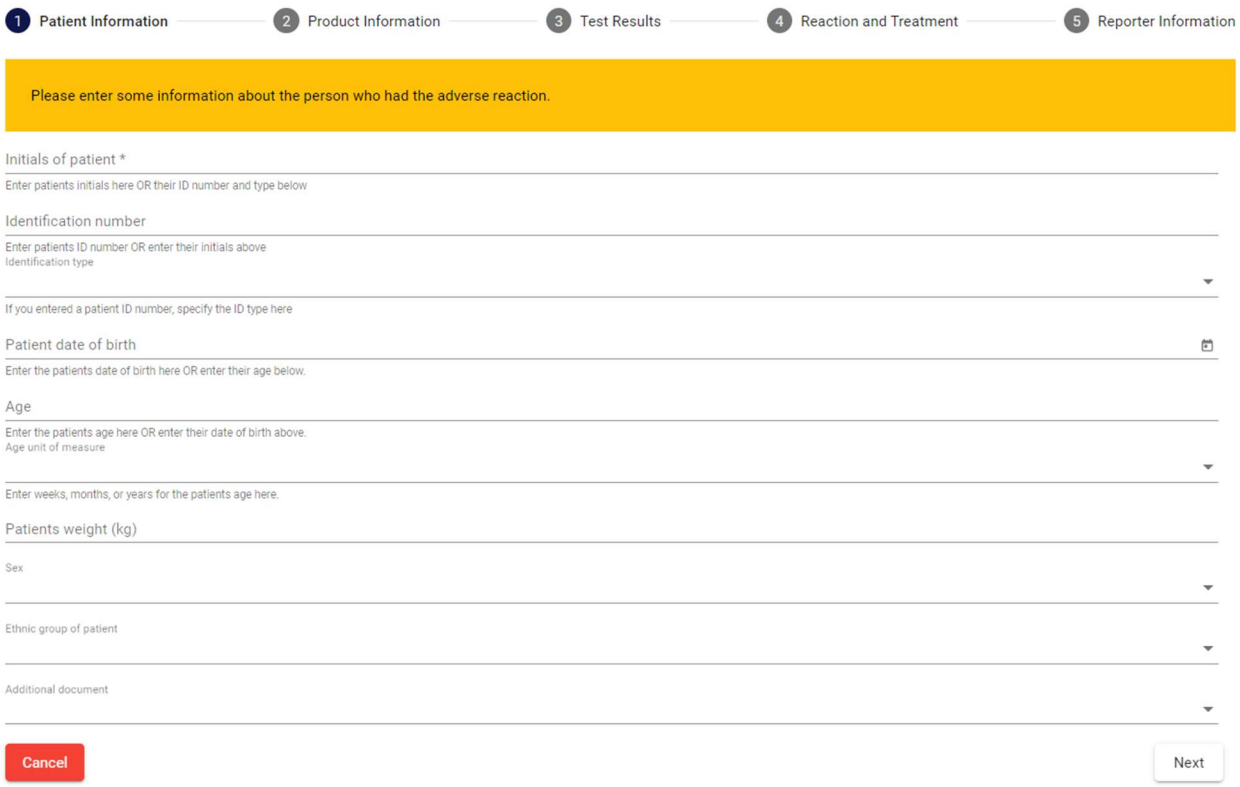
Spontaneous reports are composed of the following sections:

Patient Information	Information related to the patient who suffered the adverse event
Product Information	Information related to the medication that potentially caused the adverse event
Test Result	Any test results that are relevant to the adverse event
Reaction and Treatment	Details of the adverse event
Reporter Information	Details of the person who has logged the adverse event

8.1.1.1 Patient Information

The **Patient Information** section captures basic patient demographic information about the person who suffered the adverse event.

To enter patient information, **enter text** in the corresponding fields (e.g., **Initials of Patient**). Or click the **arrow** in a selected field to display a list of values and select one value from the list. All elements with a red asterisk are mandatory.



Fields in the **Patient Information** section are described below:

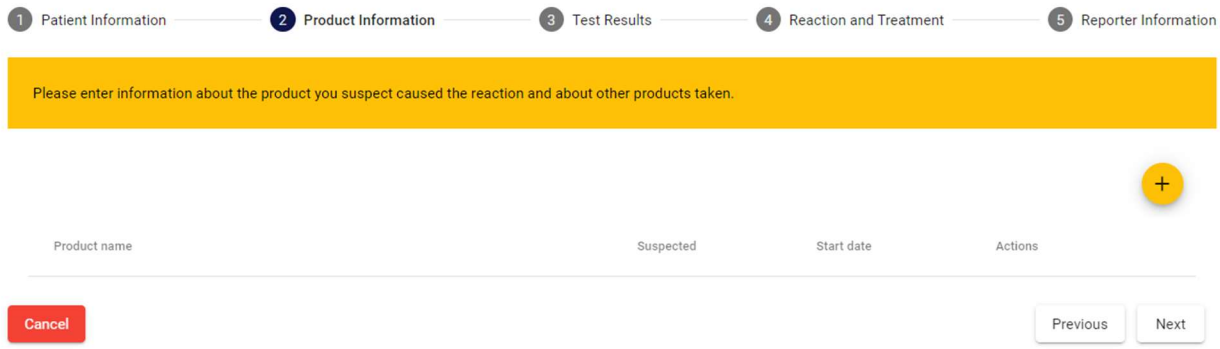
Initials of Patient	Identification of the patient is facilitated through the capturing of their initials in a text field
Identification Number	Identification of the patient is facilitated through the capturing of their ID Number in a text field
Identification Type	Dropdown list to select the patient's type of identity number specified
Patient Date of Birth	Either specify the patient's date of birth
Age	Or specify the patient's age
Age Unit of Measure	If age is specified, specify the unit type for the age (e.g. days, months etc.)
Patient Weight (kg)	The weight of the page at the time of the adverse event, in kilograms
Sex	Dropdown list to specify the gender of the patient
Ethnic Group of Patient	Dropdown list to specify the ethnic group of the patient

Click the **Next** button to navigate to the next screen or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen.

8.1.1.2 Product Information

The **Product Information** section captures a list of medications that the patient was taking at the time of the adverse event.

To enter medication information, click the **Add Product Information item** button. Once a product has been added, you are able to remove or edit the medication using the appropriate button next to the medication.



Fields in the **Product Information** Section are described below:

Product	The name of the medication that the patient was taking (generic or brand name)
Drug Strength	Free format description of the drug strength, **e.g. 250**
Drug Strength Unit	Dropdown list specifying the unit of the drug strength, e.g., mg
Product Suspected	Is this product suspected of causing the adverse event
Dose Number	Drug dosage
Dose Unit	Dropdown list specifying the unit of the drug dosage
Route of Administration	Dropdown list specifying how the drug has been administered
Start and End Date	When did the patient start the drug and if they have completed taking the drug, when was the last date of administration
Treatment Duration	How long has the patient been on the drug

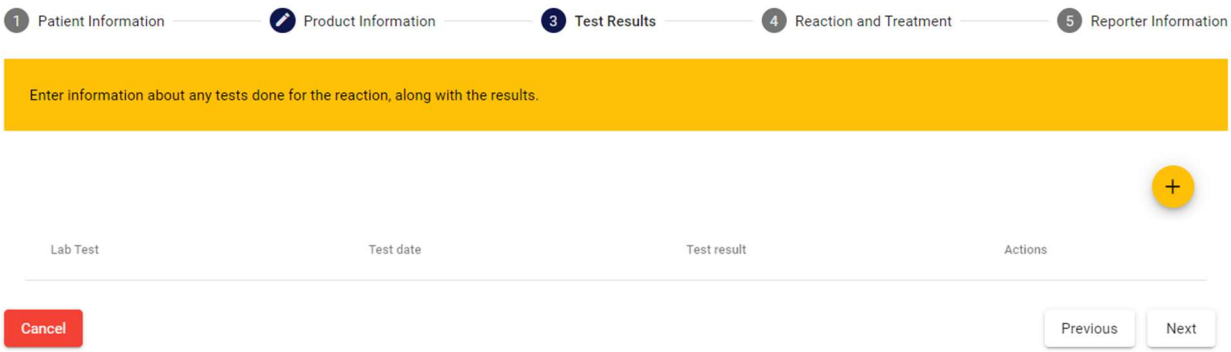
Treatment Duration Unit	Dropdown list specifying the unit for the duration
Indication	Indication for why the patient is taking this drug
Frequency	How frequently is the patient taking the drug
Batch Number	The batch number the drug forms part of
Action Taken	What action was taken when the adverse event occurred
Product Challenge	Was a challenge performed on the product when suspected of the adverse event
Product Rechallenge	Was a rechallenge performed

Click the **Next** button to navigate to the next screen or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen.

8.1.1.3 Test Results

The **Test Results** section captures a list of test results that are applicable to the adverse event.

To enter test results, click the **Add Test Results item** button. Once a test has been added, you are able to remove the result or edit the test using the appropriate button.



Fields in the **Test Results** Section are described below:

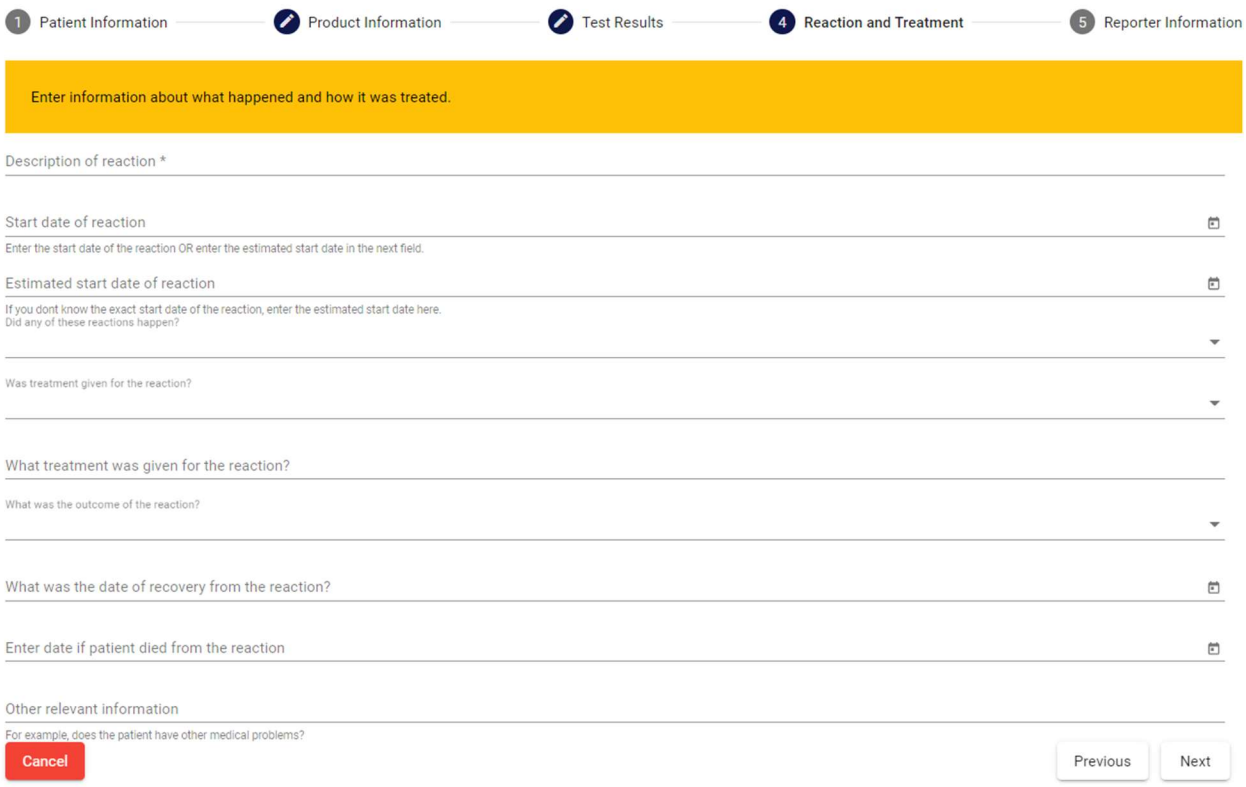
Test Date	The date the test was conducted
Test Name	The name of the test conducted
Test Result	The result of the test conducted
Test Unit	Any unit associated to the test result
Low- and High-Test Range	Test result range that is considered normal
More Information	More information associated to the test result?

Click the **Next** button to navigate to the next screen or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen.

8.1.1.4 Reaction and Treatment

The **Reaction and Treatment** section captures details of the adverse event.

To enter reaction and treatment information, **enter text** in the corresponding fields (e.g., **Description of Reaction**). Or click the **arrow** in a selected field (e.g.) to display a list of values and select one value from the list. All elements with a red asterisk are mandatory.



Fields in the **Reaction and Treatment** Section are described below:

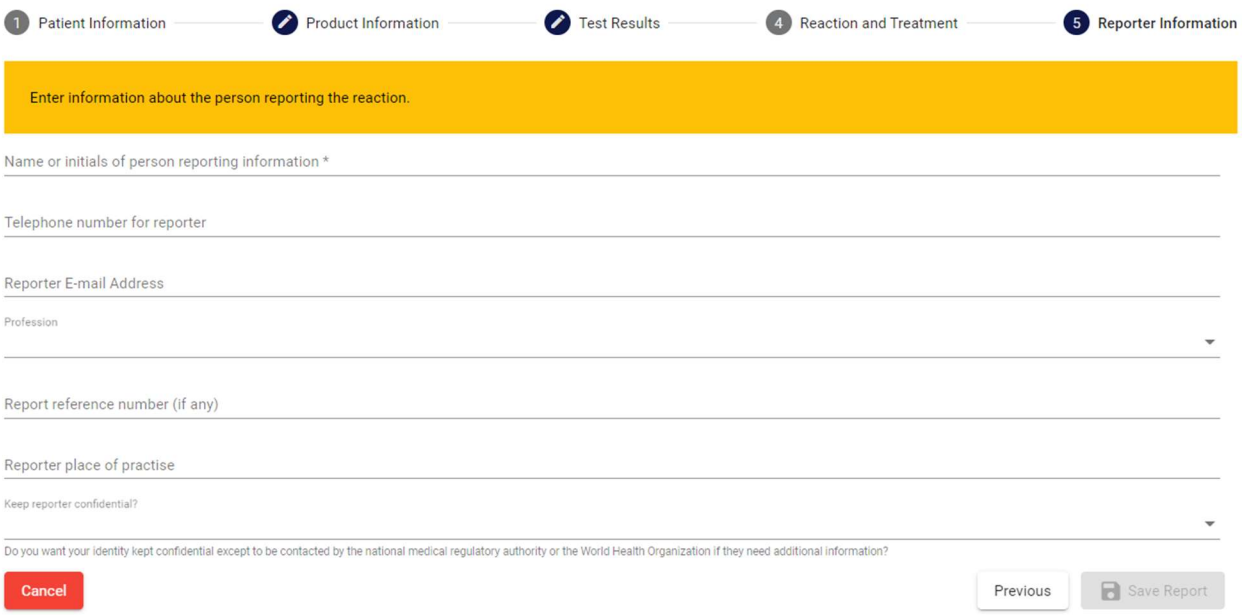
Description of Reaction	A description of the adverse event
Start Date of Reaction	The date the reaction first appeared in the patient
Estimated Start Date of Reaction	Only specify this date if the exact start date is not known
Reactions	Did the patient experience a reaction
Treatment for Reaction	Was a treatment given for the reaction itself
What Treatment Given for Reaction	If a treatment was given for the reaction, what treatment was it
Reaction Outcome	What was the outcome of the reaction
Recovery Date	If the patient has recovered, what is the date of the recovery
Deceased Date	If the patient has died from the adverse event, what was the date of death
Other Relevant Info	Is there other information relevant to the adverse event

Click the **Next** button to navigate to the next screen or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen.

8.1.1.5 **Reporter Information**

The **Reporter Information** section captures details of the person who has reported the event.

To enter reporter information, **enter text** in the corresponding fields (e.g. **Name or Initials of person reporting the event**). Or click the **arrow** in a selected field (e.g.) to display a list of values and select one value from the list. All elements with a red asterisk are mandatory.



Fields in the **Reporter Information** Section are described below:

Name or Initials	The name or initials of the person reporting the event
Telephone Number	Contact number of the person reporting the event
Email Address	Email address of the person reporting the event
Profession	The profession of the person who has reported the event
Reference Number	Reference number for the event that has been reported
Place of Practice	At which facility does the reporter work
Confidentiality	Should the report remain confidential

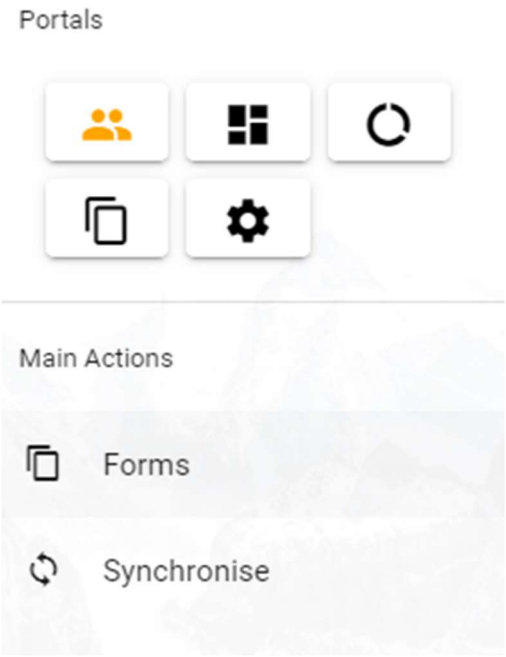
Click the **Next** button to navigate to the next screen or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen.

Click the **Save Report** button to submit the report or the **Cancel** button to cancel the registration of the spontaneous report and return to the login screen. Please note that the **Save** button will only be activated once all mandatory fields and field validations are completed successfully.

9 Appendix A: Forms Reference Guide

Step 1: Navigate to the Forms menu in the clinical portal

Once you have logged into PViMS, ensure you have the **clinical** portal icon selected. You will then be presented with a list of menu options in the left-hand pane of the system. Look for the **Forms** menu and select this menu by pressing on it.



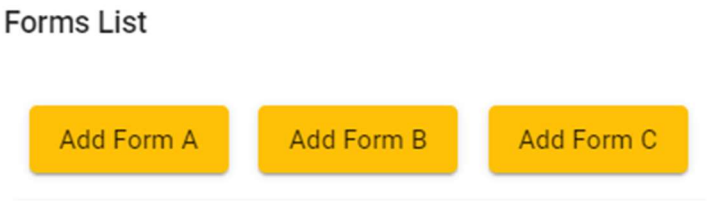
What is the clinical portal?
The clinical portal is the area of the PViMS application that focuses on the collection of longitudinal clinical data for the patient.

Step 2: Select the type of form you would like to capture

Once you have pressed on the **Forms** menu, you will be presented with a page that allows you to capture a new form and view all existing forms that have been captured. Press on the **Add Form A** button of the form to capture a new Form A.

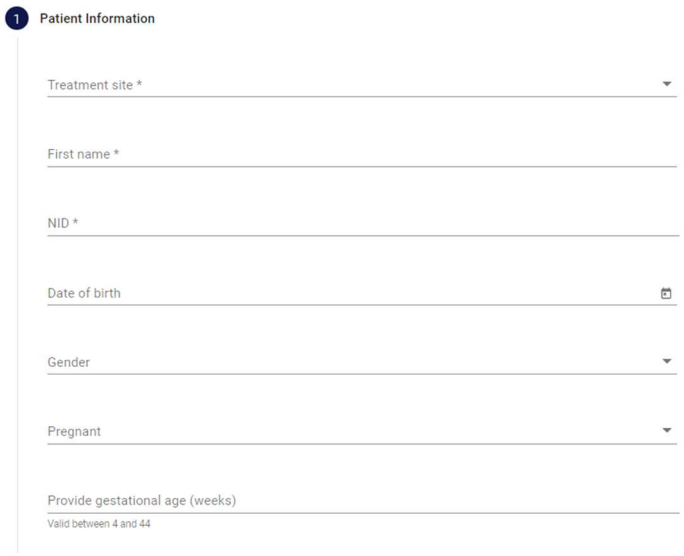
What is a button?
A button is a clickable or touchable object on a web page that allows the user to initiate a new or complete an existing action within the system.

Add Form A



Step 3: Complete the patient information section

Once you have pressed on the **Add Form** button, you will be presented with the capture screen for a new form. In all existing forms, section 1 contains patient demographic information used to identify the patient the form belongs to.

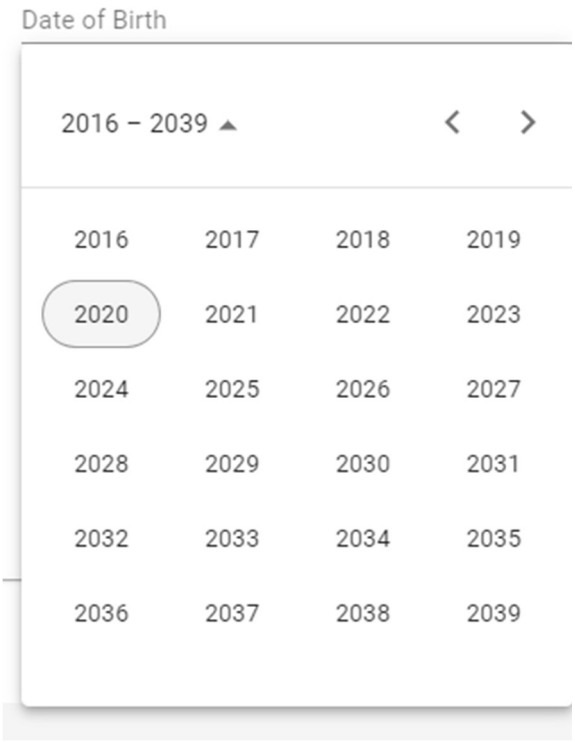


1. Press on the treatment site field to activate a drop-down list containing a list of all treatment sites you have been assigned to. Select the appropriate treatment site.
2. Press on the first name field to activate a free format text box that allows you to capture a patient's first name.
3. Press on the date of birth field to activate a date field where you can enter the patient's date of birth.
4. Proceed to capture all mandatory fields.

What is a drop-down list?
Sometimes referred to as a pull-down menu, drop-down list, or drop-down box, a drop-down menu is a list of items that appear when pressing on a button or text selection.

What is a free format field?
A free format field allows you to capture details relating to the field using any text and numeric characters. A free format field may allow text and numeric data or may be limited to numeric data only for fields such as weight. When applicable the system will also notify you of any numeric ranges applicable to the numeric field.

How do date fields work?
When pressing on the little calendar on the right of the field, you will be presented with a date control which allows you to navigate to the date you would like to select. Use the left and right arrows to change the year. Once you have selected the year, you will then be prompted to select the month followed by the day.



What is a mandatory field?
Mandatory fields are fields with an * at the end of the field name and mean that that field is required and must be captured before you are able to continue. If you do not capture this field, you will be presented with an error.

Step 4: Navigate to the next section of the form

Once you have completed the first section of the form and all mandatory fields are captured, you are able to navigate to the next section or save the form and exit.

What does it mean to save and exit the form?
When saving and exiting, the form will be captured and stored on your device including any data that has been captured up to this point. You are then able to edit this form and continue where you left off at a later point in time.
Please note, if you save and exit, you will not be presented with a unique form identifier.

Step 5: Updating a section with a table

Once you have navigated to section 2 of the form, you will be presented with a table where you can add one or many conditions to the patient.

1. Add a new row to the conditions table by clicking on the plus button and then capturing all fields on the modal form presented.
2. Once you have added a new row, note that this row appears within the conditions table.
3. Edit the newly added tow by clicking on the pencil button that appears next to the

row you would like to update. Update data in this row and click save.

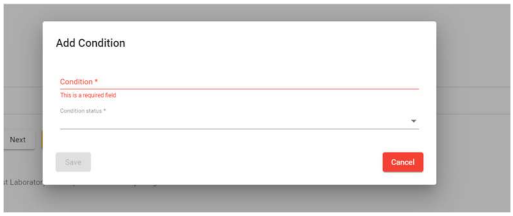
4. Once you have updated the row, note that the table now contains updated information from the newly updated row.
5. To delete a row from the table, press on the red bin icon.

What is a table and how does it work?

A table is like an MS Excel spreadsheet where it is made up of multiple rows and columns. it is used to represent a one to many relationship between the patient and the area that the table is representing. In this situation each row in the condition table represents a separate condition that the patient has.

What is a modal form and how does it work?

A modal form is a pop-up form that appears on top of the previous form you were busy on. Once you are finished completing the pop-up form or if you cancel out the pop-up form you will be returned to the exact position you were in previously.

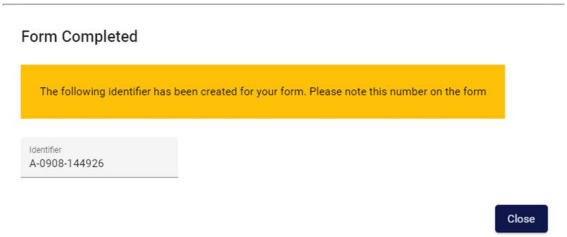


What is the plus button and how does it work?

You can add a new row to the conditions table at any point by pressing on the plus button at the top right-hand side of the table. When pressing on this button you will be presented with a modal form that allows you to capture the various details of the new condition.

Step 6: Marking a form as complete

Once you have completed updating all sections of the form, you are able to mark the form as completed. Press on the mark form as completed button that appears in the last section of the form to mark the form as completed. Once you have pressed on this button, you will be presented with a new modal form that specified what the unique identifier for the form is.



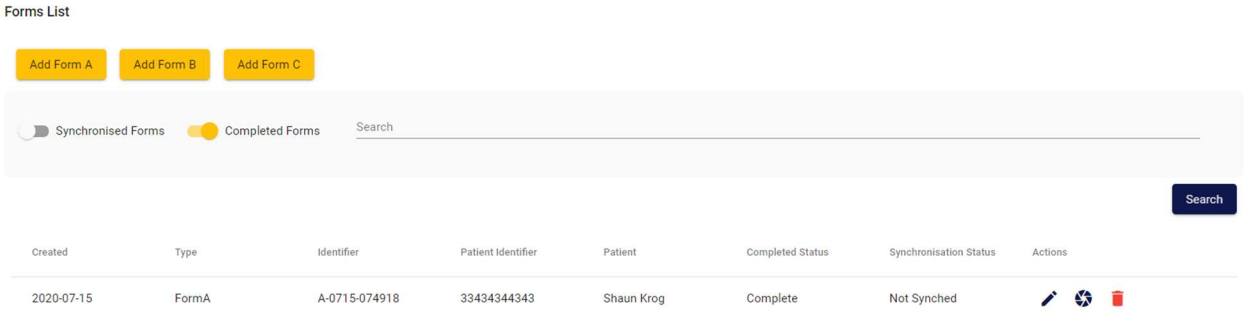
What does it mean for the form to be completed?
A form may not be synchronized to the server if you have not marked the form as complete. The action of marking a form as complete signifies to PViMS that you are happy with the status of the form and it is ready to be uploaded.

What is the unique identifier used for?
On completion of the form, you will be presented with a unique identifier that can be written onto the paper-based form. This provides a measure of quality control to ensure the correct paper-based form is allocated to the electronic record.

Step 7: Searching for an existing form

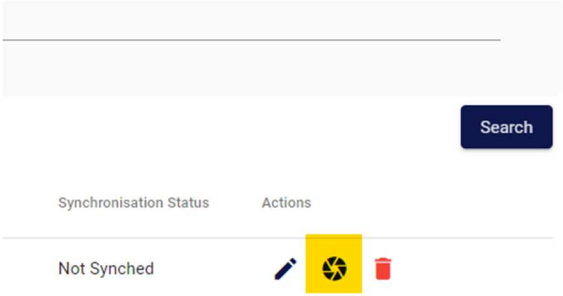
To search for a form that has been captured, you may use the form list filter criteria to search for the form.

1. Specify if you would like to search for synchronized or completed forms
2. Specify the patients name, record or identity number in search field. You are able to specify partial search criteria where you only enter a few letters of the search term you are searching for.
3. Click search to search for existing forms



Step 8: Taking a photo and attaching it to the form

Once you have searched for a form and the form is listed in the forms table, you will be presented with a camera icon that allows you take a photo for the form you have pressed on the icon for.



1. Once you have pressed on the camera icon, a modal form will appear.
2. Save the photo once you have placed the paper—based form in view of the camera.
3. Note that the camera icon has now been replaced with a picture icon, which allows you to view the newly taken picture.

What is the camera icon and how does it work?
Once you press on the camera icon, a modal form will pop up and present you with a visual stream from your device's camera. Click save to save the photo as an attachment to the form.

What is the picture icon and how does it work?
Once you have taken a picture, you will be presented with a picture icon which when pressed will pop up and present you with a picture of the form you had taken a photo of. It is possible to delete the picture in this modal form which will allow you take the photo again.

Step 9: Checking if PViMS is offline

Forms capture can take place on your device irrespective of the online status of PViMS. However, it is not possible to synchronise unless PViMS is marked as online. To determine the current status of PViMS, note the colour of the connectivity icon located in the primary toolbar of PViMS.

- A green icon indicates PViMS is online
- A red icon indicates PViMS is offline

Please note, it is not possible to synchronise your form if PViMS is offline.

Step 10: Synchronising the newly added form to the server

Once you have taken the photo, you are now able to synchronise the form to the server. To synchronise a form, look for the **Synchronise** menu and select this menu by pressing on it. Once you have pressed on the **Synchronise** menu, you will be presented with a page that allows you to synchronise completed forms with photos attached.

Please note, it is not possible to synchronise a form until the photo has been taken and attached to the form. This is to ensure any data quality issues can be verified using the source form.

1. Either select the all forms option by pressing on the checkbox next to the word all or by pressing the checkbox next to the form in the forms grid. **Please note, the system will only synchronise forms you have selected.**
2. Click on the synchronise button to synchronise the form.
3. Note any issues highlighted by the server and update the form as necessary or note that

the form has been synchronized successfully.

What is the synchronise button and how does it work?
Once you press on the synchronise icon, the system will automatically prepare the form and upload it to the server. The server will then attempt to process the form and add it as a clinical record to the PViMS database.